

OXIDATIVE DAMAGE IN NEURODEGENERATION: ROLES IN THE PATHOGENESIS AND PROGRESSION OF ALZHEIMER DISEASE

AUTHORS

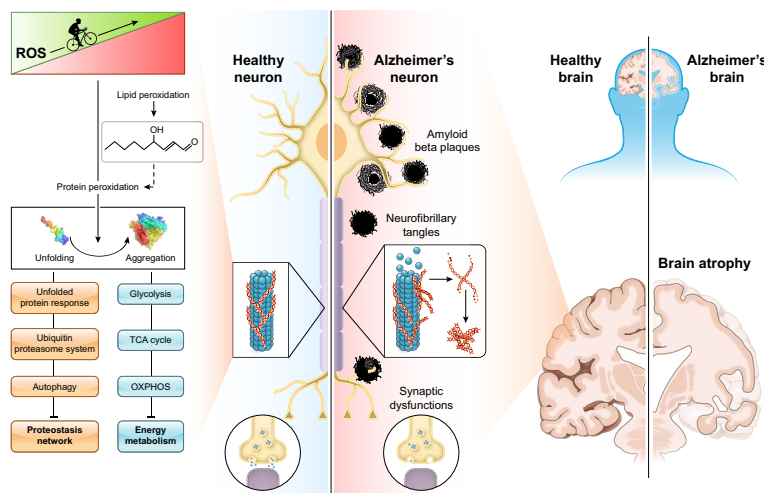
Marzia Perluigi, Fabio Di Domenico,
D. Allan Butterfield

CORRESPONDENCE

dabcns@uky.edu

KEY WORDS

A β and oxidative damage; Alzheimer disease and lipid peroxidation and protein oxidation; Alzheimer disease and translational approaches; Down syndrome and Alzheimer disease and oxidative stress; oxidative stress and energy dysmetabolism



CLINICAL HIGHLIGHTS

- Aging, characterized by the increased oxidation of proteins and lipids and by decreased antioxidant defense, is a common risk factor for the development of Alzheimer disease (AD).
- It is now clear that oxidative damage, including that associated with small A β 42 oligomers, by targeting many aspects of brain cell structure and function, is fundamental to the pathogenesis and progression of AD.
- Attempts to use antioxidants for AD treatment have been largely disappointing, in part because of the significant stage of AD at which patients have usually been involved in clinical trials.
- Clinical trials designed to take into account the inherent redox state of the patients involved (the lower the redox state, the less effective antioxidants will be), the stage of AD or mild cognitive impairment (MCI) involved, and specifically targeted antioxidants conceivably may permit a closer examination of the potential of antioxidant therapy to slow or perhaps retard the rate of cognitive decline in these disorders.
- Since AD is a multifactorial disease (see redox proteomics results that show many pathways affected because of secondary oxidative modification), the use of a single-molecule/single-target approach may result in limited efficacy.
- Multitargeted approaches and “healthy lifestyle,” including the combined administration of antioxidant compounds, antioxidant-rich dietary patterns, intellectual stimulation (formation of new synapses), and physical exercise, are strongly advised and currently under investigation in human studies.

OXIDATIVE DAMAGE IN NEURODEGENERATION: ROLES IN THE PATHOGENESIS AND PROGRESSION OF ALZHEIMER DISEASE

Marzia Perluigi,¹ Fabio Di Domenico,¹ and D. Allan Butterfield²

¹Department of Biochemical Sciences "A. Rossi Fanelli," Laboratory affiliated to Istituto Pasteur Italia-Fondazione Cenci Bolognietti, Sapienza University of Rome, Rome, Italy and ²Department of Chemistry and Sanders-Brown Center on Aging, University of Kentucky, Lexington, Kentucky, United States

Abstract

Alzheimer disease (AD) is associated with multiple etiologies and pathological mechanisms, among which oxidative stress (OS) appears as a major determinant. Intriguingly, OS arises in various pathways regulating brain functions, and it seems to link different hypotheses and mechanisms of AD neuropathology with high fidelity. The brain is particularly vulnerable to oxidative damage, mainly because of its unique lipid composition, resulting in an amplified cascade of redox reactions that target several cellular components/functions ultimately leading to neurodegeneration. The present review highlights the "OS hypothesis of AD," including amyloid beta-peptide-associated mechanisms, the role of lipid and protein oxidation unraveled by redox proteomics, and the antioxidant strategies that have been investigated to modulate the progression of AD. Collected studies from our groups and others have contributed to unraveling the close relationships between perturbation of redox homeostasis in the brain and AD neuropathology by elucidating redox-regulated events potentially involved in both the pathogenesis and progression of AD. However, the complexity of AD pathological mechanisms requires an in-depth understanding of several major intracellular pathways affecting redox homeostasis and relevant for brain functions. This understanding is crucial to developing pharmacological strategies targeting OS-mediated toxicity that may potentially contribute to slow AD progression as well as improve the quality of life of persons with this severe dementing disorder.

Aβ and oxidative damage; Alzheimer disease and lipid peroxidation and protein oxidation; Alzheimer disease and translational approaches; Down syndrome and Alzheimer disease and oxidative stress; oxidative stress and energy dysmetabolism

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1. INTRODUCTION

Alzheimer disease (AD) is the most common cause of dementia worldwide in individuals aged >65 yr. Neuropathological hallmarks of the disease include the abnormal deposition of amyloid β-peptide (Aβ) fibrils, the main component of senile plaques (SPs), and the intracellular accumulation of neurofibrillary tangles (NFTs), composed of hyperphosphorylated Tau protein (pTau). However, in addition to neurotoxicity associated with Aβ peptide and pTau, other key

determinants of the neurodegenerative process need to be included in the list of the main causative factors of AD. Among these, neuroinflammation, mitochondrial abnormalities, excitotoxicity, and oxidative stress (OS) are considered to play critical roles. OS can be considered as a bridge that connects the different hypotheses and mechanisms of AD because it enters in various pathways affecting brain homeostasis (1). Thus, several studies have contributed to the identification of redox-mediated mechanisms involved in the neurodegenerative processes of AD and other neurodegenerative disorders.

In particular, in the field of AD research, Butterfield's group dedicated decades of research to unraveling the close relationship between perturbation of redox homeostasis in the brain and AD neuropathology, by elucidating OS-mediated mechanisms potentially involved in both the pathogenesis and progression of AD. Published studies from the Butterfield laboratory and others showed that the brain of AD subjects presents a significant extent (accumulation) of oxidative damage associated with the abnormal marked accumulation of Aβ and the deposition of neurofibrillary tangles (2).

CLINICAL HIGHLIGHTS

- Aging, characterized by the increased oxidation of proteins and lipids and by decreased antioxidant defense, is a common risk factor for the development of Alzheimer disease (AD).
- It is now clear that oxidative damage, including that associated with small A β 42 oligomers, by targeting many aspects of brain cell structure and function, is fundamental to the pathogenesis and progression of AD.
- Attempts to use antioxidants for AD treatment have been largely disappointing, in part because of the significant stage of AD at which patients have usually been involved in clinical trials.
- Clinical trials designed to take into account the inherent redox state of the patients involved (the lower the redox state, the less effective antioxidants will be), the stage of AD or mild cognitive impairment (MCI) involved, and specifically targeted antioxidants conceivably may permit a closer examination of the potential of antioxidant therapy to slow or perhaps retard the rate of cognitive decline in these disorders.
- Since AD is a multifactorial disease (see redox proteomics results that show many pathways affected because of secondary oxidative modification), the use of a single-molecule/single-target approach may result in limited efficacy.
- Multitargeted approaches and “healthy lifestyle,” including the combined administration of antioxidant compounds, antioxidant-rich dietary patterns, intellectual stimulation (formation of new synapses), and physical exercise, are strongly advised and currently under investigation in human studies.

Because of its membrane composition, the brain is particularly vulnerable to free radical insults that easily attack polyunsaturated fatty acids (PUFA), initiating lipid peroxidation reactions (see below). However, in addition to lipids, the oxidation of proteins by free radicals or free radical-derived products is significant in AD, as the oxidative modifications of brain proteins can affect enzymes critical to neuronal and glial functions. Proteins are the major effectors of cellular functions; thus, any crucial alteration of protein structure/activity is fundamental to understanding how molecular events translate into clinical phenotypes.

To this aim, several experimental approaches have been utilized to identify which critical modifications of a protein affect its structure, stability, and function and how it may be possible to prevent/slow the rate of these modifications by modulating OS levels. It is well recognized that proteins are highly susceptible to oxidative damage that inevitably affects secondary and tertiary structure, resulting in irreversible modification of protein shape and, consequently, function. These changes include dissociation of subunits, unfolding, exposure of hydrophobic residues, aggregation, and backbone fragmentation among others. Proteins lose their ability to perform their physiological functions and, if not efficiently replaced, may trigger additional toxic events such as protein aggregation and deposition (3).

In addition, OS in the central nervous system (CNS) may seriously damage the brain via several interacting mechanisms, including an increase in intracellular free Ca²⁺, release of excitatory amino acids, and neurotoxicity (4).

Other important sources or modulators of OS include reactive nitrogen species (RNS), including nitric oxide (NO) and peroxynitrite, which can be extremely reactive with proteins, lipids, nucleic acids, and other molecules in further altering structure and/or functions, leading to detrimental effects for the brain (5). Cells with an accumulation of oxidized products such as unsaturated aldehydes and isoprostanes, protein carbonyls (PC), and base adducts from DNA oxidation can be seriously altered (6). Consequently, the considerable reactive oxygen species (ROS) formation increased by the electron transport system within the mitochondria under stressful conditions and in aging constitutes a risk for developing AD when no efficient antioxidant system is available. Thus, mitochondria function as both the source and target of toxic ROS since mitochondrial dysfunction and OS are important in aging and neurodegenerative diseases, including AD (7).

The present review traces the OS hypothesis of AD by encompassing the A β -peptide-associated OS mechanism, the role of protein oxidation unraveled by redox proteomics, and the antioxidant strategies that have been explored to prevent/slow the progression of AD. Furthermore, considering that AD involves multiple pathological mechanisms, this review aims to discuss the widespread effect of OS on several cellular processes that are essential to maintain neuronal homeostasis. Thus, understanding the cascade of events initiated by OS is crucial to potentially prevent the “amplified cascade” of oxidative damage that ultimately leads to the failure of proteins, lipids, and nucleic acids. Moreover, we describe recent findings obtained by the study of Down syndrome (DS) neuropathology, which represents a unique opportunity to analyze the combined scenario of a genetic form of both OS and AD. Indeed, chromosome 21 encodes several genes associated with A β production (*APP*) and perturbation of redox homeostasis (*SOD1*, *BACH1*, *CBS* among others). In DS individuals, Alzheimer-like neuropathology and dementia routinely develop at a defined age (40–50 yr of age), in marked contrast to AD, in which symptoms appear much later in life, though neuropathology often occurs up to 20 yr before development of symptoms. Comparison of persons with AD in DS (DSAD) versus AD in the normal population may offer the opportunity to identify the sequelae of pathological events that are central to neurodegeneration.

2. OXIDATIVE STRESS IN NEURODEGENERATION

2.1. Central Nervous System and Oxidative Stress: Membrane Lipid Peroxidation

OS and nitrosative stress (NS) generally refer to the consequences of excess oxygen- or nitrogen-containing

free radical production compared to the cell's ability to mount a sufficient antioxidant defense. Two major targets of OS and NS are proteins and lipids. As discussed in more depth below, protein oxidation is characterized by excess levels of protein posttranslational modifications (PTMs) (8). Arguably, the most neurotoxic marker of lipid peroxidation is the protein addition of bound 4-hydroxynonenal (HNE), which forms Michael adducts with Cys, His, and Lys residues and changes the conformation of such bound proteins in a way that causes loss of activity (FIGURE 1). Protein oxidation resulting from OS is often characterized by excess protein carbonyls, which occur on many different amino acids, and by elevated levels of 3-nitrotyrosine, which involves nitric oxide and superoxide radical reaction producing NO₂ addition to the 3-position of tyrosine residues as discussed below. Another involvement of NO is in modification of Cys residues following several reaction steps to

form R-S-NO (FIGURE 1). Protein modification by nitrosylation is discussed further in sect. 2.2.

As the level of reactive oxygen species (ROS) increases and increased oxidative modification of protein occurs, proteins tend to change their conformations, with consequent loss of function, and, furthermore, expose hydrophobic domains to water (8). As discussed below, such domains will spontaneously aggregate together with similar or different proteins, increasingly destabilizing the function of such oxidatively modified proteins and eventually leading to cell death. Neurons are particularly vulnerable to oxidative damage in the presence of OS and NS because of several factors involving, among others, the presence of large amounts of polyunsaturated fatty acids in their lipid bilayers, the high dependence on paramagnetic oxygen for neuronal survival, and the high metabolic rate of neurons and glia with attendant free radical leak from mitochondria (8).

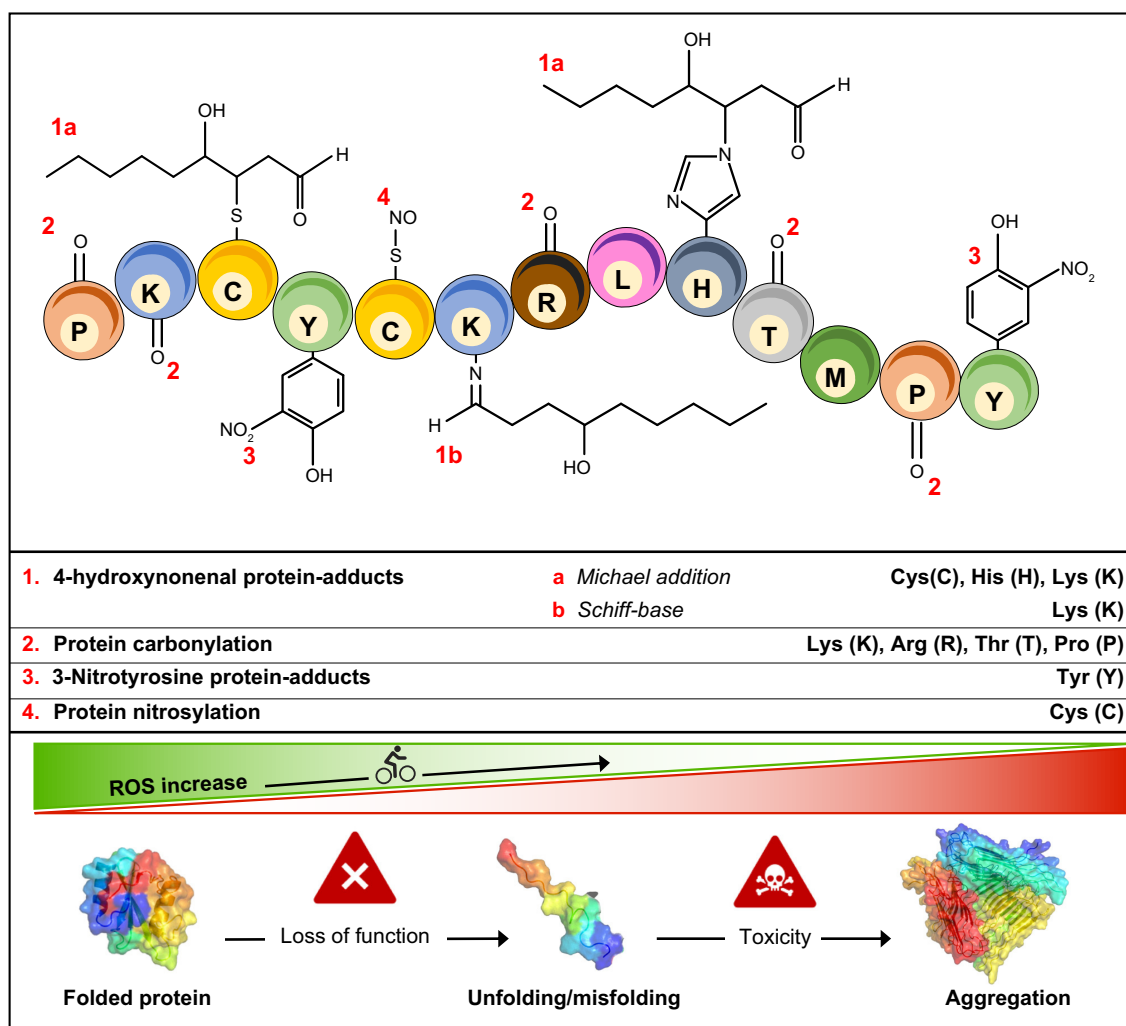


FIGURE 1. Different types of oxidative modification to proteins, based on the specific amino acid residue involved. For example, 4-hydroxynonenal can bind to Cys (C), His (H), Lys (K); protein carbonylation detected on K, Arg (R), Thr (T), Pro (P); 3-nitrotyrosine modification on Tyr (Y) and protein nitrosylation on C. Depending on the extent of these modifications, proteins undergo unfolding and eventually aggregation, with loss of their function, and ultimately accumulate in cells. ROS, reactive oxygen species.

This oxidative damage and consequent protein aggregation is significantly present in the brains of persons with AD, which contributes to the neuronal loss in this disorder (FIGURE 2).

2.1.1. Lipid peroxidation.

The brain is rich in unsaturated phospholipids and sphingolipids. FIGURE 3A shows the general structure of phospholipids and some of the major phospholipids that are named by their polar or (in some cases) charged head groups (9). The fatty acid acyl chains form esters with two of the OH functionalities on glycerol. In the carbon-1 position, fatty acids nearly always are fully saturated, i.e., contain no double bonds, whereas the acyl chains esterified to the carbon-2 position of glycerol nearly always are unsaturated. One can speculate that this degree of unsaturation was evolutionarily controlled to provide fluidity to neurons and glial cells. The fluid nature of the lipid bilayer increases as the distance from the ester bond of the fatty acid on carbon-2 of glycerol

increases, which would provide flexibility of functionally important transmembrane proteins to adopt various conformations to perform their respective molecular tasks (10). FIGURE 3B presents the names and structures of major unsaturated fatty acids in brain found to be esterified to carbon-2 of the glycerol backbone of phospholipids. Note that arachidonic acid, for example, is rich in allylic H-atoms, i.e., those H-atoms bonded to carbon atoms one carbon away from a C=C double bond. Because of the electronics associated with C=C double bonds, allylic H-atoms are inherently labile and therefore subject to abstraction as H-atoms (H[•]) by a bilayer-resident free radical, which satisfies the needed electron pair formation of the free radical and results in a carbon-centered free radical on the acyl chain from which the allylic H-atom was abstracted. This is the key initiation step in the mechanism of lipid peroxidation (FIGURE 4). Note that free radicals residing outside the bilayer are highly unlikely to be involved in mechanisms of lipid peroxidation, since they generally are so reactive that they would immediately react with a nearby substrate and

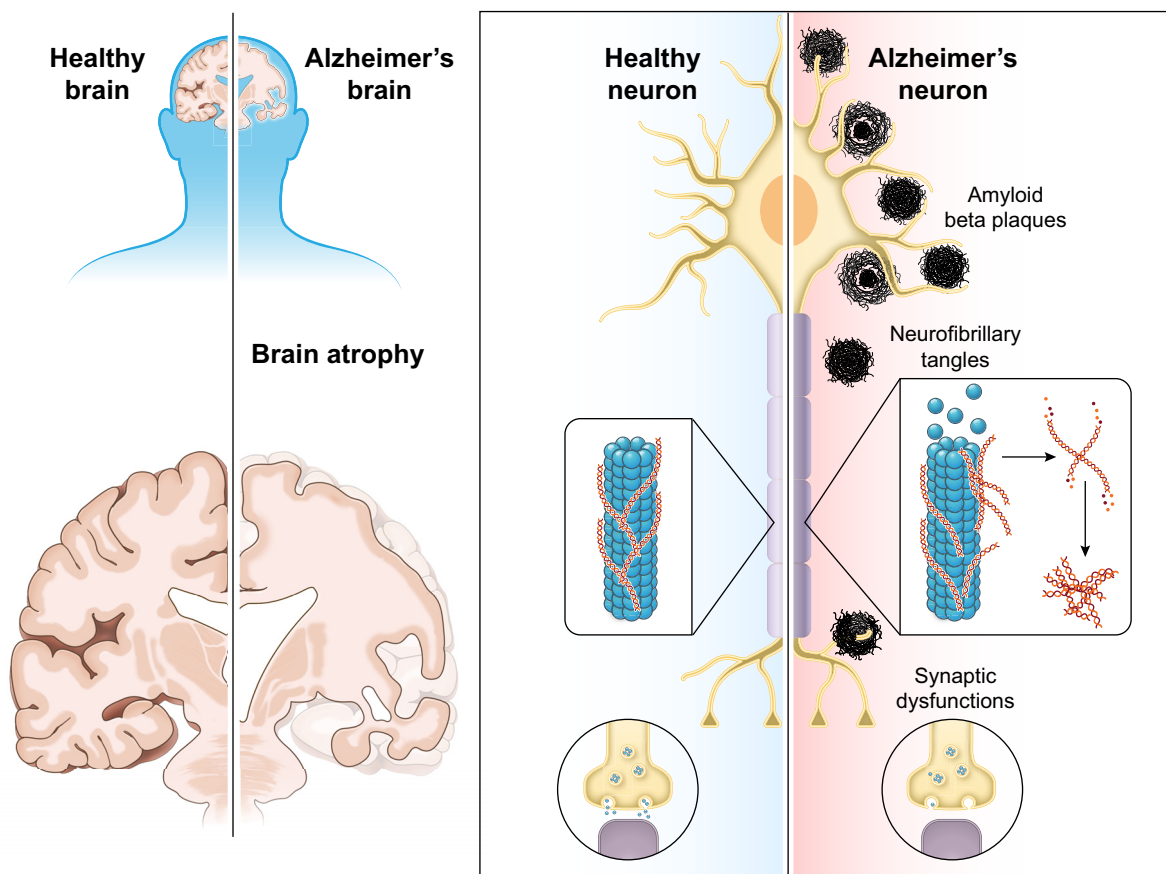


FIGURE 2. Neuropathological signatures of Alzheimer disease compared to control brain include deposition of senile plaques (SPs), composed mainly of amyloid beta-peptide fibrils and dystrophic neurites, and neurofibrillary tangles (NFTs), composed by hyperphosphorylated tau as paired helical filaments, in association with loss of synapses. The spread of tangles and plaques through the Alzheimer brain advances in a predictable pattern with marked atrophy, including widened sulci, shrinkage of gyri, and enlarged ventricles.

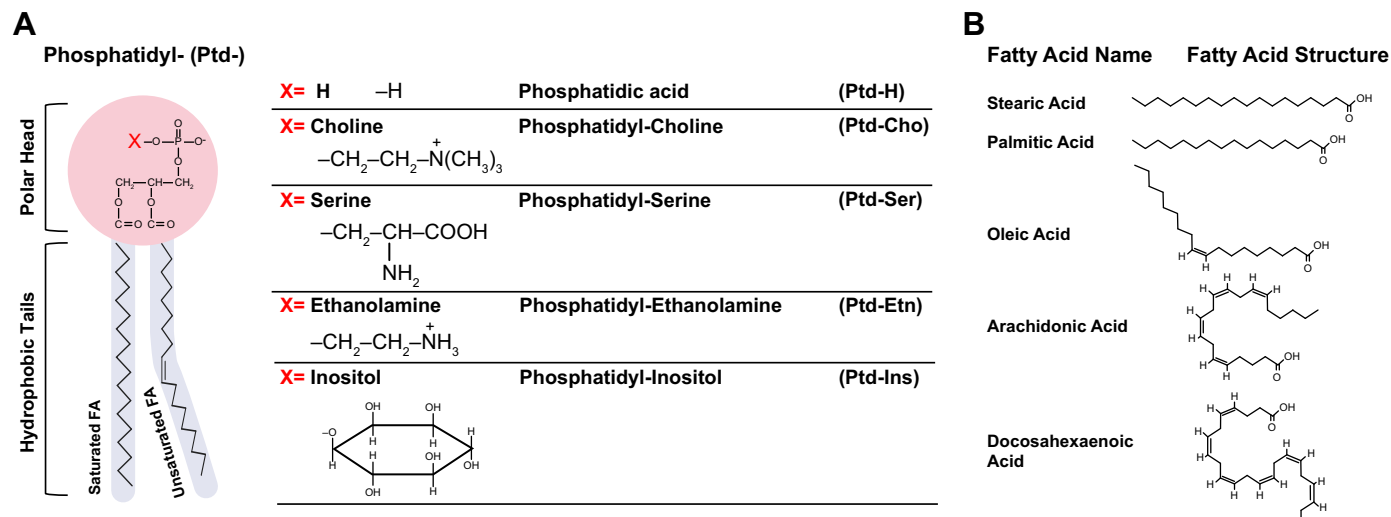


FIGURE 3. A: general structure of phospholipids and names of a select group of neuron-resident phospholipids. B: structures of fatty acids that comprise selected lipids in the brain. Note the unsaturated fatty acids and the significant amount of labile allylic H-atoms that can be abstracted by free radicals during processes of lipid peroxidation.

not be able to live long enough to penetrate the lipid bilayer and diffuse to within a van der Waals distance of labile allylic H-atoms on the acyl chains of the phospholipids.

As shown in **FIGURE 4**, the next step in the mechanism of lipid peroxidation is binding of molecular oxygen to the carbon-centered free radical produced in *step 1*. Why is molecular oxygen soluble in the lipid bilayer, and why does this reaction occur so rapidly? Molecular oxygen (O_2) is a homonuclear diatomic molecule and therefore has no dipole moment, i.e., it is nonpolar and therefore can easily solubilize in the low-dielectric medium of the hydrophobic acyl chains of phospholipids. Estimates of oxygen solubility in lipid bilayer reach millimolar levels (11). Molecular orbital considerations of molecular oxygen show that this nonpolar molecule has

two unpaired electrons and is therefore paramagnetic. This electronic nature of molecular oxygen explains the rapid formation of lipid peroxyl free radical ($\text{LOO}^\bullet$) from reaction of molecular oxygen with the C-centered free radical on the unsaturated acyl chain of phospholipids: radical-radical recombination reactions are among the fastest reactions known. In *step 3* of the mechanistic process of lipid peroxidation, the lipid peroxyl free radical abstracts another labile allylic H-atom from the acyl chain resident on carbon-2 of glycerol backbone of a phospholipid to form a lipid hydroperoxide, LOOH , and produces another carbon-centered free radical on this acyl chain, which immediately will be attacked by another molecule of oxygen, again producing a lipid peroxyl free radical, $\text{LOO}^\bullet$, i.e., a chain reaction. *Steps 2 and 3* are propagating steps of the chain reaction that will continue

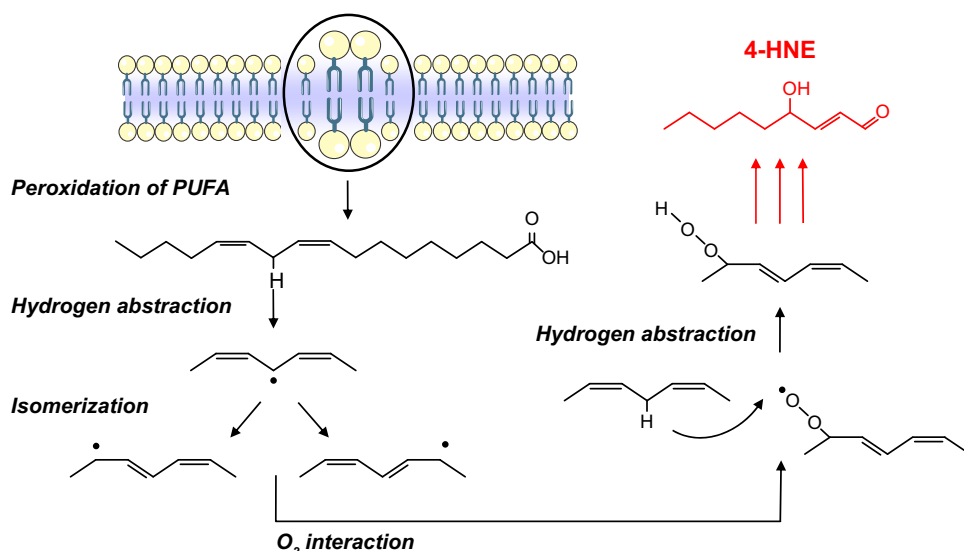


FIGURE 4. Lipid peroxidation mechanism leading to formation of neurotoxic 4-hydroxynonenal (HNE). See text for description of steps involved. PUFA, polyunsaturated fatty acids.

as long as there are allylic H-atoms and molecular oxygen present. This is an important component of chain reactions, since only a small amount of initiating free radical is necessary in the lipid bilayer to greatly amplify the amount of lipid hydroperoxide formed. This point is critical to understanding the neurodegeneration associated with lipid peroxidation. The reason is that the lipid hydroperoxide breaks down into several products, one of the most neurotoxic of which is the molecule 4-hydroxy-2-trans-nonenal (HNE), a 9-carbon alkenal (a molecule with an aldehyde and a C=C double bond). HNE has an aldehyde on carbon-1, an adjacent C=C carbon-carbon double bond between carbon-2 and carbon-3, and a hydroxyl group bound to carbon-4. Therefore, the more LOOH formed by the mechanisms of lipid peroxidation, the more highly reactive and neurotoxic HNE is produced.

The high reactivity of HNE results from the diminished electron density around carbon-3 of HNE as a consequence of two factors: 1) the resonance between the C=C of carbons-2 and -3 and the C=O bond of the aldehyde functional group on carbon-1 and 2) the inductive effect of the more electronegative O atom on the OH group of carbon-4 with the electrons around carbon-3. This dearth of electron density makes this carbon-3 highly susceptible to nucleophilic reactions by electron-rich atoms on proteins, i.e., the S-atom of Cys, the ϵ -amino N-atom of lysine, and the more labile H-atom on His. The reaction is known as Michael addition and is shown for Cys in **FIGURE 5**. The Michael adduct of HNE to Cys on proteins causes considerable changes in conformations of the proteins (12). Moreover, HNE is diffusible to other parts of the cell, such as mitochondria (8), and can even be carried from one brain region to another (13). If the Cys to which HNE is bonded is located in a more hydrophobic region of the protein, the significant dipole moment of the aldehyde functionality of HNE will be a driving force to

move that Cys closer to the surface of the protein, thereby exposing hydrophobic regions of the protein to the aqueous phase. To increase the entropy of water following exposure of HNE-bound hydrophobic regions of the proteins (14), other exposed hydrophobic regions of the same or different proteins will interact, i.e., this leads to protein aggregation of similar or different HNE-bound proteins (15).

The consequences of HNE binding to proteins in neurons and glia are to change their conformation and lead to significant diminution if not loss of functions (16–21). As discussed in sect. 3, redox proteomics analyses have identified a significant number of HNE-modified brain proteins in AD and mild cognitive impairment (MCI) brains, all of which had diminished function (22–29). The resultant neurotoxicity that is discussed below contributes significantly to neuronal loss and consequent cognitive decline in brains of individuals with AD and MCI.

Other indexes of lipid peroxidation, such as F₂-isoprostanes, are easier to detect by mass spectrometry (30–32) and result from peroxidation of arachidonic acid [20:4], which, as mentioned above, is rich in brain.

2.2. Protein Oxidation

As noted, protein oxidation is a posttranslational modification of proteins that often results in functional diminution (8). This diminished function is a consequence of altered protein conformation based on a similar principle as that noted for HNE-bound proteins discussed above. Namely, a major index of protein oxidation is the introduction of carbonyl groups to the protein, and the dipole moment of the C=O bond leads to rearrangement of the protein conformation to allow the polarity of the carbonyl bonds to be close to the surface of the protein and in contact with the cell's various aqueous phases.

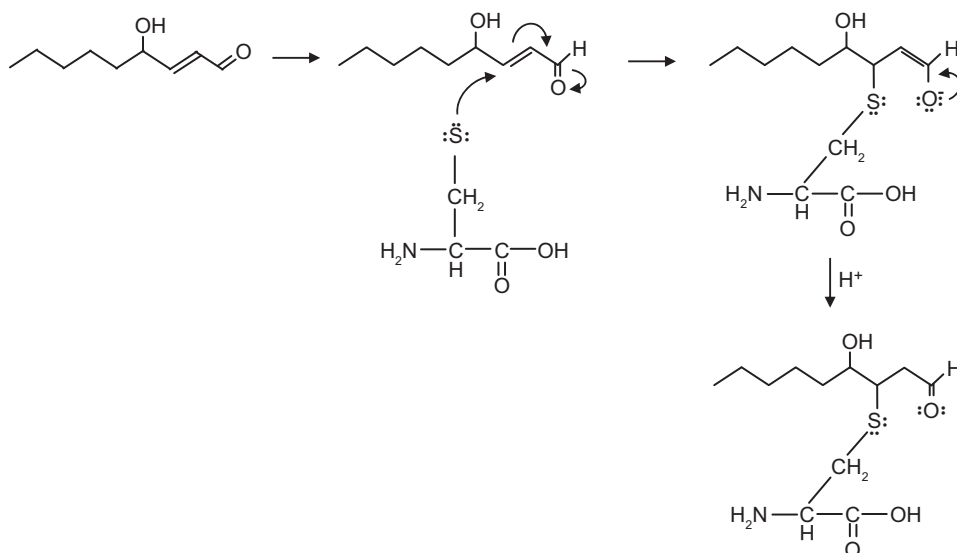


FIGURE 5. Michael addition reaction. The lipid peroxidation product 4-hydroxy-nonenal (HNE) covalently binds Cys, His, or Lys amino acids in proteins via the Michael addition reaction. Carbon-3 of HNE is partially positive because of resonance of the double bond with carbon-2 and carbon-3 with carbon-1 and because of the inductive effect of the OH moiety on carbon-4 of HNE for electrons surrounding carbon-3. A nucleophilic reaction by electrons on the S-atom of Cys, for example, will covalently bind to partially positive carbon-3, leading to a covalent adduct on the protein, which alters its structure and function.

Aggregation of proteins results from interaction of exposed hydrophobic domains of proteins as discussed above.

Protein carbonyls arise upon free radical attack on the protein by four processes (8): 1) cleavage of the primary amino acid chain with resultant transient free radicals on both sides of the cleavage point that immediately binds molecular oxygen via radical-radical recombination reactions like those noted above for lipid peroxidation; 2) oxidation of amino acid side chains, one example of which is 4-oxo-histidine; 3) binding of reactive alkenals such as HNE to proteins, which brings aldehydic carbonyl moieties to the protein; and 4) following Amadori chemistry, which begins with lysine ϵ -amino groups reacting with the carbonyls of reduced sugars to form the Schiff base (a.k.a. imine), C=N. Subsequently, less well-defined reactions occur that lead to introduction of carbonyls into the protein. Protein carbonyls are often detected by reaction with 2,4-dinitrophenylhydrazine to form the hydrazone and the protein-bound hydrazone detected by immunochemistry (8).

Protein carbonyls are elevated in brains that have been exposed to free radical insults resulting from ionizing radiation that splits water into reactive OH radicals, metabolism of certain chemicals/pharmaceuticals, e.g., rotenone, that leads to inhibition of complex I of the mitochondrial electron transport chain, aging, and certain neurodegenerative disorders such as Parkinson disease, Huntington disease, amyotrophic lateral sclerosis, and AD (among other disorders) (33–37). Elevated protein carbonyls in AD and MCI are discussed in more detail in sect. 3.

Another index of formal oxidation in proteins is the introduction of NO₂, a free radical, to the 3-position of Tyr to form 3-nitrotyrosine (3-NT). (For the nonexpert: counting the number of valence electrons in the NO₂ molecule results in 5 for N and 6 each for the two O atoms, totaling 17, an odd number, meaning that there is 1 electron remaining after accounting for all 2-electron chemical bonds, i.e., a free radical.) **FIGURE 6** shows the mechanisms for formation of 3-nitrotyrosine, a biomarker of OS or NS. Basically, the superoxide free radical (for example from leak from complex I in mitochondria and biochemically by reactions involving xanthine/xanthine oxidase or NADPH oxidase) and nitric oxide [NO, formed by catalytic action of nitric oxide synthase (NOS) on arginine] react in a radical-radical recombination to form peroxynitrite, a non-radical (38, 39). In the presence of carbon dioxide, nitroso-peroxycarbonate is formed. This intermediate undergoes rearrangement to form nitrocarbonate, on which homolysis occurs to form the free radical NO₂. A free radical removes the H-atom of the phenolic OH group in the 4-position of the aromatic ring of Tyr. The resulting single electron on the O atom delocalizes within the aromatic

ring, directed to the 3-position since the OH moiety on a 6-member aromatic ring is an ortho- and para-directing substituent. Only the ortho 3-position is an option since the para carbon is already bonded within the Tyr amino acid chemical structure. Consequently, the free radical on the 3-position of Tyr is attacked by the free radical on NO₂ by radical-radical recombination (16, 40).

Since the OH group on Tyr is the phosphorylation target catalyzed by receptor tyrosine kinases (RTKs), having the steric hinderance of NO₂ on the 3-position of Tyr greatly hinders the effective phosphorylation of the 4-OH group on Tyr. This steric hinderance means that important RTK-mediated phosphorylation of Tyr is hindered, with the consequence that neuronal survival could be affected (5, 40). Further discussion of 3-NT in AD and MCI brains is found in sect. 3.

2.3. Thiol Oxidation

Normal physiological functions and cellular metabolism are finely regulated by the cellular redox status, which is continuously exposed to a dynamic fluctuation (41). In most eukaryotic cells, protein cysteine thiols (Cys) are predominantly found in functionally or structurally crucial regions, where they act as stabilizing, catalytic, metal-binding, and/or redox-regulatory entities. Cys are the elements most sensitive to fluctuations of the cellular redox environment, undergoing different types of chemical modifications in response to such variations. In the presence of increased oxidant levels, as occurs, for example, in response to growth factors, cytokines, and thiol-disulfide exchange reactions, specific proteins could act as redox switches that regulate the conformation and activity of different proteins. These reversible posttranslational modifications (PTMs) enable redox-sensitive dynamic changes in cell signaling and function to maintain redox homeostasis in various cellular compartments, protect organisms from oxidative insults, and actively participate in redox-regulatory and signaling processes (41).

Reactive thiols of redox-regulated proteins usually have distinctive chemical properties that define reactivity toward oxidants and reversibility. Such proteins can be modified in various ways, and reactivity is greatly enhanced for Cys whose thiol side chain is in the thiolate form, i.e., deprotonated at physiological pH (S[−]): sulfenylation, nitrosylation, glutathionylation, persulfidation, and disulfide formation, responding to different oxidants (**FIGURE 1**). This diversity of modification, both reversible and irreversible, is unique to Cys residues. Because of their high nucleophilicity, thiol residues in peptides and proteins are extremely susceptible to direct oxidation by ROS that causes alterations in protein structure and function (42). For example, progressive H₂O₂-induced

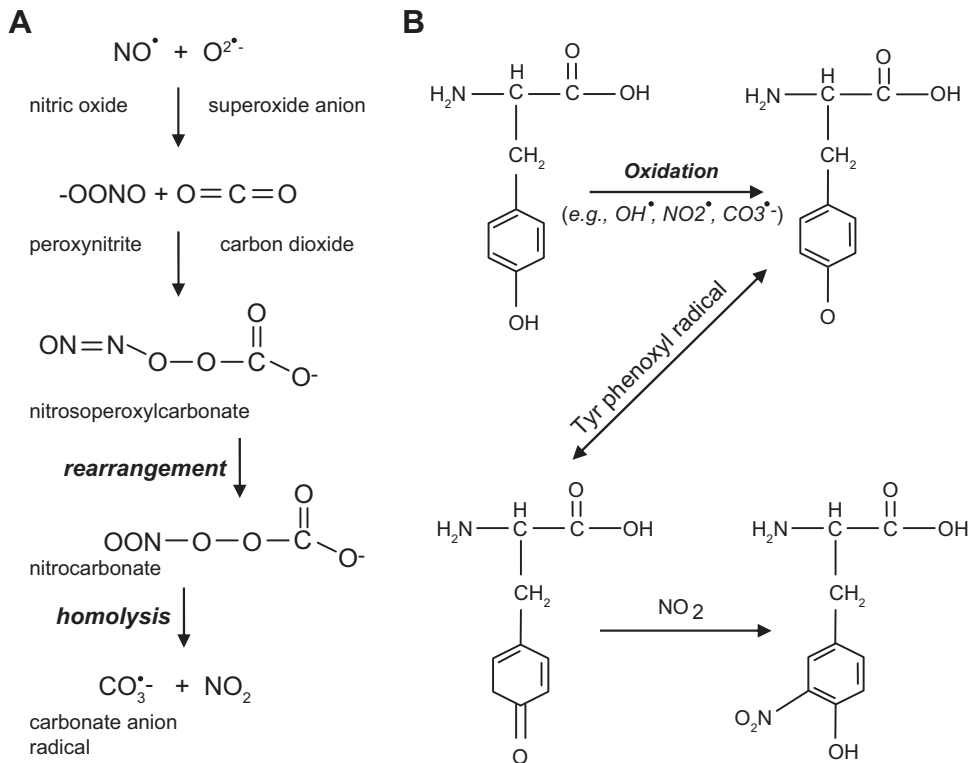


FIGURE 6. 3-Nitrotyrosine formation reactions. **A:** formation of nitrogen dioxide, a free radical, from reaction of superoxide free radical and nitric oxide. **B:** formation of 3-nitrotyrosine from NO₂ and a radical on the 3-position of Tyr. See text for description of steps involved.

Cys oxidation leads to Cys sulfenylation (SOH), sulfinylation (SO₂H), and sulfonylation (SO₃H). Among these, oxidation to SO₃H is considered irreversible. S-sulfhydration (also called persulfidation) can occur after reactions between derivatives of hydrogen sulfide (H₂S) and thiols (43).

Reactive nitrogen species (RNS), including nitric oxide (NO), react with some Cys, causing S-nitrosylation (44, 45). Cys can also undergo lipid modifications such as palmitoylation and prenylation or bind metal ions (Zn²⁺, Fe²⁺, and Cu²⁺). This latter property is crucial for formation of zinc fingers and iron-sulfur clusters. Because of their nucleophilicity, thiolate groups also participate in nonredox reactions as in the catalytic groups of cysteine-proteases and ubiquitin ligases (42).

Consequently, cells spend considerable effort to prevent nonspecific thiol oxidation while allowing the formation of protein disulfide bonds or other oxidative thiol modifications that play physiological roles. The formation of structural disulfide bonds in proteins, for instance, is a specialized process catalyzed by oxidoreductases and thiol oxidases and occurs mostly in the endoplasmic reticulum (ER) and the mitochondrial intermembrane space (IMS) (46).

In most aerobic cells, the tripeptide γ-Glu-Cys-Gly, known as glutathione (GSH), is the major nonprotein thiol that undergoes oxidation to its corresponding disulfide form (GSSG), and GSH is highly abundant in cells, with its concentration ranging from 0.1 to 10 mM. GSH acts to maintain protein thiols in a reduced state, thereby regulating the overall reduced state of the cell and

supporting a variety of redox reactions including ROS scavenging and xenobiotic detoxification and facilitating cell signaling (41). In addition to the high reduced state of the cytosol, with high GSH levels different compartments and organelles within the cell are at different redox potentials and are not at equilibrium with each other. In contrast to the cytosol, the ER maintains a more oxidized redox state (47) to assist protein folding and maintenance of disulfide bridges in secreted proteins in which the latter confer structural rigidity and prevent oxidation in the more oxidizing extracellular environment. Similarly, the mitochondria maintain their own distinct intracellular redox state (48).

It is not surprising that the human genome contains >20,000 cysteine residues that are distributed on protein surfaces or within globular domains (49). Some of these Cys residues can be oxidized to disulfides even under reduced and nonoxidizing conditions. As a result, the formation and breakdown of such disulfide bonds within globular domains lead to conformational changes and regulation of the protein as well as its function.

Considering that these chemical modifications are responsible for specific changes of function and signaling, the formation of disulfide bonds in the cytosol is tightly regulated, and uncontrolled thiol oxidation is efficiently reversed by the thioredoxin-thioredoxin reductase system (Trx/Trx reductase). Protein disulfides are also reduced by glutaredoxin (Grx), requiring GSH as a cofactor. However, both the Trx and Grx systems are not general mechanisms for maintaining protein thiols in the

reduced state but preserve reduced thiols in specific proteins such as protein kinases, transcriptional regulators, and ribonucleotide reductase. Trx in particular has been shown to regulate several signal transduction pathways involving ASK-1, NF- κ B, PTEN, and AP-1 and the NLRP-3 inflammasome (41), suggesting that the switch between thiol/disulfide is a critical component of intracellular signaling machinery that turns protein targets on/off to activate or deactivate specific signaling pathways. Such mechanisms are important in regulating cell growth, proliferation, and death. Moreover, such reactions include reduction of hydrogen peroxide or lipid hydroperoxides by glutathione peroxidases, detoxification of lipid peroxidation products by glutathione S-transferases, reduction of dehydroascorbate, and glutaredoxin-mediated removal of protein disulfides. In addition, glutathione can directly prevent the oxidation of protein thiols by thiol-disulfide exchange and by forming mixed disulfides. During the formation of mixed disulfides, the binding of GSH to protein thiols results in the addition of a glutathione molecule to the protein, leading to the formation of glutathionylated proteins.

2.4. Nucleic Acid Oxidation

2.4.1. DNA oxidation.

As we have discussed for protein and lipids, ROS also are extremely dangerous for the integrity of nucleic acids. Oxidative modifications of DNA result in base alterations, formation of single- and double-strand breaks, sister chromatid exchanges and translocation in replicating cells, as well as formation of DNA-DNA and DNA-protein cross-linking (50). All four bases (purines and pyrimidines), and their respective deoxynucleosides, undergo oxidative modifications (>20 oxidized bases can be generated); however, guanine has the lowest oxidation potential among bases and is the most susceptible to ROS attack, especially to the highly reactive OH radical (51). Oxidative damage on guanosine (C8) leads to the formation of 8-OHdG and its deoxynucleoside equivalent, 8OHdG. In addition, under conditions of reduced oxygen tension guanosine can be modified to 2,6-diamino-4-hydroxy-5-formamidopyrimidine (FapydG), which together with 8-OHdG are the most abundantly formed oxidized forms of DNA bases. Because of normal oxygen tension, 8-OHdG is considered the standard marker of DNA oxidation that can be detected by a variety of analytical techniques, based on the type of samples to be investigated (52). For example, immunoassays are the selected tools for the analysis of urine, comet assays for the analysis of blood cells, and high-pressure liquid chromatography (HPLC) coupled with electrochemical detection (ECD) or mass spectrometry (MS) or gas chromatography (GC) coupled

with MS for the analysis of tissue specimens (53). In addition to oxidative base modifications, ROS attack on DNA can result in the formation of DNA strand breaks (DSBs). Increased levels of DSBs were detected in AD brain neurons as well as in mice transgenic for human APP, which mimics several key aspects of AD neuropathology (54).

Formation of modified DNA bases induces mutagenic properties that could result in binding of transcription factors, alterations in replication of DNA, or appropriate base pairing producing mutations that could lead to altered protein synthesis (55). For example, it has been suggested that 8-OHdG can lead to damage of promoter elements, thus affecting the binding of transcription factors including methyl-CpG binding protein 2 (MeCP2). The authors showed that the presence of 8-OHdG in methyl-CpG sequences reduces the binding of the methyl-CpG binding domain (MBD) of MeCP2 (56). If not repaired, these oxidative lesions to DNA would interrupt the sequence of events involved in the propagation of epigenetic signals, thus leading to dysregulation of transcriptional activity.

Cellular defenses are activated in response to insults to oxidized DNA and include four major repair mechanisms: 1) nucleotide excision repair (NER) and 2) base excision repair (BER), both of which excise bulky helix-distorting DNA lesions; 3) mismatch repair (MMR), to correct mismatches of the normal bases; and 4) double-strand break repair (DSBR), to intervene when both strands of the DNA backbone break (whereas the breakage of only 1 strand is handled by the BER pathway) (57). Among all these repair mechanisms, BER is the first line of defense for repairing oxidative DNA damage, involving substrate-specific DNA glycosylase enzymes to remove the damaged base. These enzymes hydrolyze the N-glycosidic bond between the modified base and the sugar to release the base and generate an abasic site DNA phosphodiester backbone, and this site is, in a second step, cleaved by a lyase or endonuclease (58).

2.4.2. RNA oxidation (coding and noncoding RNAs).

Because of its single-stranded state, RNA is likely more vulnerable to oxidative modifications than DNA (59). However, considering that nucleic acids have similar structures, the same will be for oxidized form of both nucleotides, especially for guanosine. Thus, 8OHdG is, similarly to DNA, the most studied oxidized RNA marker. RNA is mainly present in cells as rRNAs and tRNAs, in addition to different types of noncoding RNAs (ncRNAs), including microRNAs (miRNAs), small nucleolar RNAs (snoRNAs), and small nuclear RNAs (snRNAs) (60). Increasingly, studies have demonstrated well that noncoding RNAs play a crucial role in mRNA splicing

regulation, nonspliced RNA modification, and mRNA translation. There are many forms of oxidative damage to RNA: 1) direct RNA strand breaks (61), as a result of the production of nucleic acid bases or their peroxy groups, with the abstraction of hydrogen atoms from adjacent ribose molecules, which in turn results in the breakage of the RNA strand, and 2) translation errors, caused by oxidized mRNA, that may lead to premature cessation of translation or degradation of peptide chains, resulting in short-chain polypeptides and protein changes (62). In addition, oxidative changes of bases on mRNAs cause mismatches with the bases on tRNAs during translation and can result in protein variation. Among different forms, mRNA is more likely to be oxidized; if this happens during transcription, the presence of oxidized mRNA chain will cause transcription errors, coupled with subsequent translational errors. Accordingly, Keller and coworkers (63) demonstrated that once RNA is oxidized, protein synthesis significantly decreases over time in primary and the decrease is directly correlated with the oxidation time.

Defense mechanisms to repair oxidatively damaged RNA under normal physiological conditions remain largely unknown (60). Among putative scenarios the first event would be premature termination of the translational processes at the oxidized site, thus producing short polypeptides. Another event is the production of mutated proteins in response to disturbance of base-pairing capacity with tRNA by oxidized mRNA. Alternatively, oxidized RNA may be recognized as mutated forms, thereby activating surveillance mechanisms that lead to its degradation. Finally, oxidized bases may be repaired enzymatically; none of these enzymes has yet been identified in cells. However, considering that mRNA oxidation occurs with as much frequency as methylation, repair enzyme for oxidized mRNA might be identified in future studies. Collected studies have demonstrated that RNA oxidation, due to the altered profile of oxidized mRNA or impaired function of some ncRNA, such as tRNA and rRNA, causes the impairment of transcriptional/translational integrity. The accumulation of 8-OHdG is associated with aberrant mRNA decoding ability, resulting in protein misfolding and short polypeptide formation. Furthermore, relatively recent genetic studies showed an expanding landscape of RNA beyond its traditional function, i.e., the crucial role for the transfer of genetic information from DNA to proteins (64). Oxidative damage to ncRNAs, coupled with dysregulation of gene expression, might contribute to degenerative phenomena affecting a variety of brain functions. Of particular interest in geroscience, some abundant cellular ncRNAs, such as microRNAs (miRNAs), might play understudied functions that are central in controlling signaling dynamics. Accumulating evidence revealed that miRNAs, acting as posttranscriptional regulators of gene expression, are critical for neuronal functions both during development

and in the adult brain (65). miRNAs are small, noncoding RNA molecules that are 18–25 nucleotides in length known to regulate several cellular processes such as cell growth and proliferation, embryonic development, tissue differentiation, and cell death. miRNAs can be directly oxidized and this, in turn, causes aberrant recognition of specific targets (65). Besides direct oxidation of miRNAs, evidence suggests that oxidative stress affects expression levels of multiple miRNAs; conversely, miRNAs target several genes involved in redox homeostasis, including antioxidant enzymes (66). Furthermore, a number of miRNAs involved in the regulation of oxidative stress are also related to degenerative pathways in many age-associated brain disorders (67).

For example, age- and disease-related downregulation of the miRNA biogenesis pathway in adult neurons reportedly can lead to changes in their survival, functions, and connectivity (68). Interestingly, the dysregulation of many miRNAs in neurons is related to increased ROS production by mitochondria (69). Mitochondrial miRNAs also were demonstrated to modulate the function of antioxidant enzymes that contribute to the maintenance of redox homeostasis (70). For example, miR-98, miR-204, miR-34a, miR-15b, miR-375, miR-140, and miR-335 are involved in the cross talk between mitochondria and oxidative stress (69). Changes in the location and distribution of mitochondrial miRs affect mitochondrial dynamics, biogenesis, and mitophagy (70). Oxidative stress can modulate miRNAs both positively and negatively, and, conversely, several miRNAs can affect cellular responses to oxidative stress. Therefore, oxidative stress and microRNA regulatory networks may significantly impact neuronal processes leading to neurodegeneration, including mitochondrial defects, loss of proteostasis, and increased neuroinflammation among others. Modulating the levels of a relatively small number of microRNAs may conceivably greatly alleviate pathological oxidative damage and have neuroprotective activity (71).

2.5. Antioxidant Systems in the Brain

OS indicates a condition where prooxidant species overwhelm cellular antioxidant defense systems by an increase of ROS production and/or by a decrease in the antioxidant response. Therefore, brain cells are endowed with a selection of endogenous antioxidant systems, including antioxidant enzymes, antioxidant proteins, and antioxidant molecules, to protect these cells from increased ROS and to maintain their physiological redox balance. Aging, characterized by the increased oxidation of proteins and lipids and by decreased antioxidant defense, is a common risk factor for the development of AD, which is characterized by severe redox imbalance and by robust increased OS. The potentiation of antioxidant systems

has gained considerable importance as a feasible therapeutic strategy for the prevention of AD-related cognitive decline (72, 73).

2.5.1. Antioxidant enzymes.

Antioxidant enzymes catalyze the breakdown of free radicals and ROS, usually in the intracellular environment. One of the most abundant antioxidant enzymes is superoxide dismutase (SOD), which catalyzes the dismutation of O_2^- to H_2O_2 and molecular oxygen. Three types of SOD isoenzymes have been identified in mammals, and they are characterized by their prosthetic metal ion and cellular localization. The constitutively expressed copper- and zinc-containing SOD (Cu/Zn-SOD or SOD1) is a dimer that exists intracellularly in the cytoplasm but also in the nucleus and in peroxisomes. The inducible manganese-containing SOD (Mn-SOD or SOD2) is a tetrameric isoenzyme located in mitochondria. A second isoform of Cu/Zn-SOD (EC-SOD or SOD3) is located mainly in the extracellular space. Paradoxically, the activity of SOD leads to the generation of a new harmful oxidant ROS, hydrogen peroxide, which needs to be detoxified. Organisms contain two other enzymes, glutathione peroxidase (GPx), of which there are several isoforms, and catalase (CAT), which are involved in the decomposition of hydrogen peroxide to oxygen and water. Therefore, to avoid the accumulation of hydrogen peroxide, which in the presence of Fe(II) or Cu(I) leads to hydroxyl radical formation that damages membrane lipids, proteins, and nucleic acids, it is very important to maintain the right ratio between SOD and GPx + CAT activities. In brain, SOD1 and SOD2 are differentially distributed in the neuronal and glial compartments. SOD1 is located primarily in astrocytes throughout the CNS, but the enzyme is also detectable in neurons. SOD2, on the other hand, is localized predominantly in neurons throughout the brain and spinal cord, whereas in astroglial cells MnSOD is observed at a lower level than in neurons (74). Microglia are highly resistant to oxidative challenges because they possess high amounts of GPx and can detoxify hydrogen peroxide (75). Astrocytes also are able to upregulate GPx expression, whereas neurons lack this ability.

2.5.2. Glutathione.

Glutathione (GSH) is the most abundant endogenous nonenzymatic antioxidant and oxyradical scavenger, critical to maintaining a reducing environment in the cell and protecting against oxidative damage by ROS (76). GSH maintains the cellular redox balance depending upon the pH of the cellular compartment and is involved in various biosynthetic processes as well. GSH is present

in high concentration throughout the brain, with higher expression in glial cells of the cortex (77). The consecutive actions of two cytosolic enzymes, γ -glutamyl cysteinyl ligase (GCL) and GSH synthase (GCS), catalyze the synthesis of GSH. The first step involves ligation of γ -glutamate to cysteine to form γ -glutamylcysteine, a reaction catalyzed by GCL. This step is the rate-limiting step in GSH production, and GCL is feedback inhibited by GSH itself (78). GSH, in conjugation with glutathione reductase (GR), GPx, glutathione S-transferase (GST), and NADPH provides protection against various toxic electrophiles and hydrogen peroxide. GSH is oxidized to GSSG by the GPx-specific isoform during detoxification of H_2O_2 or organic hydroperoxide, and GSSG can also be converted back to GSH by GR, with NADPH serving as electron donor. The basal and inducible levels of GCL, GCS, and GR are regulated by the nuclear transcription factor erythroid 2-related factor 2-antioxidant response element (Nrf2-ARE) pathway (79). GST participates in the detoxification of xenobiotic compounds by catalyzing their conjugation with GSH to form nontoxic products that are often eliminated by cells through the multidrug resistance protein MRP1 (73).

2.5.3. Nrf2 signaling.

The transcription factor Nrf2 is a master regulator of antioxidant defenses and drug-metabolizing enzymes, controlling genes that contain an antioxidant response element (ARE) sequence. These genes code for a group of cytoprotective and antioxidant proteins that include multiple components of the thioredoxin pathway, antioxidant enzymes, i.e., NADPH quinone oxidoreductase 1 (NQO1) and heme oxygenase 1 (HO-1), and numerous enzymes that regulate glutathione biosynthesis and utilization (80). Under normal conditions, Keap1, a cysteine-rich protein that senses redox changes in the cell, binds to Nrf2, leading to its retention in the cytosol and causing its proteasomal degradation (81). Under OS, conformational changes in Keap1 following its Cys oxidation lead to its dissociation from the Nrf2-Keap1 complex and to the translocation of free Nrf2 into the nucleus. Subsequently, nuclear Nrf2 binds to small Maf proteins increasing the transcription of ARE-driven genes, which encode for phase II detoxification enzymes. The release of Nrf2 from Keap1 is also achieved by its phosphorylation on Ser40 by several kinases such as protein kinase C (PKC), casein kinase-2 (CK2), phosphoinositide-3-kinase (PI3K), c-Jun NH₂-terminal kinase (JNK), extracellular signal-regulated kinase (ERK), and protein kinase RNA-like ER kinase (PERK) (82). Antioxidant enzyme systems regulated by Nrf2 include MnSOD, CAT, sulfiredoxin (Srx), thioredoxin, peroxiredoxin, GPx, GR, GCL, GCS, NADPH quinone oxidoreductase 1 (NQO1), and heme oxygenase 1 (HO-1) (83).

Nrf2 is ubiquitously expressed and, as mentioned above, in the brain is an important defense against increased prooxidant species (81).

2.6. Oxidative Stress and Intracellular Signaling

Redox signaling mainly includes reversible modification, either oxidation or covalent adduct formation, of specific target proteins, allowing further translation of signals that ultimately determine cellular fate. Accumulating evidence suggests that ROS are not only toxic by-products of cellular metabolism but also essential participants in cell signaling and regulation (84). Although this role for ROS is a relatively novel concept in vertebrates, there is strong evidence of a physiological role for ROS in several nonmammalian systems. The apparent paradox in this dual-face role of ROS, regulation of cellular functions versus toxic by-products of metabolism, can be reconciled considering differences in the amount of ROS production. Among signaling cascades, below we discuss major evidence on the most relevant signaling cascades such as PI3K/Akt, MAPK, and JNK in response to ROS stimulation (85). Moreover, in sect. 4 we discuss the implications among ROS, OS, and the regulation of autophagy by the mammalian (a.k.a. mechanistic) target of rapamycin complex 1 (mTORC1) complex.

2.6.1. The phosphoinositide-3-kinase/Akt pathway.

The phosphoinositide-3-kinase (PI3K)/Akt axis participates in a multitude of intracellular processes, such as those involved in protein synthesis, cell proliferation and death, and autophagy, and also in the development of drug tolerance upon stimulation by cytokines and growth factors (GFs) (84). For example, activation of GF receptors activates PI3K, either through the stimulation via the regulatory subunit bound to the receptor or through adapter molecules such as the insulin receptor substrate (IRS) proteins. As a result, PI3K catalyzes the phosphorylation of phosphatidylinositol 4,5-bisphosphate (PIP₂), forming phosphatidylinositol 3,4,5-trisphosphate (PIP₃) (86), a docking site that recognizes proteins with a specific domain, the pleckstrin homology (PH) domain, including the phosphoinositide-dependent protein kinase (PDK) and protein kinase B (Akt) kinases (Ser/Thr kinases). PDK and Akt activation further stimulates the transcription of target genes such as *GSK3*, *FOXO*, *mTOR*, and *p53*, among others. ROS are able to directly stimulate PI3K, thereby promoting the downstream effects, as described above, but also may lead to the inhibition of the phosphatase and tensin homolog (PTEN). This latter enzyme is critical for the regulation of PIP₃

levels because it dephosphorylates PIP₃ to PIP₂, thereby reducing PIP₃ accumulation. PTEN has a cysteine residue in its phosphatase domain; this Cys in the active site can be oxidized by ROS, especially hydrogen peroxide, thereby inactivating the PTEN phosphatase enzyme and Akt signaling. Furthermore, ROS can modulate PTEN activity by promoting the casein kinase II-dependent phosphorylation that marks PTEN for proteolysis (87). Another additional regulatory mechanism involves protein phosphatase 2A (PP2A) that, if eventually inhibited by ROS, is no longer able to dampen Akt/PKB signaling (84).

2.6.2. The mitogen-activated protein kinase cascades.

The mitogen-activated protein kinases (MAPKs) occur as a large kinase complex playing important roles in a variety of physiological processes such as cell growth and differentiation, cell development, cell survival, and cell death (88). This large complex has four major arms that involve the extracellular signal-related kinases (ERK1/2), p38 kinase (p38), c-Jun NH₂-terminal kinases (JNKs), and MAP kinase 1 (BMK1/ERK5) pathway. ERK, p38, JNK, and BMK1 belong to the Ser/Thr kinase family and share many common motifs. Different extracellular/intracellular stimuli phosphorylate and activate 1) a MAP kinase kinase kinase (MAPKKK); 2) a MAP kinase kinase (MAPKK); and 3) a MAP kinase (MAPK). These MAPKs phosphorylate a variety of substrates, leading to modulation of several cellular processes (89). Growth factors and cytokines are the main activators of the ERK pathway as a result of ligand-stimulated tyrosine kinase receptors (RTKs). Activation of these RTKs is coupled with GDP/GTP exchange to Ras, the latter becoming activated only upon GTP binding. GTP-bound Ras induces the recruitment of cytoplasmic MAPKKK to the membrane, thus activating it. The active form of MAPKKK phosphorylates MEK1/2 (MAPKK), which in turn phosphorylates ERK1/2 (MAPK), favoring its nuclear translocation and activation of target transcription factors (90). Although the mechanisms are not completely clarified, EGF and PDGF receptors are activated in the presence of ROS, stimulating Ras and the consequent activation of the ERK pathway (91). Supporting this sequence of events is the finding that ROS production by commensal bacteria-inactivated dual-specific phosphatase 3 leads to ERK activation (92). However, treatment with H₂O₂ stimulates the phosphorylation and consequent activation of phospholipase C- γ , which catalyzes the formation of inositol 3,4,5-trisphosphate (IP₃) and formation of diacylglycerol (DAG) (93). IP₃ is known to increase intracellular calcium levels through Ca²⁺ release from the endoplasmic reticulum. This released Ca²⁺ activates the ERK pathway and leads to release of DAG. Furthermore, elevation of intracellular calcium triggers the activation of

the components of the protein kinase C (PKC) family, promoting activation of Ras and Raf (94).

2.6.3. The c-Jun NH₂-terminal kinase pathway.

The c-Jun NH₂-terminal kinase (JNK) pathway is activated by OS and cytokines (tumor necrosis factor- α and FAS) and, similar to the ERK pathway, stimulates a cascade of phosphorylation reactions involving MAPKKK, MAPKK, and ultimately phosphorylating JNK on threonine and tyrosine residues. The activation of JNK results in its translocation to the nucleus and regulation of the activity of several transcription factors. The JNK pathway involves different types of MAPKKK such as MEKK1, 2, 3, and 4, MLK, and ASK1 as well as different types of MAPKK, i.e., MKK3, 4, 6, and 7 (95). ROS may impact Trx and Grx, which are redox-sensitive proteins that if dissociated from ASK-1 lead to activation of the latter. ASK-1 activation, in turn, activates JNK (96). Furthermore, ROS are proposed to lead to dissociation of JNK from GSTp; the latter normally binds to JNK to suppress its activation, so release of GSTp from JNK causes this kinase to become activated. ROS also can promote the oligomerization and autophosphorylation of ASK1 through the oxidation of thioredoxin that, in its oxidized form, disassociates from ASK1, permitting oligomerization and activation of the latter (84). This mechanism is confirmed by the finding that TNF- α receptor-associated JNK activation is inhibited by administration of antioxidant compounds. Moreover, it is likely that ROS at low levels do not change the phosphatase activity, resulting in a transient activation of JNK. In contrast, increased levels of ROS can inhibit activity of phosphatases, leading to a prolonged activation of JNK. Furthermore, persistent neuroinflammation is associated with pathological ROS production, likely due to NF- κ B activation by redox signaling, including involvement of NOX2 (97). In the absence of ROS, NF- κ B is repressed by the protein inhibitor of κ B (I κ B). However, redox signaling leads to the increased activity of I κ B kinase (IKK), which phosphorylates and targets I κ B for degradation; as a result, redox signaling activates NF- κ B.

3. ALZHEIMER DISEASE

3.1. Pathological Hallmarks of Alzheimer Disease

A formal diagnosis of AD requires the presence of both senile plaques, which are composed of aggregated fibrils of A β (mostly 42 and 40 amino acids in length) together with dystrophic neurites and neurofibrillary tangles, which are composed of aggregates of hyperphosphorylated Tau protein (98, 99). Other pathological signatures of AD, among others, include loss of synapses, shorter dendritic

lengths, and large cortical sulci due to loss of neurons compared with control brains (100–102).

3.2. Amyloid β -Peptide-Induced Oxidative Stress in Alzheimer Disease

3.2.1. A β and oxidative stress.

Amyloid β -peptide is produced by the actions of two proteases on amyloid precursor protein (APP), a type I transmembrane protein of undetermined function. The two relevant proteases are β -site APP amyloid cleaving enzyme 1 (BACE1; there also is a BACE2 enzyme) and γ -secretase, a multicomponent membrane bilayer-resident protease (103, 104). The amino acid sequences of A β 42 and A β 40, as well as sequences of other A β peptides discussed in this review, are shown in **FIGURE 7**. Both A β 42 and A β 40 are largely hydrophobic, with one redox center, i.e., methionine residue 35. In an early paper (105), Näslund and coworkers reported that a large percentage of senile plaques (described above) in AD brain had A β chains with methionine sulfoxide, i.e., the peptide was oxidized.

The Butterfield laboratory reported in 1994 (106) that A β 40, in the presence of a nonparamagnetic nitron-based spin trap, phenyl-*N*-t-butyl nitron (PBN), led to the production of a nitroxide, i.e., a stable free radical that could be detected by electron paramagnetic resonance (EPR) (**FIGURE 8**). Three concerns about this work arose: 1) there may have been some redox-active metal contamination or spin trap impurity that led to the EPR spectrum; 2) the single tyrosine at residue 10 of A β 42 could leak a free radical and thereby be the source of the spin-trapped nitroxide formation, and evidence of ditryrosine formation from A β peptides was reported (107); and 3) Cu²⁺ bound to histidine residues at 6, 13, and 14 could take part in an electron transfer reaction leading to Fenton chemistry to form hydroxyl free radical from ubiquitously present H₂O₂ (108, 109). The *K*_D of Cu²⁺ bound to A β was determined to be at attomolar level (10⁻¹⁸), which could be removed from aggregated A β in brains of AD mice in vivo by a Cu²⁺ chelator (110) of known nanomolar (10⁻⁹) *K*_D. (This point is addressed further below). We address each of these criticisms as discussed in the following paragraphs.

The PBN nitron spin trap was repeatedly recrystallized, which led to a highly purified molecule (111), to show that the purified trap did not lead to any nitron-trapped radical EPR spectrum, but this ultrapure trap did capture a free radical when neurotoxic A β 42 or A β 40 was present (**TABLE 1**). The ultrapure water used to make buffers in the spin trapping experiments was treated with various metal ion-chelating agents. In each case of using A β 42 or A β 40, an EPR spectrum was



FIGURE 7. Amino acid sequences of Aβ peptides mentioned in this review.

observed. However, if methionine residue-35 of Aβ were substituted by norleucine (TABLE 1) in which the S-atom of Met was replaced by a CH₂-group, giving the molecule essentially the same side-chain length and dipole moment as the native peptide, no EPR spectrum was observed (109), even though any putative metal ions suggested by others would have been present. Moreover, in contrast to protein oxidation, lipid peroxidation, and cell death following Aβ40 or Aβ42 addition to neuronal cultures, none of these occurred with the norleucine-substituted Aβ peptide (112–114), i.e., the Met35 residue of Aβ42 was hinted at as being important in the free radical associated with this neurotoxic peptide. Consistent with this notion, others showed that Aβ42-mediated inhibition of cytochrome-c oxidase was dependent on both Met35 residue being reduced (and thus able to undergo a 1-electron oxidation) and Aβ42's ability to bind and reduce Cu²⁺ to Cu⁺ (117). As noted above, a chelator reportedly was able to remove weakly bound Cu²⁺ from aggregated Aβ42, leading to its absence in brains of an AD mouse model (110). Rather than binding to His residues with a very low K_D in Aβ42, Cu²⁺ bound to a high-K_D, low-affinity site, i.e., Met, coupled to an oxidation-reduction one-electron transfer from the Met S-atom to the Cu(II) ion, is consonant with the mechanism of forming a sulfuranyl free radical on the Met35 residue posited by the Butterfield laboratory as key to initiating lipid peroxidation in AD as described above.

To address the second concern, i.e., that electron leak via the Tyr residue-10 of Aβ42 was associated with the

nitron spin trapping to result in an EPR-detectable nitroxide and oxidative damage to and death of neurons, we repeated these experiments, comparing Aβ42 with Aβ42Y10F (113, 114). Aβ42 with its Tyr-10 substituted by phenylalanine would keep the aromatic nature of the side chain at residue 10 of Aβ but lack the OH group of Tyr that was purported by others to be the source of the free radical of Aβ42. This modified peptide that lacked Tyr at residue 10 would be predicted not to lead to cell death or protein oxidation of neuronal cultures or be able to activate the nitron spin trap to form the EPR-measurable nitroxide free radical if the Tyr residue of Aβ42 was critical to these effects of Aβ42. In fact, the opposite occurred: not only was an EPR spectrum observed, but Aβ42Y10F was as toxic to rat hippocampal neuronal cultures and led to their oxidative damage to a similar extent as observed with native human Aβ42 (111, 112, 115). In another study to address this same question of the necessity of having Tyr10 present in Aβ42 for oxidative damage and neurotoxicity, we examined the short peptide Aβ1–16, which has all three His residues to which Cu²⁺ would bind as well as Tyr10 (115). We also examined a component peptide of Aβ42, namely Aβ17–42, which has the critical Met residue at position 35 and is present in AD brain. In contrast to the presence of a spin-trapped radical observed by EPR, neuronal protein oxidation, and neurotoxicity observed with Aβ17–42, none of these findings was observed with Aβ1–16 (TABLE 1) (115). Confirming the significant role of Met35 of Aβ17–42 in these observations, when the Met35

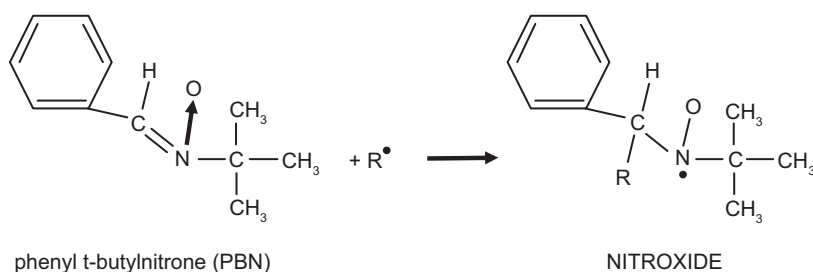


FIGURE 8. Production of an electron paramagnetic resonance (EPR)-detectable nitroxide by a free radical in the presence of a nonparamagnetic nitron-based spin trap such as phenyl-*N*-*t*-butylnitron (PBN).

Table 1. Role of methionine residue of Aβ peptides in leading to an EPR spectrum with PBN, neuronal oxidative damage, and neuronal cell death

Peptide ^a	EPR Spectrum Present with PBN ^b Spin Trapping?	Hippocampal Neuronal Cell Cultures	
		Excess oxidative damage compared to PBS control?	Cell death compared to PBS control?
PBS buffer	No		
Aβ ₄₂	Yes	Yes	Yes
Aβ ₄₀	Yes	Yes	Yes
Aβ ₄₂ M35Nle ^c	No	No	No
Aβ ₄₂ Y10F ^d	Yes	Yes	Yes
Aβ ₁₋₁₆	No	No	No
Aβ ₁₇₋₄₂	Yes	Yes	Yes
Aβ ₁₇₋₄₂ M35Nle ^c	ND	No	No
Aβ ₄₂ H13L ^e	ND	Yes	Yes
Aβ ₄₂ I31P ^f	ND	No	No

See Refs. 111–116 for more details. EPR, electron paramagnetic resonance; ND, not determined. ^aEach peptide was dissolved in highly chelexed PBS buffer prepared with ultrapure water. ^bThe phenyl-*t*-butyl nitron (PBN) spin trap was repeatedly recrystallized (3 times) before use. ^cMethionine side chain (-CH₂-CH₂-S-CH₃) replaced by the norleucine (Nle) side chain (-CH₂-CH₂-CH₂-CH₃). ^dTyrosine at residue-10 replaced by another amino acid with an aromatic side chain, phenylalanine. ^eHistidine side at residue-13 replaced by leucine. ^fIsoleucine residue-31 replaced by helix-breaking proline.

residue was substituted by norleucine (as noted above, CH₂ in place of the S-atom of Met), no oxidative damage or neurotoxicity to hippocampal neurons was found (115). Taken together, these observations constitute highly consistent evidence supporting the notion that free radical-associated effects of Aβ₄₂ require Met in position 35 of this neurotoxic peptide, and if Tyr10 is involved, we suggest that its role is likely relatively limited.

With respect to the third concern of our initial findings regarding the spin trapping of a radical associated with Aβ₄₂, this peptide has His moieties at residues 6, 13, and 14. Being a soft base, His would bind a soft metal, for example Cu²⁺. Others suggested that Cu²⁺ binding to the His residues 13 and 14 of Aβ₄₂ would provide a strong impetus for pulling a radical from the tyrosine residue at residue 10 (108, 109). The adjacent His residues at positions 13 and 14 of Aβ₄₂ conceivably could provide bidentate ligand binding of Cu²⁺. The use of Aβ₄₂H13L (TABLE 1), in which one of the adjacent His residues was replaced by Leu, a substitution that would preclude any bidentate Cu²⁺ ligand binding, led to results consistent with the notion that bidentate Cu²⁺ ligand binding of these His residues was not essential for the oxidative damage and cellular toxicity of the native peptide (115).

In addition to the involvement of Met35 of Aβ₄₂ in oxidative damage observed in *in vitro* studies, the

requirement of Met35 of Aβ₄₂ for oxidative damage also was demonstrated *in vivo*. When the DNA of human Aβ₄₂ was inserted into *Caenorhabditis elegans*, elevated protein oxidation was observed and the worm became paralyzed since a muscle wall promoter was used to insert the DNA of human Aβ₄₂; however, if the codon for Met35 in the DNA of Aβ₄₂ was replaced by that of cysteine and the resulting DNA inserted into *C. elegans*, no elevated protein oxidation was observed and the worms were not paralyzed, although approximately the same level of Aβ deposition was observed (113). Moreover, in a separate study using *C. elegans*, a temperature-sensitive mutation also was inserted into the worm DNA along with the DNA of human Aβ₄₂ such that any foreign DNA would be destroyed but at elevated temperature the foreign DNA could be translated and the mRNA transcribed. In that way, one could time the production of human Aβ₄₂ to determine the time at which the paralysis phenotype occurred and its relation to protein oxidation. The results showed that paralysis occurred in the worm after 24 h of DNA transcription of human Aβ₄₂, the same time at which elevated protein oxidation in the worm reached its zenith (118).

The requirement for Met35 of Aβ₄₂ for oxidative damage to brain cells *in vivo* was later demonstrated for the first time in a mammalian species (119). The *APP^{swe}/Ind* (J20) mouse, which has two mutations in the human

APP gene, was modified further to make a third mutation by replacing Met631 by leucine (M631L) in the *APP* gene, which corresponded to replacing Met35 of A β 42 with Leu. Both the J20 and modified J20 mice were aged to 1 yr of age, a time when OS and A β deposition occurred in the J20 mouse brain, and showed in a mammalian species that no elevated OS in brain occurred in the Met631L-modified J20 mouse (119). This result solidified the notion that the Met35 residue of A β 42 is critical to the oxidative damage and neuronal death associated with this neurotoxic peptide. Moreover, in brain of this APPM631L-modified J20 mouse, in contrast to the regular J20 mouse, A β 42 deposition was not plaquelike but rather punctate in nature, from which we speculate that Met35 of A β 42 conceivably could be important in oligomer and subsequent plaque formation as well as in oxidative damage associated with this neurotoxic peptide (119).

Small oligomers of A β 42, being relatively hydrophobic (TABLE 1), insert into the lipid bilayer with each monomer of the oligomer aligned in the same orientation as elegantly demonstrated by Glabe and coworkers (120). Like other type I proteins that traverse the lipid bilayer, the lipid-spanning portion of the peptide would be predicted to adopt an alpha-helical secondary conformation, a secondary structure characterized by every amino acid interacting with the 4th amino acid beyond, the so-called *i* + 4 rule of alpha-helical protein conformations. In this case, the Ile31 peptide carbonyl oxygen, being within a van der Waals distance from the S-atom of Met35 (121), coupled to the greater electronegativity of oxygen compared to sulfur, the lone-pair electrons around the S-atom of Met35 would be drawn toward the Ile31 oxygen and away from the attraction of the protons in the S-nucleus in Met35. As discussed in sect. 3.2.2, such a displacement of a lone pair of electrons on the S-atom of Met35 would prime the S-atom to a one-electron oxidation of S to form a sulfuranyl free radical (122) (see further discussion of sulfuranyl free radical on A β 42 below).

To test the notion that the helical secondary structure of A β 42 in the lipid bilayer contributes to the reactivity of this peptide and consequent oxidative damage and neuronal death, the Ile31 residue was substituted by the helix-breaking amino acid proline. This substitution prevented the interaction that occurs in wild-type (WT) A β 42 of the peptide carbonyl oxygen of Ile31 with the S-atom of Met35, which, as noted above, were shown to be within a van der Waals distance of each other (121). Addition of A β 42I31P to rat primary hippocampal neuronal cultures showed no cell death nor any elevated oxidative damage, in marked contrast to what was observed with addition of human A β 42 (116). These results demonstrated that the secondary structure of membrane bilayer-resident A β 42 oligomers contributes to the oxidative damage, as described in more detail below.

3.2.2. Lipid-resident oligomeric A β and its sulfuranyl free radical.

The seminal studies of Schöneich (121–123) demonstrated that peptides containing a Met residue can form a sulfuranyl free radical (S⁺) in the presence of a suitable oxidant. Given that A β 42 peptide oligomers are resident in the lipid bilayer (17, 18, 21, 120), coupled with the requirement that a lipid-resident free radical is necessary to initiate the processes of lipid peroxidation (free radicals outside the bilayer would be too reactive to survive entering the bilayer in the first place), consistent with the location of the Met35 residue of A β 42, we posit that the sulfuranyl free radical of A β 42 is the initiator of lipid peroxidation associated with this peptide (FIGURE 8). The sulfuranyl radical would be stabilized by the dipole moment of the alpha-helix secondary structure of A β 42 to survive long enough to abstract a labile allylic H-atom in reactions and begin the processes of lipid peroxidation (FIGURE 4).

Four important considerations arise from the reaction involving a sulfuranyl free radical:

- 1) The free radical formed on S-atom of Met35 of A β 42 is likely due to one or both of two sources: i) As discussed above, because of its zero-dipole moment, molecular oxygen is present in large concentrations in the lipid bilayer (FIGURE 4). Therefore, the sulfuranyl free radical on the S-atom of Met35 could be formed by molecular oxygen interaction with the S-atom of Met35, leading to an electron transfer to oxygen from the S-atom resulting in the reduction of oxygen to superoxide free radical and the one-electron oxidation of the S-atom to the sulfuranyl free radical and/or ii) Cu²⁺ weakly bound to the S-atom of Met35 conceivably would lead to a one-electron reduction of Cu²⁺ and oxidation of the S-atom to the sulfuranyl free radical as discussed above. In the case of nonpolar molecular oxygen as the primary oxidant, both superoxide free radical and Cu⁺ are formed. Superoxide free radical rapidly is converted chemically or enzymatically by superoxide dismutase to H₂O₂, which in the presence of Cu⁺ leads to production of the highly reactive hydroxyl free radical that is highly capable of initiating lipid peroxidation (1, 8). Similarly, if Cu²⁺ weakly bound to the sulfur atom of Met35 is the recipient of an electron from this S-atom to form the reduced Cu⁺ (see above for discussion of a relatively weaker electron pair interaction with the S-atom protons in a polypeptide with α -helical structure), this latter moiety can lead to the highly reactive hydroxyl free radical by Fenton chemistry as noted, initiating lipid peroxidation.
- 2) As shown in FIGURE 4, the reaction involving initiation of lipid peroxidation is a catalytic reaction. Therefore, only a small fraction of A β 42 oligomers need to undergo this formation of sulfuranyl free radicals to greatly increase the extent of lipid peroxidation and

therefore significantly amplify the production of neurotoxic HNE (20, 22–25, 28, 30–32, 124). This key product of lipid peroxidation leads to significantly elevated protein-bound HNE with notable diminution, if not loss, of protein function and consequent neuronal death in brains from individuals with AD or amnesic MCI and models thereof (23, 27, 125).

- 3) The third consideration of consequences of sulfuranyl radical formation on the Met35 residue of oligomeric Aβ42 concerns the intrinsic dipole moment of the helical structure of this lipid bilayer-resident oligomer. As opined above, this α-helix structure dipole moment could stabilize the sulfuranyl free radical long enough to allow the radical to abstract labile allylic H-atoms from allylic C-atoms of acyl chains of phospholipids, thereby initiating the chain reaction of lipid peroxidation. And, because Aβ42 is longer than Aβ40, the former has a greater dipole moment than the latter, suggesting that Aβ42 would stabilize the sulfuranyl radical longer than would Aβ40 to permit more facile abstraction of labile allylic H-atoms, i.e., promote formation of neurotoxic HNE. Such considerations conceivably may contribute to the known greater neurotoxicity of Aβ42 small oligomers relative to those of Aβ40 (126).
- 4) The fourth consideration of the formation of sulfuranyl free radical on Aβ42 oligomers in the lipid bilayer of neuronal membranes deals with lipid peroxidation having a fundamental importance in the pathogenesis and progression of AD. As shown in **FIGURE 4**, the initiation step of lipid peroxidation induced by sulfuranyl free radicals on Aβ42 oligomers involves allylic H-atom abstraction from allylic C-atoms of unsaturated fatty acid components of phospholipids. Others reported studies using AD mouse models fed a diet including deuterated 11,11-D₂-linoleic acid ethyl ester (D₂-LA) and 11,11,14,14-D₄-α-linolenic acid ethyl ester (D₄-ALA) in their diets for a sufficient period such that the phospholipids in brain became rich in deuterated fatty acids with allylic deuterium atoms in the indicated unsaturated fatty acids of phospholipids (127, 128). Deuterium is an isotope of hydrogen, and the C-D bond is considerably stronger than the C-H bond (129). As the mouse with deuterium-enriched PUFA aged, examination of the brain showed a marked reduced formation of F₂-isoprostanes or HNE in cortical and hippocampal brain regions, likely due to the stronger C-D bond of allylic carbons compared with the bond strength of the C-H bond, that prevented cognitive decline in one of these investigations. These results are consistent with the notion that lipid peroxidation, including that initiated by Aβ42-resident sulfuranyl free radicals, is a key step in formation of HNE and other products of lipid peroxidation and consequent neuronal death in AD and more broadly in the pathogenesis and progression of AD. Indeed, earlier in our studies of AD, we observed that Aβ peptide added to synaptosomal membranes led to

lipid peroxidation at a rate faster than the rate for protein oxidation (130). Reverse or scrambled sequences of Aβ peptides did not cause lipid peroxidation or protein oxidation.

How does HNE binding to neuronal proteins lead to cell death? There are many responses to this question, several of which are discussed in this review article. As noted above, one major source of neuronal death in AD and MCI brains is associated with Aβ-mediated, free radical-induced lipid peroxidation. Specifically, redox proteomics identified oxidative modification by HNE of many enzymes associated with glucose metabolism, including those associated with glycolysis, the Krebs cycle, and the electron chain of mitochondria (6, 15, 25, 29, 124, 125, 131–134). HNE binding to proteins almost always results in diminished or complete loss of protein activity (18, 21, 26, 135–142). In the case of proteins associated with glucose metabolism, loss of activity following HNE oxidative modification would lead to loss of ATP production. This, in turn, would affect ion-motive ATPases (6, 21, 131, 135, 138–141), which would cause diminution of the neuronal cell potential with corresponding loss of the ability of the neuron to participate in neurotransmission, which would be expected in a disease associated with loss of cognition. The decreased neuronal cell potential would lead to opening of voltage-gated Ca²⁺ channels that would allow Ca²⁺, at 1.5 × 10⁴ greater concentration outside the neuron compared to intraneuronal Ca²⁺ concentration, to rush into the neuron, overwhelming intracellular storage sites and activating necrotic and apoptotic pathways (6, 139). Neuronal death would ensue. We also showed that HNE can detrimentally affect proteins and cellular pathways other than those related to glucose metabolism as well as lead to loss of bilayer lipid asymmetry, with both factors leading to neuronal death as well (135–137, 140, 141). The topic of oxidative damage in AD brain is further discussed in greater detail in sect. 4.

3.2.3. Aβ and oxidative stress in selected studies involving mammalian models of Alzheimer disease.

Studies of Aβ and OS in *C. elegans* were discussed above. To learn more about Aβ's putative role in oxidative damage in AD and MCI brains, studies in mammalian models of this devastating dementing disorder were required. Because of the extensive literature on mice models of AD, a review article on this topic alone could be justified. Therefore, in the interest of space and the need to cover more than just mice, this review is limited to selected key studies related to Aβ and oxidative damage in several mammalian models of AD. It is important to note that the absence of citation of specific studies of

OS in these models of AD does not imply anything with respect to the significance of those studies; rather, space limitations forced selection of specific studies and not others.

3.2.3.1. A β 42-ASSOCIATED OXIDATIVE STRESS IN MOUSE MODELS OF ALZHEIMER DISEASE. Selected studies of the role of A β 42-associated OS in mouse models of AD are shown in **TABLE 2**. When searched in PubMed with the search terms “A β -peptide and oxidative stress” and “Alzheimer disease mouse models,” 467 articles were retrieved. This number was lowered substantially by requiring only studies associated with live mouse models, not cell cultures; OS as end point or implied based on the known literature of the mouse model used; and A β 42 with results described. With the caveat stated at the beginning of sect. 3.2.3, **TABLE 2** shows a wide range of preclinical mouse models of AD selected.

Taken together, the results of investigations of preclinical mouse models of AD (**TABLE 2**) give rise to several commonalities of relevance to the theme of this review article. Among these are:

- No single model has 100% fidelity to human AD. That stated, the results of these various studies provide remarkable insights into specific protein oxidative modifications observed in AD and MCI brains, often by the lipid peroxidation product HNE.
- A β 42 oligomers invariably are associated with oxidative damage to brain cell bilayer lipids and cellular proteins, including membrane proteins. Blockage of in vivo A β 42 production, i.e., using antisense oligonucleotides to genes of key proteins involved in the production of A β 42, leads to significant diminution of oxidative damage to brain, a conclusion that strengthens the notion that A β 42 oligomer-associated oxidative damage and subsequent neurotoxicity is a fundamental process in mouse models of AD and MCI brains.
- Some antioxidants are not protective to neurons in these mice, but many antioxidant treatments or elevation of antioxidant enzymes are protective.

3.2.3.2. A β 42-ASSOCIATED OXIDATIVE STRESS IN RAT MODELS OF ALZHEIMER DISEASE. There are nearly an order of magnitude fewer publications on in vivo rat models of AD associated with A β 42 than is the case for mouse models. Therefore, this section of this review article briefly highlights a few rat models involving this neurotoxic peptide. Rats are higher on the phylogenetic tree of evolution than mice, and their larger brain size makes it easier to inject drugs or to sample cerebrospinal fluid (CSF) than for mice. Finally, rats present behavioral characteristics of a more sophisticated nature than mice.

One such model involves the direct injection of A β 42 into rat brain parenchyma. Injection of A β 42 into rat basal forebrain followed a week later by examination of

oxidative damage in hippocampus demonstrated a significant elevated level of HNE-modified proteins (13), consistent with the known diffusion of HNE from its site of origin in cell or organelle membranes to other areas within the cell (8, 34) and consistent with the connection between the forebrain and outer molecular layer of hippocampus (98). Treatment of A β 42-injected rats with an inhibitor of dipeptidylpeptidase-4, sitagliptin, together with quercetin, a flavonol from the flavonoid group of polyphenols, significantly diminished the level of A β 42 compared with either treatment alone but increased antioxidant protein levels (i.e., SOD, CAT) and the level of GSH and a diminution of histopathologically determined neuronal damage. The authors suggested that activation of the Nrf2/HO-1 pathway was responsible for these effects (179). A study employing a different dipeptidylpeptidase-4 inhibitor, linagliptin, following A β 42 injection to brain ameliorated cognitive impairment, attenuated levels of A β 42 oligomers, decreased the degree of insulin resistance (by decreasing levels of phosphorylation of the inhibitory Ser307 site on IRS-1), and diminished levels of GSK-3 β and proinflammatory cytokines and, importantly, decreased levels of oxidative and nitrosative damage in hippocampus. Immunohistochemistry showed that this drug provided neuroprotective and anti-A β 42 aggregation effects. The authors suggest that their studies provide insights into A β 42 and brain insulin resistance as a significant contributing component of the pathophysiology of AD (180).

A significant caveat of this model of AD is the lack of progressive A β accumulation with age of the rat, but snapshots in time can be obtained in a relatively simple and straightforward manner. However, when a single dose of protofibrillar A β 42 was delivered intracerebroventricularly bilaterally to dorsal hippocampus, even 6 wk later the rats retained long-term memory deficits and increased anxiety levels (181). Moreover, the amyloid deposits were still present at this time, and OS and NS indexes were elevated compared with the sham-treated control.

Cuello and colleagues developed the McGill-R-Thy1-APP rat model of AD (182). This transgenic rat is rich in human oligomeric A β 42, and at 6–9 mo senile plaques are observed in brain regions seen in AD patients and not in cerebellum. The APP transgene used has Swedish and Indiana mutations. NP03 is a lithium-based microdose preparation that given to the McGill-R-Thy1-APP rat diminishes AD-like A β neuropathology (183). Our laboratory determined in a blinded fashion that NP03-treated McGill-R-Thy1-APP rats had decreased OS and NS (184). This was the first demonstration that this unique rat model of AD with A β pathology similar to humans had associated OS and that NP03 treatment could significantly reduce this damage.

A different type of AD rat model was developed by Harvey and Syková (185). Heterotransplantation of fetal

Table 2. *Aβ42 and oxidative stress in selected mouse models of Alzheimer disease*

Mouse Model of AD (sex)	Major Conclusions	Ref(s)
APP ^{NL-G-F/NL-G-F} × Keap1 ^{FA/FA} (males)	Elevated GSH prevented increased oxidative stress and ameliorated cognitive loss compared to APP mice. Plasmalogen PtdEtn unsaturated fatty acids were decreased, which would be predicted if these were involved in lipid peroxidation.	(143)
Review and Tg2576 mice (males and females)	Reviews of protective effects of curcumin (found in the spice turmeric) against Aβ42 in AD mouse models. Note the need to increase bioavailability of curcumin to the CNS. F ₂ -isoprostanes were observed in brain of Tg2576 mice prior to deposition of Aβ42. This result suggests that lipid peroxidation precedes Aβ deposition and is consistent with Aβ oligomer-facilitated lipid peroxidation. First demonstration in Tg2576 mice that repetitive mild brain injury episodes drive deposition of AD-like Aβ deposition, elevation of lipid peroxidation, and cognitive decline	(144–146)
APP/PS1 human double-mutant knockin mice, 3xTg-AD mice (males and females)	In brains of human double-mutant APP/PS1 knockin mice, elevated oxidative stress markers in proteins related to metabolism, synaptic plasticity, proteostasis control, signal transduction, and neuroinflammation were observed. In these mice, oxidative damage to brain and to specific brain proteins was significantly diminished by providing <i>N</i> -acetylcysteine (NAC) in the drinking water. NAC provides the rate-limiting substrate (Cys) for GSH biosynthesis in brain. Brains from human double-mutant APP/PS1 knockin mice demonstrated loss of phospholipid asymmetry that was correlated with elevation of lipid peroxidation. NADPH oxidase (NOX) catalyzes production of superoxide free radicals. NOX activity and expression of NOX4 subunit were significantly increased in these APP/PS1 double-mutant knockin mice in an age-dependent manner. This elevated oxidative stress was correlated with diminished cognitive performance. In 3xTg mice, loss of GSH and decreased levels of intrinsic Vit E were observed, all indicative of elevated oxidative damage.	(147–152)
APP-overexpressing mice; mitochondrially expressed catalase Tg human mutant APP mice (males and females)	Elevated oxidative stress, but chronic α-lipoic acid did not correct cognitive decline. Mice with mitochondrially expressed catalase had significantly decreased oxidative damage, decreased Aβ neuropathology, and increased life span compared to Tg human mutant APP mice.	(153, 154)
Tg19959 mice with 2 human mutations in APP gene (males and females)	Thiamine deficiency induces oxidative stress and exacerbates plaque pathology. Nicotinamide forestalls pathology and cognitive decline in this AD mouse model by improving bioenergetics for neurons. Coenzyme Q10 supplementation to this mouse decreases oxidative stress and amyloid pathology.	(155–158)

Continued

Table 2.—Continued

Mouse Model of AD (sex)	Major Conclusions	Ref(s)
J20 mice (APP _{SWE/IND}); J20 mice (APP _{SWE/IND})M631L (males and females)	Methionine at residue 631 of human mutant APP (location of residue 35 of Aβ42) was demonstrated to be necessary for oxidative stress in brain in vivo. Several key brain proteins identified by redox proteomics as being oxidatively modified in J20 mice but not in J20 mice in which the key Met residue is absent	(119, 159, 160)
Tg human mutant APP/PS-1 mice (males and females)	Mitochondrial methionine sulfoxide reductase B2 normally protects brain from oxidative damage, but this enzyme is significantly decreased in this model of AD and likely contributes to AD-like neuropathology of Aβ42. Elevated 3-nitrotyrosine and HNE levels on brain proteins that correlated with Aβ neuropathology Vitamin D-deficient diet accelerated cognitive impairment, a result associated with elevated levels of APP and BACE1 hyperphosphorylation; results related to oxidative stress by decreased levels of Cu,Zn-SOD and GPx4 p38 was elevated in hippocampus, but if mice were fed a diet enriched in Vit E elevated p38 did not occur. Vit E is a chain-breaking antioxidant that inhibits lipid peroxidation.	(161–164)
SAMP8 age-accelerated mouse model of AD (males and females)	Less oxidative damage to specific brain proteins in age-accelerated mice treated with antisense oligonucleotides against APP or PS-1, thereby leading to less Aβ42 Lipoic acid or NAC treatment of age-accelerated mice led to decreased oxidative damage to specific brain proteins associated with improved cognitive performance.	(165–170)
Hemizygous human APP _{SWE} /PS1dE9 mice (males and females)	Highly novel electron paramagnetic resonance imaging methods noninvasively demonstrated elevated oxidative damage in brain at 7 mo of age, 2 mo sooner than shown in earlier studies by others. Mice had significantly lower vitamin C transporters. This observation correlated with lower cognitive performance on various tests, elevated lipid peroxidation indexed by F ₂ -isoprostanes, and significantly more Aβ42 deposits.	(171, 172)
Thy1-APP(SL) mice; PDGF-APP695(SDL) mice (males and females)	Thy1-APP(SL) mice have high levels of Aβ and demonstrated significantly increased lipid peroxidation that correlated with low activity of Cu,Zn-SOD. In contrast, PDGF-APP695(SDL) mice have relatively low levels of Aβ, and no elevated lipid peroxidation in brain was observed in these latter mice.	(173)
Tg mice with human APP _{ARC} mutation (males and females)	First report that mitochondrial dysfunction and ROS and oxidative damage in brain occur early in a transgenic mouse containing human APP with an Arctic mutation	(174)
Review of various mouse models of AD (males and females)	Essentially all mouse models of AD show that human Aβ42 is related to brain mitochondrial dysfunction with consequent impairment of axonal transport and energy starvation at synapses.	(175)
Gerbils treated with D609 and isolated synaptosomes or brain mitochondria treated with Aβ42 (males)	Tricyclodecan-9-yl-xanthogenate (D609), a GSH mimetic, or saline was delivered intraperitoneally to gerbils, and subsequently isolated synaptosomes were treated with Aβ42. D609-treated rodents had improved cognitive function, increased levels of brain GSH, decreased brain protein oxidation and lipid peroxidation, and protected mitochondria after Aβ42 treatment vs. saline-treated gerbils.	(176–178)

AD, Alzheimer disease; APP, amyloid precursor protein; CNS, central nervous system; GSH, glutathione; HNE, 4-hydroxynonenal; PtdEtn, phosphatidylethanolamine; ROS, reactive oxygen species; SOD, superoxide dismutase.

cortical tissue from one rat was transplanted into the mid-brain of a different, neonatal rat. Chronic astrogliosis and microgliosis occurred as a result. Moreover, pathology reminiscent of AD brain was observed. Using this unique rat model of AD, Bates et al. (186) demonstrated that prolonged OS indexed by protein carbonyls and the lipid peroxidation biomarker HNE were present. If the rat chow diet was supplemented by *N*-acetylcysteine, noted elsewhere in this article to provide the rate-limiting reagent for glutathione synthesis and protect against Aβ42-associated OS in vivo (147), OS was significantly reduced. The authors conclude that dietary antioxidants may be useful as preventative agents in some of the pathological pathways associated with AD. Taken together, these three different rat models of AD are united in their findings that AD-like Aβ 42 pathology is present, as is OS.

3.2.3.3. Aβ42-ASSOCIATED OXIDATIVE STRESS IN THE BEAGLE DOG MODEL OF ALZHEIMER DISEASE. With increased aging, dogs exhibit AD-like neuropathology in brain including elevated levels of Aβ42 (of the same amino acid sequence as that in humans), amyloid plaque formation, and hyperphosphorylated Tau protein (187). Decreased cognition and elevated OS are observed in aged beagle dogs (188). These observations led to studies to test the hypothesis that high-antioxidant diets given to 12-yr-old beagle dogs for 3 yr would lead to decreased OS in brain of 15-yr-old dogs that was associated with decreased AD-like neuropathology and improved cognition. Cotman and Head led these studies, whose results were consistent with the hypothesis (188). The Butterfield laboratory, in collaboration with Cotman and Head, confirmed the elevated OS in aged beagle dogs and confirmed that a diet enriched with antioxidant-rich food, coupled with a program of behavioral enrichment [to make new synaptic

connections], decreased AD-like neuropathology and, further, led to decreased oxidative damage to specific brain proteins identified by redox proteomics in brains of 15-yr-old dogs (189). **TABLE 3** shows the brain proteins that were identified by redox proteomics as being protected against oxidative modification by this combined intervention of high-antioxidant diets and a program of behavioral enrichment compared with aged beagles fed regular dog chow and not subjected to behavioral enrichment. The protein carbonyl levels of these brain proteins were significantly decreased relative to those in the control group of dogs (189). The redox proteomics-facilitated identified proteins are among those we had identified in brains of persons with AD and MCI (124). Consistent with our earlier observations that the structure and function of oxidatively modified brain proteins are significantly altered (6, 12, 15–19, 21, 26, 124), the enzymatic activities of these brain proteins were found to be significantly elevated in the dogs that had high-antioxidant diets and a program of behavioral enrichment. Among these protected proteins in aged beagle brains were those whose functions are related to improved learning and memory (189). These results with aged beagle dogs, whose brains, as noted above, have Aβ42 of identical amino acid sequence as Aβ42 in human brains and therefore the observations are likely applicable to humans, support the notion that translational intervention in MCI and AD individuals involving these two strategies, i.e., high-antioxidant-containing diets (such as one would experience with a Mediterranean diet) coupled with deliberative efforts to produce new synaptic connections (such as learning a new language, to play a new musical instrument, and/or to solve challenging puzzles) conceivably would decrease the neuropathology and rate of cognitive decline associated with both disorders. As discussed in sect. 6

Table 3. Brain proteins from aged beagle dogs identified by redox proteomics as having less oxidative modification when dogs were fed a high-antioxidant diet coupled to a program of behavioral enrichment for a 3-yr period*

Identified Protein	Main Function in Brain
α-Enolase	Glucose metabolism—glycolysis
Glyceraldehyde 3-phosphate dehydrogenase	Glucose metabolism—glycolysis
Neurofilament triplet L	Cytoskeletal protein (brain protein found in plasma) appears to be a reliable biomarker for neurodegeneration, including in AD.
Glutamate dehydrogenase	Converts Glu back to α-ketoglutarate to keep TCA cycle intermediates available.
Glutathione S-transferase P	Conjugates GSH to moieties that can be removed from neurons via multidrug resistance protein-1 (MRP-1).
Fascin actin bundling protein	Involved in synaptic remodeling important in processes associated with learning and memory

*Compared with similarly aged beagles fed regular dog chow and not subjected to a program of behavioral enrichment. AD, Alzheimer disease; GSH, glutathione.

below, exercise added to these two approaches also potentially would contribute to an improved outcome. A value-added aspect of this proposed translational approach to MCI and AD persons, based on the strikingly positive results in beagle dogs, is that indexes of heart health likely would also increase. Decline in these heart-healthy indexes in the vasculature is reportedly a risk factor for AD (98, 99, 102).

3.3. *Tau Pathology and Oxidative Stress*

Tau protein, known as neuronal microtubule-associated protein tau, is a highly soluble, natively unfolded, and phosphorylated protein predominantly located in axons of mature neurons. In the adult human brain, tau consists of six isoforms, produced by alternative mRNA splicing of the *MAPT* gene located on chromosome 17q21.31, and the *tau* gene contains 16 exons. These isoforms are generated by alternative splicing of exons 2, 3, and 10 of its pre-mRNA. The six tau isoforms differ from each other by the presence/absence of one or two inserts in the NH₂-terminal region and the presence/absence of the second microtubule-binding repeat in the COOH-terminal portion. The alternative splicing of exon 10 leads to tau isoforms, termed 4R (4 microtubule-binding domains, with exon 10) or 3R (three microtubule-binding domains, without exon 10), respectively. Adult human brains express both 3R-tau and 4R-tau, whereas fetal human brains express only 3R-tau (190). Although in the normal adult human brain the level of 3R-tau is similar to that of 4R-tau, immunocytochemistry and biochemical analysis indicate that the ratio of 3R- to 4R-tau is altered in AD and other neurodegenerative disorders (191).

Moreover, posttranslational modifications (PTMs) are a major determinant behind the diversity of tau. Besides phosphorylation, the crucial PTM in tau, Tau can be also nitrated, acetylated, ubiquitinated, proteolytically cleaved, SUMOylated, deaminated, oxidized, glycosylated, gly-cated, methylated, and demethylated on numerous residues. Importantly, the patterns, co-occurrences, and interactions between PTMs may play key roles in Tau functions and in the aggregation processes (192).

Tau is critical for neuronal morphology, by promoting microtubule assembly, stabilizing microtubules, affecting the dynamics of microtubules in neurons, and inhibiting apoptosis, particularly in axons (190). The pathogenesis of tau-mediated neurodegeneration is unclear, but hyperphosphorylation, oligomerization, fibrillization, and propagation of tau pathology have been proposed as the likely pathological processes that induce the loss of function or gain of tau toxicity, which cause neurodegeneration.

According to current AD hypotheses, tau becomes abnormally phosphorylated, dissociates from microtubules,

and aggregates into NFTs (193). Tau has >45 phosphorylation sites, most of which are located in the proline-rich region (P-region) (residues 172–251) and the COOH-terminal tail region (C-region) (residues 368–441) (194). Tau phosphorylation at both of these regions affects its capacity to interact with microtubules. Importantly, it is well documented that phosphorylation could contribute and enhance tau polymerization (194). Tau phosphorylation is regulated through the balance between the activation of various protein kinases and phosphatases, and its disruption leads to abnormal tau phosphorylation, as observed in AD. Tau phosphorylation is increased in both physiological and pathological conditions, but it is still unclear whether the mechanisms involved in physiological and pathological tau hyperphosphorylation overlap. Tau phosphorylation is markedly increased during embryonic development, likely in response to increased need for neuronal plasticity. Several cellular stress conditions such as OS, heat stress or hypothermia, and even starvation modulate tau phosphorylation (195). Tau hyperphosphorylation may result from OS-mediated interaction with tau protein kinase and phosphatase, particularly GSK-3 β and PP2A, the predominant enzymes regulating Tau metabolism. A recent study indicated that GSK-3 β activity is upregulated under OS (196).

In human embryonic kidney 293/tau cells treated with H₂O₂, GSK-3 β activity is increased and tau hyperphosphorylation occurs at Ser396, Ser404, and Thr231. However, besides GSK3 β , OS affects other signaling pathways and/or kinases mediating tau hyperphosphorylation. In particular, several tau kinases belong to the family of stress-activated protein kinases, which are activated in response to OS (197). Specifically, HNE reportedly directly activates two members of the stress-activated kinase family, JNK and p38, in NT2 neuronal cells (198). Another possible link between OS and pathological tau phosphorylation is peptidyl prolyl *cis-trans* isomerase 1 (PPI-ase1 or Pin1). This enzyme is significantly downregulated and oxidized in AD hippocampus (199). Because Pin1 has been implicated in dephosphorylation of tau protein, it can be hypothesized that in vivo oxidative modification of Pin1, as found in AD hippocampus, reduces Pin1 activity, leading to increased tau phosphorylation (199).

Intriguingly, insulin may also play a role in OS-induced tau phosphorylation. Under OS conditions decreased insulin secretion and sensitivity are observed (199). Although insulin is highly sensitive to OS, it plays an important regulatory role in tau phosphorylation in neuronal cell cultures, and abnormal insulin levels in mice lead to tau hyperphosphorylation in brain neurons (200). Experimental evidence shows that chronic OS in vitro increases the levels of tau phosphorylation at paired helical filaments (PHF-1) epitope (serine 396/404)

via the inhibition of glutathione synthesis by buthionine sulfoximine (201). In primary rat cortical neuronal cultures stimulated by $\text{Fe}^{2+}/\text{H}_2\text{O}_2$, tau phosphorylation is significantly increased by the elevated activity of GSK-3β (202). Furthermore, treatment of rat hippocampal cells and SHSY5Y human neuroblastoma cells with H_2O_2 at the early stages of oxidative stress exposure results in tau dephosphorylation at the tau1 epitope by CDK5 via Pin1 activation (203).

In addition to phosphorylation, a critical role is also played by enzymes that are involved in Tau dephosphorylation, among which the protein phosphatase PP2A is the most relevant. Studies have suggested a link between PP2A and OS. Treatment of rat cortical neurons with okadaic acid reportedly inhibits PP2A activity, resulting in an abnormal increase in mitochondrial ROS and mitochondrial fission (204). Other findings reveal that ROS inhibits PP2A and PP5, leading to the activation of JNK and ERK1/2 pathways. The potential involvement of Pin1 oxidative modification and increased tau phosphorylation in AD and MCI brains was discussed above (199), and Pin1 regulates the activity of PP2A (199). Taken together, the above discussion is consistent with the notion that although further studies are needed to better understand the cross talk between OS and tau phosphorylation/dephosphorylation, a close interplay exists that suggests that these events are key components of a vicious circle that plays a crucial role in tau pathology in AD.

3.4. Apolipoprotein E Genetic Variants

Apolipoprotein E (apoE), which has three isoforms (apoE2, apoE3, apoE4) coded for by three different gene alleles (*ApoE2*, *ApoE3*, *ApoE4*), is a pleiotropic protein with several major roles among which are cholesterol transport and efflux of key molecules from brain to blood, one of which is Aβ42 (205). Each of these three isoforms of apoE has a different composition of two specific amino acid residues, 112 and 158: apoE2 has Cys in both these residues, whereas apoE3 has Cys in residue 112 and positively charged Arg in residue 158 and apoE4 has Arg in both residues 112 and 158.

There are two principal hypotheses by which the presence of apoE4 confers significantly elevated risk of developing AD. In the first model, the composition and structure of apoE4, which is markedly different from the structure of apoE2 and apoE3, is proposed to constitute this elevated risk. The structure of apoE is dependent on the isoform being considered, and the respective structure governs its interactions with many molecules, including Aβ42 and Tau, which conceivably could contribute to the increased risk of developing AD by persons who inherit *APOE4* (206–208). The second model involves the lack of critical Cys amino acids in residues

112 and 158 and consequent elevated damage to neurons by the lipid peroxidation product HNE (209). In this scenario, highly reactive and neurotoxic HNE would bind to key membrane and cellular proteins in brain, with greater extent in persons with apoE4, which lacks Cys residues, compared to persons who inherit apoE2 and apoE3, which do have two and one Cys residues, respectively. Consistent with the above, elevated lipid peroxidation, protein oxidation, and DNA oxidation occur to a greater extent in AD brains from persons carrying the *APOE4* gene relative to those carrying genes for *APOE2* or *APOE3* (210–212). Similar conclusions were reached with human *APOE* targeted-replacement mice in which mouse *ApoE* genes were knocked out and the human *APOE* allele gene knocked in. For example, elevated levels of lipid peroxidation and protein oxidation occurred to the greatest amount in ApoE4 targeted-replacement mice, followed by levels of lipid peroxidation and protein oxidation in ApoE3 mice and then in ApoE2 mice (213–215).

In support of the hypothesis that Aβ42 would damage neuronal membranes in persons carrying the *APOE4* gene due to absence of Cys in residues 112 and 158 of apoE4 (206, 209), decreased HNE binding to ApoE4 compared to ApoE3 was observed (216), and ApoE3 was more cross-linked by HNE than was ApoE4 (217). Further supporting this hypothesis, when Aβ42 was added to synaptosomes isolated from brains of *APOE* targeted-replacement mice, the degree of lipid peroxidation assessed by protein-bound HNE and protein oxidation indexed by protein carbonyls and protein-resident 3-nitrotyrosine was greatest in synaptosomes from ApoE4 mice followed by ApoE3 mice and then ApoE2 mice (218), which is the reverse order of the number of Cys residues in the respective apoE proteins in brains of the respective ApoE mice.

3.5. Mitochondrial Network: Energy, Dynamics, and Retrograde Signaling

Mitochondria are complex organelles that serve as the powerhouses of the eukaryotic cell by playing an extraordinary coordinating action among a plethora of cellular functions. This functional diversity is evidenced by the fact that most mitochondrial genes reside in the nucleus but also by the functional recombination through which mitochondria participate in various cellular processes. Thus, a finely tuned regulatory network orchestrates the biogenesis, maintenance, and turnover of mitochondria.

The most prominent metabolic process carried out by mitochondria is oxidative phosphorylation (OXPHOS) to generate ATP. Mitochondria also play fundamental roles in multiple cellular processes such as apoptosis, steroid

biosynthesis, calcium homeostasis, intermediary metabolism, and cell signaling (219, 220). Mitochondria also are cellular sources of superoxide ($O_2^{\bullet-}$) and hydrogen peroxide (H_2O_2), which are involved in the redox regulation of cytosolic and nuclear signaling pathways, together with other messengers originating from mitochondria such as ATP/(AMP + ADP) and the fine tuning of the $NAD^+/NADH$ redox couple (5). Cellular metabolic states dictate mitochondrial size, shape, function, and positioning. Mitochondrial shape varies from discrete isolated organelles to large, interconnected reticular networks within cells. These morphological adaptations are highly responsive to metabolic states and are facilitated by dynamic events including transport, fusion, fission, and quality control (219, 221). By changing their dynamics and strategic positioning within the cytoplasm, mitochondria carry out critical functions and also participate actively in interorganelle cross talk, assisting metabolite transfer, degradation, and biogenesis (222).

3.5.1. Mitochondrial biogenesis.

Mitochondrial biogenesis is a complex process that requires the synthesis, import, and incorporation of proteins and lipids to the existing mitochondrial reticulum as well as the replication of the mitochondrial DNA (mtDNA). The mitochondrion is a unique organelle that contains components of its own self-replicating genome. The mtDNA encodes 13 essential components of the electron transport chain (ETC) as well as all rRNAs and tRNAs necessary for translation of the mtDNA-encoded proteins (223). However, the vast majority of mitochondrial proteins (>1,000) are encoded by the nuclear genome (224, 225). Therefore, the biogenesis of mitochondria requires exquisite coordination of both mitochondrial and nuclear genomes. The control mechanisms for the orchestrated control of mitochondrial biogenesis are achieved largely through factors encoded by the nuclear genome, as described below (226). Indeed, all the machinery needed for replication and transcription of the mitochondrial genome is contained within the nuclear-encoded mitochondrial proteins. The expression of these genes is directly or indirectly regulated by Nrf-1 and Nrf-2 and coactivators such as peroxisome proliferator-activated receptor γ coactivator-1 α (PGC-1 α) (227). As a mechanism to optimize the mitochondrial number and their metabolic capacity, mitochondria biogenesis is activated in response to the imbalance between cellular energy demand and mitochondrial energy transduction induced by a variety of intracellular signals and extracellular stimuli, such as inflammation (222). Acute inflammation induces mitochondrial dysfunction and OS; conversely, impaired mitochondria lead to further inflammation by releasing various compromising molecular patterns

and by activating the assembly of the inflammasome. Depending on the duration and severity of the inflammatory signal, mitochondrial biogenesis may be activated as a restoration mechanism and correlates significantly with the clinical recovery of sepsis. Signaling pathways (e.g., NF- κ B, Akt, and MAPKs) and small signaling molecules (e.g., nitric oxide and carbon monoxide) are involved in inflammation-dependent activation of mitochondrial biogenesis by connecting inflammatory signals to the biogenesis machinery, centered on transcriptional factors and coactivators (220).

3.5.2. Mitochondrial dynamics.

Mitochondria are highly dynamic organelles and undergo fusion and fission continuously in their interconnected network. These processes regulate not only mitochondrial morphology but also their biogenesis, transportation and localization, quality control and degradation, and apoptotic cell death (219). A coordinated balance between fusion and fission serves to maintain the quality of the mitochondria network, whereas perturbations to this balance are associated with pathologies. Zorzano et al. (228) reviewed the interactions between mitochondrial dynamics and metabolic function, focusing on one of the key GTPases required for mitochondrial fusion, mitofusin-2 (Mfn2). Mfn2 is regulated by metabolic factors such as obesity and type II diabetes, inflammatory factors such as TNF- α and interleukin 6, as well as other oxidative and hormonal signals. Conversely, Mfn2 deficiency leads to insulin resistance and metabolic dysfunction through OS- and ER stress-related signals. Therefore, Mfn2 is proposed as a sensor and regulator of metabolic homeostasis in multiple tissues.

3.5.3. Mitophagy.

Homeostasis in mitochondrial mass and tissue energetic demands requires biogenic synthesis of new mitochondrial components to be spatiotemporally matched with the removal of old/damaged mitochondria (229). The most completely understood mechanism for targeted removal of damaged mitochondria is through the PINK1 (PTEN-induced kinase 1)-Parkin mitophagy pathway (230). When mitochondrial damage accumulates above a certain threshold, for example in response to energy deprivation, OS, or hypoxia, cells activate a mechanism to selectively target and remove damaged mitochondria via the autophagic machinery to renew the mitochondrial pool (231). Mitochondrial autophagy, or mitophagy, includes selective sequestration of damaged mitochondria by the autophagosome and their degradation in the lysosome. Mitophagy is closely linked to mitochondrial fission and fusion processes (220).

3.5.4. Retrograde signaling.

Retrograde signaling is activated in altered mitochondria, which communicate with the nucleus to reprogram nuclear gene expression. Upon changes in mitochondrial function, retrograde signaling pathways are activated to coordinate mitochondrial protein synthesis during biogenesis but also to communicate eventual mitochondrial malfunctions, triggering compensatory responses in the nucleus (232). Among causative events, altered mitochondrial proteostasis, ATP, ROS, and Ca²⁺ are able to trigger mitochondrial retrograde signaling that in turn activates well-defined signal transduction pathways. These include the mitochondrial unfolded protein response (UPR^{mt}), the integrated stress response (ISR), mTOR signaling, and the endoplasmic reticulum unfolded protein response (UPR). Together, these transcriptional responses act to restore mitochondrial function and maintain cellular homeostasis (233).

3.6. Oxidative Stress and Synaptic Dysfunction

Synapse loss is an early event in AD (234), and AD has been called a disease caused by synaptic failure (235). Synaptic membranes and cellular and organelle proteins, pre- and postsynaptic, are responsible for neurotransmission, learning and memory, and several other functions. Such processes require synaptic remodeling (i.e., synaptic plasticity). Accordingly, when synaptic membranes, their components, or binding partners are damaged by OS or NS, cognitive functions are impaired. Prolonged oxidative and nitrosative damage to these membranes, lipids, and proteins, as is the case in AD and its earlier stage amnesic MCI, leads to synaptic dysfunctions and eventual synaptic loss, which underlies the most telling characteristic of this devastating dementing disorder: cognitive diminution (236).

Neurotoxic A β 42 oligomers contribute significantly to these effects, including OS and NS (as described in detail in sect. 3.2.1) (6, 15, 124, 132), with consequent oxidative and nitrosative modification of key proteins involved in maintenance of neuronal cell potentials (138), Ca²⁺ signaling (139, 237), neurotransmission and associated synaptic remodeling (124, 238), and long-term potentiation (LTP) (239) and survival of synapses themselves (234, 235, 240). Researchers isolated A β 42 oligomers from AD brains and separated these by size from dimers to larger oligomers. These researchers then investigated the effect of the AD brain-isolated oligomeric size on synaptic toxicity and dysfunction. Large A β 42 oligomers were found to be far less neuroactive than small oligomers (241, 242). Specifically, A β 42 dimers caused synaptic dysfunction, i.e., diminished LTP, learning, and memory, whereas larger A β oligomers did not

(238). Similarly, AD patient-derived cerebrospinal fluid containing A β 42 dimeric oligomers caused disruption of synaptic plasticity, which as noted above is key to learning and memory (243). These researchers further demonstrated that this synaptic dysfunction was caused by these human A β 42 dimeric oligomers by completely blocking the disruption of synaptic plasticity by passive immunization with antibodies to A β 42 dimers.

By redox proteomics, compared with normal brains several oxidatively modified, and therefore likely dysfunctional, proteins were identified in AD and MCI brains that are involved in 1) loss of synaptic membrane-mediated neurotransmission or synaptic remodeling (syntaxin-binding protein-1, fascin-1, β -actin, α -internexin); 2) decreased neurotransmitter vesicle and mitochondrial anterograde and retrograde transport along microtubules that would lead to decreased neurotransmission and to diminished levels of needed ATP to maintain synaptic membrane ion gradients (α -tubulin, neurofilament protein medium); and 3) decreased synaptic connections formed by dendrites because of decreased extension of dendritic lengths [dihydropyrimidinase-related protein-2 (a.k.a. collapsin response-mediated protein-2, CRMP-2)] (124). Each of these dysfunctional synaptic-related functions due to oxidatively modified key proteins is observed in AD and MCI brains, and we propose that these oxidatively modified proteins contribute to the cognitive dysfunction and related clinical observations in AD and MCI.

3.7. Sexual Dimorphism in Alzheimer Disease and the Involvement of Oxidative Stress

Several lines of evidence have demonstrated the influence of sex differences in cognitive function in adulthood and aging (244, 245). In addition, sexual dimorphism has been recognized as a critical determinant in neurodegenerative disorders for its impact on etiology, severity, incidence, progression, and treatment outcomes (246–248). However, the reasons and the molecular mechanisms underlying the effect of sex in these disorders are still awaiting clear explanations. Sex-related differences are evident at early stages of brain development and continue to dynamically guide brain structure and function throughout the human life cycle (249–251). Regional effects of sex hormones with specific expression profiles of sex hormone receptors could (partially) account for sexual dimorphism in brain development and for structural and functional network connectivity (252, 253). Studies also suggest that the changes in biological sex- and sex hormone-mediated memory circuitry become evident in midlife and during aging (247). Aging is generally characterized by the decline of sex hormone production; although in men it occurs as a slow progressive drop

resulting in a gradual decrease of testosterone, women experience, at around 50 yr of age, a precipitous reduction of estrogen with preserved androgen production (244). The switch in sex hormone production during menopause may play a significant role in cognitive decline in aged women. Indeed, postmenopausal women reportedly demonstrated distinctive hippocampal responses during verbal memory tasks compared with premenopausal women (247). Reduction in estrogen levels throughout aging and menopausal periods is suggested to be associated with increased oxidative stress and mitochondrial dysfunction, exacerbated proinflammatory responses, and reduced synaptic plasticity (249, 254, 255).

Accordingly, prominent sex differences exist in oxidative stress, as females are proposed to have overall lower levels of ROS-induced damage, reduced mitochondrial DNA damage, and higher antioxidant enzyme levels, thus being less vulnerable to oxidative insults than males (246, 249, 256). Sex differences in oxidative stress have been observed in numerous basic and clinical studies focusing on the NADPH oxidase (NOX) enzyme and on the levels of homocysteine (246, 257). NOX serves as a prominent source of ROS in the central nervous system, and its isoforms are widely distributed across brain, especially in key structures involved in learning and memory. Studies have revealed sex differences in NOX in vitro and in vivo, with males having overall higher NOX levels (249). Furthermore, NOX signaling can be modulated by androgens, suggesting that NOX-induced oxidative stress may be a potential mechanism in sex differences observed in neurodegenerative diseases. Homocysteine, a nonprotein α -amino acid, has been associated with high levels of oxidative stress (246). Reportedly, homocysteine can increase NOX-mediated superoxide production. Like NOX, sex differences in homocysteine levels were observed with higher levels in men than in women (246, 257). Sex hormones can influence homocysteine levels, implying that their changes may underlie the observed sex differences in oxidative stress generation. In addition, a sex-driven expression of both SOD and GPx was found in the brain, with males having lower levels of these antioxidant enzymes than females (256).

Sex differences in oxidative stress persist and worsen with aging in both animal and clinical samples, where males continue to have higher oxidative stress than females. However, aging, and specifically menopause, plays a significant role in the increase of oxidative stress in females (249, 257). Intriguingly, a study postulated that a moderate increase of ROS levels may play a role in redox homeostasis by protecting the cell through a preconditioning process, which posits that exposure to a small insult allows better tolerance of a subsequent larger insult. In this scenario, physiological levels of

testosterone can slightly increase oxidative stress and be neuroprotective against damage from subsequent exposures to oxidative stress (246).

Although major progress has been made in understanding AD pathogenesis, there has been a lack of attention until recently on sex differences. Unlike the healthy aging population, several lines of data strongly indicate that female AD subjects are more affected by disease processes than male patients (248, 251, 258, 259). Women with AD showed a greater decline in essentially all cognitive domains than men with AD. Data from the Alzheimer's Disease Neuroimaging Initiative (ADNI) study suggested that the rate of cognitive decline in women is twice as fast as that of men even after correction for the ApoE genotypes (247, 260). Multimodal imaging of brain indicates that in females biomarkers of the preclinical phase of AD, including failures in cerebral glucose metabolism and the decrease in neuron mitochondrial function, appear early and overlap the endocrine transition of perimenopause (253, 261). Neuropathological studies evidenced a more severe A β accumulation in women and increased levels of tau pathology in men (253, 262, 263). These pathological hallmarks are reflected in most AD animal models (249, 262, 264–266). Sex differences have also been observed in treatment responses of AD patients in clinical trials (265, 267, 268). The major biological factors that drive the observed sex differences in AD include differences in age-related sex hormone reduction, the incidence of genetic risks (ApoE, etc.), the impact from risks of other diseases (diabetes, depression, cardiovascular disease), and the age-related declines in brain volume and brain glucose metabolism (247, 252, 258, 263). In addition, sex differences also were observed to influence levels of OS and brain redox regulation during the development of neurodegenerative diseases. Estrogen and testosterone changes may directly or indirectly account for differences between sexes in the increase of OS during neurodegenerative diseases (246, 249). Indeed, estrogens decrease the generation and secretion of A β peptide by inhibiting the gene expression of β -secretase1, by regulating APP processing on cell membranes, and by modulating γ -secretase and Notch signaling (253). Therefore, the loss of estrogen production may explain the more prominent A β neuropathology observed in women but also the switch toward an A β -driven prooxidant state, which underlies AD development. Furthermore, estrogens may facilitate the turnover of oxidized and damaged proteins through upregulation of microglia proteasome activity and by preventing the induction of nitric oxide (253). Indeed, increased NOX activity has been linked with AD progression and individuals converting from cognitively intact to dementia status (148).

In addition, the association between homocysteine and AD has been reported (246). Homocysteine can increase oxidative stress and cell loss in the hippocampus, and

elevated homocysteine has been shown to contribute to dementia and AD progression (246). Homocysteine levels reportedly are higher in men than women, supporting the increased levels of OS. However, homocysteine levels increase with age, and further with menopause; therefore studies on men and women over the age of 50 yr demonstrated no differences for homocysteine levels between healthy women and men (269). Sexually dimorphic factors linking OS with AD development and progression also include the levels of glutathione, alteration of mitochondria, and the levels of metal ions in the brain (254, 255, 260, 270). Furthermore, epidemiological studies pointed to an association between metabolic syndrome and AD in women but not in men. The control of cholesterol and lipid metabolism is sexually dimorphic, and the postmenopausal reduction of circulating estrogens alters liver lipid metabolism, with severe systemic consequences (253). Excessive circulating lipids cause neuroinflammation and oxidative damage, both associated with cognitive impairment.

4. OXIDATIVE DAMAGE IN ALZHEIMER DISEASE

4.1. Lipid Peroxidation Studies

Lipids and proteins are major targets of ROS, which, by altering their conformation and functional role, promote in the brain the development of neurodegenerative processes. Lipids are fundamental to essentially every living system, and their role extends from membrane structure and energy storage to intercellular signaling, growth regulation, cellular differentiation and commitment, and cellular homeostasis (271). Specifically, in the brain, lipids take part in a wide variety of vital tasks such as neuron myelinating and signal transduction via lipid mediators as first or second messengers (272). Recently, the interest of researchers in lipids has grown considerably because of their wide applications that range from the development of novel systems for lipid-mediated drug delivery to the brain to the identification of reliable biomarkers for neurodegenerative diseases (273).

In the central nervous system (CNS), the most representative lipid subtypes are glycerophospholipids, sphingolipids, and sterol lipids. A disturbance in the dynamic equilibrium of lipids has been associated with damage to cell structure and increased inflammation and OS leading to neural impairment (274). Lipidomics is a method for the identification and quantification of lipids across cell types, organs, or full organisms, and its application in diseased brain allowed unveiling mechanisms that associate lipids with neurodegeneration (275). Lipid dysregulation has been largely associated with AD and other neurological

disorders (276, 277). Lipidomic analyses highlighted the similar presence of an aberrant lipid profile in the brain and in the peripheral tissues of diseased individuals (278). The principal mechanisms that link lipid metabolism with neurodegeneration in AD include neuronal signaling pathways, blood-brain barrier (BBB) disruption, mitochondrial dysfunction, OS, and inflammation, which ultimately lead to synaptic loss and memory impairment (279, 280).

A major component of the evidence that implicates lipids in AD pathogenesis is the presence of the *APOE* $\epsilon 4$ genetic allele, which represents the highest genetic risk factor for late-onset AD (281–283). In the brain, ApoE is the primary cholesterol carrier protein that facilitates delivery of cholesterol from astrocytes, where it is synthesized, to neurons. It is still not clear how possession of the *APOE* $\epsilon 4$ allele increases AD risk, but ApoE appears to be required for A β aggregation and ApoE $\epsilon 4$ has been shown to be the most efficient at promoting such oligomerization (284). To date, studies of cholesterol levels in AD have produced conflicting results. Some studies have found increased levels of cholesterol in AD, whereas other reports have not (285, 286). Despite such reports, hypercholesterolemia is still considered an early risk factor for developing AD and has been associated with impaired memory recall in the elderly (287, 288). Additional strong evidence concerning the involvement of lipids in the pathogenesis of AD includes the alteration of phospholipid, plasmalogens, ceramide, ganglioside, and sulfatide brain composition (274, 289). However, the alteration of phospholipid metabolism in AD is also observed in the blood, thus encouraging discovery studies for blood-based lipid markers (273, 290).

Moreover, increased OS leads to lipid peroxidation (LPO), and increased LPO products have been detected in the brain of AD patients. LPO generates the formation of acrolein, 4-HNE, and malondialdehyde (MDA), which are harmful toxic species for the brain (274). Acrolein, which is predominantly localized within NFTs in AD brain, directly attacks DNA, reacts with the DNA base guanine, and forms acrolein-deoxyguanosine, which is excessively present in the AD brain (291). Increased levels of HNE-protein adducts, which alter protein structure and functionality, have been detected in the diseased brain regions of AD patients (27, 125). However, MDA is the most abundant LPO product, and significantly higher levels of MDA are found in AD patients compared with healthy subjects (292). Additional LPO products include F_2 -isoprostanes (F_2 -iPs), which are a group of prostaglandin $F_2\alpha$ -like compounds derived from the nonenzymatic oxidation of arachidonic acid. F_2 -iPs are elevated in AD brain regions rich in the typical lesions such as frontal/temporal cortex and hippocampus but not in cerebellum, an area typically lacking senile plaques or NFT

(293, 294). Similarly increased levels of F₂-isoprostanes and F₄-neuroprostanes (derived from neuron-resident docosahexaenoic acid) are observed in postmortem cerebrospinal fluid (CSF) from AD and MCI patients compared with age-matched control subjects and in brain tissues from MCI patients (294). CSF F₂-iPs, in combination with other molecules, have been proposed as predictive markers of AD onset and progression because of their ability to discriminate AD from other dementing disorders and among AD staging. Increased levels of LPO products were observed also in serum/plasma from AD subjects, supporting their use as biomarker for disease diagnosis, prognosis, and therapy (294, 295).

The presence of HNE protein adducts has been demonstrated in blood-derived samples from AD patients (296). Furthermore, increased levels of circulating oxysterols, oxidized derivatives of cholesterol, such as 24-hydroxycholesterol and 27-hydroxycholesterol, have been implicated in the pathogenesis of AD. Both oxysterols have been proposed as clinically relevant biomarkers (297). In addition, elevated levels of circulating hydroxyoctadecadienoic acids, which are released from phospholipids by phospholipases, were found in the plasma and erythrocytes of patients with AD, suggesting a potential role in disease pathogenesis and/or diagnosis (292).

4.2. Redox Proteomics Studies

Proteomics techniques offer the simultaneous detection of hundreds to thousands of proteins in a single experiment and can provide important information regarding protein identification, quantification, and cellular localization as well as protein interactions and structures (298). Proteomics may give insight into posttranslational events that occur in the cell resulting in posttranslational modification (PTM) of proteins (298). Proteins are broadly targeted by ROS/RNS, which leads to a change in the structure and/or function of the oxidized proteins (12). Investigations of oxidative PTMs (ox-PTMs) that occur in vitro or in vivo are currently being performed with focused redox proteomics techniques (299–301). Redox proteomics is a subset of proteomics methods used to identify oxidized proteins and determine the extent and location of ox-PTMs in the proteomes of interest. Initial studies, pioneered in the Butterfield laboratory, aimed to provide new knowledge and better comprehension of the role of OS in the onset and progression of selected neurodegenerative disorders (298, 301).

Redox proteomics studies led to the identification of a large number of oxidatively modified proteins from brain tissue, biofluids, or other biological samples, potentially involved in the pathogenesis and/or progression of neurodegenerative diseases characterized by increased OS (29, 296). Redox proteomics adopts many of the methods of expression proteomics (i.e., separation of proteins of

interest) and employs mass spectrometry as the major platform to achieve the goals of identifying the target proteins (298, 302). Once identified, oxidatively modified proteins can be placed in specific molecular pathways to provide insights into protein oxidation and human disease. Thus, the use of redox proteomics allowed the recognition of specific patterns of oxidative damage during the progression of neurodegenerative disorders often common between different diseases or pathological stages (6, 124, 133, 303). The most common ox-PTMs that can be easily identified by the redox proteomics approach involve 1) carbonyl formation, 2) covalent adducts of 4-HNE to Cys, Lys, or His residues, 3) nitration of Tyr residues, and 4) nitrosylation of Cys residues.

Accumulating data have provided compelling evidence for the involvement of perturbations of redox homeostasis and the resulting oxidative damage to proteins in physiological aging and degenerative processes occurring in age-related diseases (132, 298, 301). However, definitive evidence for the links among oxidative damage and pathophysiological mechanisms has often been lacking because of the recognized shortcomings associated with available experimental methods. The redox proteomics analysis of postmortem brains from subjects affected by AD or earlier pathological stages allowed delineation of the involvement of the ox-PTMs of proteins involved in specific cellular functions in the cause-and-effect relationship between increased OS and brain degeneration (5, 6, 299). Furthermore, the results obtained in human AD brains provide a map of specific oxidized proteins, which could be useful as biomarkers for diagnostic and therapeutic purposes. Overall the analysis of hippocampus, inferior parietal lobule (IPL), and frontal cortex regions led to the identification of a similar set of oxidatively modified proteins (by PC, 3-NT, or HNE) in AD, EAD (a transitional stage between MCI and late-stage AD), and MCI subjects, proteins that play important roles in regulating energy metabolism (e.g., Eno 1, pyruvate kinase, ATP synthase), antioxidant and detoxification systems (e.g., peroxiredoxin, MnSOD), proteasomal activity (e.g., UCH-L1), cytoskeletal integrity (e.g., DRP-2, β -tubulin), protein folding (e.g., HSP70), and cellular communication and signaling (e.g., neuropeptide h3) (18, 19, 28, 124, 199, 304–309) (TABLE 4). The oxidative dysfunctions of the listed proteins correlate with the clinical symptoms, pathology, and/or biochemistry of this disorder, supporting the significance of the obtained data and the utility of the redox proteomics technique.

The in-depth analyses of the results revealed that the proteins enolase, neuropeptide h3, ATP synthase, and fructose bis-aldolase (FBA) were identified as oxidized in brain from subjects at each stage of AD, late-stage AD, EAD, and MCI, consistent with the concept

that their oxidation is an early event in and may contribute to progression of AD. The identification of common targets of ox-PTMs is consistent with the notion that the loss of function of these proteins represents a key mechanism in both pathogenesis and progression of AD (133, 298). In agreement with this concept, the analysis of brains from preclinical AD (PCAD) subjects, who represent an early phase of AD before clinical signs and symptoms appear, led to the identification of oxidized proteins such as HSP90 and Enolase 1, whose alterations might contribute to the transition toward MCI and memory loss (317). Within this context, the results collected in the brain from different AD stages contributed to the identification of selective metabolic networks that are impaired by oxidative damage and how these possibly translate into clinical symptoms. Furthermore, data obtained by the analysis of progressive stages of the disease strongly confirm that OS is an early event in the pathogenesis of AD, which plays a relevant role in the conversion toward later stages. Thus, in the ongoing discussion regarding the involvement of OS as a critical mechanism leading to AD or as a consequence of AD neuropathology, redox proteomics results support both possibilities.

In the last decade, the redox proteomics approach was applied also to biological fluids with the intent of identifying detectable biomarkers of OS that may reflect brain damage and provide insights into the onset of AD pathology and follow its progression (296, 311, 313). However, the analysis of protein oxidation in biofluids represents a significant experimental challenge because of 1) the lack of a direct link between tissue-specific oxidation and systemic oxidative damage; 2) the difference of proteome profile composition between brain and blood; and 3) the concomitant presence of confounding events at the peripheral level associated with complications of the pathology or comorbidities present. Indeed, all these conditions could contribute to a biased detection of oxidative biomarkers in the periphery, limiting the specificity and the accuracy of the analysis (296, 302). Therefore, reports concerning the analysis of protein oxidation in biofluids from living patients are few. Initial reports on protein oxidation in CSF samples demonstrated increased protein glycation, oxidation, and nitration in subjects with AD. The specific analysis of CSF samples from amnesic MCI and AD subjects identified the early carbonylation of extracellular chaperones, suggesting that dysfunction of these pathways is initiated many years before severe dementia is diagnosed and might represent a reliable biomarker for pathology onset and progression (311, 318, 319). Regarding protein oxidation analysis in blood-related samples from AD subjects, different studies defined a limited array of proteins increasingly oxidized, including Grp78, hemopexin,

transferrin fibrinogen γ -chain precursor, α 1-antitrypsin precursor, haptoglobin, and α 2-macroglobulin (313, 314, 320). Data from peripheral biofluids suggest on one side the systemic derangements in heme/iron/redox homeostasis and activation of the acute-phase response in sporadic AD, and on the other side the alterations in proteins acting as extracellular chaperones, which may contribute to exacerbating A β neurotoxicity.

Recently, the redox proteomics approach was applied to 3xTg-AD mice showing increased protein carbonylation for proteins crucial in glucose metabolism, protein folding, cell structure, signal transduction, and excitotoxicity, as observed in AD brain (321). In addition, markers of oxidative damage also were found to be elevated in mitochondria isolated from lymphocytes from AD patients compared with age-matched control subjects (322, 323).

4.3. Protein Quality Control Systems and Protein Oxidation

Protein homeostasis or “proteostasis” is the processes that regulate the homeostasis of the intracellular pool of functional proteins. Indeed, to maintain cell health, proteins must be accurately synthesized, folded with high fidelity, assembled, correctly localized, and properly degraded. Nonetheless, protein folding is intrinsically error prone since $\sim$ 30% of newly synthesized proteins could be affected by misfolding and aggregation. If a protein fails to fold correctly, the cell utilizes extensive security measures to maintain cellular function. Chaperones attempt to remedy the unfolded protein first and, if unsuccessful, activate cellular programs, including the UPR, heat shock response (HSR), and ER-associated degradation (ERAD), to induce processes to either fix the problem or destroy the unfolded proteins (324, 325). When the UPR fails, misfolded proteins are targeted for degradation through the ubiquitin-proteasome system (UPS) and autophagy (326, 327). These specialized mechanisms are known as protein quality control (PQC) systems and maintain or reestablish proteostasis in case malfunction signaling networks are triggered by a variety of stress sensor molecules. The preservation of proteostasis plays a crucial role in cell survival and becomes even more important for nondividing cells such as neurons, whose proteostasis machineries are reduced with aging, causing an accumulation of damaged organelles and misfolded proteins (327–329).

The initiation and progression of AD is characterized by disturbances in intracellular proteostasis due to the alteration of protein synthesis, folding, surveillance, and degradation, which trigger the accumulation of altered proteins and toxic aggregates, widely recognized to be implicated in this phenomenon (327, 330–332). The induction and functionality of PQC systems and the

Table 4. Oxidatively modified proteins from brain and other tissues from Alzheimer disease subjects at different stages of the disease revealed by redox proteomics

Pathology Stage	Tissue	Oxidative Modification	Proteins*	Reference(s)
Late Alzheimer disease (LAD)	Brain (hippocampus, inferior parietal lobule, frontal cortex)	Carbonylation	CKBB, GS, UCH-L1, DRP-2, ENO1, HSC71, Pin1, PGM1, TPI, γ -SNAP, CA, MDH, GDH, 14-3-3 ζ/δ , FBA A/C	(18,19, 199, 305, 309)
		Nitration	ENO1, LDH, neuropolypeptide h3, TPI, β -actin, GAPDH, ATP synthase β -chain, CAII, VDAC	(308, 310)
		HNE bound	ATP synthase, GS, MnSOD, DRP-2, ENO1, Aco, ALDO, PRX6, α - tubulin	(306)
	CSF	Carbonylation	Gelsolin, serotransferrin, VDBP, α -1-antitrypsin, α -1B-glycoprotein, ApoE, prostaglandin-H2 D-isomerase, λ chain	(311, 312)
	Plasma	Carbonylation	Hemopexin, transferrin, fibrinogen γ chain precursor, α 1 antitrypsin precursor, Hp β -chain, α 2 macroglobulin	(313–315)
Early Alzheimer disease (EAD)	Brain (inferior parietal lobule)	Carbonylation	PGM1, FBA-C, and GFAP	(307)
		HNE bound	MnSOD, DRP2, ENO1, MDH, TPI, and F1 ATPase, α -subunit	(27)
Mild cognitive impairment (MCI)	Brain (hippocampus, inferior parietal lobule, frontal cortex)	Carbonylation	ENO1, GS, PK/M2, Pin1, HSP 70, CAII, SBP1, MAPK I	(304, 307)
		Nitration	MDH, ENO1, GRP precursor, aldolase, GST Mu, MRP-3, 14-3-3 γ , PRX 6, DRP-2, fascin 1, HSPA8	(316)
		HNE bound	LDH B, PGK, HSP 70, ATP syn, PK, β -actin, EF-Tu, eIF α	(26)
	CSF	Carbonylation	Gelsolin, VDBP, α -1-antitrypsin, α -1B-glycoprotein, ApoE, Prostaglandin-H2 D-isomerase	(311)
	Plasma	Carbonylation	Fibrinogen γ chain precursor, Retinol-binding protein 4, Inter- α trypsin inhibitor.	(313)

*Protein acronyms are explained in the text. CSF, cerebrospinal fluid; HNE, 4-hydroxynonenal.

preservation of proteostasis are closely associated with the redox balance of the brain. Low amounts of ROS can activate adaptive cellular machinery to increase the organism's stress resistance. This response involves the enhancement of antioxidant strategies and the induction of UPR, UPS, and autophagy. Conversely, a chronic increase of ROS can either overwhelm the responsive capacity of these pathways or induce the oxidation of

specific components of UPR, UPS, or autophagy, thus exacerbating the accumulation of unfolded/ misfolded proteins (333, 334) (FIGURE 9). Summarizing the above and as discussed more below, a link between protein homeostasis and redox balance illustrates that increased OS, protein aggregation, and the failure of protein degradation pathways have mutual pathological roles.

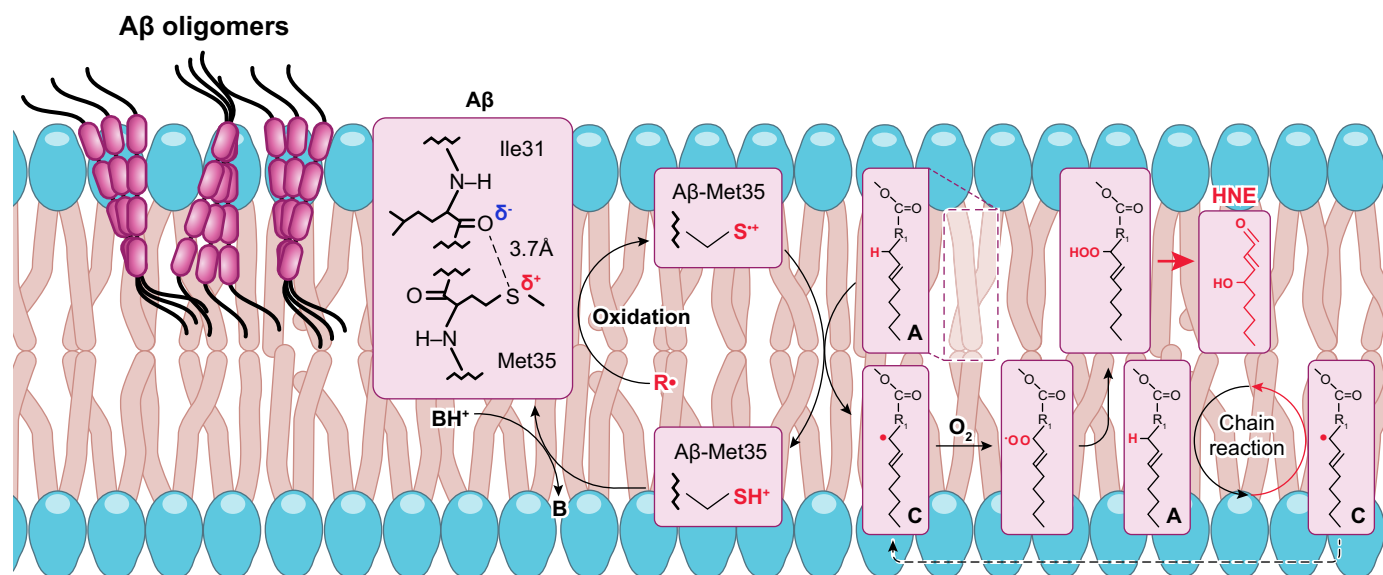


FIGURE 9. Aβ42-resident sulfuranyl free radical formation coupled to lipid peroxidation and 4-hydroxynonenal (HNE) formation in neuronal membranes. A 1-electron oxidation of the S-atom on Met35 by, for example, paramagnetic oxygen [which would lead to the known production of superoxide free radical (that would easily convert to hydrogen peroxide) associated with Aβ42 (85)] leads to a sulfuranyl free radical formation on the Met35 residue of Aβ42. This sulfuranyl free radical initiates processes of lipid peroxidation in neuronal membranes that, in turn, lead to formation of neurotoxic HNE from the lipid hydroperoxide, LOOH. HNE is an α,β-unsaturated aldehyde that can covalently bind to and alter structure of proteins by a Michael addition reaction, causing diminished or loss of protein function and neuronal death. Note that a carbon-centered free radical on a lipid unsaturated fatty acid is formed after abstraction of a labile H-atom both initially by the sulfuranyl free radical and again during the propagation steps of a chain reaction, thereby continuing the processes of lipid peroxidation and greatly amplifying the level of HNE formation. Note also that the SH⁺ moiety on Met35 is an acid whose pK_A is −5, meaning that any base B can easily remove the H⁺ to reform Met itself. That is, this scenario is also a catalytic reaction. A J20 Alzheimer disease (AD) mouse model with the Met631 of amyloid precursor protein (APP) (which is the Met35 residue of Aβ42) substituted by Leu no longer shows any evidence of elevated lipid peroxidation in brain at 1 yr of age, in marked contrast to the J20 mouse itself (95), consistent with this proposed mechanism. See text for further discussion.

4.3.1. The unfolded protein response.

The UPR is an adaptive reaction that aims to reestablish homeostasis under stress conditions by restoring the cell's capacity to produce properly folded proteins. If the homeostatic balance is not restored after UPR induction, i.e., if acute UPR remains active for a prolonged time, the cell undergoes apoptosis and the UPR becomes chronically activated. The activation of the UPR begins by the dissociation of glucose-regulating proteins (GRPs) from three types of ER transmembrane anchors, namely inositol-requiring protein 1 (IRE1), activating transcription factor 6 (ATF6), or protein kinase RNA-like ER kinase (PERK) (6). Once disconnected from the membrane, GRPs associate with nascent proteins to assist their folding and secretion from the ER (335). Meanwhile, each anchor, IRE1, ATF6, and PERK is free to initiate its own signaling pathways.

Specific markers of UPR activation are increased in AD brain tissue compared with nondemented control brain. BiP/Grp78 expression levels are increased in the hippocampus and temporal cortex of AD, and several studies have reported the increased presence of phosphorylated PERK, IRE1, and eIF2α in AD neurons (82, 336, 337). Overall, the levels of BiP/Grp78 and the occurrence of

pPERK in AD neurons correlate well with the presence of abnormally phosphorylated Tau and Braak staging of NFTs (133). ER stress-mediated cytotoxicity was also observed when cells were exposed to aggregated proteins of different sources. Supporting this view, Aβ-mediated cytotoxicity was exacerbated in cell lines compromised in specific UPR activation pathways, including PERK or XBP1 (338). These observations imply that the UPR may play a prominent role in the development of pathogenic AD mechanisms from early stages of the disease.

The adaptive response induced by UPR can modulate ROS production within the ER by reducing the folding demand and upregulating the expression of antioxidant factors (339). The control of the redox balance by UPR is essentially linked to IRE1 and PERK pathways through the induction of Nrf2-related antioxidant response and ATF4, which play key roles in GSH synthesis (340, 341). However, studies on AD brain have highlighted that under a condition of chronic UPR activation an uncoupling between PERK and Nrf2 signals occurs (82). These data support the notion of the inability of UPR to induce stress responses, favoring the buildup of prooxidant species and the damage of biological components. Consistent with these ideas, redox proteomic studies reported increased HNE modification of

BiP/Grp78 during AD pathology (342), which may cause the failure to bind to misfolded proteins, thereby contributing to the dysfunction of the UPR. In addition, the oxidation of eIF2 α , observed in MCI IPL, might lead to its impaired activity leading to UPR defects and exacerbating the accumulation of unfolded/misfolded proteins (26) (FIGURE 2).

4.3.2. The ubiquitin proteasome system.

The UPS represents one of the main pathways of protein clearance in eukaryotic cells, which not only digests misfolded, oxidized, or damaged proteins but also eliminates proteins involved in a plethora of intracellular processes (3). Most of the proteins are targeted for proteasomal degradation after being covalently modified with ubiquitin, a small protein with 76 amino acids, which is conjugated through the formation of an isopeptide bond between the ϵ -amino group of a lysine residue of the substrate and the COOH-terminal carboxylate (343). The polyubiquitin chains are recognized by the proteasome, a multicatalytic complex indicated as the 26S proteasome.

The 26S proteasome is composed of a 20S catalytic core and two 19S regulatory caps on both ends of the 20S core. The 20S proteasome contains three proteolytic activities including chymotrypsin-like, caspase-like, and trypsin-like activities. The 19S proteasome binds and unfolds ubiquitinated proteins and opens the entry gate of the 20S proteasome to allow protein in the central cavity (344). Whereas the 26S proteasome degrades polyubiquitinated proteins, the 20S proteasome by itself seems to be sufficient to degrade nonubiquitinated oxidatively modified proteins in an ATP-independent manner (345). OS and stimulants can lead to the assembly of an inducible form of the proteasome, also called the immunoproteasome, and accumulating evidence supports the idea that the immunoproteasome preferentially degrades oxidized proteins with high activity and selectivity (344, 346, 347). The activity of the proteasome is significantly decreased in the hippocampus, middle temporal gyri, and inferior temporal lobe of AD subjects (331, 348, 349). Defective UPS proteolysis induces synaptic dysfunction. Furthermore, disruption of the UPS correlates with several AD pathological features, such as A β accumulation, pTau, and autophagy impairment (350).

The formation of pTau oligomers at pre- and postsynaptic terminals reportedly is associated with increased ubiquitinated substrates in AD brain, and pTau oligomers mediate UPS proteotoxicity, thus impairing synaptic function (351). Subsequently, studies conducted in vitro indicate that defective proteasome activity disturbs A β degradation consistent with accumulation of this neurotoxic peptide and plaque formation (352). In turn, increased A β inhibits

the proteasome functionality (353). Nonetheless, the proteasome is involved in APP and APP-COOH-terminal fragment (CTF) metabolism and affects APP amyloidogenic and nonamyloidogenic processing. Furthermore, increased OS/NS levels were suggested as one of the major factors contributing to the physiological and pathological alteration of the UPS. Indeed, if moderately oxidized proteins are recognized by the UPS and it stimulates their clearance, severely oxidized proteins cannot be easily degraded and may inhibit the proteolytic activity of the proteasome, as observed in AD pathology. Accordingly, experimental data indicate that proteins severely oxidized are more resistant to proteasome degradation (133). Furthermore, a significant reduction in proteasome-mediated degradation of oxidized proteins in brains from MCI and AD subjects was associated with the increase of ox-PTMs such as carbonylation, HNE modification, and neuroprostane conjugation, thereby confirming a role for OS as a contributing factor (354). In addition, in AD brain both 20S and 19S proteasomal complexes are targets of ox-PTMs, including the formation of HNE adducts, carbonylation, S-glutathionylation, glycoxidation, as well as ADP-ribosylation and phosphorylation (346, 355, 356). The proficiency of the UPS also depends on the activity of deubiquitylation enzymes that play an essential role in determining the rate of protein clearance in cells. Among deubiquitylating enzymes, ubiquitin carboxy-terminal hydrolase-L1 (UCH-L1) was found to be oxidatively modified in AD brain, and its activity was also found to be reduced (18, 305, 342, 357). Overall, the overload of oxidized substrates together with the oxidative damage to proteasome lead to the accumulation of abnormal proteins and to selective degeneration of neurons, which characterize the AD condition (FIGURE 10).

4.3.3. Autophagy.

Autophagy is a cellular mechanism that removes degraded/dysfunctional/misshaped components and plays a key role in preserving cell metabolic balance and in cell survival (358). Autophagy is categorized into microautophagy, chaperone-mediated autophagy (CMA), and macroautophagy, which differ mainly in the manner of cargo delivery to the lysosome. Among the autophagic mechanisms macroautophagy is the most well-characterized process, by which bulk cytoplasmic components are sequestered in a double-membrane structure known as autophagosome, which is successively trafficked to the lysosome for degradation. In contrast, in microautophagy and CMA cargo is directly taken up by the lysosome, either through the invagination of the lysosomal membrane or by unfolding and translocation of proteins with a specific signal sequence (327, 332, 359).

The autophagic process is primarily controlled by two crucial signaling proteins: the mammalian (a.k.a.

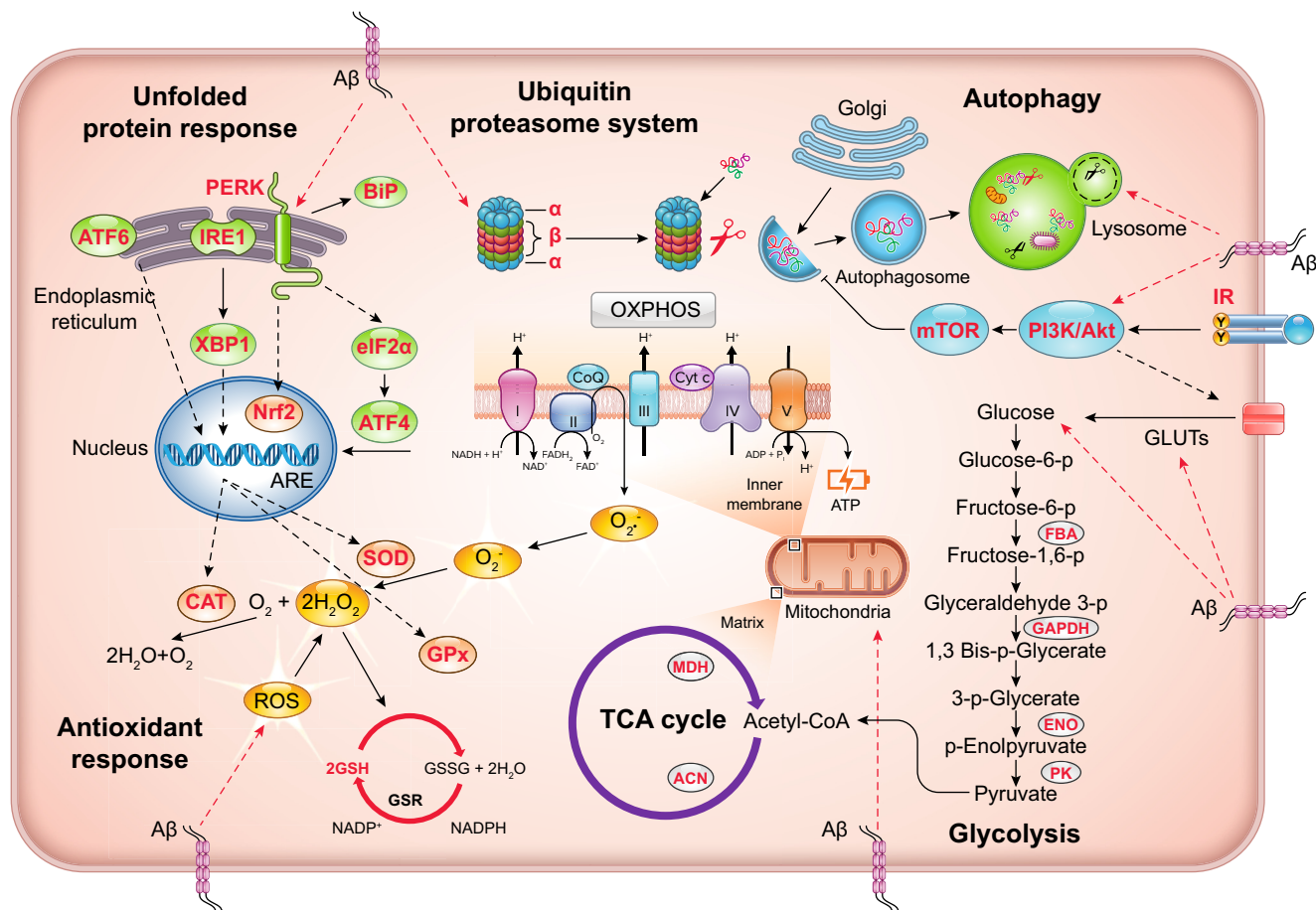


FIGURE 10. Intracellular pathways impaired by protein oxidation and Aβ increase in Alzheimer disease. Among these functionally altered proteins, those associated with the impairment of energy metabolism, including glycolysis, Krebs cycle (TCA), and oxidative phosphorylation (OXPHOS), the reduction of antioxidant responses (AORs), and the failure of protein quality control [unfolded protein response (UPR), ubiquitin-proteasome system (UPS), and autophagy] have been identified to be oxidatively modified. Furthermore, all these pathways are closely connected, as, for example, ATP levels are crucial for the efficiency of protein quality control that, if impaired, exacerbates redox reactions resulting in the accumulation of oxidized/toxic materials. ARE, antioxidant response element; CAT, catalase; GLUT, glucose-specific transporter; GPx, glutathione peroxidase; IR, insulin receptor; mTOR, mammalian target of rapamycin; PERK, protein kinase RNA-like ER kinase; PI3K, phosphoinositide-3-kinase; ROS, reactive oxygen species; SOD, superoxide dismutase.

mechanistic) target of rapamycin complex 1 (mTORC1) and the AMP-activated protein kinase (AMPK). Under normal nutrient circumstances, active mTORC1 phosphorylates and sequesters Ulk1 in a complex with Atg13 and FIP200, thereby inhibiting autophagy. Starvation, amino acid deprivation, or growth factor release, by removing the mTORC1 restraint, allows Ulk1 to promote autophagy. AMPK is a major positive regulator of autophagy, which is activated by a high AMP-to-ATP ratio or by upstream kinases, such as LKB1 or CamKII (360, 361). Once activated, Ulk1 initiates autophagosome nucleation through the class III phosphatidylinositol 3-kinase (PtdIns3K) complex. PtdIns3K, together with other Atg-related proteins, recruits Atg12-Atg5-Atg16L1 and the light-chain 3 (LC3) ubiquitin-like conjugation systems necessary for the expansion, elongation, and maturation of autophagosomes (362). Completing the process, the autophagosome fuses with the lysosome,

forming the autophagolysosome in which the acidic environment for cargo degradation is maintained by the activity of the lysosome-resident vacuolar [H⁺]-ATPase (V_o-ATPase) proton pump. Lysosomal hydrolases, such as cathepsin B, D, and L, are involved in the cleavage of autophagic substrates, whereas the resulting molecules are transported back to the cytosol for nutrient recycling (363).

Defective autophagy has been observed during aging and neurodegenerative disorders, supporting its fundamental role in advancing the progression of brain damage and cognitive decline (359). Nixon and colleagues (364, 365) first reported the pathological evidence of defective macroautophagy in EM images in the AD brain, suggesting that defective autophagy represents one of the pathogenic features of AD. Indeed, robust autophagic vesicle accumulation in dystrophic neurites from biopsy tissues of patients with AD indicated a compromised state of autophagic flux.

The accumulation of autolysosomes in the brain from AD patients mirrors the condition of compromised lysosomal proteolysis obtained via genetic knockdown of specific cathepsins or use of pharmacological inhibition of lysosomes (364–366). Additional studies suggested that reduced autophagy in AD brain and animal models is likely caused by the hyperactivation of the PI3K/Akt/mTOR axis. Additional lines of evidence support the alteration of macroautophagy in AD as demonstrated by the reduction of Beclin 1 and of LC3II/I levels (367–369). Consistent with this result, the pharmacological inhibition of mTOR activity in the brain of AD mice was shown to restore autophagy and to reduce A β and Tau levels (370–373). In AD the autophagic flux reportedly is impaired in neurons at the final stages of the process of fusion of autophagosomes with lysosomes (374, 375). ROS induce autophagy and autophagy, which, in turn, operates to reduce oxidative damage (329, 373, 376). Mild OS conditions redox-mediated signaling or moderate oxidation of macromolecules upregulate autophagic flux, leading to the degradation of impaired proteins and affected organelles. In contrast, persistent increased OS might become toxic for functionality by targeting autophagic components and impairing the process at different steps (FIGURE 10). Redox proteomic analysis of brain samples demonstrated that increased OS affects the final degradative process of autophagy, which relies on the activity of the vacuolar [H⁺]-ATPase (V₀-ATPase) and cathepsin D, both oxidized in AD individuals and in DS persons at risk of developing AD (233, 298, 342, 357). In addition, V₀-ATPase is necessary for amino acids to activate mTOR, thus supporting the view that V₀-ATPase is an indirect, but integral, component of the mTORC1 pathway (133). Apart from the initial and rapid increase in the autophagic flux mediated by posttranslational protein modifications, a delayed and extended autophagic response also relies on the activation of specific transcription factors, such as, Nrf2, p53, or FOXO3a, which respond to increased OS (358, 377). Notably, defects in mitophagy also were reported at the onset and progression of neurodegenerative diseases. Indeed, the reduced expression of mitophagy proteins, such as SIRT1, SIRT3, PGC1, and Nrf2, was observed in AD, and this effect was associated with defects in mitochondrial biogenesis and response to OS (229, 231).

4.4. Antioxidant Response: Endogenous Enzymes, Glutathione, and Nrf2 Response

4.4.1. Endogenous enzymes.

As noted above, SOD1/2 (Cu/Zn and Mn, respectively), CAT, and GPx catalyze the dismutation of O^{•−} to O₂ and H₂O₂ and the subsequent decomposition of the latter to O₂ and H₂O. SODs are among the most studied

antioxidant enzymes in AD and may be induced or consumed by increased OS. Studies of SOD activity in AD have resulted in conflicting reports. Immunoreactivity of both SOD1 and SOD2, as well as catalase, is reportedly increased in regions of neurofibrillary tangles and plaques. High levels of SOD1 protein also were observed in large pyramidal neurons of the association cortex and hippocampus in normal subjects (74). In contrast, a 25–35% reduction of SOD in frontal cortex, hippocampus, and cerebellum from AD subjects was reported (1). However, other studies find no difference in SOD expression between AD and control subjects (378). As observed in the brain, also in peripheral AD samples, the analysis of SOD is controversial, and some authors found significantly lower erythrocyte and plasma/serum levels whereas others found significantly higher erythrocyte or plasma levels (378, 379). Contradictory studies observed increased CAT activity in the amygdala of AD and later the reduction of its activity in the parietotemporal cortex, amygdala, and basal ganglia in AD. The analysis of GPx in AD initially reported elevated GPx activity in AD hippocampus; however, other authors reported no difference in GPx level between AD and control subjects or lower GPx activity in AD (378, 380). Taken together, the studies on antioxidant enzymes in AD demonstrated no consistent major deficiencies in their detoxifying processes, consistent with the idea that the mechanism of neuron death in AD is likely not a failure of these defenses against free radicals.

4.4.2. Glutathione.

Glutathione (GSH) imbalance/depletion has been reported to be involved in many brain disorders, including AD (72)]. Deficit in GSH may precede the neuropathology of these diseases, and neuronal GSH depletion may be one of the primary causes of initial brain damage in AD. The GSH redox imbalance during the onset and progression of AD and a linear correlation between the GSH-to-GSSG ratio and cognitive performances, assessed by Mini-Mental Status Examination, was observed in patients (73). Furthermore, the GSH-to-GSSG blood ratio was proposed as a reliable biomarker of increased OS during the onset and progression of AD (378). In MCI hippocampus, the ratio of GSH to GSSG was reported decreased, consonant with increased OS. Furthermore, GST activity was found to be reduced, but no significant differences in GPx or GR enzyme activity were noted (73).

Brain GSH monitoring with magnetic resonance spectroscopy (MRS) also is considered an important tool to support the diagnosis of AD. The analysis of GSH contents in different brain regions of AD patients showed the reduction in the right frontal cortex of women and in the left frontal cortex of men compared with matched

control subjects (72). Redox proteomics studies highlighted also that GST is a target of oxidation, with this enzyme being increasingly carbonylated in *C. elegans* expressing A β 42 and in a canine model of aging. GST was also found to be nitrated in the IPL from MCI patients and significantly elevated in the AD hippocampus, causing a decrease of its activity (124, 189, 381). Since GSTs catalyze the conjugation of HNE to GSH, the decline of its activity consequently leads to a compromised detoxification process of HNE and an accumulation of HNE-modified proteins. A further family of antioxidants particularly affected by OS and nitrosative stress are the peroxiredoxins (Prxs), which reduce H₂O₂ (124) and present a redox-regulated chaperone activity. Prx2 and Prx6 oxidation was found in MCI brains and associated with A β 42 in SAMP8 mice, thus supporting the idea of their reduced antioxidant activity in the development of AD (298).

4.4.3. Nrf2 response.

Nrf2 is ubiquitously expressed and in the brain is an important defense against increased prooxidant species (76). The analysis of Nrf2 response in AD brain demonstrated decreased Nrf2 nuclear localization as well as its decreased expression and phosphorylation on Ser40, thus supporting the overall reduced antioxidant activity (82). Moreover, in AD brain neurons positive for LPO markers do not contain Nrf2 in the nucleus, indicating dysfunctional Nrf2 signaling (382). In agreement, glycogen synthase kinase-3 β , which demonstrates increased kinase activity in AD, is able to directly phosphorylate Nrf2, thereby inducing its nuclear efflux. A meta-analysis of microarray datasets identified downregulated ARE genes in AD patients that may be associated with reduced Nrf2 nuclear localization (383). Similarly, in transgenic mouse models of AD, reduced Nrf2, NQO1, GCL catalytic subunit (GCLC), and GCL modifier subunit (GCLM) mRNA and Nrf2 protein levels were demonstrated (384). Loss of Nrf2 was also shown to increase levels of A β and pTau, increase glial activation, increase OS, and exacerbate cognitive decline (81). In contrast, significantly increased HO-1 expression was reported in plasma and brain from AD subjects compared with age-matched control subjects (385, 386). Increased NQO1 activity and expression were observed in astrocytes and neurons of AD brain (76).

4.5. Oxidative Stress and Dysfunction of Energy Metabolism

Despite the human brain constituting only 2% of the body's mass, its metabolism accounts for ~20% of the total glucose uptake. The high demand for energy

compared with other tissues is largely attributed to the extremely active and complex processes involved in neuronal transmission (387). This situation is reflected by the large metabolic rate of neurons and by the comparatively higher sensitivity of brain tissues to oxygen and glucose deprivation. Glucose is the principal source of energy for the brain, and it is crucial for the maintenance of neuronal survival, synaptic connections, and BBB integrity. Consequently, glucose-related functions depend on glucose transport mechanisms, regulated by insulin and insulin receptors, and on glucose utilization through cytoplasmic and mitochondrial metabolic pathways. The decreased availability and utilization of glucose leads to the development of a hypometabolic state characterized by a condition of energy crisis within the brain that may progress to atrophy of major areas and neuronal and synaptic disconnection (388, 389).

With increased age, brain glucose utilization declines to different degrees in different brain regions, and such decline is accelerated in AD, for which aging is the most important risk factor. A broad number of studies have demonstrated the reduction of glucose uptake in the brain of AD patients, and, in agreement, positron emission tomography (PET) scanning shows a consistent reduction of cerebral glucose consumption in the brain of AD patients (390, 391). Furthermore, glucose hypometabolism is present before any measurable cognitive dysfunction in AD and therefore may be an early event and marker of AD. These observations support the notion that AD is a degenerative metabolic disease in which the impairment of glucose metabolism, mediated by reduced brain glucose uptake and lowered insulin responsiveness, is closely associated with increased OS and inflammation (392). In agreement with the above, altered energy metabolism in the brain correlates with increased ER stress, impaired protein degradation pathways, altered proteostasis, and the accumulation of toxic protein products in the brain (393, 394). In addition, decreased glucose uptake is strongly associated with reduced protein O-GlcNAcylation that directly links altered brain metabolism with AD pathological hallmarks (395, 396).

Several studies on AD specimens and/or animal and cell culture models thereof describe the interaction between the alteration of key metabolic pathways and OS. On one hand altered metabolic function in the brain might be characterized by mitochondrial defects and proton leakage, thus promoting the formation of ROS. On the other hand, increased levels of OS impair key players of the glucose metabolic pathway, which further feed glucose hypometabolism (6, 133, 377). Considering the latter condition, redox proteomic analyses of AD, EAD, or MCI brain samples revealed the ox-PTMs of key proteins that belong to the energy-related metabolic

pathways and that contribute to their failure (298). Studies from the Butterfield laboratory reported the increased oxidation of several enzymes involved in glucose catabolism, as follows: fructose biphosphate aldolase (FBA), triose-phosphate isomerase (TPI), phosphoglucomutase 1 (PGM1), glyceraldehyde 3-phosphate dehydrogenase (GAPDH), alpha-enolase (Eno1), and pyruvate kinase (PK) (TABLE 4 and FIGURE 10). In agreement with this view, Eno1, PGM1, and GAPDH enzyme activities were also reported to be decreased in AD brain, thus confirming the deleterious effect exerted by oxidation. Among the glycolytic enzymes found to be increasingly oxidized, TPI, PGM1, and GAPDH were observed in late AD patients, whereas increased FBA, Eno1, and PK oxidation were found in EAD and MCI subjects as well. These results suggest the OS-induced initial impairment of a few glycolytic processes in the early stages of the disease and the subsequent widespread alteration of most of the glycolytic pathway in the late stage. Furthermore, the inhibition of glycolytic enzymes would cause the cell to shift its reliance on glycolysis to the pentose phosphate shunt, thus uncoupling the production of ATP and pyruvate from glycolysis and contributing to the growing anaerobic environment found in AD brain (133). Similar to what is observed in brain, mitochondria isolated from peripheral AD and MCI lymphocytes support the oxidation glycolytic enzymes, such as Eno1, suggesting its importance as a potential biomarker for MCI and AD (323). In addition, the alteration of GAPDH and PK in AD was further described by an immunoproteomic approach that demonstrated the presence in the CSF of autoantibodies against these enzymes (397), consistent with the notion that proteins undergoing ox-PTMs become toxic to neurons, which induce clearance systems and immune responses.

In addition to glycolysis, subsequent pathways involved in glucose metabolism, and therefore in eventual ATP formation, demonstrate the structural and functional alteration of protein components due to increased oxidation. In particular, the ox-PTMs of lactate dehydrogenase (LDH), aconitase, malate dehydrogenase (MDH), and ATP synthase were found in brains from MCI, EAD, and AD subjects (19, 27, 304–306, 398). Whereas the oxidation and likely dysfunction of ATP synthase leads to reduced ATP formation, as observed in AD brain and in AD mouse models, the oxidation of MDH is associated with increased enzyme activity in AD but also during the normal aging processes (399). Furthermore, the dysfunction of ATP synthase on the inner mitochondrial membrane is frequently accompanied by loss of mitochondrial membrane potential (MMP), decreased ATP levels, and consequent electron leakage and increased ROS production (317). Redox proteomics analysis in AD brain also led to the identification of increased oxidation of non-energetic-related proteins

in the mitochondria, such as the superoxide radical scavenger MnSOD and voltage-dependent anion channels (VDACs) (27, 306, 310).

Beyond affecting energy metabolic pathways, increased OS levels were demonstrated to promote brain insulin resistance (BIR) (200). During AD, an uncoupling between the insulin receptor (IR) and the insulin receptor substrate 1 (IRS1) occurs as a consequence of the downregulation of IR expression and/or the inactivation of IRS proteins leading to the inability of insulin to promote its normal cellular downstream outcomes. The increased ROS levels inhibit the cellular production of ATP and decrease insulin secretion and sensitivity. In turn, BIR promotes the dysfunction of mitochondria and leads to ROS overproduction, which enhances the accumulation of A β peptides and induces oxidative damage to proteins, lipids, and nucleic acids (400, 401).

Oxidative damage affects a variety of signaling pathways related to the UPR and protein degradation, which could exacerbate BIR (6). However, whether OS precedes the onset of BIR or vice versa is still under evaluation in AD, and the exact molecular mechanisms underlying the development of BIR have not been completely elucidated. Recent data collected in human brain and in AD mouse models support the concept that increased OS/NS levels precede the molecular events responsible for BIR in the development of AD (402). In addition, the progressive worsening of insulin resistance with the progression of AD correlates with increased OS levels, DNA damage, and protein oxidation demonstrated by accumulation of protein-bound HNE, PC, and 3-nitrotyrosine (3-NT) (133, 403). Moreover, insulin degrading enzyme (IDE) and biliverdin reductase-A (BVR-A) are the two enzymes associated with insulin signaling that appear to be increasingly oxidized during progression of AD, thus promoting BIR (402, 404–406). The experimental results demonstrating that A β plaque size inversely correlates with IDE expression and activity suggests that IDE deficiency could mediate plaque buildup and possibly cognitive impairment in AD (407, 408). In agreement with this notion, IDE oxidation might lead to its inactivation, reducing its ability to degrade insulin and A β . Furthermore, BVR-A is a serine/threonine/tyrosine (Ser/Thr/Tyr) kinase directly involved in the regulation of the insulin signaling (409). BVR-A is a direct target of the IR kinase activity, and as part of a regulatory loop BVR-A phosphorylates IRS1 on inhibitory Ser residues to prevent IRS1 aberrant activation in response to IR. Studies on BVR-A in postmortem MCI and AD brain demonstrated its inactivation, as a consequence of increased nitration, in association with the inhibition of IRS1 induction (402). Subsequent data collected on AD mice (3xTg-AD) support the concept that the oxidative damage of BVR-A is one of the earliest events observed during the development of BIR (410, 411). Intriguingly, BVR-

A-related unresponsiveness of insulin signaling was associated with the aberrant regulation of downstream pathways such as the mTOR/autophagy axis (412).

4.6. The Epigenetic Oxidative Redox Shift Hypothesis of Aging

To unify the free radical theory of aging with the decline of metabolism associated with reduced insulin signaling, as discussed above, Brewer proposed the epigenetic oxidative redox shift (EORS) hypothesis of aging (413–415). This hypothesis aimed to decipher the reasons behind the failure of preclinical and clinical studies designed on individually boosting antioxidant responses or improving the metabolic status of the brain (416–418). According to the EORS theory, aging is associated with sedentary behavior, which leads to an oxidized redox shift characterized by an extracellular decrease in the ratio of cysteine to cystine and an intracellular decrease in the ratio of GSH to GSSG and NAD(P)H to NAD(P) (419). Such an oxidized redox shift is initiated by low demand for bursts of energy produced by mitochondria, which accompanies low levels of physical and mental activity. This initiates a vicious cycle of oxidized membrane receptors, signaling molecules, transcription factors, and epigenetic transcriptional regulators (420, 421). Histone deacetylases, and the sirtuin family, histone acetylases, and DNA methyltransferases are the main epigenetic factors modulated by aging, which, together with redox-sensitive transcription factors, trigger the metabolic shift toward the use of energy from aerobic glycolysis instead of mitochondria (422). This metabolic shift is further mediated by insulin resistance that reinforces a downward spiral of more sedentary behavior leading to accelerated aging, stress-associated organ failure, impaired immune and vascular functions, and brain aging (423, 424). Therefore, the EORS hypothesis of aging consists of the inability to respond to energy demands or stress that leads to stress-induced catastrophic initiation of cellular death pathways and brain function decline. However, amending different steps of the pathways directly involved in the above-described vicious cycle of EORS conceivably could allow improvement in redox homeostasis and the metabolic regulation of brain cells during aging (425–427).

4.7. Nucleic Acid Oxidation in Alzheimer Disease

The first evidence on DNA damage and its association with AD neuropathology was reported by Mullaart et al. (428) by demonstrating a twofold increase in DNA strand breaks in the brain from AD subjects. Furthermore, increased strand breaks generation caused activation of poly(ADP-ribose) polymerase (PARP), a zinc finger DNA-binding protein, known to deplete intracellular NAD⁺

reserves as well as energy stores, thus ultimately resulting in cell death (429). Mecocci and colleagues (430) later reported for the first time the oxidation of specific DNA bases by using HPLC/ECD to measure the levels of 8-OHdG in nDNA and mtDNA specimens from cerebral cortex in the AD brain versus age-matched control subjects. In a follow-up study, the authors measured the levels of both nDNA and mtDNA in frontal, temporal, and parietal lobes and cerebellum from AD and age-matched normal control subjects. A significant threefold increase in 8-OHdG was observed only for mtDNA isolated from parietal lobe of AD, whereas no differences were evidenced in levels of 8-OHdG in nDNA from AD subjects (431). Studies from other groups using GC/MS showed increased 8-OHdG, 8-hydroxyadenine (8-OHA), and 5-hydroxycytosine (5-OHC) by GC/MS in total DNA from AD parietal lobe compared with age- and sex-matched control subjects (432). Oxidative DNA damage was also found in mild cognitive impairment (MCI), the stage of AD between normal aging and early dementia. Mecocci et al. (433) also showed significant elevations of 8-OHdG in lymphocytes isolated from AD subjects compared with age-matched control subjects, which negatively correlated with plasma antioxidant levels. Additional studies performed in AD lymphocytes by Kadioglu et al. (434) showed by comet assays significantly increased levels of oxidized purines and pyrimidines.

Levels of oxidized bases in AD brains were higher in frontal, parietal, and temporal lobes compared with control subjects, whereas cerebellum was only slightly affected (435). Both nuclear and mitochondrial DNA oxidation, respectively, likely contribute to neurodegeneration, including AD (50, 58). mtDNA showed much higher levels of oxidized bases than nDNA, in agreement with the fact that mitochondria are the major site for free radical production, thereby showing higher levels of oxidative stress. Thus, oxidative damage to mtDNA may play a critical role in neurodegeneration in AD brain, including its prodromal stages. According to this view, mtDNA showed more oxidative damage as indexed by levels of 8-OHdG isolated from mouse liver, heart, and brain than nDNA from the same tissues (50, 58). Mutations of somatic mtDNA control region (CR) increase with age in postmitotic tissues including the brain, and the level of mtDNA mutations is significantly elevated in the brains of AD patients.

The increased mtDNA CR mutations in AD brains are coupled with decreased mtDNA copy number and mtDNA L-strand transcript levels. In addition, although mtDNA-mediated CR mutations increase with aging in control healthy brains, the elevation is higher in the brains of AD, Down syndrome (DS), and DS with Alzheimer (DSAD) patients (436). Increased mtDNA CR mutations have also been detected in peripheral blood

DNA and in lymphoblastoid cell DNAs of AD and DSAD patients, and distinctive somatic mtDNA mutations are seen in AD and DSAD DNA from brain and cells. Related to epigenetic considerations, the level of mtDNA mutations positively correlates with BACE1 activity in aging, DS, and DSAD specimens, whereas mtDNA copy number is inversely correlated with insoluble A β 40 and A β 42 levels (437). Taken together, these findings suggest that increased BACE1 activity, i.e., increased A β production, leads to the significant elevation of mtDNA mutation, which might be relevant in the onset of age-related dementia and neuropathological changes in both AD and DSAD.

In addition to A β stimulation of ROS production, which directly causes DNA damage, it has been shown that A β 42 has DNA nicking activities similar to nucleases. Furthermore, magnesium ion was shown to enhance the DNA nicking activity of A β , and A β oligomers showed more DNA nicking activity compared with monomers and fibrillar forms. These data support a role for A β in causing direct DNA damage (431). Nevertheless, Nunomura et al. (438) showed, by comparative density measurements of immunoreactivity, that levels of intraneuronal A β 42 were inversely correlated with levels of 8-OHG. Together with recent evidence suggesting that A β peptide can act as an antioxidant, these results suggest that intraneuronal accumulation of nonoligomeric A β may be a compensatory response in neurons to oxidative stress in AD (438).

As noted above, RNA is more susceptible to oxidative damage than DNA. Nunomura et al. (439), by using in situ immunohistochemistry, were able to identify oxidized nucleosides in tissue from AD patients. RNA oxidation was detected in postmortem brains of subjects at early stage of AD, a presymptomatic phase with familial AD mutation, Down syndrome subjects with early-stage AD pathology, and subjects with MCI (440, 441). It is conceivable that neuronal RNA oxidation occurs before significant A β or tau pathology in AD. In addition, levels of RNA oxidation in CSF were higher in subjects with shorter duration of AD (59), as well as with higher cognitive scores. Among the oxidized nucleosides the majority of RNA oxidized species are present in the cytosol, not the nucleus or mitochondria.

Increasing evidence suggests that miRNAs contribute to the development of AD, regulating overproduction of A β peptides and Tau phosphorylation (67, 442). In AD, several miRNAs are downregulated, leading to increased ROS production and oxidative damage to neurons, which in turn, lead to cell death. For example, miR-204, miR-34a, miR-375, miR-140, and miR-335 were found to promote ROS production and inhibit the function of antioxidant enzymes in mitochondria (443). Decreased levels of miR-107 are associated with early stages of AD progression.

This microRNA directly targets BACE1 mRNA encoding β -secretase enzyme that processes APP to A β peptides (444). In AD patients with the APOE4 genotype, decreased levels of miR-107 were reported along with the increased production of A β peptides (444). Others showed that accumulation of A β -induced oxidative stress in APOE4 leads to the deregulation of the TP53 gene, and in addition to its role in cancer p53 protein (encoded by a TP53 gene) can be involved in cell death in AD patients, with upregulation at the early stages of the disease and downregulation during neurodegeneration (445). Previously, p53 mutations that may be associated with oxidative stress were observed in AD patients and AD animal models (446). Since miR-107 is downregulated in cell lines with mutated p53 (447), p53 mutations and accumulation of A β may result in the decrease of miR-107 levels in AD patients. Moreover, 8-oxo-2'-deoxyguanosine RNA modifications caused by oxidative stress can serve as an additional factor causing decreased miR-107 levels (448). Levels of another microRNA, miR-186, also are decreased through aging, and this microRNA targets 3'-untranslated region (UTR) of BACE1 and is implicated in the mitigation of the oxidative stress effects in AD pathogenesis by suppression of BACE1 expression with resultant lower levels of A β (449).

5. DOWN SYNDROME: A HUMAN PRODROMAL FORM OF ALZHEIMER DISEASE

5.1. Down Syndrome

Down syndrome (DS) is the most common genetic form of intellectual disability that results from the triplication of the entire portion or part of chromosome 21 (Chr21). DS occurs at a rate of 1 in every 800/1,000 live births, and it is a well-recognized syndrome with variable phenotypic expressions. Among common, more severe traits, DS individuals are characterized by a number of dysmorphic features, cardiac defects, accelerated aging, and cognitive impairment; a number of mostly nonspecific manifestations show high variability, for severity and frequency, among different individuals (450).

The "gene dosage hypothesis" states that the increased dosage of Chr21 genes is the direct cause of the phenotypic alterations of the DS population. However, the presence of trisomic genes also affects the expression of disomic genes, which in turn may gain aberrant expression and contribute to several clinical manifestations. In this dysregulated scenario, the effects caused by some dosage-sensitive genes are amplified and result in heterogeneous phenotypic traits according to the "number and dose" of genes involved (451). Thus, it is interesting

to note the case of monozygotic twins with trisomy 21 but with discordant phenotypes, as well as the case of subjects with DS phenotype but carrying a partial trisomy of a very restricted region (34 kb on distal 21q22) (452). Both examples reinforce the view that other mechanisms beyond solely triplication of Hsa21 are involved to explain the wide phenotypic variation occurring in DS individuals. For example, epigenetic histone modification and DNA methylation as well as microRNA regulation of gene expression also have been proposed to play a causal role in the etiology of DS (453). Studies found that several disomic genes present higher expression variances in human trisomic tissues compared with normal ones, and the number of disomic genes with high variance was significantly higher in trisomic tissues versus normal ones. These data suggest that the genetic imbalance observed in DS leads to greater instability in transcriptional control (454, 455).

Numerous developmental defects are associated with DS, with brain development and intellectual disabilities being the most striking features of trisomy 21, and language, learning, and memory also may be affected (456, 457). Life expectancy in the DS population is shorter compared with non-DS individuals, but improvements in medical care and drug treatments have significantly contributed to ameliorate the quality of life of people with DS (458). However, this increase in the life span poses critical issues associated with the “aging phenotypes,” among which early-onset AD neuropathology and dementia represent the most disabling features with social and economic impacts.

DS represents a unique population for studying changes of brain aging across the life span. By age 40 yr, there is a ubiquitous and significant occurrence of neuropathological hallmarks of AD, as indicated by widespread deposition of senile plaques and neurofibrillary tangles, but dementia is variable. Several studies support the link between the DS phenotype and an increased risk of development of AD (459). The incidence of dementia among DS patients is 10% in the age range 35–50 yr and 55% in the age range 50–60 yr and becomes 75% above the age of 60 yr, but AD neuropathology is present in virtually all adults with DS older than 40 yr (460). However, there is a subset of aged DS persons who do not develop clinical signs of dementia at any age (461). Although plaques have been detected in young DS autopsy samples, some as young as in the fetus, it is only in late middle age that people with DS develop AD neuropathology (457).

Consequently, the DS research field has gained much attention in the last decades as being a unique chance to understand which key dosage-sensitive genes are crucial in driving pathological phenotypes. In particular, understanding the degenerative mechanisms beyond

DS neuropathology may lead to answers to a number of open research questions: 1) What is the “genetic risk profile” for the development of early-onset AD? 2) Can DS be considered a model of premature aging, and how does this contribute to neurodegeneration? 3) Which genetic defects are (key determinants) responsible for pathological phenotypes? 4) Is DS neuropathology a consequence of neurodevelopmental abnormalities?

5.2. Alzheimer Disease in Down Syndrome: Common Pathways

Most of the above-mentioned questions represent current and future research challenges, mostly in the field of AD research. Individuals with DS are the largest population under 60 yr of age characterized by the early appearance of AD neuropathological features and are currently classified as early-onset AD (EOAD). Based on this clinical definition, DS is now considered the leading genetic risk factor for EOAD (462) to which the accelerated aging phenotype is closely linked even in the presence of other age-associated comorbidities (incidence of mixed pathology in this population). Overlapping with common traits of AD, the neuropathological changes associated with AD in DS are characterized by the initial formation of A β plaques within the cerebral cortex and then progressing into the hippocampus, striatum, and cerebellum (456). To understand the strong association between DS and AD neuropathology, first it is necessary to consider the trisomy of Chr21, in which the overexpression of selected genes already has been identified to be involved in the development of AD neuropathology in the general population (457) (FIGURE 11). By mapping Chr21, the *APP* gene encodes the amyloid precursor protein and the *EST2* gene encodes a transcription factor that further promotes the expression of *APP*, resulting in increased production of A β oligomeric toxic peptides, which form the backbone of amyloid plaques (463). APP undergoes enzymatic cleavage within the brain to form numerous fragments, including A β , which accumulates in AD-associated amyloid plaques and COOH-terminal fragments (CTFs) that may impair intracellular processes (464). However, how trisomy of Chr21 genes, other than *APP*, impacts APP biology and subsequent neurodegeneration and dementia is not well understood.

The first evidence showing APP overexpression in DS was demonstrated within enlarged endosomes and lysosomes, where APP accumulation could be initiated at the fetal stage, as early as 28 wk of gestation (465). These enlarged endosomes were more abundant than normal ones and appeared to form clusters (466, 467). The accumulation of intracellular A β within endosomes can promote mitochondrial dysfunction and OS, thereby

triggering neuronal damage (468). In adults with DS at age between 20 and 30 yr, A β deposits are in the form of diffuse plaques, although plaque deposition already is evidenced at younger ages and is followed by neuritic plaques in the fourth decade of life (469, 470). Deposition of NFTs is a secondary molecular event following deposition of A β ; however, with a pattern consistent with Braak staging, the density of NFTs significantly increases with age in DS brain (471). Interestingly, this abnormal Tau aggregation is associated with clinical observations of cognitive impairment, as assessed by Tau PET, implying a crucial aspect in the onset of dementia in DS individuals (472–474). Other protein aggregates also were detected in DS with AD (DSAD), including TDP-43, with lower frequency in DSAD compared with late-onset AD (LOAD) but comparable to the rate observed in early-onset AD (EOAD). α -Synuclein-positive Lewy bodies also were observed in the amygdala of people with DS, although at a lower level compared with EOAD (475).

A striking feature of DS population is the long gap (up to 20 yr) between the initial appearance of AD pathological hallmarks and the onset of the prodromal AD (457, 476). Diagnosing symptomatic AD in people with DS is quite complex because early symptoms can be mistaken as part of the individual's lifelong intellectual disability or masked by coexisting medical comorbidities that might affect cognition, such as obstructive sleep apnea, hypothyroidism, and depression (456). There are no validated or widely accepted clinical diagnostic criteria for DS-associated dementia. In most specialized clinics and research settings, a physician and a neuropsychologist make the diagnosis independently and, in case of discrepancy, a consensus-based protocol is used.

By crossing a mouse model of DS (the Tc1 mouse) that does not have an additional copy of *APP* to a transgenic mouse overexpressing APP with AD-causing mutations (J20 mouse), the additional copy of Hsa21, independent of an extra copy of *APP*, exacerbated A β aggregation and deposition, enhanced *APP* transgene-associated mortality, altered behavior, and reduced cognitive performance in double-mutant progeny that model DSAD (477). Different studies have shown that the deposition of A β can occur from early age in persons with DS, but after age 30–40 yr A β deposition in brain is observed in nearly all DS individuals, and accumulation is exponential, meaning not only an early onset of the disease but also an acceleration in the deposition of A β plaques with age, compared with the general population (457).

In addition to *APP*, the overexpression of *RCAN1* and *DIRK1A* contributes to the hyperphosphorylation of Tau driven by different families of kinases (FIGURE 11). Moreover, there are many other components that still need to be identified. So far, 233 genes are reported to

be encoded on Chr21 (478). In addition, noncoding RNAs, RNA molecules that do not translate into a protein, are found on Chr21 and are involved in the regulation of gene transcription. Therefore, other candidate genes or regulatory sequences on Chr21 may interfere with APP metabolism, A β production, and aggregation. Accumulating evidence shows that >600 genes, outside Chr21, are overexpressed as a consequence of trisomy 21 (457).

5.2.1. Early- vs. late-onset AD: commonalities and differences.

The different forms of AD, which include genetic and nongenetic forms, have many similarities and differences, understanding of which can provide a better understanding of AD mechanisms (479). The genetic forms include, in addition to DSAD, autosomal dominant AD (ADAD), both of which are classified as EOAD that are usually characterized by clinical onset before 60 yr of age and Mendelian inheritance, whereas sporadic AD is the well-recognized form of LOAD, with a clinical onset after 60 yr. ADAD is caused by rare and fully penetrant mutations in one of the three genes, *APP* and *presenilin* (*PSEN*) 1 and 2, which follow a Mendelian autosomal dominant inheritance pattern. ADAD represents <1–5% of all AD cases, and the relative frequencies due to mutations in *PSEN1*, *PSEN2*, and *APP* are 69%, 2%, and 13%, respectively (479). Moreover, ADAD phenotypes are different between carriers of the specific genetic mutations, as well as between variants of the same gene (480). Considering the frequency, APP mutations are the best-recognized genetic cause of AD; these mutations include either duplication of APP or missense mutations and are directly implicated in A β overproduction. To better understand the mechanistic link between AD and APP, studying the effects of aging on the DS population has offered, and will continue to do so, a unique opportunity in both the field of DS and AD research and care.

Generally, both DSAD and ADAD show age-dependent neuropathological changes that are not present in the general population until the early stages of sporadic AD. In the majority of cases in all three groups (DS, EOAD, and LOAD), the clinical appearance of dementia usually starts with memory deficits (479). However, in ADAD and DS there are more nonamnestic phenotypes. Moreover, in those with DS and most forms of ADAD, there also is more severe cerebral amyloid angiopathy (CAA) and increased neurological symptoms compared with LOAD (480, 481). Both people with DS and carriers of ADAD mutations show a higher and earlier brain A β production, as well as an increased accumulation of A β 42- and A β 40-containing senile plaques as well as

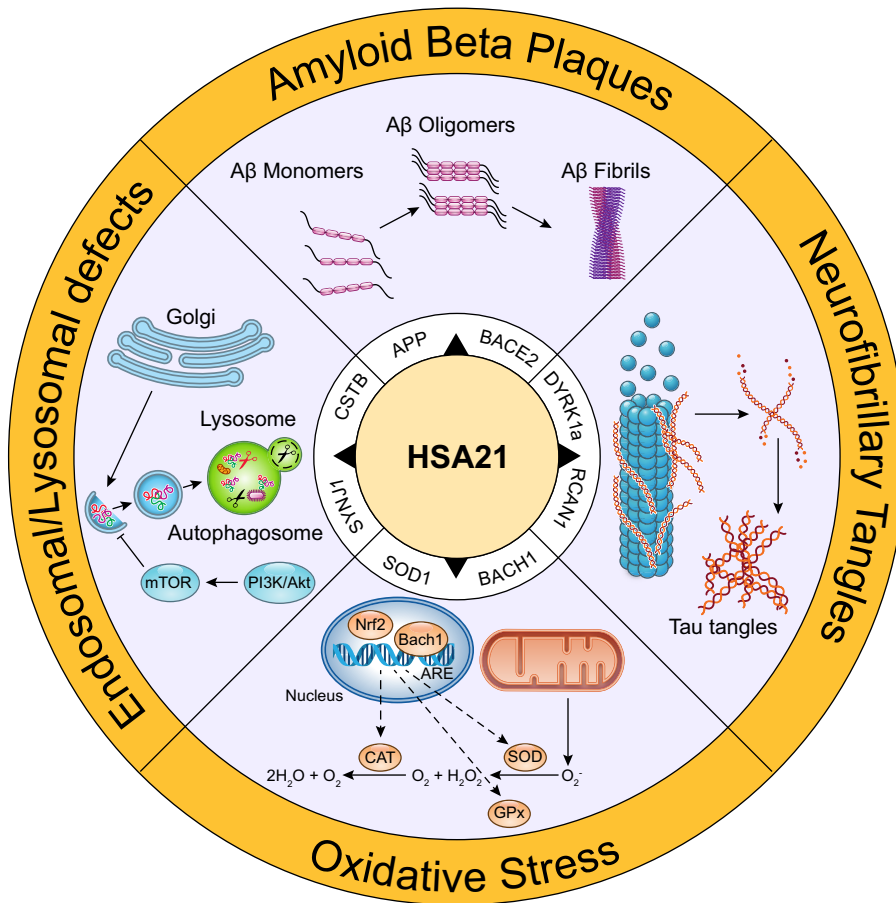


FIGURE 11. HSA21-encoded proteins and their role in the onset of Alzheimer disease (AD) neuropathology in Down syndrome (DS) individuals. Oxidative stress, endosomal/lysosomal dysfunction, and aberrant amyloid precursor protein (APP) and Tau metabolism, leading to overproduction of Aβ and Tau hyperphosphorylation, respectively, are common molecular features of DS and AD neuropathology. Moreover, compared to AD, these damaging effects occur early in brains of DS populations and accelerate onset of AD-like neuropathology and dementia with increasing age. See text for more details.

NFTs. It is important to highlight that overexpression of APP, associated with increased Aβ deposition, most likely accounts for the early age of onset described in genetic cases. Although the magnitude and direction of changes in the three conditions are generally similar, there are also some important differences. Furthermore, ADAD and DSAD show an increased initial Aβ accumulation in the subcortical regions, particularly in the striatum, compared with LOAD (479–481). Moreover, biochemical changes, the amyloid deposition map, and the hypometabolism map, as well as Tau distribution, are very similar among the three AD forms. However, studies from our group have extensively demonstrated that the accumulation of oxidative damage occurs very early in the DS population, thus suggesting that several abnormalities, including in protein activity and signaling, may accelerate aging and neurodegeneration (see sect. 5.3). Defining the pathology across the different forms of AD, molecular and clinical differences and similarities, will allow a deeper understanding of AD and development of new targeted and personalized therapies. We opine that, because of the restricted cases with ADAD, DS remains the selected population on which further research should be focused to address many of the remaining gaps between genes and phenotypes.

5.3. Oxidative Stress in Down Syndrome

Several studies have demonstrated that OS contributes to several DS pathological phenotypes, including neurodevelopmental defects, neuronal dysfunction, as well as accelerated aging and EOAD (481, 482). As noted above, OS is a physio/pathological condition that results from either overproduction of ROS/RNS or a reduction of the antioxidant responses to ROS/RNS or both. The CNS is particularly sensitive to the damaging effects of oxidative reactions because of the composition of neuronal membranes, rich in unsaturated fatty acids, and to high aerobic metabolic activity, both events acting in concert to favor free radical production (342, 483). Among these highly reactive species, $O_2^{\cdot-}$, H_2O_2 , and $HO^{\cdot}$ are constantly produced as by-products of aerobic respiration in parallel with other catabolic/anabolic processes (484). Also mentioned above, a major source of free radicals is mitochondrial oxidative phosphorylation (OXPHOS), in which electron leakage starts with the formation of $O_2^{\cdot-}$ that, in turn, is neutralized by mitochondrion-resident MnSOD into H_2O_2 and O_2 (485). In line with this evidence, studies have shown that in skin fibroblasts isolated from both fetal and adult DS individuals, dysfunction of complex I occurs, coupled with increased ROS release (486).

In DS, increased OS is directly related to the triplication of some Hsa21 genes, as explained below, together with a deregulation of gene/protein expression of disomic genes (487). In addition, the increased production of ROS also is closely linked to mitochondrial abnormalities, which occur in DS cells as early as the embryonic stage (488). Several reports have demonstrated the involvement of OS-mediated “loss of function” in DS phenotypes (489, 490), although a direct cause-and-effect relationship between the accumulation of oxidative damage and clinical manifestation of DS has not yet been clarified. It is reasonable to speculate that OS is a chronic feature in DS brain that is initiated already during embryonic development and further progresses with aging. This feature represents a strong risk factor for neurodegeneration in adult life in the DS population (491, 492). Notably, if one maps the Chr21, a number of genes including *SOD1*, *APP*, *BACH1*, *ETS2*, *CBS*, *CR*, and *S100B*, among others, are considered to play a causative role in increasing OS levels as demonstrated both in DS individuals and in mouse models of the disease (493). Among trisomic genes, much attention has been given to *SOD1*.

SOD1 represents a first line of antioxidant defense, as described above (132). However, it has been shown that trisomic cells display an imbalance in the ratio of *SOD1* to *CAT* and *GPX*, resulting in the accumulation of H_2O_2 (494). Recent evidence also points to the fact that expression levels of *SOD1*, found to be ~50% higher than normal in a variety of DS cells and tissues, do not necessarily correlate with enzyme activity (493). Indeed, researchers followed the same patients for 4 yr and found that lower *SOD1* activity was predictive of a decrease in memory performance over time (495). Moreover, a proteomic study reported that OS in fetal DS is mainly caused by reduction of a class of antioxidant enzymes responsible for neutralizing hydrogen peroxide, such as glutathione transferases and thioredoxin peroxidases, thus revisiting exclusively the role of *SOD1* itself (496).

Another key player in the regulation of antioxidant response is the trisomic gene *BACH1* (483). *BACH1* is a transcription repressor that acts as a key regulator of the expression of genes involved in the cell stress response. In physiological conditions, *BACH1* is assembled in the form of a heterodimer with small Maf proteins, which bind AREs of DNA, thus negatively regulating the expression of specific target proteins. When OS levels are elevated, the function of *BACH1* is suppressed by promoting *BACH1* nuclear export and thus enhancing the expression of its gene targets. For example, when the intracellular heme levels rise, as may occur under a prooxidant insult, nuclear *BACH1* binds heme and dissociates from the AREs, thereby allowing the transcription of genes such as

NQO1, *GST*, *GCL*, and *HO-1*. In DS, it is likely that upregulation of *BACH1* could block the induction of antioxidant genes, therefore promoting increased OS in the cell (483). Increased expression of *BACH1* likely disturbs Nrf2-induced antioxidant response, thus contributing to impaired redox homeostasis in DS cells and tissues (82, 83, 497).

Cystathionine β -synthase (*CBS*) is one of the key mammalian enzymes that is responsible for the biological production of the gaseous transmitter hydrogen sulfide (H_2S). When H_2S is overproduced, it can exert harmful cellular effects, mostly resulting from inhibition of mitochondrial complex IV activity. Increased expression of *CBS* and the consequent overproduction of H_2S are well documented in individuals with DS (498). Accordingly, increased expression of *CBS* coupled with overproduction of H_2S was observed in the brain of DS rats, with *CBS* primarily localizing to astrocytes and the vasculature. These molecular alterations were associated with neurobehavioral defects (499). DS rats treated with an inhibitor of *CBS* (aminooxyacetate) showed increased ability to generate ATP in brain tissue together with a recovery of both the electrophysiological and neurobehavioral alterations. Several studies support the idea that overproduction of H_2S is closely related to mitochondrial abnormalities, not only in DS but also in other pathological conditions (500). In agreement with this hypothesis, an increasing number of reports posit that DS mitochondria are characterized by reduced ability to produce ATP through OXPHOS, decreased respiratory capacity, and disruption of mitochondrial membrane potential, all associated with loss of mitochondrial dynamics. These mitochondrial defects are present in all DS cell types, from peripheral tissues to the brain. Therefore, mitochondrial dysfunction is considered an inherent feature of DS, associated with a condition of increased OS (501), likely closely linked to triplication of some Hsa21 genes including *CBS*.

Elevated levels of OS also could be caused by increased production of $A\beta$, the cleavage product of *APP*. Several studies have extensively demonstrated that $A\beta_{40}$ and $A\beta_{42}$ are able to induce OS, as shown elsewhere in this review. As described above, Butterfield's group and others proposed that $A\beta_{42}$ inserts as small oligomers in the lipid bilayer and serves as mediator of ROS, thus initiating LPO (2, 6, 431). Deposition of senile plaques is observed in postmortem brain from DS individuals (502), and levels of both $A\beta_{42}$ and $A\beta_{40}$ in plasma are higher in DS compared with non-DS control subjects (503). In addition, $A\beta$ plays a role in the modulation of metal homeostasis through coordination of metal ions, Zn^{2+} , Cu^{2+} , and Fe^{2+} , that participate in both production and defense against ROS and are required to regulate the neuronal activity in the synapses and other biological

functions in the brain. Furthermore, studies from Anandatheerthavarada et al. (504) pointed out that full-length APP itself also may have toxic effects, in particular damaging mitochondria. The idea is that in the presence of increased APP expression levels, a progressive accumulation of transmembrane-arrested APP impaired mitochondrial function, which in turn resulted in disturbance of energy metabolism. Moreover, mice overexpressing wild-type human APP show cognitive defects and neuronal pathology similar to those observed in AD models, although these mice do not show significant A β deposition in the hippocampus (505). Nonetheless, although in this case APP processing was nonamyloidogenic, increased levels of phosphorylated Tau were observed. These findings support the notion that, conceivably, trisomy of APP may promote mitochondrial dysfunction in DS independent of A β aggregation or deposition.

5.4. Mitochondrial Defects in Down Syndrome

A significant alteration of mitochondrial structure and function and the increased production of ROS are closely linked events in DS pathological phenotypes and contribute to intellectual disability (501). This intersection of OS, mitochondrial status and functions, and cognitive loss is relevant for not only DS but also other neurodevelopmental diseases, including Rett's syndrome and autism, and in neurodegenerative diseases, including AD and Parkinson disease (226).

Several neural developmental processes such as cellular proliferation and differentiation, axonal and dendritic growth, and formation of synaptic spine and presynaptic compartments require elevated mitochondrial ATP production coupled with maintenance of redox homeostasis in brain mitochondria (506). Neuronal function, survival, and CNS development are significantly perturbed by deficits in energy metabolism associated with mitochondrial dysfunctions, and both intellectual disability and several forms of dementia result from these combined events (501). The reduction of mitochondrial energy metabolism coupled with increased production of radical species are critically associated with several pathological phenotypes in DS, with evidence also showing a direct correlation with defective neurogenesis and impaired neural plasticity, ultimately resulting in cognitive decline (507). Furthermore, abnormalities of mitochondrial function are well-known markers of accelerated aging and early neurodegenerative processes occurring in DS, as well as in the general population. One of the key mitochondrial phenotypes of DS cells, both in the CNS and in peripheral tissues, is a reduced efficiency to produce ATP through OXPHOS, a decreased respiratory capacity and ability to maintain mitochondrial membrane potential, together with defects of mitochondrial dynamics (486).

Therefore, mitochondrial dysfunction is an inherent feature of DS (508).

Some mitochondrial transporters such as adenine nucleotide transporters (ANTs), which exchange ATP from mitochondria with cytosolic ADP, and adenylate kinase (AK), which catalyzes the reversible interconversion of ATP and AMP into two ADP molecules, are metabolically interconnected with the mitochondrial respiratory chain (MRC), contributing to the mitochondrial synthesis of ATP (509). In DS the OXPHOS machinery is selectively affected at the molecular level. For example, genes encoding for MRC subunits (complexes I and III) are downregulated whereas certain enzymes of the Krebs cycle (i.e., aconitase and NADP⁺-linked isocitrate dehydrogenase) are increased in the heart of DS fetuses and in brain regions of subjects with DS (487, 510). Furthermore, by analyzing mitochondrial function of both fibroblasts and lymphoblastoid cells isolated from DS subjects, several defects in the OXPHOS machinery have been identified. Specifically, the MRC complex I, ATP synthase, the ADP/ATP translocator, and the AK enzyme are impaired, ultimately leading to an energy deficit and increased free radical production from mitochondria. A severe bioenergetic deficit was also observed in neural progenitor cells (NPCs) together with a reduction in cell proliferation in hippocampal tissue isolated from Ts65Dn mice, a commonly utilized DS murine model (511). Despite an apparent activation in the glycolytic flux, as marked by increased levels of L-lactate in NPCs, a significant decrease of cellular ATP content was observed. It is likely that reduction of NPC proliferation might be a consequence of the drop in cellular ATP levels. The impairment of the MRC complex activity, particularly at the level of complexes I and V, can affect both respiration-mediated and general mitochondrial ATP production (501).

Recent published work by Vacca and Bartesaghi showed further evidence of defective functional activities of MRC complexes, complex I, and ATP synthase in the brains of neonate Ts65Dn mice similar to what was already demonstrated in fibroblasts and lymphoblastoid cells from DS subjects as well as Ts65Dn-derived hippocampal NPCs and brain cortex of Ts16 mice (512). The level of ATP measured in the mouse brain was significantly lower in postnatal day (P)15 mice compared to euploid control mice, thus suggesting that inefficient mitochondrial ATP production found in Ts65Dn mice is an early event that translates into an alteration of the whole brain energy status. Paradoxically, the mitochondrial energy deficit in Ts65Dn mice at P3 was not associated with a decrease in the brain ATP content, suggesting that tentative compensatory events might occur early, such as an enhancement of glycolysis, as detected in fetal human Hsa21 fibroblasts (513).

Aberrant mitochondrial energy metabolism and OS could result in the increased susceptibility of individuals with DS to a large spectrum of diseases such as AD, cardiomyopathy, and autism spectrum disorders (514). Coskun et al. (515) tested the hypothesis that peripheral cells from elderly subjects with DS show dementia-specific and disease-specific metabolic features. Using lymphoblastic cell lines derived from individuals with DS and DS with dementia, the study showed that DS cells exhibited a slower growth rate under minimum feeding and reduced expression of the autophagy marker LC3-II. Taken together, these findings underscore the close relationship between metabolic dysfunction and impaired autophagy in DS. Notably, several studies by our groups and others contributed to investigations of the cross talk among OS and defects of protein quality control systems, including UPS and autophagy, as discussed in sect. 5.5.

5.5. Redox Proteomics Studies in Down Syndrome

As discussed above, oxidation of proteins affects different processes including protein expression, protein turnover, and cell signaling, eventually leading to cell death (334, 516). A detailed description of the effect of oxidative modifications to proteins during early and late stages of AD is provided in sect. 4. Studies from our laboratories identified several oxidatively modified proteins in DS brain before and after development of AD-like neuropathology and dementia (342, 357). We analyzed frontal cortex from postmortem brains from a cohort of DS individuals, before and after development of AD neuropathology (DS >40 yr) versus respective age-matched control subjects. Among the OS/NS markers, DS patients (<40 yr) were characterized by increased total levels of protein-bound HNE together with increased levels of SDS- and formic acid (FA)-extracted A β 40 and A β 42 fractions in the frontal cortex (342, 482). The comparison between these four groups allowed identification of 1) the proteins oxidized/dysfunctional in the DS population (DS vs. young control < 40 yr) that indicate which pathway is altered as a direct consequence of the trisomy of Chr21; 2) the effect of AD development in DSAD versus DS that suggests which molecular pathways contribute to early-onset AD in DS individuals; 3) the “EOAD phenotype” by comparing DSAD versus old control subjects, which permitted comparison of EOAD in DS with normal AD; and 4) old versus young control subjects to consider the effect of normal aging in the general population. Among the proteins identified by redox proteomics to be irreversibly oxidatively modified, by either increased carbonylation or HNE modification, we found proteins involved in 1) neuronal trafficking; 2) proteostasis network; 3) energy metabolism, and 4) mitochondrial function (342). Notably, all of these cellular processes directly or indirectly rely on ATP

consumption to occur efficiently. Reduced ATP levels, increased ROS, impaired calcium homeostasis, and altered mitochondrial permeability are characteristic mitochondrial defects of degenerating neurons in many neurodegenerative disorders. These studies demonstrated that a number of oxidized proteins are in common between DS and AD subjects (517), namely Grp78, UCH-L1, HSC71, and GFAP. Among the components of the proteostasis network, Grp78, UCH-L1, cathepsin D, V₀-ATPase, and GFAP were increasingly carbonylated in the frontal cortex of DS individuals at ~20 yr of age compared with age-matched control subjects (357, 363). These initial findings suggest the hypothesis that younger subjects with DS may already show disturbance of the proteostasis network possibly linked to increased OS, many years before appearance of AD neuropathology (376). However, dementia associated with AD in DS occurs at an earlier age than that of sporadic AD, yet there are common oxidative alterations in the proteostasis network and glucose metabolism between these two conditions, implying that these processes are intimately associated with dementia and potentially targets for therapeutic intervention. In addition to oxidatively modified proteins associated with proteostasis and glucose metabolism, redox proteomics studies led to the identification of oxidized proteins belonging to several dysfunctional pathways, among which were detoxification systems, excitotoxicity, and synapse function that highly correlate with DS and AD pathological features, supporting the role of protein oxidative damage in neuronal degeneration and cognitive decline (303, 493). This scenario suggests that any pharmacological intervention may be more effective if it engages more than one molecular pathway and drugs that target not only specific mechanisms but also the interplay among them might be beneficial. Current therapies are moving toward the use of formulations containing compounds able to modify several oxidative aspects of the disease (518).

As mentioned above, impairment of energy metabolism is a key pathological feature of DS brain (519). Glucose metabolism is essential for brain health, and dysfunction of glucose utilization in the brain represents a crucial event in the development of neurodegenerative disorders (6, 389). Furthermore, epidemiological studies indicate that hallmarks of metabolic disorders, such as glucose intolerance and/or impairment of insulin secretion, are associated with a higher risk of developing dementia, including AD (391, 403). Accordingly, redox proteomics studies revealed that FBA-A/C, MDH, α -enolase, glyceraldehyde-3-phosphate dehydrogenase, cytochrome *b*-c1 complex, aconitate hydratase, and pyruvate kinase isozymes M1/M2 are HNE modified in DS and DSAD patients (124, 342). These enzymes, which are critical components of the glycolytic pathway and

the TCA cycle and are oxidatively modified, may be toxic stressors within neurons by reducing energy production. Similarly, reduced glucose utilization resulting from OS-induced impairment of glycolytic enzymes would be further associated with increased glucose levels as occur in the early phases of AD (492). It is likely that increased glucose levels would trigger extracellular glutamate release that leads to NMDR activation and increased neuronal $[Ca^{2+}]$, neuronal NOS (nNOS) stimulation, as well as increased OS/NS levels. This initial phase, characterized by increased glucose levels, can evoke rapid changes in neuronal excitability through inhibition of K_{ATP} channels, which result in increased A β production. Increased levels of A β oligomers, in turn, likely mediate similar effects, amplifying this vicious cycle and promoting mitochondrial damage via OS.

The picture that emerges from these studies unravels a pathological metabolic phenotype of DS resulting from either trisomy or several additional molecular events. Defects of mitochondrial function contribute to a general loss of cellular functions, most of which necessarily depend on ATP availability. Indeed, ATP production and redox homeostasis in brain mitochondria are essential to sustain neural developmental processes including cellular proliferation and differentiation, axonal and dendritic growth, generation of synaptic spine and presynaptic compartments, and neurotransmission (520). Mitochondrial deficits in DS are mainly the result of reduced efficiency to produce ATP through OXPHOS, together with decreased respiratory capacity and disruption of membrane potential and mitochondrial dynamics. As such, OS and mitochondrial dysfunction create a vicious cycle that further sustains the accumulation of oxidative damage in the presence of dysfunctional mitochondria. Any treatment able to block or attenuate this self-sustaining neurotoxic cycle has the potential to be effective to prevent or at least slow early-onset AD in DS but also may be effective in the general population as well. Section 6 discusses antioxidant strategies that have been employed in both preclinical and clinical AD studies.

6. ANTIOXIDANT STRATEGIES AGAINST OXIDATIVE STRESS

6.1. Overview of Multiple Antioxidant Strategies

The discovery of OS implication in the pathogenesis of CNS disorders led to a considerable growth in the daily use of antioxidant supplements, especially in western countries (521). Antioxidant supplements may exert a beneficial role in the brain by the induction of different protective mechanisms. These range from the direct inhibition of chain reactions involving the formation of free

radicals to the induction/improvement of endogenous antioxidant response and to the restoration/removal of damaged cell membranes and other biological components (522). As stated above, endogenous antioxidant defense systems tend to decline during natural aging and with the development of neurodegenerative diseases (see sect. 4.4). Thus, it seems reasonable to provide the brain with the adequate amount of qualitative exogenous, nonenzymatic defenses, such as vitamins, plant components, and minerals (523, 524).

Data collected by preclinical studies on AD animal models demonstrated promising results concerning the use of antioxidant molecules in halting or slowing brain damage and cognitive decline. However, contradictory results have been observed in human studies, and meta-analyses regarding antioxidant supplementation trials have shown only limited efficiency (525–527). It is noteworthy that the antioxidant potential of exogenous antioxidant molecules depends on several variables, including absorption and bioavailability of the compounds, the achievement of an efficient plasma concentration, and the nature, location, and mechanisms by which free radicals were produced. Within this context, the main factors that limit the translation of antioxidant supplements from bench to bedside may be related to the difficulty to reach the ideal pharmacological concentrations in the targeted brain region. Consistent with this notion, recent research projects in drug therapy have focused on innovative delivery mechanisms and formulations to increase the bioavailability of antioxidant supplementation for CNS disorders. Additional confounding variables in some clinical trials involving antioxidants were represented by the heterogeneity of study designs, which did not help delineate a consistent profile of efficacy, and the use of a single-molecule/single-target approach, which was repeatedly demonstrated to be unsuccessful for neurodegenerative diseases such as AD because of the multifactorial triggers of this dementing disorder (525, 527–529). Therefore, multitargeted approaches, including the combination of antioxidant compounds as well as antioxidant-rich dietary patterns and/or physical exercise, have been strongly advised and are currently under investigation in human studies. Accordingly, “healthy lifestyle” intervention strategies might offer potential synergistic and neuroprotective effects, modulate brain plasticity, and build an “epigenetic memory” able to promote neuronal resilience against stressors (530, 531).

Below in this section, we report the latest preclinical studies and clinical trials concerning the use of antioxidant supplementation, such as natural antioxidants, synthetic antioxidants, inorganic molecules, and vitamins, in the context of AD. Furthermore, we take into consideration recent research studies promoting the use of diet-based nutritional approaches and/or physical exercise.

6.2. Preclinical Studies

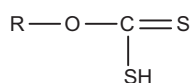
Many preclinical studies performed in the past years in cellular and animal models of AD have been focused on testing antioxidant properties of several natural or synthetic compounds, with the aim of translating their use to human trials (TABLE 5).

6.2.1. N-acetyl-L-cysteine.

N-acetyl-L-cysteine (NAC) is a derivative of cysteine containing an acetyl group that is attached to the nitrogen atom of cysteine. Preclinical studies on NAC showed its ability to protect against protein oxidation in the APP/PS1 human double-mutant knockin mouse model of AD (147, 149). In addition to preventing OS damage to proteins, NAC was able to ameliorate alterations concerning energy-related pathways, excitotoxicity, cell cycle signaling, synaptic abnormalities, cellular defense, and cellular structure and to prevent social isolation-induced accelerated impairment of contextual fear memory and loss of hippocampal LTP through reduction of A β levels via inhibition of γ -secretase (532). Several in vitro and in vivo studies employed NAC in the formulation of cocktails, containing further antioxidant molecules (e.g., curcuminoids), which were able to some extent to reduce A β toxicity and ameliorate neuronal damage (522, 525).

6.2.2. Tricyclodecan-9-YL-xanthogenate.

Tricyclodecan-9-YL-xanthogenate (D609) is a tricyclodecanol derivative of xanthic acid, which has a xanthate moiety as its key structure:



Studies concerning pretreatment of primary hippocampal cells with D609 significantly attenuated A β 42-induced cytotoxicity, intracellular ROS accumulation, protein oxidation, LPO, and apoptosis. D609, when oxidized, formed a disulfide bond exactly as does GSH, and like GSH D609 is a substrate for glutathione reductase (GR) (176, 614, 615). Synaptosomes isolated from gerbils were challenged by Fe²⁺/H₂O₂, 2,2-azobis-(2-amidinopropane) dihydrochloride (AAPH), or A β (1–42) and treated with D609, demonstrating the ability of this antioxidant molecule to protect cells against OS damage, including against loss of phospholipid asymmetry, by acting as glutathione mimetic (177, 616). Indicating that it was the GSH-like characteristics of D609 that provided this protection against OS damage, methylated D609, with the thiol functionality no longer able to form the disulfide upon oxidation, did not protect neuronal cells against A β 42-induced OS.

6.2.3. Selenium.

Selenium (Se) is known to provide protection from free radical-induced cell damage and is incorporated into proteins as the amino acid selenocysteine or selenomethionine (617). Preclinical studies on Se supplementation in Wistar rats injected bilaterally with intracerebroventricular administration of streptozocin (ICV-STZ) demonstrated the reduction of OS, increased GSH response, the improvement of A β toxicity, and the amelioration of memory loss. Moreover, the treatment of ICV-STZ rats and mice with sodium selenite or *p,p'*-methoxyl-diphenyl diselenide showed protection against increased ROS caused by the reduction of glutathione levels and the increase in SOD and GST activities (534, 535, 618). The administration of a Se-enriched diet (Sel-Plex) from 4 to 9 mo of age to APP/PS1 mice showed significantly lower levels of A β plaque deposition and decreased levels of DNA and RNA oxidation, associated with the increase of GPx activity (536). Sodium selenate-treated transgenic TAU441 mice demonstrated significantly lower levels of phosphorylated and total Tau in the hippocampus and amygdala and exhibited significantly improved spatial learning and memory on the Morris water maze task (537). In parallel, the chronic oral treatment of two independent Tau transgenic mouse strains with NFT pathology, the P301L mutant pR5 and the K369I mutant K3, with selenium reduced pTau, abrogated NFT formation, and improved contextual memory and motor performance (538). The treatment of 3xTg-AD mice with selenomethionine improved cognitive functions, reduced the level of total and pTau, induced autophagy, mitigated the decrease of synaptic proteins, and increased the level of reduced glutathione (539, 540).

6.2.4. Ferulic acid.

Ferulic acid (FA) is a common polyphenolic compound most abundant in vegetables. Chronic administration of ferulic acid, at different doses, to APP/PS1 mice significantly enhanced cognitive performance, reduced A β deposition, attenuated neuroinflammation, and stabilized OS (541, 543–545). The ethyl ester derivative of ferulic acid (FAEE) was shown to have both anti-inflammatory and antioxidant properties as stated for FA, but because of its lipophilic nature, it has increased ability to permeate the BBB (619–621). The analysis of synaptosomes isolated from 3-mo-old Mongolian gerbils after 1 h of FAEE treatments demonstrated the reduction of OS markers and of inducible NOS (iNOS) expression and the induction of endogenous antioxidant response (542). FA-based solid lipid nanoparticles (SLN-FA) displayed greater efficacy (EC₅₀) and potency (maximal activity) than FA or

Table 5. Preclinical studies employing antioxidant molecules in *in vitro* and *in vivo* models of AD neuropathology

Antioxidant Molecule	Formulation and Dose	Administration Route	Length of Treatment	Animal Model	Sex	Main Outcomes*	Reference(s)
N-acetyl-L-cysteine	NAC, 2 mg/kg/day	0.001% solution in drinking water	5 mo	APP/PS1 mice, 9 and 12 mo old	100% males	↓ protein oxidation; ↑ GPx and GR activity and stress response	(147, 149)
	NAC, 200 mg/kg/day	Intraperitoneal injection	Once per day for 7 days	APP/PS1 mice, 11 wk old	100% males	↓ γ-secretase activity and Aβ production; ↑ GluR1 and GluR2 surface expression	(532)
	NAC 170, mg/day; curcuminoids, 406 mg/day; EGCG, 406 mg/day; LA, 102 mg/day; piperine, 34 mg/day	Oral capsule	3 mo	Aged dogs, 98–115 mo old	n.s.	↑ spatial attention and motivation deficits	(533)
Selenium	Sodium selenite, 0.1 mg/kg/day	Intraperitoneal injection	7 days pretreatment	ICV-STZ-treated Wistar rats	100% males	↑ learning and memory, GSH, GPx, GR, and ATP; ↓ OS	(534)
	<i>p,p'</i> -Methoxy-di-phenyl diselenide, 25 mg/kg	Oral gavage	Twice	ICV-STZ-treated mice	100% males	↑ learning and memory	(535)
	Sel-Plex, 1 μg/g of Se	Diet	5 mo	APP/PS1 mice 4 mo old	100% males	↓ Aβ plaque deposition, DNA and RNA oxidation; ↑ GPx activity	(536)
	Sodium selenate, 1.2 mg/mL	Drinking water	3 mo	TAU441 mice	100% males	↑ spatial learning and memory; ↓ p-tau and tau levels.	(537)
	Sodium selenate, 12 μg/mL	Drinking water	5 wk, 4 mo	P301L mutant pR5 K369I mutant K3 mice	n.s.	↑ contextual memory and motor performance; ↓ tau phosphorylation and NFT	(538)
	Selenomethionine (Se-Met), 6 μg/mL	Drinking water	3 mo	3xTg-AD mice 8 mo old	50% males	↑ cognitive deficit, synaptic proteins, GSH levels and autophagy; ↓ total and phospho-tau;	(539, 540)
	FA, 14–19 mg/kg/day	0.006% solution in drinking water	4 wk pretreatment	Aβ1-42 ICV mice	100% males	↑ passive avoidance, Y maze and water maze tests and acetylcholine levels; ↓ inflammation;	(541)
Ferulic acid	FA ethyl ester (FAEE)	FAEE in DMSO 150 mg/kg body weight	1 h	Mongolian gerbils synaptosomes, 3-mo-old male	100% males	↓ OS, lipid peroxidation, iNOS; ↑ antioxidant response	(542)

Continued

Table 5.—Continued

Antioxidant Molecule	Formulation and Dose	Administration Route	Length of Treatment	Animal Model	Sex	Main Outcomes*	Reference(s)
Melatonin	FA, 30 mg/kg	Oral delivery	6 mo	PSAPP mice 6 mo old	50% males	↑ hyperactivity, novel object recognition, spatial working and reference memory; ↓ β-amyloid deposits and Aβ oligomers	(543)
	FA, 5.3 mg/kg/day	Oral delivery	6 mo	APP/PS1 mice 6 mo old	100% females	↑ performance in novel-object recognition task; ↓ amyloid deposition and IL-1β	(544)
	FA, 30 mg/kg; EGCG 30 mg/kg	Oral delivery	3 mo	APP/PS1 mice 12 mo old	50% males	↑ cognition; ↓ β-amyloid deposits, neuroinflammation, oxidative stress and synaptotoxicity	(545)
	Melatonin, 150 mg/day	0.5 mg/mL drinking water	7.5 mo	Tg2576 mice 4 mo old	n.s.	↓ Aβ deposition, mouse survival; ↓ protein nitration	(546)
	Melatonin, 10 mg/kg	Oral gavage	4 mo	APP 695 mice 8 mo old	n.s.	↑ behavior; ↓ Aβ deposits, apoptosis.	(547)
	Melatonin, 50 mg/kg	Intraperitoneal injection	3 days	Aβ 25–35-injected mice	100% males	↓ ROS production; ↑ antioxidant enzyme activities	(548)
	Melatonin, 0.1, 1, 10 mg/kg	Intraperitoneal injection	7–10 days	Middle-aged and elder Aβ 25–35-injected rats	n.s.	↓ learning and memory deficits, lipid peroxidation; ↑ antioxidant response	(549, 550)
	Melatonin, 0.1, 1, 10 mg/kg	Intraperitoneal injection	30 days	Rats with constant illumination	100% males	↑ behavioral and molecular impairments	(551)
	Melatonin, 1, 10 mg/kg	Intraperitoneal injection	9 days pretreatment	Calyculin A-treated mice	100% males	↓ calyculin A-induced synaptophysin loss, memory retention deficits and tau hyperphosphorylation	(552)
	Melatonin, 10 mg/kg; physical exercise	Drinking water	6 mo	3xTg-AD mice 6 mo old	100% males	↓ Aβ oligomers, phosphorylated tau; ↓ cognitive impairment, brain OS and mitochondrial DNA	(553)
Coenzyme Q10	CoQ10, 1,200 mg/kg/day	Food	60 days	L235P PS-1 mice 16–17 mo old	100% females	↓ Aβ overproduction and Aβ deposit, OS and SOD activity	(554)
	CoQ10, 0.5%	Food	9 mo	P301S tau mice 1 mo old	Both sexes	↑ survival and behavioral deficits, ETC enzymes; ↓ oxidative stress	(555)

Continued

Table 5.—Continued

Antioxidant Molecule	Formulation and Dose	Administration Route	Length of Treatment	Animal Model	Sex	Main Outcomes*	Reference(s)
MitoQ	MitoQ, 100 mM	Drinking water	5 mo	3xTg-AD mice 2 mo old	100% females	↓ cognitive decline, OS, Aβ accumulation, astrogliosis, synaptic loss, and caspase activation	(556)
	MitoQ complexed to β-cyclodextrin (1-to-4 ratio), 100 mM	Drinking water	5 mo	3xTg-AD mice 12 mo old	100% females	↑ memory retention; ↓ brain oxidative stress, synapse loss, astrogliosis, microglial cell proliferation, Aβ accumulation, caspase activation, and tau hyperphosphorylation	(557)
Curcumin	Curcumin, 160, 2,000, 5,000 ppm	Oral delivery	6 mo	Tg2576 mice 10 mo old	Both sexes	↓ inflammation, oxidative stress, Aβ levels, plaques; spatial memory impairment	(558)
	Curcumin, 500 ppm	Oral delivery	4 mo	Tg2576 mice	Both sexes	↓ plaque burden and Aβ levels	(559)
	Curcumin, 7.7 mg/kg/day	Intravenous injection	1 wk	APP/PS1 mice 7.5–8.5 mo old	Both sexes	↑ clearance of Aβ deposits	(560)
	Curcumin, 500 ppm	Oral delivery	10 mo	Tg2576 mice 5 mo old	100% females	↑ levels of Aβ monomer; ↓ Aβ oligomer content	(561)
	Curcumin, 100, 200, and 400 mg/kg	Oral delivery	6 mo	APP/PS1 mice 3 mo old	50% males	↓ Aβ levels and aggregation, the expression of pre-senilin-2; ↑ expression of β-amyloid-degrading enzymes	(562–564)
	Curcumin derivatives FMeC1 and FMeC2, 500 ppm	Oral delivery	6 mo	APP/PS1 mice 9 mo old	Both sexes	↓ Aβ toxicity, Aβ deposits and glial cell activity	(565)
	CNB-001, 500 ppm	Oral delivery	6 mo	APP/PS1 mice 3 mo old	100% males	↑ degradation of aggregated intracellular Aβ, PERK/eIF2/ATF4 pathway	(566)
	Curcumin, 150 mg/kg; PPARγ antagonist GW9662, 4 mg/kg	Intraperitoneal injection	4 wk	APP/PS1 mice 8 mo old	n.s.	↓ spatial learning and memory deficits; ↑ PPARγ activity	(567)
	Curcumin NCF, 47 mg/kg	Drinking water	2 mo	APP/PS1 mice 14 mo old	n.s.	↓ deterioration of cognitive functions	(568)
	Curcumin micelles, 500 mg/kg	Food	3 wk	Thy1-APP751SL mice 7 mo old	n.s.	↓ levels of Aβ; ↑ mitochondrial membrane potential	(569)

Continued

Table 5.—Continued

Antioxidant Molecule	Formulation and Dose	Administration Route	Length of Treatment	Animal Model	Sex	Main Outcomes*	Reference(s)
Caffeine	Curcumin-loaded nanocapsules (NLC C), 10 mg/kg	Oral gavage	12 days (alternate days)	ICV A β 25-35-treated mice 3 mo old	100% males	↑ antidepressant-like and antioxidant effects	(570)
	Curcumin, 10 mg/kg	Intraperitoneal injection	7 wk	ICV-STZ + D-galactose-treated rats	100% males	↓ OS; ↑ abilities of active avoidance and locomotor activity	(571)
	Curcumin nanostructured lipid carriers (NLCs), 4 mg/kg	Intravenous injection	4 days	ICV-A β -treated mice	100% males	↓ OS parameters; ↑ spatial memory	(572)
	Caffeine, 1.5 mg/mouse	0.3 mg/mL in drinking water	5.5 mo	APP/PS1 4 mo old	50% males	↑ spatial learning/reference memory, working memory, and recognition/identification; ↓ A β levels and PS1 expression	(573)
	Caffeine, 1 mg/mL and 30 or 80 mg/kg	Drinking water and intraperitoneal injection	12 days pretreatment and single acute injection	ICV A β 25-35-treated mice 3–4 mo old	100% males	↓ A β -induced cognitive impairment	(574)
	Caffeine, 1.5 mg/mouse	0.3 mg/mL in drinking water	4–5 wk	APP/PS1 18–19 mo old	50% males	↑ working memory; ↓ A β levels	(575)
	Caffeine, 1.5 mg/mouse	0.3 g/L in drinking water	10 mo	THY-Tau22 mice 2 mo old	100% males	↓ development of spatial memory deficits, proinflammatory and OS markers	(576)
Resveratrol	Caffeine, 1.5 g/mouse	300 mg/L in drinking water	4 mo	Tg4-42 mice and 5xFAD mice 2 mo old	Both sexes	↓ hippocampal neuron loss, learning and memory deficits, impaired neurogenesis, behavioral deficits	(577)
	<i>Trans</i> -resveratrol, 1 g/kg	Food	7 mo	SAMP8 mice 2 mo old	100% males	↑ mean life expectancy and maximal life span, AMPK and SIRT1; ↓ cognitive impairment, amyloid and tau	(578)
	Resveratrol, 4 g/kg; physical exercise	Food	5 mo	3xTg-AD mice 8 wk old	100% males	↓ toxicity of A β oligomers, suppression of neuronal autophagy apoptosis; ↑ of key growth-related proteins	(579)
	Resveratrol, 100 mM/5 mL	ICV injection	7 days	ICV A β 25-35-treated rats	100% males	↑ spatial memory; ↓ A β -induced neurotoxicity	(580)

Continued

Table 5.—Continued

Antioxidant Molecule	Formulation and Dose	Administration Route	Length of Treatment	Animal Model	Sex	Main Outcomes*	Reference(s)
Quercetin	Quercetin, 20 mg/kg/day	Food	4 mo	APP/PS1 mice 3 mo old	50% males	↑ learning and memory deficits; ↓ senile plaques, mitochondrial dysfunction,	(581)
	Quercetin, 25 mg/kg	Intraperitoneal injection	Every 48 h for 3 mo	3xTg-AD mice 18–21 mo old	n.s.	↑ spatial learning and memory; ↓ tauopathy and A β levels, astrogliosis and microgliosis	(582)
	Quercetin, 100 mg/kg	Oral delivery	Every 48 h for 12 mo	3xTg-AD mice 6 mo old	n.s.	↑ cognitive functional; ↓ A β levels, tauopathy	(583)
	Quercetin, 40 mg/kg	Oral delivery	1 mo	ICV A β 1-42-treated rats	100% males	↑ neurogenesis through the promotion of CREB BDNF, NGF, and EGR-1 gene expression	(584)
	Quercetin conjugated with dextran-coated SPIONs, 25 mg/kg	Oral delivery	35 days	IP-STZ rats 2 mo old	100% males	↑ spatial learning and memory	(585)
	PLGA-functionalized quercetin NPs, 10, 20, 30 mg/kg	Intravenous injection	30 days	APP/PS1 mice	100% females	↑ spatial memory of AD mice in a dose manner	(586)
Vitamin E	α -Tocopherol 1%	Food	12 mo	Apolipoprotein E-deficient mice 1 mo old	100% females	↑ behavioral performance in the MWM; preserved dendritic structure; ↓ levels of lipid peroxidation	(587)
	Vitamin E, 8–10 IU/day/mouse	Food (2 IU/g diet)	6–8 mo	Tg2576 mice 5 and 14 mo old	50% males	↓ OS markers in mouse young group	(588)
	α -Tocopherol, 150 mg/kg	Oral delivery 1 mL/kg	27 days (7 days pretreatment)	ICV A β 1-42-treated mice	100% males	↓ memory impairments, OS	(589)
	α -Tocopherol 100 mg/kg and α -tocotrienol, 50 and 100 mg/kg	Oral delivery	21 days	ICV STZ rats	100% males	↓ cognitive impairment, reduction in glutathione and catalase, malonaldehyde, nitrite and cholinesterase activity	(590)
	α -Tocopherol 500 mg/kg and folic acid 50 mg/kg	Oral delivery	14 days	ICV A β 1-42-treated mice 3 mo old	100% males	↓ increase in the activity of mitochondrial complexes I and IV, NO generation	(591)

Continued

Table 5.—Continued

Antioxidant Molecule	Formulation and Dose	Administration Route	Length of Treatment	Animal Model	Sex	Main Outcomes*	Reference(s)
α-Lipoic acid	N-acetylcysteine, 50 mg/100 g; α-lipoic acid, 3 mg/100 g; and α-tocopherol, 1.5 mg/100 g	Food	4–6 mo	Aged rats 18 mo old	n.s.	↓ extent of oxidative stress and proinflammatory state, synaptosomal alterations and memory and learning deficits	(592, 593)
	α-Tocopherol quinone (α-TQ), 100 mg/kg	Oral gavage	4 wk	APP/PS1 mice	100% males	↓ memory impairment, oxidative stress and Aβ oligomer, production of inflammatory mediators, microglial activation	(594)
	TRF 60 g/kg; α-tocotrienol, 196 mg/g, β-tocotrienol 24 mg/g; γ-tocotrienol, 255 mg/g; δ-tocotrienol, 75 mg/g; α-tocopherol, 168 mg/g	TRF in water	10 mo	APP/PS1 mice 5 mo old	100% males	↓ Aβ immunoreactive depositions plaques; ↑ cognitive function.	(595)
	α-Lipoic acid, 100 mg/kg	Subcutaneous injection	4 wk	SAMP8 male mice 12 mo old	100% males	↑ learning and memory; ↓ OS and lipid peroxidation	(165, 166)
	α-Tocopherol acetate, 1,000 ppm; L-carnitine, 250 ppm; α-lipoic acid, 120 ppm; ascorbic acid, 80 ppm; behavioral enrichment	Food	2.8 yr	Beagle dogs 8–12 yr old	37–47% males	↓ age-associated cognitive decline, OS, lipid peroxidation, protein oxidation; ↑ discrimination and reversal learning, antioxidant enzyme activity	(189, 596, 597)
	α-Lipoic acid, 0.1%	Food	6 mo	Tg2576 mice	100% females	↓ hippocampus-dependent memory deficits without affecting β-amyloid levels or plaque deposition	(598)
	α-Lipoic acid, 100 mg/kg/day	Subcutaneous injection	Until death and 4 wk	SAMP8 mice 11 and 18 mo old	n.s.	↑ learning and memory, glutathione; ↓ glutathione peroxidase and malondialdehyde	(599)
Sulforaphane	R-sodium lipoic acid, 0.23%	Water	1 mo	3xTg-AD mice 12 mo old	100% males	↑ glucose metabolism	(600)
	Sulforaphane, 5 mg/kg/day	Intraperitoneal injection	4 mo	PS1V97L Tg mice 6 mo old	Both sexes	↓ Aβ generation and aggregation, tau hyperphosphorylation, oxidative stress, and neuroinflammation; ↑ cognition	(601)

Continued

Table 5.—Continued

Antioxidant Molecule	Formulation and Dose	Administration Route	Length of Treatment	Animal Model	Sex	Main Outcomes*	Reference(s)
Epigallocatechin-3-gallate	Sulforaphane, 5 mg/kg/day	Intraperitoneal injection	7 days	AβOs ICV-treated rats	100% males	↓ inflammatory factors and oxidative stress; ↓ memory impairment and depressive-like behavior	(602)
	Sulforaphane, 25 mg/kg	Oral gavage	90 days	D-galactose /aluminum-treated mice	50% males	↓ behavioral deficits, amyloid beta deposits, and peroxidation	(603, 604)
	Sulforaphane, 25 mg/kg	Oral gavage	5 mo	APP/PS1 mice 4 mo old	50% males	↓ behavioral cognitive impairments and brain Aβ burden	(605)
	Sulforaphane, 30 mg/kg	Intraperitoneal injection	6 days	ICV Aβ 1-40-treated mice 5.5 wk old	100% males	↑ cognitive function	(606)
	Sulforaphane, 50 mg/kg	Oral gavage	8 wk	3xTg-AD mice 12 mo old	100% females	↑ levels of CHIP and HSP70	(607)
	Sulforaphane, 10 mg/kg	Intraperitoneal injection	Every other day for 2 mo	5xFAD mice 9 mo old and 3xTg-AD 7 mo old	56% males	↓ AD-related cognitive deficits by reducing Bace1 and Bace1-AS expression, Aβ generation	(608)
	EGCG, 50 mg/kg	Water	6 mo	Tg2576 mice 8 mo old	100% females	↓ Aβ deposition tau phosphorylation; ↑ cognition	(609)
	EGCG, 1.5, 3 mg/kg	Water	3 wk pretreatment	LPS-treated mice	100% males	↓ memory impairment and apoptotic neuronal cell death, Aβ formation, and inflammation	(610)
	EGCG, 10 mg/kg/day	Oral gavage	4 wk	ICV STZ rats	100% males	↑ cognition, reversed S100B content, glutathione peroxidase activity, reactive oxygen species content	(611)
	EGCG, 2, 6 mg/kg	Oral gavage	4 wk	APP/PS1 mice 12 mo old	100% females	↓ impairment of learning and memory, IRS-1pS636 level, and Aβ42 levels	(612)
	EGCG, 5, 15 mg/kg	Intragastric administration	60 days	SAMP8 mice	n.s.	↓ levels of the rate-limiting degradation enzyme of Aβ, neprilysin	(613)

AβO, Aβ oligomer; CHIP, C-terminus of heat shock protein (HSP)70-interacting protein; GPx, glutathione peroxidase; GR, glutathione reductase; GSH, glutathione; ICV, intracerebroventricular; LPS, lipopolysaccharide; MWM, Morris water maze; NO, nitric oxide; NP, nanoparticle; n.s., not specified; OS, oxidative stress; PERK, protein kinase RNA-like ER kinase; PPAR, peroxisome proliferator-activated receptor; SOD, superoxide dismutase; SPION, superparamagnetic iron oxide nanoparticles; STZ, streptozocin; TRF, tocotrienol-rich fraction; ↑, increase; ↓, decrease. *Discussed in text.

FAEE against AAPH- and NADPH/ADP-Fe³⁺-induced LPO (622).

6.2.5. Melatonin.

Melatonin is a lipophilic hormone that is mainly produced and secreted at night by the pineal gland. In vivo studies demonstrated that long-term administration of melatonin partially inhibited the expected time-dependent elevation of A β and reduced the abnormal nitration of proteins in Tg2576 transgenic mice (546). Similarly, long-term melatonin administration reportedly prevented the abnormal upregulation of apoptotic markers and alleviated memory impairments in APP695 transgenic mice (547). Melatonin showed improvement of learning and memory in A β (25–35)-treated elder rats, whereas it prevented LPO, induced antioxidant response, and facilitated learning and memory in A β (25–35)-treated middle-aged rats (549, 550). Two caveats of these studies employing A β (25–35) are 1) there is no evidence of this 11-amino acid peptide in AD brain and 2) the mechanisms of oxidative stress associated with this peptide with a terminal Met residue are not the same as mechanisms associated with A β 42 with Met35 in a nonterminal location within the peptide (114). In melatonin-deficient mice, supplementation of melatonin partially arrested behavioral and molecular impairments (551). In addition, in calyculin A-treated mice melatonin attenuated hippocampal memory dysfunction, pTau, and neurodegenerative damage through the modulation of the PI3K/Akt/GSK-3 β pathway (623). In 3xTg-AD male mice the combined effects mediated by physical exercise and melatonin decreased soluble A β oligomers and protected against cognitive impairment, brain OS, and decreased mitochondrial DNA (553).

6.2.6. Coenzyme Q.

Coenzyme Q10 (CoQ10) is a lipid-soluble electron carrier and a cofactor of mitochondrial uncoupling proteins that blocks apoptosis by inhibiting the mitochondrial permeability transition pore (624). Several studies have been conducted using CoQ10 in in vivo models. CoQ10 pretreatment prevented the decrease in mitochondrial transmembrane potential and reduced mitochondrial ROS generation (227). Aged PS1 transgenic mice fed with CoQ10 for 60 days showed partial reduction of A β overproduction and of intracellular A β deposits (554). The administration of CoQ10 to P301S Tau mice reduced OS, upregulated the function of ETC complexes, increased neuronal survival rates, and improved cognitive behavior (555).

Like CoQ10, MitoQ, a triphenylphosphonium-linked ubiquinone derivative that concentrates several hundredfold in mitochondria, exerted protection from toxicity

in mouse neuroblastoma cells. The 3xTg-AD mouse model supplemented in the diet with MitoQ for 5 mo showed protection against cognitive decline as well as OS, A β accumulation, astrogliosis, synaptic loss, and caspase activation (556, 557).

6.2.7. Curcumin.

Curcumin is derived from the rhizome of *Curcuma longa*. Epidemiologically, dietary curcumin intake is positively related to cognitive function in healthy elderly individuals. Curcumin and its formulations are known to be potent antioxidant, anti-inflammatory, and antiamyloidogenic compounds (625). In vitro studies on different cell lines under damaging prooxidant stimuli demonstrated the protective effects of curcumin against the inhibition of OS damage, A β -induced toxicity, and pTau (626). In vivo studies on APP/PS1 and in Tg2576 transgenic mice demonstrated that curcumin could inhibit A β formation and accumulation, reduce inflammation, activate neuronal stem cell proliferation, and ameliorate cognitive impairment (558–566, 568, 627). Similarly, the use of curcumin in an AD rat model induced by bilateral hippocampal injection of STZ showed the reduction of OS and A β load (571). Recently, to overcome the poor bioavailability and biodistribution of curcumin, curcumin-loaded nanostructured lipid core capsules (NLC C) were tested in AD mice. The results showed that NLC C given to mice treated with A β (25–35) reduced the levels of ROS and regularized the levels of antioxidant enzymes, such as SOD and CAT, in the prefrontal cortex (570). Note that the caveats noted above regarding use of A β (25–35) apply to this study as well. In addition, the evaluation of NLC C in male Sprague-Dawley rats treated with A β 42 showed a decrease in ROS levels and LPO in the hippocampal tissue (572).

6.2.8. Caffeine.

Caffeine is a methylxanthine alkaloid mainly present in coffee or tea with antioxidant, antiapoptotic, anti-inflammatory, and antiamyloidogenic properties (625). Prolonged caffeine administration in drinking water prevented spatial memory impairment in several AD models. Caffeine treatment was able to reduce proinflammatory mediators, OS, and hippocampal A β and pTau levels (573–577).

6.2.9. Resveratrol.

Resveratrol is a nonflavonoid polyphenol found in berries, peanuts, and especially red grape seeds and skin (521). Resveratrol administration was shown to increase mean life expectancy in SAMP8 mice by reducing A β burden, pTau, and cognitive impairment (578). Resveratrol supplementation in 3xTg-AD mice demonstrated a marked

reduction in proinflammatory markers in comparison with controls (579). Resveratrol supplementation in STZ intra-peritoneally injected or A β hippocampally injected Wistar rats showed improvements in cognitive function with the upregulation of antioxidant enzyme activities and of GSH levels in both hippocampus and prefrontal cortex (628). In addition, resveratrol treatment of adult male Sprague-Dawley rats exposed to A β showed a marked reduction in iNOS content and an increase in the free radical scavenger HO-1, supporting its antioxidant activity (580).

6.2.10. Quercetin.

Quercetin is a flavonoid abundant in apple skins and onions with significant antioxidant properties. In vitro studies demonstrated that low doses of quercetin protected primary neurons against A β 42-induced toxicity via modulation of OS (625), whereas high dosage demonstrated loss of protection and toxic effects (629). In addition, the treatment of PC12 cells with quercetin led to considerable neurite outgrowth and increased the complexity of the neuronal branching (630). As well, animal studies on quercetin treatment demonstrated the protective antioxidant properties of this compound (631). Long-term oral administration of quercetin ameliorated mitochondrial dysfunction and decreased A β plaques and ROS production in APP/PS1 mice (581). In A β 42 intracerebroventricularly induced rats quercetin improved neurogenesis by upregulating the CREB/brain-derived neurotrophic factor (BDNF) signaling pathway (584). Quercetin treatment exerted neuroprotection in 3xTg-AD mice via reduction of the levels of extracellular β -amyloidosis, tauopathy, and astrogliosis in both hippocampus and amygdala, and quercetin improved learning and spatial memory performance (582). A 12-mo-long treatment with quercetin by gavage in 3xTg-AD mice prevented the accumulation of A β and reduced tauopathy in the hippocampus and amygdala (583). The use of solid lipid nanoparticles (NPs) of quercetin in an AlCl₃-induced AD rat model led to the significant induction of antioxidant activities (632). Also, injection of PLGA-functionalized quercetin NPs into APP/PS1 mice ameliorated cognition and memory dysfunctions (586). Oral administration of quercetin packed in zein NPs improved cognitive functions and memory loss and reduced expression of astrogliosis markers in SAMP8 mice (633). Furthermore, treatment with quercetin-conjugated superparamagnetic iron oxide NPs (QTSPIN) improved learning and memory in STZ rats through targeting NF- κ B and Nrf2 pathway (585).

6.2.11. Vitamin E.

Vitamin E represents a group of eight lipid-soluble compounds, four tocopherols and four tocotrienols synthesized by plants. α -Tocopherol accounts for 90% of

vitamin E in human tissues and demonstrates strong antioxidant properties. In vitro studies showed that vitamin E administration was able to prevent A β 42-induced protein oxidation, ROS production, and neurotoxicity in primary rat embryonic hippocampal neuronal culture (634). Furthermore, vitamin E pretreatment in cells exposed to A β 42 increased cell viability and reduced OS by preventing reduction in phospholipid and ubiquinone levels (635). In vivo studies showed that α -tocopherol, administered before A β injection in A β 42-treated mice, significantly attenuated brain damage and ameliorated memory deficits (589). Early vitamin E administration reduced A β 40/42 levels and A β deposits in young, but not in old, Tg2576 mice. However, vitamin E counteracted OS induction in both age groups (588). Treatment with α -tocopherol quinine was able to reduce A β oligomer levels, to inhibit microglial activation by reducing the levels of the proinflammatory cytokines, to decrease OS by increasing SOD activity, and to improve memory and cognitive dysfunction (594). α -Tocopherol and tocotrienol were tested in STZ-treated rats, showing the enhancement of the cognitive function, by preventing the reduction of GSH, SOD, and catalase activity and by reducing MDA, nitrite, and cholinesterase activity (590). α -Tocopherol demonstrated its efficacy in protecting dendritic structure and improving behavioral performances in apolipoprotein E-deficient mice treated for 12 mo (587). Furthermore, a tocotrienol-rich fraction (TRF) was tested on APP/PS1 mice, showing that it was able to mitigate cortical A β deposition, thioflavin-S-positive fibrillar type plaques in the hippocampus and cortex, and cognitive function (595).

Vitamin E often has been tested in vivo in combination with different compounds (636). Tg2576 mice received a diet supplemented with α -tocopherol and indomethacin and showed suppression of brain OS and inflammation (637). APP/PS1 mice supplemented with vitamins E and C demonstrated improved spatial memory deficits and lower neuroprostanol levels (638). Combination therapy with folic acid and α -tocopherol was evaluated in mice injected in the lateral ventricle with A β 40, showing that their joint effect exerted a protective action against A β -induced cognitive decline through a reduction of synaptic dysfunction (591). The administration of a combination of *N*-acetylcysteine, α -lipoic acid, and α -tocopherol to aged rats prevented OS, inflammation, and age-dependent changes in synaptosomal parameters together with reduction of LPO and improvement in learning and memory (592, 593). Furthermore, this dietary supplementation mix showed a reduction of APP, β -secretase activity, and A β 42 levels.

6.2.12. α -Lipoic acid.

α -Lipoic acid is an endogenous antioxidant. Metabolically, α -lipoic acid serves a catalytic function in mitochondrial

pyruvate dehydrogenase and α -ketoglutarate dehydrogenase. Initial studies reported that α -lipoic acid administration in old Fischer rats increased nuclear Nrf2 levels and induced Nrf2 binding to the ARE, with consequently higher GCLC levels and GCL activity (341). SAMP8 mice that received α -lipoic acid had significantly increased GSH and decreased GPx and MDA, indicating that this antioxidant molecule improved memory, reduced protein oxidation, and reversed other indexes of OS (165,166, 599). An antioxidant dietary pattern, including α -lipoic acid, and behavioral enrichments were tested for 2.8 yr on beagle dogs with an age range of between 8 and 12 yr, demonstrating improvements in discrimination and reversal learning and the reduction of protein oxidation and lipid peroxidation (189, 596, 597). Chronic dietary α -lipoic acid administration reduced hippocampus-dependent memory deficits in Tg2576 mice without affecting A β levels or plaque deposition (598). α -Lipoic acid treatment was shown to successfully overcome the hypometabolic state associated with lower glucose-TCA cycle-related metabolite flux in the brains of 3xTg-AD mice (600).

6.2.13. Sulforaphane.

The sulforaphane molecule presents the isothiocyanate group of organosulfur compounds, which can efficiently activate Nrf2 by reacting with the cysteine residues of Keap1 (639). In vitro studies on sulforaphane demonstrated that the treatment of SH-SY5Y cells led to the overexpression of Nrf2 and its related antioxidant response and reduced the transcription level of *BACE1* involved in amyloidogenesis processes. Similarly, N2a cells harboring human mutant APP were treated with sulforaphane, leading to reduced intracellular levels of A β 40/42 and to decreased OS. Furthermore, sulforaphane administered to microglial cells pretreated with A β oligomers induced an increase of phagocytic activity (640). The in vivo testing of sulforaphane protective effects was performed in PS1V97L transgenic mice, administered via long-term intraperitoneal injection, showing improvement of cognitive deficits, inhibition of A β aggregation and pTau elevation, along with reduction of OS and neuroinflammation (601). In addition, sulforaphane treatment of male Sprague-Dawley rats injected intracerebroventricularly with A β showed reduction of neuroinflammation and OS and improvement in depressive behavior and spatial learning (602). Similarly, sulforaphane administered orally to C57BL/6 mice with induced AD-like lesions demonstrated improved cognitive and locomotor deficits, protection against the formation of A β plaques, and reduced OS compared to control (603). Furthermore, the use of sulforaphane in Kunming mice treated with neurotoxic doses of aluminum and D-galactose led to improved cognitive deficits and decreased the loss of cholinergic neurons

(604). Sulforaphane administration to A β -treated ICR mice improved cognitive and memory deficits without preventing the formation of aggregated A β (606). Oral gavage of sulforaphane reduced levels of monomeric and polymeric forms of A β as well as Tau and pTau in 3xTg-AD mice (607). In 5xTg-FAD mice sulforaphane improved cognitive deficits, reduced BACE-1 expression, and decreased A β aggregation (608). Moreover, the administration of sulforaphane via gavage in APP/PS1 mouse improved cognitive deficits and reduced A β plaque levels (605).

6.2.14. Epigallocatechin-3-gallate.

Epigallocatechin-3-gallate (EGCG) is a natural flavanol mainly found in green tea. EGCG through its antioxidant property reportedly prevents A β -induced hippocampal neuronal cell death in hippocampal cells (641). Consistent with this report, many other studies have demonstrated protective effects (605, 607–611, 641). In vivo studies on EGCG treatment demonstrated that it was able to decrease A β levels and plaque formation in Tg2576 mice when administered via drinking water (609). In mice with systemic inflammation induced by lipopolysaccharide (LPS), EGCG treatment prevented memory impairment and apoptotic neuronal cell death (610). Moreover, EGCG prevented both the activation of astrocytes and increased cytokine expression. In rats treated by ICV-STZ, EGCG abrogated cognitive deficits by influencing the activities of acetylcholinesterase (AChE) and GPx, nitric oxide metabolites, and ROS content (611). In an AD mouse model generated by D-galactose, EGCG demonstrated a protective effect by decreasing the hippocampal expression of APP and A β (642). Furthermore, EGCG given to APP/PS1 mice reduced spatial memory impairment by rescuing insulin signaling (612). In SAMP8 mice EGCG reduced A β accumulation and rescued cognitive deterioration by upregulating neprilysin expression (613). The engineering of nanolipidic EGCG particles to improve oral bioavailability of EGCG for the treatment of AD demonstrated a twofold enhancement in respect to free EGCG (643).

6.3. Clinical Trials

Clinical trials employing antioxidant molecules are summarized in TABLE 6.

6.3.1. N-acetylcysteine.

The first trial involving NAC administration was the NAC-003 P.L.U.S. Program (NCT01370954), which enrolled 204 subjects with early memory impairment administered CerefolinNAC daily for 12 wk. In the oral formulation N-acetylcysteine was associated with methycobalamin and L-methylfolate calcium. The trial ended in December 2012,

but results were not posted. Simultaneously, Memory XL, a nutraceutical formulation containing *N*-acetylcysteine added to folic acid, vitamin B12, α -tocopherol, *S*-adenosyl-methionine, and acetyl-L-carnitine, was tested in 10 subjects with MCI treated for 1 yr (NCT00903695). The trial was completed in May 2011, and the evaluation of cognitive and behavioral status demonstrated no significant differences between nutraceutical and placebo groups (644). Conceivably, the limited number of patients enrolled affected the results. Another large multisite placebo-controlled clinical trial with Memory XL (NCT01320527) conducted on 135 patients with MCI and AD ended in April 2012 and showed maintenance or improved cognitive performance and mood/behavior in the nutraceutical group (645, 646).

6.3.2. Vitamin E.

The therapeutic effects mediated by the administration of vitamin E are still unclear because of conflicting results. Studies reported a decrease in vitamin E levels in aging and dementia with a correlation to memory loss (522). Supplementation of vitamin E does increase levels of this vitamin in AD and decreases susceptibility of lipoproteins to oxidation; however, it is not clear as to whether vitamin E improves cognition (668). In 1997 Sano and colleagues (647) published results from a double-blind, placebo-controlled, multicenter randomized clinical trial (RCT) in patients with AD of moderate severity. The researchers enrolled 341 AD patients, who received selegiline, a selective monoamine oxidase inhibitor (10 mg/day), and α -tocopherol (2,000 IU/day) alone or in combination for 2 yr. The results demonstrated that in patients with moderately severe impairment from AD, treatment with selegiline or α -tocopherol slowed the progression of the disease. In another study (the Cache County study), it was demonstrated that the combined supplementation of AD patients with vitamins E (400 IU/day) and C (1 g/day) could reduce AD prevalence and incidence (648). In a further double-blind, placebo-controlled RCT (NCT00117403) that ended on September 2007, the combination of 800 IU of vitamin E with 200 mg of vitamin C and with 600 mg of α -lipoic acid (E/C/ALA) was tested on 75 AD patients. The results showed no influence on pathways related to A β and Tau pathology, whereas the formulation demonstrated antioxidant properties with the significant reduction of F₂-isoprostane levels in the CSF (649). In 2009 Lloret and colleagues published the results from a 6-mo treatment of AD patients with vitamin E (800 IU/day). After the treatment, the researchers divided patients in two groups: “respondents,” whose GSSG levels decreased after vitamin E administration and maintained the scores in cognitive tests, and “nonrespondents,” in whom OS

did not decrease after vitamin E treatment (650). Surprisingly, in nonrespondent patients vitamin E administration seemed to act as a prooxidant because in these patients cognition decreased sharply to levels even lower than those of patients taking placebo. The TEAM-AD, a large double-blinded, placebo-controlled RCT of vitamin E (2,000 IU/day) and memantine (20 mg/day) in 613 participants (NCT00235716), showed that among patients with mild to moderate AD vitamin E decreased functional decline at a rate that was superior to placebo and current AD treatment (651, 652). In the study published in 2015 from the AD Cooperative Study Group, Petersen and colleagues selected 769 subjects and performed a double-blind trial for the administration of 2,000 IU of vitamin E daily, 10 mg of donepezil daily, or placebo for 3 yr. The study showed that vitamin E had no benefit in patients with MCI, whereas donepezil therapy was associated with a lower rate of progression to AD during the first 12 mo of treatment (653). The PREADViSE trial (NCT00040378) recruited participants over 60 yr of age ($n = 7,547$) who were treated with long-term supplementation of vitamin E (400 IU/day) or selenium (200 μ g/day) and with vitamin E-selenium combination for an average time of ~ 5 yr. The study showed that neither vitamin E, nor selenium, nor even a combination of both prevented dementia (654–656). Similar observations were obtained in a study published in 2013 in which the authors assessed vitamin E supplementation (300 mg) and vitamin C (400 mg) on the cognitive function of MCI patients ($n = 256$) with an age of 60–75 yr. The intervention lasted a year and resulted in a reduction of the OS of the body, including a decrease in the level of malonic aldehyde, but did not improve cognitive abilities (657).

6.3.3. Acetyl-L-carnitine hydrochloride.

Acetyl-L-carnitine hydrochloride (ALCAR) was tested in clinical trials, alone or in association with donepezil or rivastigmine, on MCI and mild AD patients, showing beneficial effects on both clinical and psychometric tests (658, 669). A multicenter, double-blind, placebo-controlled RCT (NCT02955706) to assess the efficacy of acetyl-L-carnitine in AD patients ended in 2019, but results are not available yet.

6.3.4. Caffeine.

Kim and colleagues (659) analyzed the relationship between coffee intake and AD biomarkers in the human brain from 411 participants with MCI without a diagnosis of dementia. The findings from neuroimaging techniques demonstrated that the longer coffee intake was significantly associated with a reduced pathological cerebral A β deposition. Conceivably, these results could

Table 6. Clinical trials employing antioxidant molecules in patients ranging from early memory loss to late AD

Antioxidant Molecule	Trial Name and ID	Formulation	Administration	Length of Treatment	No. of Participants	Sex Eligibility/Participation*	Condition	Main Outcomes	Ref(s).
N-acetyl-L-cysteine	NAC-003 P.L.U.S. Program; NCT01370954	CerefolinNAC: methyl cobalamin 2 mg, N-acetylcysteine 600 mg, L-methylfolate calcium 6 mg	Oral, 1 caplet/day	12 wk	204	All/n.s.	Early memory loss	Not posted	N/A
	Memory XL; NCT00903695	Folic acid 400 mg, Vit B12 6 µg, α-tocopherol 30 IU, S-adenosyl methionine 400 mg, N-acetylcysteine 600 mg, acetyl-L-carnitine 500 mg	Oral, 2 pills/day	12 mo	10	All/100% males	MCI	No significant differences between nutritional and placebo group	(644)
	Memory XL; NCT01320527	Folic acid 400 mg, Vit B12 6 µg, α-tocopherol 30 IU, S-adenosyl methionine 400 mg, N-acetylcysteine 600 mg, acetyl-L-carnitine 500 mg	Oral, 2 pills/day	12 mo	135	All/n.s.	MCI, AD	Maintained or improved cognitive performance and mood/behavior in the nutritional group	(645, 646)
Vitamin E	N/A	Selegiline 5 mg (10 mg/day), α-tocopherol 1,000 IU (2,000 IU/day)	Oral, 2 times/day	24 mo	341	All/64.9% females	AD	Lowered disease progression	(647)
	Cache County study	vitamins E (400 IU/day) and C (1 g/day)	Supplements	12 mo	4,740 respondents	All/57.2% females	AD	Reduced AD prevalence and incidence	(648)
	Antioxidant Treatment of Alzheimer's Disease; NCT00117403	Vitamin E 800 IU, vitamin C 200 mg, and α-lipoic acid 600 mg	Oral, 3 capsules/day	4 mo	75	All/46–48% females*	AD	no influence on Aβ and tau pathology; reduction of F ₂ -isoprostanates levels in the CSF	(649)
N/A		Vitamin E 800 IU/day	Oral	6 mo	75	All/n.s.	AD	Cognitive status was maintained in some cases, but in others it was detrimental in terms of cognition.	(650)

Continued

†

Table 6.—Continued

Antioxidant Molecule	Trial Name and ID	Formulation	Administration	Length of Treatment	No. of Participants	Sex Eligibility/ Participation*	Condition	Main Outcomes	Ref(s).
Acetyl-L-carnitine	TEAM-AD; NCT00235716	α-Tocopherol 1,000 IU (2,000 IU/day); ± memantine 5 mg (10 mg/day)	Oral, 2 capsules/day	48 mo	613	All/96–98% males*	AD	Decreased functional decline	(651, 652)
	N/A	Vitamin E 2,000 IU/day; Donepezil 10 mg	Oral	3 yr	769	All/44–47% females*	MCI	No significant effects on AD progression	(653)
	PREADVISE; NCT00040378	Vitamin E 400 IU/day ± selenium 200 µg/day	Oral, 1 pill/day	5 yr	7,547	Males/100% males*	AD	No significant prevention of dementia	(654–656)
	N/A	Vitamin E 300 mg/day + vitamin C 400 mg/day	Oral	1 yr	256	All/50–56% females*	MCI	Reduced OS, no effect on cognitive function	(657)
	N/A	Donepezil 5 mg/day, rivastigmine 3 mg twice/day + acetyl-L-carnitine 2 mg/day twice/day	Oral	3 mo	23	All/60% females	AD AChE-I; nonresponders	Beneficial cognitive effects on AChE-I; nonresponders	(658)
Caffeine	NCT02955706	Nicetle: acetyl-L-carnitine 500 mg	Oral	24 wk	265	All/n.s.	AD	No results posted	N/A
	N/A	Coffee ≥ 2 cups/day	Oral	3 yr	411	All/40–65% females*	MCI	Lowered risk of AD and reduced cerebral amyloid deposition	(659)
Curcumin	CAFCA; NCT04570085	Caffeine 200 mg (400 mg/day)	Oral 2 capsules/day	27 wk	248	All/n.s.	Early AD stage	Ongoing study	N/A
	Curcumin and Ginkgo for Treating Alzheimer's Disease; NCT00164749	Curcumin 1,4 g/day + 120 mg/day ginkgo leaf extract	Oral	6 mo	36	All/29% males	AD	No significant improvements were obtained concerning cognitive testing, Aβ levels, or antioxidant responses	(660, 661)

Continued

Table 6.—Continued

Antioxidant Molecule	Trial Name and ID	Formulation	Administration	Length of Treatment	No. of Participants	Sex Eligibility/ Participation*	Condition	Main Outcomes	Ref(s).
	18-Month Study of Memory Effects of Curcumin; NCT01383161	Theracurmin: curcumin 30 mg (180 mg/day)	Oral 6 capsules/day	18 mo	46	All/53–57% females*	MCI, aging	Memory and attention benefits, decreased plaque and tan- gle accumulation, modulation of mood and memory	(662)
	Curcumin in Patients with Mild to Moderate Alzheimer's Disease; NCT00095710	Curcumin C3 complex: Curcumin 2, 4 g/day	Oral	12 mo	33	All/55–70% females*	Mild to moderate AD	No clinical or biochemical evidence of efficacy	(663)
	N/A	Longvida optimized curcumin 400 mg = curcumin 80 mg/day	Oral capsule	Acute: 1, 3 h after single dose chronic: 4 wk acute-on-chronic: 1, 3 h after a single dose following 4-wk treatment	61	All/33–40% males*	60-yr-old subjects	Improved cognitive function, reduced fatigue, and lessened the detrimental impact of psychological stress on mood.	(664)
	Curcumin and Yoga Therapy for Those at Risk for Alzheimer's Disease; NCT01811381	Longvida curcumin: curcumin 800 mg/day ± aerobic yoga	Oral 4 capsules/day	12 mo	117	All/n.s.	AD risk	No results posted	N/A
Resveratrol	Randomized Trial of a Nutritional Supplement in Alzheimer's Disease; NCT00678431	Dextrose 5 g (10 g/day), malate 5 g (10 g/day), resveratrol 5 mg (10 mg/day)	Oral in liquid grape juice twice a day	12 mo	27	All/56–61% males*	Probable AD	Reduced cognitive deterioration	(665)

Continued

Table 6.—Continued

Antioxidant Molecule	Trial Name and ID	Formulation	Administration	Length of Treatment	No. of Participants	Sex Eligibility/ Participation*	Condition	Main Outcomes	Ref(s).
Epigallocatechin-3-gallate	Resveratrol for Alzheimer's Disease; NCT01504854	Resveratrol 500, 1,000 mg/day	Oral, 1 or 2 capsules/day	12 mo	119	All/51–63% females*	Mild to moderate AD	Attenuated MMSE score decline, decreased CSF MMP9, modulated neuroinflammation, and induced adaptive immunity	(666)
	Effects of green tea consumption on cognitive dysfunction and atherosclerosis: a randomized-controlled study; JPRN-UMIN000011668	Green tea 2 g/day containing catechins 220.2 mg	Oral	12 mo	27	All/82–94% females*	AD	No significant effect on cognitive function (MMSE-J), reduced OS	(667)
	Sunphenon EGCG in the Early Stage of Alzheimer's Disease; NCT00951834	Sunphenon EGCG 200, 400, 600, 800 mg/day	Oral	18 mo	21	All/n.s.	AD	No results posted	N/A
	Prevention of cognitive decline in ApoE4 carriers with subjective cognitive decline after EGCG and a multimodal intervention; NCT03978052	EGCG + multimodal intervention: EGCG 260–520 mg, Font-up 49–98 g $\pm$ physical activity $\pm$ lifestyle recommendations	Oral	15 mo	200	All/n.s.	ApoE4 carriers	Currently recruiting	N/A

AD, Alzheimer disease; CSF, cerebrospinal fluid; MCI, mild cognitive impairment; MMSE, Mini-Mental Status Examination; N/A, not applicable; n.s., not specified; OS, oxidative stress. *percent range between groups.

herefore be linked to a lowered risk for AD or cognitive decline. Furthermore, no correlation between coffee intake and AD-associated glucose hypometabolism, atrophy, or cortical thickness and cerebral white matter hyperintensities was detected (659). An ongoing clinical trial (NCT04570085) that is supposed to end in 2024 will test the effect of caffeine on cognition in AD (CAFCA).

6.3.5. Curcumin.

A phase I/II trial (NCT00164749) evaluated the effect of curcumin (1 and 4 g/day) and ginkgo extracts (120 mg/day) over 6 mo on the progression of AD. Results from 36 participants indicated that curcumin did not seem to cause side effects in AD patients; however, no significant improvements were obtained concerning cognitive testing, A β levels, or antioxidant responses (660, 661). A phase II clinical trial (NCT01383161) tested the effect of curcumin (Theracurmin 180 mg/day) on 46 participants with age-associated cognitive impairment and MCI for 18 mo. The results showed that daily oral Theracurmin led to significant memory and attention benefits, associated with decreases in plaque and tangle accumulation in brain regions and with the modulation of mood and memory (662). The effect of 24-wk curcumin supplementation (curcumin C3 complex) was evaluated in a further phase II double-blind, placebo-controlled RCT (NCT00099710) on 33 participants. Results showed that curcumin was generally well tolerated, but it did not demonstrate clinical or biochemical evidence of efficacy in AD patients (663). Cox et al. (664) showed that supplementation with solid lipid curcumin formulation (Longvida 80 mg) in ~60-yr-old subjects improved cognitive function, reduced fatigue, and lessened the detrimental impact of psychological stress on mood, which may improve the quality of life for the growing elderly population. A phase II clinical trial ended in December 2020 (NCT01811381) evaluated the effect of curcumin and yoga in patients with MCI. Results from the study are not available yet.

6.3.6. Resveratrol.

A RCT completed in 2011 (NCT00678431) evaluated the effect of dietary supplementation with resveratrol, glucose, and malate on 27 participants with the aim of slowing the progression of AD. Results from this study demonstrated that after 12 mo the Alzheimer's Disease Assessment Scale–cognitive subscale, the Mini-Mental State Examination, the Alzheimer's Disease Cooperative Study–Activities of Daily Living Scale, or the Neuropsychiatric Inventory showed less deterioration in the treatment group but without statistical significance

(665). A placebo-controlled, double-blind, multicenter phase II RCT (NCT01504854) of resveratrol (500–1,000 mg/day) was performed in 119 individuals with mild to moderate AD for 52 wk. Data demonstrated that resveratrol markedly reduced CSF MMP9 and increased macrophage-derived chemokines (MDCs), interleukin (IL)-4, and fibroblast growth factor (FGF)-2. Furthermore, resveratrol increased plasma MMP10 and decreased IL-12P40, IL-12P70, and RANTES. In addition, resveratrol treatment attenuated declines in Mini-Mental Status Examination (MMSE) scores, change in ADL (ADCS-ADL) scores, and CSF A β 42 levels but did not alter Tau levels (666).

6.3.7. Epigallocatechin-3-gallate.

A phase IV RCT (NCT00981292) tested EGCG (135 and 270 mg) in 27 healthy human adults and reported that EGCG was able to modulate cerebral blood flow parameters without affecting cognitive performance or mood (670). Scholey and colleagues (671) showed that 300-mg administration of EGCG to 31 participants resulted in reduced stress, increased calmness, and increased electroencephalographic activity in the midline frontal and central brain regions. A 1-yr-long double-blind RCT (UMIN000011668) was conducted on 27 participants to assess the effects of green tea consumption (2 g/day) on cognitive dysfunction. Results suggested that green tea consumption may not significantly affect cognitive function assessed by MMSE-J but prevented the increase of OS in the elderly population (667). A phase II/III RCT (NCT00951834) for the treatment of 21 patients with early AD already under donepezil with EGCG for 18 mo was completed in 2015, but no data are available yet. A double-blind personalized RCT (NCT03978052), using EGCG on ~200 participants, is currently recruiting. The study aims to evaluate the efficacy of a multimodal intervention (dietary, physical activity, and cognition) combined with EGCG in slowing cognitive decline in ApoE4 carriers.

6.4. Nutritional Approaches

About one-third of AD cases worldwide might be attributable to potentially modifiable risk factors, and it is well established that AD conditions can be improved by modifying lifestyle aspects (521, 527). Several lines of evidence showed that lifestyle factors such as education, controlled dietary regimens, cognitive activity, and physical activity may delay the onset of dementia (672). The role of diet in controlling AD risk is critically important. Nutrients including fats, sugars, minerals, and vitamins have been conclusively associated with AD risk and disease progression via a range of mechanisms. Recent data have suggested a direct role for diet in modulating OS, inflammation, synaptic plasticity, amyloidogenesis,

and neurogenesis. Specific dietary patterns have been definitively linked to AD risk, and a complex mutual association between diet and AD may exist (531, 673). Furthermore, unbalanced diets, beyond playing a role in the outcomes of AD, might alter drug metabolism and modify the perceived clinical efficacy of therapeutic agents. Finally, because prevention must occur when the patient starts being at risk and treatment depends on lifestyle changes, a well-balanced diet is the most reasonable approach to optimize the maintenance of a good health status (674).

6.4.1. Multidomain strategies and dietary patterns.

The use of multidomain strategies by nutritional and lifestyle intervention showed that these strategies may alleviate some of the uncertainties raised by single-nutrient trials. They offer the advantage of pointing at multiple targets instead of modulating single entities. In the FINGER study, a RCT (NCT01041989) on 1,200 participants, researchers employed a 2-yr-long multidomain approach using balanced nutrition, exercise, cognitive activity, and general control of health in patients with MCI at risk of dementia (675). The results of the study showed that treated patients had a better outcome in terms of cognition after 2 yr. A recently published study of a 2-yr treatment (NTR1705) with a specific multinutrient approach in individuals with prodromal AD has shown some encouraging results (676). Although multinutrient intervention had no significant effect on the neuropsychological test battery, nutritional treatment had better results on the cognitive-functional measure CDR-SB and less brain atrophy assessed by magnetic resonance imaging. In contrast, some multidomain approaches did not show clinical benefits. A 6-yr multidomain vascular intervention (ISRCTN29711771) in 3,526 participants aged 70–78 yr did not result in a reduced incidence of dementia (677).

Among dietary patterns, the Mediterranean diet (MeDi) demonstrated significant reductions in mortality and morbidity in different disease contexts (678, 679). The MeDi can be considered a nutritional model for healthy dietary habits since it contains all the essential nutrients including monounsaturated fatty acids, polyunsaturated fatty acids, antioxidants, vitamins, and minerals (680). Moreover, the diet is low in saturated fats, cholesterol, high-fat dairy products, and red meats. It therefore excludes many of the known foods associated with AD risk and includes several protective nutrients (681, 682). Thus, it is not surprising that cross-sectional studies, prospective studies, and RCTs demonstrated that adherence to the MeDi was correlated with a reduced risk of AD (682–685). Two cross-sectional studies showed an inverse correlation between the MeDi and AD in older American and

Australian adults (681, 684). In prospective studies published in the last 15 years, greater adherence to the MeDi was associated with a reduced risk of AD, a lower risk of developing AD in patients with MCI, and lower mortality in AD patients, suggesting a possible role of MeDi in modulating not only the pathogenetic pathways but also the subsequent course of AD (685–688). However, other prospective studies found no association between MeDi adherence and the risk of developing AD (689). The PREDIMED-NAVARRA trial was the first randomized study to evaluate the effects of long-term MeDi intervention on cognitive function. The nutritional intervention of PREDIMED consisted of a typical MeDi supplemented with extra virgin olive oil or mixed nuts compared with a control low-fat diet. The study was conducted on 522 participants at high vascular risk and assessed the overall cognitive performance at study completion, after 6.5 yr. The researchers reported a significant difference in mean MMSE and clock drawing test (CDT) scores in both intervention groups versus the low-fat control group (690). Subsequently, a MeDi intervention trial (ISRCTN35739639) similar to PREDIMED was conducted on 334 participants, demonstrating a positive effect on cognition (691). The MedLey study was a RCT (ACTRN12613000636752) conducted in 137 older non-Mediterranean adults for 6 mo. This study did not find any significant beneficial effects of the MeDi intervention on cognitive functions (692). The NU-AGE RCT (NCT01754012), was carried out in both Mediterranean and non-Mediterranean European countries on 1,279 relatively healthy older adults aged 65–79 yr (693). The NU-AGE diet consisted of a culturally adapted and individually tailored Mediterranean-like diet. After 1 yr, all participants showed improvements in their cognitive performance, but the differences between the two groups did not reach statistical significance. Nonetheless, the authors highlighted that the participants in the intervention group with the highest adherence to the NU-AGE diet showed a significant improvement in episodic memory (693).

The “Dietary Approaches to Stop Hypertension” (DASH) diet was a nutritional pattern developed to identify dietary factors affecting blood pressure (BP) (694, 695). The DASH diet is high in fruits, vegetables, nuts, whole cereal products, low-fat dairy products, fish, and poultry, all of which are rich in blood pressure-reducing nutrients. This type of dietary pattern has been shown to protect against many cardiovascular risk factors that play a role in the development of dementia and AD (such as high blood pressure or LDL cholesterol), at least in part by modulating the pathological processes that characterize the neuropathological mechanisms (OS, inflammation, and BIR). As observed for the MeDi, higher adherence to the DASH diet was associated with slower rates of cognitive decline and reduced incidence of AD

(695, 696). The ENCORE trial (NCT00571844), a RCT that examined the potential effects of the DASH diet on neurocognitive functioning, was performed on 124 subjects with high blood pressure (697). Participants were randomized to DASH diet alone, DASH combined with a behavioral weight management program including exercise and calorie restriction, or a “usual diet” control group. After 4 mo of intervention, only the group who underwent a combination of DASH diet with aerobic exercise and calorie restriction showed a significant improvement in neurocognitive function (698). Changes in dietary habits, weight, and BP persisted for 8 mo after completion of the 16-wk ENCORE program, with some attenuation of the benefits (699). The MIND diet (Mediterranean-DASH Intervention for Neurodegenerative Delay) represents a hybrid of the Mediterranean-DASH diet. Two papers published by Morris and colleagues demonstrated the superiority of the MIND diet compared to both the MeDi and the DASH diet in slowing the rates of cognitive decline and in reducing the risk of incident AD or dementia (687, 696). Two RCTs testing the effects of an intervention with the MIND diet on cognitive decline and brain neurodegeneration are currently ongoing (NCT02817074; NCT03585907).

6.4.2. Caloric restriction and fasting.

Caloric restriction (CR) and fasting are well-recognized nutritional dietary strategies to reduce the risk and incidence of AD in the normal population. These approaches produce a “metabolic switch” where at first the main source of energy for neurons is glucose obtained from glycogen stored in the liver and then ketone bodies (KBs) are produced by the liver from the fatty acids released by adipose tissue. In addition, reduced calorie intake can enhance a complex series of adaptive responses to limited food availability, which are to some degree the same endogenous stress-response systems activated by foods and other bioactive compounds (700). CR/fasting is a reliable method of increasing life span, decreasing ROS levels, and stimulating hormesis, subtoxic levels of OS molecules that protect against more severe stressors (701). Hormetic conditions are normally present in the healthy brain and lead to the upregulation of antioxidants, DNA repair enzymes, and antiapoptotic proteins (702).

Mechanistically, the benefits of CR/fasting seem to be related to reduced IGF1/insulin signaling and to the repression of mTORC1, thus allowing the cell to induce degradation pathways (e.g., autophagy) entering a “conservative” energy mode (see sect. 4). Studies on animal models of AD and on humans demonstrated that daily 20–40% CR can protect against aging, OS, and neurodegenerative disorders and can extend longevity (703). Studies in rodents showed that fasting can improve

motor functions and hippocampus-dependent tasks like learning and memory (704–707). Moreover, CR/fasting has been associated with reduced OS and brain structural improvements (708). Other research groups have also documented a role of CR/fasting regimens in reducing A β deposition and pTau in the hippocampus and cerebral cortex of a transgenic mouse model of AD (709–712).

Regarding human studies, several reports showed that decreased energy intake can improve glucose and lipid metabolism, reduce blood pressure, and decrease blood biomarkers of inflammation, thus improving protection from the development of neurodegenerative diseases such as AD (713, 714). Short periods of CR were able to improve cognitive function in elderly subjects, and 1 mo of low-glycemic diet in MCI patients resulted in an improvement in delayed visual memory, cerebrospinal fluid biomarkers of A β metabolism, and brain bioenergetics (715, 716). Currently, no studies concerning CR/fasting have been conducted in human subjects with established AD, since the practicability and long-term tolerability of this diet regimen is low in elderly subjects and it might further decrease quality of life in patients with AD (531).

6.5. Physical Exercise

Lifestyle habits that do not include sufficient exercise training may increase the risk of developing neurodegenerative diseases including AD. Daily exercise reportedly improves learning and memory in older adults (717). The mechanisms by which exercise improves cognition are associated with 1) reduction of chronic OS along with the upregulation of autophagy and with stimulation of mitochondria biogenesis and 2) stimulation of the synthesis of neurotransmitters and trophic factors (e.g., IGF-1) (718). Several reports highlighted that the length of an exercise program is crucial in determining its efficacy against OS. Indeed, whereas short-term exercise manages to increase ROS, long-term exercise induces antioxidant enzymes (719). Female 12-mo-old Wistar rats exercised for 15 wk on a treadmill at moderate intensity showed decreased hippocampal levels of ROS and increased levels of PGC1 α , AMPK, SOD1, SOD2, and GPX (720). Exercise reportedly also accelerates DNA repair: a recent study of sedentary and active male volunteers found that although all participants had radiation-induced DNA damage, trained individuals had more rapid repair of radiation-induced DNA strand breaks after exhaustive exercise. In addition, exhaustive exercise only produced DNA strand breaks in the lymphocytes of sedentary individuals (721). Long-term exercise also was shown to reduce OS in models of AD. Female 12-mo-old 3xTg-AD mice having access to a running wheel for 3 mo demonstrated reduced levels

of lipoperoxide, GSSG, GPX, and GR, whereas CuZn-SOD was increased (722). A network analysis showed that the reduction of OS was followed by behavioral and pathological changes, including improved spatial memory, reduced A β and pTau, and reduced anxiety.

In STZ-induced AD rats it was reported that 4 wk of exercise on a treadmill reduced OS, mitochondrial dysfunction, A β , and pTau levels (723). Recent studies suggest that long-term exercise makes the brain more resilient to stressors. Indeed, in 8- to 9-mo-old Lewis rats treated with rotenone, 6 wk of exercise reduced hydrogen peroxide levels and increased the activity of GPX (724). In the presence of age, toxins, or neurodegeneration, long-term exercise reduced OS and helped restore homeostatic conditions in the damaged brain. For APP/PS1 mice, Lin and colleagues (725) suggested that exercise was able to delay the onset of AD, because 10 wk of treadmill training increased their dendritic arbor of CA1 and CA3 neurons and hippocampus-associated memory and restored the amygdala-associated memory and dendritic arbor of amygdala basolateral neurons. Furthermore, the authors reported that exercise increased the levels of BDNF/TrkB signaling molecules and reduced the levels of soluble A β . In humans, physical exercise is opined to be a nonpharmacological strategy that may help in protecting against cognitive decline and decrease the risk of AD (726). Physical exercise helps stabilize and improve the cognitive function in AD patients and reduces and delays the onset of severe neuropsychiatric symptoms like apathy, confusion, and depression (727). Exercise has also been shown to induce anti-inflammatory effects and neurotrophic factors (728, 729).

In the last two decades several clinical trials have tested the effects of physical exercise on cognition in MCI and/or AD patients. However, since not all of the trials posted trial results and because the number of participants in trials was often small, a clear picture on the benefits of physical exercise in the AD context has not yet been reached. A pilot study (NCT01264614) on 22 participants to assess the neuropsychological and neurophysiological effects of strengthening exercise for early dementia demonstrated increased cognitive efficiency following 10 wk of strengthening exercise (730). The combination of physical exercise and rivastigmine was tested by a phase III clinical trial (NCT01183806) on 40 participants, showing improvements for quality of life but no changes concerning cognition in patients with AD (731). In a pilot RCT (NCT02196545), 30 participants, including AD patients, were trained for 12 wk, leading to benefits in activities of daily living (ADL) and cognitive and physical skills (732). However, a phase III clinical trial (NCT01681602) to test physical exercise on AD patients was performed on 200 participants and demonstrated no significant association between physical performance parameters and ADL (733). A further pilot RCT

(NCT011283610 on 76 participants tested the effects of aerobic exercise and stretching on AD for 26 wk (734). Dissection of the results led to the tentative conclusion that aerobic exercise in early AD is associated with benefits in functional ability, such as improved memory performance and reduced hippocampal atrophy. In a subset of 16 participants from a RCT (NCT03034746) to observe the impact of physical activity on aging, the authors showed that exercise training was effective on patients' locomotion but not on patients' gait parameters (735). Findings from a RCT on 23 participants (NCT02384993) demonstrated that executive function, but not episodic memory, was significantly improved after enhanced physical activity relative to usual physical activity (736). After 26 wk of aerobic exercise training, improvement in executive function correlated with increased peak oxygen consumption ($\text{VO}_{2\text{peak}}$), whereas favorable cardiorespiratory fitness adaptation was associated with improvements in posterior cingulate cortex, glucose metabolism, and executive function. Furthermore, the metabolomic profiles related to brain health were beneficially altered after 26 wk of aerobic exercise training in late middle-aged adults at risk for AD, and such alterations were related to systemic biomarkers such as BDNF, CTSB, and klotho (737). A RCT (NCT02708485) performed on 10 participants to evaluate the effect of a 3-mo-duration walking program on brain energy metabolism in patients with mild AD suggested that aerobic training increases both ketone availability to the brain and the brain's capacity to metabolize ketones, while maintaining brain glucose metabolism (738). The effects of a long-term exercise program were studied in a RCT (NCT02444078) over a 6-mo duration in 91 people with dementia living in nursing homes. Analyses of the results suggested that exercise effects did not differ from social intervention effects on neuropsychiatric symptoms, pain, and medication consumption (739). A RCT (NCT02968875) performed on 52 participants with mild to moderate AD tested the effect of 9-wk continuous versus interval aerobic training on plasma BDNF levels, aerobic fitness, cognitive capacity, and quality of life (740). The data showed no significant change in all groups for plasma BDNF level and cognitive performance, but after 9 wk of continuous and interval aerobic training significantly improved aerobic fitness parameters and quality of life were found. A RCT (NCT02000583) for AD prevention through exercise demonstrated that, in 44 Apo e4 carriers, 52-wk intervention led to inverse relationships of changes in systolic blood flow and hippocampal blood flow suggesting beneficial outcomes (741).

Taken together, the exercise studies noted above suggest that more positive outcomes were observed with movement exercises, and we opine that to the extent possible aged individuals should try to incorporate a

supervised physical exercise program to lower risk of developing AD.

7. GAPS IN KNOWLEDGE AND FUTURE PERSPECTIVES

The brain is particularly vulnerable to OS, and accumulation of oxidative damage may disrupt redox homeostasis and reduce the ability of antioxidant/degradative systems to eliminate oxidized materials. Perturbation of redox homeostasis may in turn exacerbate ROS production that will result in the modification of several intracellular targets, including nucleic acids, lipids, and proteins.

AD involves multiple, complex pathogenic mechanisms, most of which are highly connected and/or share common signaling pathways. Among these, redox-regulated processes appear to act synergistically to accelerate neurodegeneration, ultimately leading to AD. OS acts as a bridge connecting different pathogenic mechanisms of AD and at the same time is an intrinsic molecular aspect of all these mechanisms. This characteristic creates a feedforward loop that further sustains redox reactions that ultimately lead to neuronal damage and CNS neurodegeneration. It is likely that lipid peroxidation is among the initiating toxic events, with small A β 42 oligomers reacting with membrane lipids, the products of which, in turn, target other moieties within the cell, i.e., proteins and nucleic acids. Oxidative modification of proteins, lipids, and nucleic acids mainly results in irreversible modifications and consequent impairment of function leading to disturbance of a plethora of cellular processes (6, 8). However, considering the major role played by A β , some questions remain partially unsolved:

- 1) What is the radical source in the lipid bilayer that initiates the one-electron oxidation of the S-atom of Met on residue 35 of A β 42? In our view, the radical source is highly likely to be only one of two moieties: i) oxygen molecules, which have a higher electron affinity than does sulfur, abstract one electron from the S-atom's two lone pairs of electrons, forming the S $^{+\cdot}$ sulfuranyl free radical on A β 's Met35 residue and superoxide free radical or ii) Cu $^{2+}$ weakly bound to the S-atom of Met35. Electron transfer from the S-atom to Cu $^{2+}$ would lead to reduced to Cu $^{+}$ (which then can react with omnipresent and lipid-soluble H $_2$ O $_2$ to produce the highly reactive OH radical), while the S-atom of Met35 of A β 42 is oxidized to S $^{+\cdot}$. The first possibility is consistent with the absence of any free radical associated with systems that have been thoroughly saturated with nitrogen or argon, highly diluting any oxygen present (111) and, since superoxide readily reacts with water to produce H $_2$ O $_2$, with the reported formation of H $_2$ O $_2$ from A β peptide (108). The latter possibility is consistent with published studies that showed the drug

clioquinol, a nanomolar K_D drug, for the chelation and removal of Cu $^{2+}$ from a site such as Met with a micromolar K_D for Cu $^{2+}$ but would not be predicted to remove Cu $^{2+}$ from a much lower K_D binding site (such as His residues), with consequent diminution of senile plaques and improved cognition (111, 112). If A β peptides, whose key amino acids are posited to be critical in initiating a free radical on A β 42, are substituted by other amino acids of similar properties but known not to bind Cu $^{2+}$ readily, are utilized to determine indexes of oxidative stress and cell death in primary neuronal cultures, the importance, or lack thereof, of these A β 42 amino acids can be determined. Such studies have, in fact, been performed as reviewed in the present article. Importantly, substituting the Met codon by the codon for Leu in the human *APP* gene already carrying two mutations (Swe/Ind-J20 mouse) corresponding to the Met35 residue of A β 42 also showed that without Met35 there is no elevated oxidative stress in brain compared with WT littermates, demonstrating in vivo that Met35 of A β 42 is critical to the oxidative damage associated with oligomers of this neurotoxic peptide in mammalian systems (119).

- 2) Does Cu $^{2+}$ binding to His lead to electron flow from the OH group of Tyr on residue 10 of A β 42 leading to oxidative damage? We showed that when Phe is substituted for Tyr at residue 10 (thereby keeping the aromaticity of Tyr but having no OH group in the 4-position of the benzene ring of Tyr) there would be no path for electron flow with Phe, yet the free radical-based OS production was the same as that of the native A β 42 (111, 112, 115).

The present review provides important information for the in-depth understanding of the pathogenesis and progression of AD by focusing on the central role of OS and its relationship to A β 42. For example, brains from persons with amnesic MCI showed significant oxidative damage to lipids and proteins and metabolic changes well before any dementia occurred (5, 6, 742).

In our opinion, OS should not be considered as a single event itself but within the complexity of reactions and signaling cascades in which oxidative damage takes part. This integrated view emerges from extensive studies that led to the identifications of the most relevant redox-regulated brain dysfunctions that strongly contribute to the neurodegenerative process and how these alterations translate into a clinical phenotype, i.e., cognitive decline, including MCI or late-stage AD. To describe the cellular underpinning of cognitive decline, various aspects come into the picture that seem to drive the age-dependent reduction of cognitive abilities: oxidative stress and its associated damage, changes in energy metabolism, epigenetic alterations, the gradual loss in repair and recovery mechanisms, all of which accompany brain dysfunction, should be considered.

We suggest a direct link between irreversible oxidative modifications of proteins, particularly those induced secondary to lipid peroxidation, with loss of the functions of these modified proteins and translation into clinical phenotypes. The significant roles of small oligomers of A β 42 in these changes were emphasized. Consistent with our thesis promoted in this review that small oligomers of A β 42 are associated with lipid peroxidation and secondary or direct protein oxidation are critical to the pathogenesis and progression of AD, recent studies demonstrated that small A β 42 oligomers, rather than large oligomers, either isolated from AD brain or in vivo in brains of persons with AD or MCI caused synaptic dysfunction and cognitive loss, respectively (241). Moreover, in addition to the identified oxidatively modified and dysfunctional brain proteins across different stages of AD, many other changes occurring either at an epigenetic level or at an expression level as well as other PTMs should be considered in the progression of AD. We are aware that the present review mainly focuses on the role of lipid peroxidation and protein oxidation in AD progression by showing the implication of several identified proteins across different stages of Alzheimer neurodegeneration, suggesting the critical cellular functions that are affected, ultimately resulting in neuronal death and clinical presentations. In addition, we have addressed the fact that epigenetic changes in response to OS, lipid oxidation, as well as protein oxidation may change the level/activity of many other proteins, independently of oxidative damage. Several studies have extensively addressed changes of proteome network in AD and the proteotoxic nature of AD (124, 131–134, 743–745).

For example, a recent proteomics study by Johnson et al. (746) demonstrated that changes in protein expression association with AD are not always reflected in changes in RNA levels. Nevertheless, alteration of protein levels results from the combination and the cross-regulation of genetic, epigenetic, and proteostatic processes. Proteins undergo complex posttranslational modifications that extend protein function beyond that dictated by gene transcripts (747). These posttranslational modifications can also act as retrograde signals to regulate gene expression, so that aberrant accumulation of a protein in the cell is likely to interfere with biosynthesis of this protein (748).

We suggest that the complexity and heterogeneity (genetic and nongenetic forms) of AD require the integration of genomic, epigenomic, and proteomic data to fully understand the molecular networks that contribute to the onset and progression of this disorder (749). Collectively, studies from our groups and others have provided valuable insights into the molecular mechanisms involved in both the pathogenesis and progression of AD, demonstrating that OS contributes to the impairment of numerous cellular processes such as energy production, cellular structure, signal transduction, synaptic function, mitochondrial function, cell

cycle progression, and degradative systems. Each of these cellular functions normally contributes to maintenance of healthy neuronal homeostasis, so the deregulation of one or more of these functions could contribute to the pathology and clinical presentations of AD (6, 124, 744, 745, 749).

Despite profound evidence supporting the role of oxidative stress in AD, none of the currently available treatments addressing oxidative stress has shown to be particularly effective. However, in the present review we outline preclinical and clinical studies related to AD that provide hope that such approaches coupled to others are beginning to show promise. One set of such preclinical studies are those with the beagle dog, which has A β 42 of the same amino acid sequence as humans. The conclusions of these studies do provide hope that such approaches may lower risk of developing AD. Specifically, these studies (132, 177, 617) demonstrated that even at old age a 3-yr program of a high-antioxidant diet, similar to the MeDi, coupled with behavioral enrichment to produce new synapses and exercise to produce brain-derived nerve growth factor, significantly caused elderly beagles to have in their brains much lower levels of A β 42, much lower phosphorylated Tau protein, much lower indexes of oxidative damage, and significantly improved cognitive status (188, 189, 596, 597). In our opinion, these three factors—consumption of a diet rich in antioxidants, continuously learning new things by challenging one's brain, and, to the extent possible, staying active more days than not—are likely to significantly reduce risk of persons not carrying autosomal dominant mutations in the *APP* or *PSEN1* or *PSEN2* genes and not carrying the epsilon 4 allele of the *APOE* gene from developing AD. Note that lower risk is not the same as certainty, but in our view it is better to do what one can to lower risk of AD, knowing that such approaches do not come with a guarantee.

Although preclinical studies utilizing cellular and animal models suggest that antioxidants have therapeutic promise, limitations exist regarding their ability to treat AD in clinical trials. These limitations pose a number of questions that should be addressed in the future that take into account 1) challenges with dosage, appropriate time points, and intervals for intervention; 2) the lack of reliable OS biomarkers that provide insights to the OS status of each individual; 3) the probability that other factors currently not understood may play a significant role in neurodegeneration; and 4) the fact that one antioxidant compound may be not sufficient alone to counteract oxidative stress and have an impact on disease progression. This latter consideration supports the need to investigate the combination of antioxidant therapies, where more than one antioxidant is utilized, as a more effective and disease-modifying approach to treating AD and other neurogenerative conditions that have oxidative stress as a main contributing factor (750).

The present review has shown that oxidative damage is a strong contributor to the pathogenesis and progression of AD, and, as noted above, oxidative damage can influence dysfunction of many pathways of relevance to AD. Consequently, we believe that ways to mitigate against production or effects of small oligomers of A β 42 or prevent the significant brain damage that small oligomers of A β 42-associated OS engender are two likely means of slowing or, in the ideal case, stopping this highly detrimental disorder in individuals with the disease and thereby increase the quality of life of the patients and their families. We were highly encouraged in this regard by the recent studies of Daggett and co-workers (751). These scientists identified neurotoxic small oligomers of A β that had a unique structure called an α -sheet and developed a detection system for these small oligomers based on this structural element. This soluble oligomer binding assay (SOBA) demonstrated that plasma from nearly 380 human specimens could discriminate with 99% specificity neurotoxic small oligomers of AD patients from control subjects. In at least one case, a control sample demonstrated these neurotoxic small oligomers, but it was later determined that this subject developed mild cognitive impairment, consistent with the observation that AD pathology occurs often two decades before the appearance of symptoms of this disorder. From our perspective, the importance of this work is that it validates the major premise of our laboratory that small oligomers of A β 42 are the neurotoxic agent that we showed facilitates lipid peroxidation that ultimately leads to HNE formation and subsequent neuronal death. Loss of cognition would result. Continued research to exploit this new validation is in progress.

CORRESPONDENCE

D. A. Butterfield (dabcns@uky.edu).

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AUTHOR CONTRIBUTIONS

D.A.B. conceived and designed research; F.D.D. prepared figures; M.P., F.D.D., and D.A.B. drafted manuscript; M.P. and D.A.B. edited and revised manuscript; M.P., F.D.D., and D.A.B. approved final version of manuscript.

REFERENCES

1. Markesbery WR. Oxidative stress hypothesis in Alzheimer's disease. *Free Radic Biol Med* 23: 134–147, 1997. doi:10.1016/s0891-5849(96)00629-6.
2. Butterfield DA, Swomley AM, Sultana R. Amyloid beta-peptide (1-42)-induced oxidative stress in Alzheimer disease: importance in disease pathogenesis and progression. *Antioxid Redox Signal* 19: 823–835, 2013. doi:10.1089/ars.2012.5027.
3. Chondrogianni N, Petropoulos I, Grimm S, Georgila K, Catalgol B, Friguet B, Grune T, Gonos ES. Protein damage, repair and proteolysis. *Mol Aspects Med* 35: 1–71, 2014. doi:10.1016/j.mam.2012.09.001.
4. Halliwell B. Reactive oxygen species and the central nervous system. *J Neurochem* 59: 1609–1623, 1992. doi:10.1111/j.1471-4159.1992.tb10990.x.
5. Butterfield DA, Boyd-Kimball D. Mitochondrial oxidative and nitrosative stress and Alzheimer disease. *Antioxidants (Basel)* 9: 818, 2020. doi:10.3390/antiox9090818.
6. Butterfield DA, Halliwell B. Oxidative stress, dysfunctional glucose metabolism and Alzheimer disease. *Nat Rev Neurosci* 20: 148–160, 2019. doi:10.1038/s41583-019-0132-6.
7. Misrani A, Tabassum S, Yang L. Mitochondrial dysfunction and oxidative stress in Alzheimer's disease. *Front Aging Neurosci* 13: 617588, 2021. doi:10.3389/fnagi.2021.617588.
8. Butterfield DA, Stadtman ER. Protein oxidation processes in aging brain. In: *Advances in Cell Aging and Gerontology*, edited by Timiras PS, Bittar EE. Amsterdam: Elsevier, 1997, vol. 2, p. 161–191.
9. Albers RW. Cell membrane structures and functions. In: *Basic Neurochemistry: Principles of Molecular, Cellular, and Medical Neurobiology* (8th ed.). Waltham, MA: Academic Press, 2012, p. 26–39.
10. Hubbell WL, McConnell HM. Molecular motion in spin-labeled phospholipids and membranes. *J Am Chem Soc* 93: 314–326, 1971. doi:10.1021/ja00731a005.

11. Subczynski WK, Widomska J, Feix JB. Physical properties of lipid bilayers from EPR spin labeling and their influence on chemical reactions in a membrane environment. **Free Radic Biol Med** 46: 707–718, 2009. doi:[10.1016/j.freeradbiomed.2008.11.024](https://doi.org/10.1016/j.freeradbiomed.2008.11.024).
12. Subramaniam R, Roediger F, Jordan B, Mattson MP, Keller JN, Waeg G, Butterfield DA. The lipid peroxidation product, 4-hydroxy-2-trans-nonenal, alters the conformation of cortical synaptosomal membrane proteins. **J Neurochem** 69: 1161–1169, 1997. doi:[10.1046/j.1471-4159.1997.69031161.x](https://doi.org/10.1046/j.1471-4159.1997.69031161.x).
13. Boyd-Kimball D, Sultana R, Poon HF, Lynn BC, Casamenti F, Pepeu G, Klein JB, Butterfield DA. Proteomic identification of proteins specifically oxidized by intracerebral injection of amyloid beta-peptide (1-42) into rat brain: implications for Alzheimer's disease. **Neuroscience** 132: 313–324, 2005. doi:[10.1016/j.neuroscience.2004.12.022](https://doi.org/10.1016/j.neuroscience.2004.12.022).
14. Tanford C. The hydrophobic effect and the organization of living matter. **Science** 200: 1012–1018, 1978. doi:[10.1126/science.653353](https://doi.org/10.1126/science.653353).
15. Butterfield DA. Brain lipid peroxidation and Alzheimer disease: synergy between the Butterfield and Mattson laboratories. **Ageing Res Rev** 64: 101049, 2020. doi:[10.1016/j.arr.2020.101049](https://doi.org/10.1016/j.arr.2020.101049).
16. Butterfield DA. Ubiquitin carboxyl-terminal hydrolase L-1 in brain: Focus on its oxidative/nitrosative modification and role in brains of subjects with Alzheimer disease and mild cognitive impairment. **Free Radic Biol Med** 177: 278–286, 2021. doi:[10.1016/j.freeradbiomed.2021.10.036](https://doi.org/10.1016/j.freeradbiomed.2021.10.036).
17. Butterfield DA, Castegna A, Lauderback CM, Drake J. Evidence that amyloid beta-peptide-induced lipid peroxidation and its sequelae in Alzheimer's disease brain contribute to neuronal death. **Neurobiol Aging** 23: 655–664, 2002. doi:[10.1016/s0197-4580\(01\)00340-2](https://doi.org/10.1016/s0197-4580(01)00340-2).
18. Castegna A, Aksenov M, Aksenova M, Thongboonkerd V, Klein JB, Pierce WM, Booze R, Markesbery WR, Butterfield DA. Proteomic identification of oxidatively modified proteins in Alzheimer's disease brain. Part I: creatine kinase BB, glutamine synthase, and ubiquitin carboxy-terminal hydrolase L-1. **Free Radic Biol Med** 33: 562–571, 2002. doi:[10.1016/s0891-5849\(02\)00914-0](https://doi.org/10.1016/s0891-5849(02)00914-0).
19. Castegna A, Aksenov M, Thongboonkerd V, Klein JB, Pierce WM, Booze R, Markesbery WR, Butterfield DA. Proteomic identification of oxidatively modified proteins in Alzheimer's disease brain. Part II: dihydropyrimidinase-related protein 2, alpha-enolase and heat shock cognate 71. **J Neurochem** 82: 1524–1532, 2002. doi:[10.1046/j.1471-4159.2002.01103.x](https://doi.org/10.1046/j.1471-4159.2002.01103.x).
20. Higdon AN, Landar A, Barnes S, Darley-Usmar VM. The electrophile responsive proteome: integrating proteomics and lipidomics with cellular function. **Antioxid Redox Signal** 17: 1580–1589, 2012. doi:[10.1089/ars.2012.4523](https://doi.org/10.1089/ars.2012.4523).
21. Lauderback CM, Hackett JM, Huang FF, Keller JN, Szweda LI, Markesbery WR, Butterfield DA. The glial glutamate transporter, GLT-1, is oxidatively modified by 4-hydroxy-2-nonenal in the Alzheimer's disease brain: the role of A beta 1-42. **J Neurochem** 78: 413–416, 2001. doi:[10.1046/j.1471-4159.2001.00451.x](https://doi.org/10.1046/j.1471-4159.2001.00451.x).
22. Butterfield DA. The 2013 SFRBM discovery award: selected discoveries from the Butterfield laboratory of oxidative stress and its sequela in brain in cognitive disorders exemplified by Alzheimer disease and chemotherapy induced cognitive impairment. **Free Radic Biol Med** 74: 157–174, 2014. doi:[10.1016/j.freeradbiomed.2014.06.006](https://doi.org/10.1016/j.freeradbiomed.2014.06.006).
23. Butterfield DA, Bader Lange ML, Sultana R. Involvements of the lipid peroxidation product, HNE, in the pathogenesis and progression of Alzheimer's disease. **Biochim Biophys Acta** 1801: 924–929, 2010. doi:[10.1016/j.bbalip.2010.02.005](https://doi.org/10.1016/j.bbalip.2010.02.005).
24. Butterfield DA, Reed T, Perluigi M, De Marco C, Coccia R, Cini C, Sultana R. Elevated protein-bound levels of the lipid peroxidation product, 4-hydroxy-2-nonenal, in brain from persons with mild cognitive impairment. **Neurosci Lett** 397: 170–173, 2006. doi:[10.1016/j.neulet.2005.12.017](https://doi.org/10.1016/j.neulet.2005.12.017).
25. Perluigi M, Coccia R, Butterfield DA. 4-Hydroxy-2-nonenal, a reactive product of lipid peroxidation, and neurodegenerative diseases: a toxic combination illuminated by redox proteomics studies. **Antioxid Redox Signal** 17: 1590–1609, 2012. doi:[10.1089/ars.2011.4406](https://doi.org/10.1089/ars.2011.4406).
26. Reed T, Perluigi M, Sultana R, Pierce WM, Klein JB, Turner DM, Coccia R, Markesbery WR, Butterfield DA. Redox proteomic identification of 4-hydroxy-2-nonenal-modified brain proteins in amnesic mild cognitive impairment: insight into the role of lipid peroxidation in the progression and pathogenesis of Alzheimer's disease. **Neurobiol Dis** 30: 107–120, 2008. doi:[10.1016/j.nbd.2007.12.007](https://doi.org/10.1016/j.nbd.2007.12.007).
27. Reed TT, Pierce WM, Markesbery WR, Butterfield DA. Proteomic identification of HNE-bound proteins in early Alzheimer disease: insights into the role of lipid peroxidation in the progression of AD. **Brain Res** 1274: 66–76, 2009. doi:[10.1016/j.brainres.2009.04.009](https://doi.org/10.1016/j.brainres.2009.04.009).
28. Sultana R, Perluigi M, Butterfield DA. Protein oxidation and lipid peroxidation in brain of subjects with Alzheimer's disease: insights into mechanism of neurodegeneration from redox proteomics. **Antioxid Redox Signal** 8: 2021–2037, 2006. doi:[10.1089/ars.2006.8.2021](https://doi.org/10.1089/ars.2006.8.2021).
29. Sultana R, Perluigi M, Butterfield DA. Lipid peroxidation triggers neurodegeneration: a redox proteomics view into the Alzheimer disease brain. **Free Radic Biol Med** 62: 157–169, 2013. doi:[10.1016/j.freeradbiomed.2012.09.027](https://doi.org/10.1016/j.freeradbiomed.2012.09.027).
30. Montine TJ, Quinn J, Kaye J, Morrow JD. F-2-isoprostanes as biomarkers of late-onset Alzheimer's disease. **J Mol Neurosci** 33: 114–119, 2007. doi:[10.1007/s12031-007-0044-1](https://doi.org/10.1007/s12031-007-0044-1).
31. Praticò D. F₂-isoprostanes: sensitive and specific non-invasive indices of lipid peroxidation in vivo. **Atherosclerosis** 147: 1–10, 1999. doi:[10.1016/S0021-9150\(99\)00257-9](https://doi.org/10.1016/S0021-9150(99)00257-9).
32. Milne GL, Musiek ES, Morrow JD. F-2-isoprostanes as markers of oxidative stress in vivo: An overview. **Biomarkers** 10, Suppl 1: S10–S23, 2005. doi:[10.1080/13547500500216546](https://doi.org/10.1080/13547500500216546).
33. Wong CM, Marcocci L, Liu L, Suzuki YJ. Cell signaling by protein carbonylation and decarbonylation. **Antioxid Redox Signal** 12: 393–404, 2010. doi:[10.1089/ars.2009.2805](https://doi.org/10.1089/ars.2009.2805).
34. Levine RL, Stadtman ER. Oxidative modification of proteins during aging. **Exp Gerontol** 36: 1495–1502, 2001. doi:[10.1016/s0531-5565\(01\)00135-8](https://doi.org/10.1016/s0531-5565(01)00135-8).
35. Beal MF. Oxidatively modified proteins in aging and disease. **Free Radic Biol Med** 32: 797–803, 2002. doi:[10.1016/s0891-5849\(02\)00780-3](https://doi.org/10.1016/s0891-5849(02)00780-3).
36. Poon HF, Hensley K, Thongboonkerd V, Merchant ML, Lynn BC, Pierce WM, Klein JB, Calabrese V, Butterfield DA. Redox proteomics analysis of oxidatively modified proteins in G93A-SOD1 transgenic mice—a model of familial amyotrophic lateral sclerosis. **Free Radic Biol Med** 39: 453–462, 2005. doi:[10.1016/j.freeradbiomed.2005.03.030](https://doi.org/10.1016/j.freeradbiomed.2005.03.030).
37. Ren X, Hinchie A, Swomley A, Powell DK, Butterfield DA. Profiles of brain oxidative damage, ventricular alterations, and neurochemical metabolites in the striatum of PINK1 knockout rats as functions of age and gender: relevance to Parkinson disease. **Free Radic Biol Med** 143: 146–152, 2019. doi:[10.1016/j.freeradbiomed.2019.08.008](https://doi.org/10.1016/j.freeradbiomed.2019.08.008).
38. Fridovich I. Superoxide dismutases. **Annu Rev Biochem** 44: 147–159, 1975. doi:[10.1146/annurev.bi.44.070175.001051](https://doi.org/10.1146/annurev.bi.44.070175.001051).

39. Forman HJ. On "Biological effects of the superoxide radical" by Irwin Fridovich. **Arch Biochem Biophys** 726: 109117, 2022. doi:[10.1016/j.abb.2022.109117](https://doi.org/10.1016/j.abb.2022.109117).
40. Calabrese V, Mancuso C, Calvani M, Rizzarelli E, Butterfield DA, Stella AM. Nitric oxide in the central nervous system: neuroprotection versus neurotoxicity. **Nat Rev Neurosci** 8: 766–775, 2007. doi:[10.1038/nrn2214](https://doi.org/10.1038/nrn2214).
41. Baba SP, Bhatnagar A. Role of thiols in oxidative stress. **Curr Opin Toxicol** 7: 133–139, 2018. doi:[10.1016/j.cotox.2018.03.005](https://doi.org/10.1016/j.cotox.2018.03.005).
42. Fra A, Yoboue ED, Sitia R. Cysteines as redox molecular switches and targets of disease. **Front Mol Neurosci** 10: 167, 2017. doi:[10.3389/fnmol.2017.00167](https://doi.org/10.3389/fnmol.2017.00167).
43. Mishanina TV, Libiad M, Banerjee R. Biogenesis of reactive sulfur species for signaling by hydrogen sulfide oxidation pathways. **Nat Chem Biol** 11: 457–464, 2015. doi:[10.1038/nchembio.1834](https://doi.org/10.1038/nchembio.1834).
44. Evangelista AM, Kohr MJ, Murphy E. S-nitrosylation: specificity, occupancy, and interaction with other post-translational modifications. **Antioxid Redox Signal** 19: 1209–1219, 2013. doi:[10.1089/ars.2012.5056](https://doi.org/10.1089/ars.2012.5056).
45. Nakamura T, Oh CK, Liao L, Zhang X, Lopez KM, Gibbs D, Deal AK, Scott HR, Spencer B, Masliah E, Rissman RA, Yates JR 3rd, Lipton SA. Noncanonical transnitrosylation network contributes to synapse loss in Alzheimer's disease. **Science** 371: eaaw0843, 2021. doi:[10.1126/science.aaw0843](https://doi.org/10.1126/science.aaw0843).
46. Ulrich K, Jakob U. The role of thiols in antioxidant systems. **Free Radic Biol Med** 140: 14–27, 2019. doi:[10.1016/j.freeradbiomed.2019.05.035](https://doi.org/10.1016/j.freeradbiomed.2019.05.035).
47. Hudson DA, Gannon SA, Thorpe C. Oxidative protein folding: from thiol-disulfide exchange reactions to the redox poise of the endoplasmic reticulum. **Free Radic Biol Med** 80: 171–182, 2015. doi:[10.1016/j.freeradbiomed.2014.07.037](https://doi.org/10.1016/j.freeradbiomed.2014.07.037).
48. Shen D, Dalton TP, Nebert DW, Shertzer HG. Glutathione redox state regulates mitochondrial reactive oxygen production. **J Biol Chem** 280: 25305–25312, 2005. doi:[10.1074/jbc.M500095200](https://doi.org/10.1074/jbc.M500095200).
49. Requejo R, Hurd TR, Costa NJ, Murphy MP. Cysteine residues exposed on protein surfaces are the dominant intramitochondrial thiol and may protect against oxidative damage. **FEBS J** 277: 1465–1480, 2010. doi:[10.1111/j.1742-4658.2010.07576.x](https://doi.org/10.1111/j.1742-4658.2010.07576.x).
50. Markesbery WR, Lovell MA. DNA oxidation in Alzheimer's disease. **Antioxid Redox Signal** 8: 2039–2045, 2006. doi:[10.1089/ars.2006.8.2039](https://doi.org/10.1089/ars.2006.8.2039).
51. Cadet J, Wagner JR. DNA base damage by reactive oxygen species, oxidizing agents, and UV radiation. **Cold Spring Harb Perspect Biol** 5: a012559, 2013. doi:[10.1101/cshperspect.a012559](https://doi.org/10.1101/cshperspect.a012559).
52. Moreira PI, Nunomura A, Nakamura M, Takeda A, Shenk JC, Aliev G, Smith MA, Perry G. Nucleic acid oxidation in Alzheimer disease. **Free Radic Biol Med** 44: 1493–1505, 2008. doi:[10.1016/j.freeradbiomed.2008.01.002](https://doi.org/10.1016/j.freeradbiomed.2008.01.002).
53. Chao MR, Evans MD, Hu CW, Ji Y, Møller P, Rossner P, Cooke MS. Biomarkers of nucleic acid oxidation—a summary state-of-the-art. **Redox Biol** 42: 101872, 2021. doi:[10.1016/j.redox.2021.101872](https://doi.org/10.1016/j.redox.2021.101872).
54. Madabhushi R, Pan L, Tsai LH. DNA damage and its links to neurodegeneration. **Neuron** 83: 266–282, 2014. doi:[10.1016/j.neuron.2014.06.034](https://doi.org/10.1016/j.neuron.2014.06.034).
55. Ghosh R, Mitchell DL. Effect of oxidative DNA damage in promoter elements on transcription factor binding. **Nucleic Acids Res** 27: 3213–3218, 1999. doi:[10.1093/nar/27.15.3213](https://doi.org/10.1093/nar/27.15.3213).
56. Valinluck V, Tsai HH, Rogstad DK, Burdzy A, Bird A, Sowers LC. Oxidative damage to methyl-CpG sequences inhibits the binding of the methyl-CpG binding domain (MBD) of methyl-CpG binding protein 2 (MeCP2). **Nucleic Acids Res** 32: 4100–4108, 2004. doi:[10.1093/nar/gkh739](https://doi.org/10.1093/nar/gkh739).
57. Jeppesen DK, Bohr VA, Stevnsner T. DNA repair deficiency in neurodegeneration. **Prog Neurobiol** 94: 166–200, 2011. doi:[10.1016/j.pneurobio.2011.04.013](https://doi.org/10.1016/j.pneurobio.2011.04.013).
58. Santos RX, Correia SC, Zhu X, Lee HG, Petersen RB, Nunomura A, Smith MA, Perry G, Moreira PI. Nuclear and mitochondrial DNA oxidation in Alzheimer's disease. **Free Radic Res** 46: 565–576, 2012. doi:[10.3109/10715762.2011.648188](https://doi.org/10.3109/10715762.2011.648188).
59. Liu Z, Chen X, Li Z, Ye W, Ding H, Li P, Aung LH. Role of RNA oxidation in neurodegenerative diseases. **Int J Mol Sci** 21: 5022, 2020. doi:[10.3390/ijms21145022](https://doi.org/10.3390/ijms21145022).
60. Kong Q, Lin CL. Oxidative damage to RNA: mechanisms, consequences, and diseases. **Cell Mol Life Sci** 67: 1817–1829, 2010. doi:[10.1007/s00018-010-0277-y](https://doi.org/10.1007/s00018-010-0277-y).
61. Görg B, Qvartskhava N, Keitel V, Bidmon HJ, Selbach O, Schliess F, Häussinger D. Ammonia induces RNA oxidation in cultured astrocytes and brain in vivo. **Hepatology** 48: 567–579, 2008. doi:[10.1002/hep.22345](https://doi.org/10.1002/hep.22345).
62. Tanaka M, Chock PB, Stadtman ER. Oxidized messenger RNA induces translation errors. **Proc Natl Acad Sci USA** 104: 66–71, 2007. doi:[10.1073/pnas.0609737104](https://doi.org/10.1073/pnas.0609737104).
63. Ding Q, Dimayuga E, Keller JN. Oxidative stress alters neuronal RNA- and protein-synthesis: implications for neural viability. **Free Radic Res** 41: 903–910, 2007. doi:[10.1080/10715760701416996](https://doi.org/10.1080/10715760701416996).
64. Haberer V, Stark A. Eukaryotic core promoters and the functional basis of transcription initiation. **Nat Rev Mol Cell Biol** 19: 621–637, 2018. doi:[10.1038/s41580-018-0028-8](https://doi.org/10.1038/s41580-018-0028-8).
65. Nunomura A, Moreira PI, Castellani RJ, Lee HG, Zhu X, Smith MA, Perry G. Oxidative damage to RNA in aging and neurodegenerative disorders. **Neurotox Res** 22: 231–248, 2012. doi:[10.1007/s12640-012-9331-x](https://doi.org/10.1007/s12640-012-9331-x).
66. Carbonell T, Gomes AV. MicroRNAs in the regulation of cellular redox status and its implications in myocardial ischemia-reperfusion injury. **Redox Biol** 36: 101607, 2020. doi:[10.1016/j.redox.2020.101607](https://doi.org/10.1016/j.redox.2020.101607).
67. Reddy PH, Williams J, Smith F, Bhatti JS, Kumar S, Vijayan M, Kandimalla R, Kuruva CS, Wang R, Manczak M, Yin X, Reddy AP. MicroRNAs, aging, cellular senescence, and Alzheimer's disease. **Prog Mol Biol Transl Sci** 146: 127–171, 2017. doi:[10.1016/bs.pmbts.2016.12.009](https://doi.org/10.1016/bs.pmbts.2016.12.009).
68. O'Carroll D, Schaefer A. General principals of miRNA biogenesis and regulation in the brain. **Neuropsychopharmacology** 38: 39–54, 2013. doi:[10.1038/npp.2012.87](https://doi.org/10.1038/npp.2012.87).
69. John A, Kubosumi A, Reddy PH. Mitochondrial microRNAs in aging and neurodegenerative diseases. **Cells** 9: 1345, 2020. doi:[10.3390/cells9061345](https://doi.org/10.3390/cells9061345).
70. Gowda P, Reddy PH, Kumar S. Deregulated mitochondrial microRNAs in Alzheimer's disease: Focus on synapse and mitochondria. **Ageing Res Rev** 73: 101529, 2022. doi:[10.1016/j.arr.2021.101529](https://doi.org/10.1016/j.arr.2021.101529).
71. Walgrave H, Zhou L, De Strooper B, Salta E. The promise of microRNA-based therapies in Alzheimer's disease: challenges and perspectives. **Mol Neurodegener** 16: 76, 2021. doi:[10.1186/s13024-021-00496-7](https://doi.org/10.1186/s13024-021-00496-7).

72. Gu F, Chauhan V, Chauhan A. Glutathione redox imbalance in brain disorders. **Curr Opin Clin Nutr Metab Care** 18: 89–95, 2015. doi:10.1097/MCO.0000000000000134.
73. Pocernich CB, Butterfield DA. Elevation of glutathione as a therapeutic strategy in Alzheimer disease. **Biochim Biophys Acta** 1822: 625–630, 2012. doi:10.1016/j.bbadis.2011.10.003.
74. Maier CM, Chan PH. Role of superoxide dismutases in oxidative damage and neurodegenerative disorders. **Neuroscientist** 8: 323–334, 2002. doi:10.1177/107385840200800408.
75. Lindenau J, Noack H, Possel H, Asayama K, Wolf G. Cellular distribution of superoxide dismutases in the rat CNS. **Glia** 29: 25–34, 2000. doi:10.1002/(SICI)1098-1136(20000101)29:1<25::AID-GLIA3>3.0.CO;2-G.
76. Joshi G, Johnson JA. The Nrf2-ARE pathway: a valuable therapeutic target for the treatment of neurodegenerative diseases. **Recent Pat CNS Drug Discov** 7: 218–229, 2012. doi:10.2174/157488912803252023.
77. Sies H. Glutathione and its role in cellular functions. **Free Radic Biol Med** 27: 916–921, 1999. doi:10.1016/s0891-5849(99)00177-x.
78. Schulz JB, Lindenau J, Seyfried J, Dichgans J. Glutathione, oxidative stress and neurodegeneration. **Eur J Biochem** 267: 4904–4911, 2000. doi:10.1046/j.1432-1327.2000.01595.x.
79. Wild AC, Moinova HR, Mulcahy RT. Regulation of gamma-glutamyl-cysteine synthetase subunit gene expression by the transcription factor Nrf2. **J Biol Chem** 274: 33627–33636, 1999. doi:10.1074/jbc.274.47.33627.
80. Hardingham GE, Lipton SA. Regulation of neuronal oxidative and nitrosative stress by endogenous protective pathways and disease processes. **Antioxid Redox Signal** 14: 1421–1424, 2011. doi:10.1089/ars.2010.3573.
81. Brandes MS, Gray NE. NRF2 as a therapeutic target in neurodegenerative diseases. **ASN Neuro** 12: 1759091419899782, 2020. doi:10.1177/1759091419899782.
82. Lanzillotta C, Zuliani I, Tramutola A, Barone E, Blarmino C, Folgiero V, Caforio M, Valentini D, Villani A, Locatelli F, Butterfield DA, Head E, Perluigi M, Abisambra JF, Di Domenico F. Chronic PERK induction promotes Alzheimer-like neuropathology in Down syndrome: insights for therapeutic intervention. **Prog Neurobiol** 196: 101892, 2021. doi:10.1016/j.pneurobio.2020.101892.
83. Perluigi M, Tramutola A, Pagnotta S, Barone E, Butterfield DA. The BACH1/Nrf2 axis in brain in Down syndrome and transition to Alzheimer disease-like neuropathology and dementia. **Antioxidants (Basel)** 9: 779, 2020. doi:10.3390/antiox9090779.
84. Zhang J, Wang X, Vikash V, Ye Q, Wu D, Liu Y, Dong W. ROS and ROS-mediated cellular signaling. **Oxid Med Cell Longev** 2016: 4350965, 2016. doi:10.1155/2016/4350965.
85. Manning BD, Toker A. AKT/PKB signaling: navigating the network. **Cell** 169: 381–405, 2017. doi:10.1016/j.cell.2017.04.001.
86. Sarbassov DD, Guertin DA, Ali SM, Sabatini DM. Phosphorylation and regulation of Akt/PKB by the rictor-mTOR complex. **Science** 307: 1098–1101, 2005. doi:10.1126/science.1106148.
87. Leslie NR, Downes CP. PTEN: the down side of PI 3-kinase signaling. **Cell Signal** 14: 285–295, 2002. doi:10.1016/s0898-6568(01)00234-0.
88. Pimienta G, Pascual J. Canonical and alternative MAPK signaling. **Cell Cycle** 6: 2628–2632, 2007. doi:10.4161/cc.6.21.4930.
89. Kyriakis JM, Avruch J. Mammalian mitogen-activated protein kinase signal transduction pathways activated by stress and inflammation. **Physiol Rev** 81: 807–869, 2001. doi:10.1152/physrev.2001.81.2.807.
90. Chen Z, Gibson TB, Robinson F, Silvestro L, Pearson G, Xu B, Wright A, Vanderbilt C, Cobb MH. MAP kinases. **Chem Rev** 101: 2449–2476, 2001. doi:10.1021/cr000241p.
91. Lei H, Kazlauskas A. Growth factors outside of the platelet-derived growth factor (PDGF) family employ reactive oxygen species/Src family kinases to activate PDGF receptor alpha and thereby promote proliferation and survival of cells. **J Biol Chem** 284: 6329–6336, 2009. doi:10.1074/jbc.M808426200.
92. Wentworth CC, Alam A, Jones RM, Nusrat A, Neish AS. Enteric commensal bacteria induce extracellular signal-regulated kinase pathway signaling via formyl peptide receptor-dependent redox modulation of dual specific phosphatase 3. **J Biol Chem** 286: 38448–38455, 2011. doi:10.1074/jbc.M111.268938.
93. Banan A, Fields JZ, Zhang Y, Keshavarzian A. Phospholipase C-gamma inhibition prevents EGF protection of intestinal cytoskeleton and barrier against oxidants. **Am J Physiol Gastrointest Liver Physiol** 281: G412–G423, 2001. doi:10.1152/ajpgi.2001.281.2.G412.
94. Dann SG, Golas J, Miranda M, Shi C, Wu J, Jin G, Rosfjord E, Upeslaci E, Klippel A. p120 catenin is a key effector of a Ras-PKC epsilon oncogenic signaling axis. **Oncogene** 33: 1385–1394, 2014. doi:10.1038/ncr.2013.91.
95. Davies C, Tournier C. Exploring the function of the JNK (c-Jun N-terminal kinase) signalling pathway in physiological and pathological processes to design novel therapeutic strategies. **Biochem Soc Trans** 40: 85–89, 2012. doi:10.1042/BST20110641.
96. Katagiri K, Matsuzawa A, Ichijo H. Regulation of apoptosis signal-regulating kinase 1 in redox signaling. **Methods Enzymol** 474: 277–288, 2010. doi:10.1016/S0076-6879(10)74016-7.
97. Mittal M, Siddiqui MR, Tran K, Reddy SP, Malik AB. Reactive oxygen species in inflammation and tissue injury. **Antioxid Redox Signal** 20: 1126–1167, 2014. doi:10.1089/ars.2012.5149.
98. Nelson PT, Braak H, Markesbery WR. Neuropathology and cognitive impairment in Alzheimer disease: a complex but coherent relationship. **J Neuropathol Exp Neurol** 68: 1–14, 2009. doi:10.1097/NEN.0b013e3181919a48.
99. Markesbery WR. Neuropathologic alterations in mild cognitive impairment: a review. **J Alzheimers Dis** 19: 221–228, 2010. doi:10.3233/JAD-2010-1220.
100. Scheff SW, Price DA. Alzheimer's disease-related alterations in synaptic density: neocortex and hippocampus. **J Alzheimers Dis** 9: 101–115, 2006. doi:10.3233/jad-2006-9s312.
101. Coleman PD, Flood DG. Neuron numbers and dendritic extent in normal aging and Alzheimer's disease. **Neurobiol Aging** 8: 521–545, 1987. doi:10.1016/0197-4580(87)90127-8.
102. Selkoe DJ. Alzheimer's disease: genes, proteins, and therapy. **Physiol Rev** 81: 741–766, 2001. doi:10.1152/physrev.2001.81.2.741.
103. Tolia A, De Strooper B. Structure and function of gamma-secretase. **Semin Cell Dev Biol** 20: 211–218, 2009. doi:10.1016/j.semcdb.2008.10.007.
104. Cole SL, Vassar R. BACE1 structure and function in health and Alzheimer's disease. **Curr Alzheimer Res** 5: 100–120, 2008. doi:10.2174/156720508783954758.

105. Näslund J, Schierhorn A, Hellman U, Lannfelt L, Roses AD, Tjernberg LO, Silberring J, Gandy SE, Winblad B, Greengard P. Relative abundance of Alzheimer A beta amyloid peptide variants in Alzheimer disease and normal aging. **Proc Natl Acad Sci USA** 91: 8378–8382, 1994. doi:[10.1073/pnas.91.18.8378](https://doi.org/10.1073/pnas.91.18.8378).
106. Hensley K, Carney JM, Mattson MP, Aksenova M, Harris M, Wu JF, Floyd RA, Butterfield DA. A model for beta-amyloid aggregation and neurotoxicity based on free radical generation by the peptide: relevance to Alzheimer disease. **Proc Natl Acad Sci USA** 91: 3270–3274, 1994. doi:[10.1073/pnas.91.8.3270](https://doi.org/10.1073/pnas.91.8.3270).
107. Atwood CS, Perry G, Zeng H, Kato Y, Jones WD, Ling KQ, Huang X, Moir RD, Wang D, Sayre LM, Smith MA, Chen SG, Bush AI. Copper mediates dityrosine cross-linking of Alzheimer's amyloid-beta. **Biochemistry** 43: 560–568, 2004. doi:[10.1021/bi0358824](https://doi.org/10.1021/bi0358824).
108. Huang X, Atwood CS, Hartshorn MA, Multhaup G, Goldstein LE, Scarpa RC, Cuajungco MP, Gray DN, Lim J, Moir RD, Tanzi RE, Bush AI. The A beta peptide of Alzheimer's disease directly produces hydrogen peroxide through metal ion reduction. **Biochemistry** 38: 7609–7616, 1999. doi:[10.1021/bi990438f](https://doi.org/10.1021/bi990438f).
109. Streltsov VA, Titmuss SJ, Epa VC, Barnham KJ, Masters CL, Varghese JN. The structure of the amyloid-beta peptide high-affinity copper II binding site in Alzheimer disease. **Biophys J** 95: 3447–3456, 2008. doi:[10.1529/biophysj.108.134429](https://doi.org/10.1529/biophysj.108.134429).
110. Cherny RA, Atwood CS, Xilinas ME, Gray DN, Jones WD, McLean CA, Barnham KJ, Volitakis I, Fraser FW, Kim Y, Huang X, Goldstein LE, Moir RD, Lim JT, Beyreuther K, Zheng H, Tanzi RE, Masters CL, Bush AI. Treatment with a copper-zinc chelator markedly and rapidly inhibits beta-amyloid accumulation in Alzheimer's disease transgenic mice. **Neuron** 30: 665–676, 2001. doi:[10.1016/s0896-6273\(01\)00317-8](https://doi.org/10.1016/s0896-6273(01)00317-8).
111. Varadarajan S, Yatin S, Aksenova M, Butterfield DA. Review: Alzheimer's amyloid beta-peptide-associated free radical oxidative stress and neurotoxicity. **J Struct Biol** 130: 184–208, 2000. doi:[10.1006/jsbi.2000.4274](https://doi.org/10.1006/jsbi.2000.4274).
112. Varadarajan S, Yatin S, Kanski J, Jahanshahi F, Butterfield DA. Methionine residue 35 is important in amyloid beta-peptide-associated free radical oxidative stress. **Brain Res Bull** 50: 133–141, 1999. doi:[10.1016/s0361-9230\(99\)00093-3](https://doi.org/10.1016/s0361-9230(99)00093-3).
113. Yatin SM, Varadarajan S, Link CD, Butterfield DA. In vitro and in vivo oxidative stress associated with Alzheimer's amyloid beta-peptide (1-42). **Neurobiol Aging** 20: 325–330, 1999. doi:[10.1016/s0197-4580\(99\)00056-1](https://doi.org/10.1016/s0197-4580(99)00056-1).
114. Varadarajan S, Kanski J, Aksenova M, Lauderback C, Butterfield DA. Different mechanisms of oxidative stress and neurotoxicity for Alzheimer's A beta(1-42) and A beta(25-35). **J Am Chem Soc** 123: 5625–5631, 2001. doi:[10.1021/ja010452r](https://doi.org/10.1021/ja010452r).
115. Butterfield DA, Boyd-Kimball D. The critical role of methionine 35 in Alzheimer's amyloid beta-peptide (1-42)-induced oxidative stress and neurotoxicity. **Biochim Biophys Acta** 1703: 149–156, 2005. doi:[10.1016/j.bbapap.2004.10.014](https://doi.org/10.1016/j.bbapap.2004.10.014).
116. Kanski J, Aksenova M, Schöneich C, Butterfield DA. Substitution of isoleucine-31 by helical-breaking proline abolishes oxidative stress and neurotoxic properties of Alzheimer's amyloid beta-peptide. **Free Radic Biol Med** 32: 1205–1211, 2002. doi:[10.1016/s0891-5849\(02\)00821-3](https://doi.org/10.1016/s0891-5849(02)00821-3).
117. Crouch PJ, Barnham KJ, Duce JA, Blake RE, Masters CL, Trounce IA. Copper-dependent inhibition of cytochrome c oxidase by Abeta(1-42) requires reduced methionine at residue 35 of the Abeta peptide. **J Neurochem** 99: 226–236, 2006. doi:[10.1111/j.1471-4159.2006.04050.x](https://doi.org/10.1111/j.1471-4159.2006.04050.x).
118. Drake J, Link CD, Butterfield DA. Oxidative stress precedes fibrillar deposition of Alzheimer's disease amyloid beta-peptide (1-42) in a transgenic *Caenorhabditis elegans* model. **Neurobiol Aging** 24: 415–420, 2003. doi:[10.1016/s0197-4580\(02\)00225-7](https://doi.org/10.1016/s0197-4580(02)00225-7).
119. Butterfield DA, Galvan V, Lange MB, Tang H, Sowell RA, Spilman P, Fombonne J, Gorostiza O, Zhang J, Sultana R, Bredesen DE. In vivo oxidative stress in brain of Alzheimer disease transgenic mice: Requirement for methionine 35 in amyloid beta-peptide of APP. **Free Radic Biol Med** 48: 136–144, 2010. doi:[10.1016/j.freeradbiomed.2009.10.035](https://doi.org/10.1016/j.freeradbiomed.2009.10.035).
120. Sokolov Y, Kozak JA, Kaye R, Chanturiya A, Glabe C, Hall JE. Soluble amyloid oligomers increase bilayer conductance by altering dielectric structure. **J Gen Physiol** 128: 637–647, 2006. doi:[10.1085/jgp.200609533](https://doi.org/10.1085/jgp.200609533).
121. Pogocki D, Schöneich C. Redox properties of Met(35) in neurotoxic beta-amyloid peptide. A molecular modeling study. **Chem Res Toxicol** 15: 408–418, 2002. doi:[10.1021/tx0101550](https://doi.org/10.1021/tx0101550).
122. Schöneich C. Methionine oxidation by reactive oxygen species: reaction mechanisms and relevance to Alzheimer's disease. **Biochim Biophys Acta** 1703: 111–119, 2005. doi:[10.1016/j.bbapap.2004.09.009](https://doi.org/10.1016/j.bbapap.2004.09.009).
123. Schöneich C, Pogocki D, Hug GL, Bobrowski K. Free radical reactions of methionine in peptides: mechanisms relevant to beta-amyloid oxidation and Alzheimer's disease. **J Am Chem Soc** 125: 13700–13713, 2003. doi:[10.1021/ja036733b](https://doi.org/10.1021/ja036733b).
124. Butterfield DA, Boyd-Kimball D. Redox proteomics and amyloid beta-peptide: insights into Alzheimer disease. **J Neurochem** 151: 459–487, 2019. doi:[10.1111/jnc.14589](https://doi.org/10.1111/jnc.14589).
125. Di Domenico F, Tramutola A, Butterfield DA. Role of 4-hydroxy-2-nonenal (HNE) in the pathogenesis of Alzheimer disease and other selected age-related neurodegenerative disorders. **Free Radic Biol Med** 111: 253–261, 2017. doi:[10.1016/j.freeradbiomed.2016.10.490](https://doi.org/10.1016/j.freeradbiomed.2016.10.490).
126. Roychaudhuri R, Yang M, Deshpande A, Cole GM, Frautschy S, Lomakin A, Benedek GB, Teplow DB. C-terminal turn stability determines assembly differences between Abeta40 and Abeta42. **J Mol Biol** 425: 292–308, 2013. doi:[10.1016/j.jmb.2012.11.006](https://doi.org/10.1016/j.jmb.2012.11.006).
127. Elharram A, Czegledy NM, Golod M, Milne GL, Pollock E, Bennett BM, Shchepinov MS. Deuterium-reinforced polyunsaturated fatty acids improve cognition in a mouse model of sporadic Alzheimer's disease. **FEBS J** 284: 4083–4095, 2017. doi:[10.1111/febs.14291](https://doi.org/10.1111/febs.14291).
128. Raefsky SM, Furman R, Milne G, Pollock E, Axelsen P, Mattson MP, Shchepinov MS. Deuterated polyunsaturated fatty acids reduce brain lipid peroxidation and hippocampal amyloid beta-peptide levels, without discernable behavioral effects in an APP/PS1 mutant transgenic mouse model of Alzheimer's disease. **Neurobiol Aging** 66: 165–176, 2018. doi:[10.1016/j.neurobiolaging.2018.02.024](https://doi.org/10.1016/j.neurobiolaging.2018.02.024).
129. Shchepinov MS. Polyunsaturated fatty acid deuteration against neurodegeneration. **Trends Pharmacol Sci** 41: 236–248, 2020. doi:[10.1016/j.tips.2020.01.010](https://doi.org/10.1016/j.tips.2020.01.010).
130. Butterfield DA, Hensley K, Harris M, Mattson M, Carney J. beta-Amyloid peptide free radical fragments initiate synaptosomal lipoperoxidation in a sequence-specific fashion: implications to Alzheimer's disease. **Biochem Biophys Res Commun** 200: 710–715, 1994. doi:[10.1006/bbrc.1994.1508](https://doi.org/10.1006/bbrc.1994.1508).
131. Butterfield DA, Boyd-Kimball D. Oxidative stress, amyloid-beta peptide, and altered key molecular pathways in the pathogenesis and progression of Alzheimer's disease. **J Alzheimers Dis** 62: 1345–1367, 2018. doi:[10.3233/JAD-170543](https://doi.org/10.3233/JAD-170543).

132. Butterfield DA, Perluigi M, Reed T, Muharib T, Hughes CP, Robinson RA, Sultana R. Redox proteomics in selected neurodegenerative disorders: from its infancy to future applications. **Antioxid Redox Signal** 17: 1610–1655, 2012. doi:[10.1089/ars.2011.4109](https://doi.org/10.1089/ars.2011.4109).
133. Di Domenico F, Barone E, Perluigi M, Butterfield DA. The triangle of death in Alzheimer's disease brain: the aberrant cross-talk among energy metabolism, mammalian target of rapamycin signaling, and protein homeostasis revealed by redox proteomics. **Antioxid Redox Signal** 26: 364–387, 2017. doi:[10.1089/ars.2016.6759](https://doi.org/10.1089/ars.2016.6759).
134. Tramutola A, Lanzillotta C, Perluigi M, Butterfield DA. Oxidative stress, protein modification and Alzheimer disease. **Brain Res Bull** 133: 88–96, 2017. doi:[10.1016/j.brainresbull.2016.06.005](https://doi.org/10.1016/j.brainresbull.2016.06.005).
135. Bader Lange ML, Cenini G, Piroddi M, Abdul HM, Sultana R, Galli F, Memo M, Butterfield DA. Loss of phospholipid asymmetry and elevated brain apoptotic protein levels in subjects with amnesic mild cognitive impairment and Alzheimer disease. **Neurobiol Dis** 29: 456–464, 2008. doi:[10.1016/j.nbd.2007.11.004](https://doi.org/10.1016/j.nbd.2007.11.004).
136. Castegna A, Lauderback CM, Mohammad-Abdul H, Butterfield DA. Modulation of phospholipid asymmetry in synaptosomal membranes by the lipid peroxidation products, 4-hydroxynonenal and acrolein: implications for Alzheimer's disease. **Brain Res** 1004: 193–197, 2004. doi:[10.1016/j.brainres.2004.01.036](https://doi.org/10.1016/j.brainres.2004.01.036).
137. Cenini G, Sultana R, Memo M, Butterfield DA. Elevated levels of pro-apoptotic p53 and its oxidative modification by the lipid peroxidation product, HNE, in brain from subjects with amnesic mild cognitive impairment and Alzheimer's disease. **J Cell Mol Med** 12: 987–994, 2008. doi:[10.1111/j.1582-4934.2008.00163.x](https://doi.org/10.1111/j.1582-4934.2008.00163.x).
138. Mark RJ, Hensley K, Butterfield DA, Mattson MP. Amyloid beta-peptide impairs ion-motive ATPase activities: evidence for a role in loss of neuronal Ca^{2+} homeostasis and cell death. **J Neurosci** 15: 6239–6249, 1995. doi:[10.1523/JNEUROSCI.15-09-06239.1995](https://doi.org/10.1523/JNEUROSCI.15-09-06239.1995).
139. Mattson MP, Chan SL. Neuronal and glial calcium signaling in Alzheimer's disease. **Cell Calcium** 34: 385–397, 2003. doi:[10.1016/s0143-4160\(03\)00128-3](https://doi.org/10.1016/s0143-4160(03)00128-3).
140. Owen JB, Sultana R, Aluise CD, Erickson MA, Price TO, Bu G, Banks WA, Butterfield DA. Oxidative modification to LDL receptor-related protein 1 in hippocampus from subjects with Alzheimer disease: implications for Abeta accumulation in AD brain. **Free Radic Biol Med** 49: 1798–1803, 2010. doi:[10.1016/j.freeradbiomed.2010.09.013](https://doi.org/10.1016/j.freeradbiomed.2010.09.013).
141. Sultana R, Butterfield DA. Alterations of some membrane transport proteins in Alzheimer's disease: role of amyloid beta-peptide. **Mol Biosyst** 4: 36–41, 2008. doi:[10.1039/b715278g](https://doi.org/10.1039/b715278g).
142. Sultana R, Butterfield DA. Oxidatively modified, mitochondria-relevant brain proteins in subjects with Alzheimer disease and mild cognitive impairment. **J Bioenerg Biomembr** 41: 441–446, 2009. doi:[10.1007/s10863-009-9241-7](https://doi.org/10.1007/s10863-009-9241-7).
143. Urano A, Matsumaru D, Ryoike R, Saito R, Kadoguchi S, Saigusa D, Saito T, Saido TC, Kawashima R, Yamamoto M. Nrf2 suppresses oxidative stress and inflammation in App knock-in Alzheimer's disease model mice. **Mol Cell Biol** 40: e00467-19, 2020. doi:[10.1128/MCB.00467-19](https://doi.org/10.1128/MCB.00467-19).
144. Reddy PH, Manczak M, Yin X, Grady MC, Mitchell A, Tonk S, Kuruva CS, Bhatti JS, Kandimalla R, Vijayan M, Kumar S, Wang R, Pradeepkiran JA, Ogunmokin G, Thamarai K, Quesada K, Boles A, Reddy AP. Protective effects of Indian spice curcumin against amyloid-beta in Alzheimer's disease. **J Alzheimers Dis** 61: 843–866, 2018. doi:[10.3233/JAD-170512](https://doi.org/10.3233/JAD-170512).
145. Praticò D, Uryu K, Leight S, Trojanowski JQ, Lee VM. Increased lipid peroxidation precedes amyloid plaque formation in an animal model of Alzheimer amyloidosis. **J Neurosci** 21: 4183–4187, 2001. doi:[10.1523/JNEUROSCI.21-12-04183.2001](https://doi.org/10.1523/JNEUROSCI.21-12-04183.2001).
146. Uryu K, Laurer H, McIntosh T, Praticò D, Martinez D, Leight S, Lee VM, Trojanowski JQ. Repetitive mild brain trauma accelerates Abeta deposition, lipid peroxidation, and cognitive impairment in a transgenic mouse model of Alzheimer amyloidosis. **J Neurosci** 22: 446–454, 2002. doi:[10.1523/JNEUROSCI.22-02-00446.2002](https://doi.org/10.1523/JNEUROSCI.22-02-00446.2002).
147. Huang Q, Aluise CD, Joshi G, Sultana R, St Clair DK, Markesbery WR, Butterfield DA. Potential in vivo amelioration by N-acetyl-L-cysteine of oxidative stress in brain in human double mutant APP/PS-1 knock-in mice: toward therapeutic modulation of mild cognitive impairment. **J Neurosci Res** 88: 2618–2629, 2010. doi:[10.1002/jnr.22422](https://doi.org/10.1002/jnr.22422).
148. Bruce-Keller AJ, Gupta S, Knight AG, Beckett TL, McMullen JM, Davis PR, Murphy MP, Van Eldik LJ, St Clair D, Keller JN. Cognitive impairment in humanized APPxPS1 mice is linked to Abeta(1-42) and NOX activation. **Neurobiol Dis** 44: 317–326, 2011. doi:[10.1016/j.nbd.2011.07.012](https://doi.org/10.1016/j.nbd.2011.07.012).
149. Robinson RA, Joshi G, Huang QZ, Sultana R, Baker AS, Cai J, Pierce W, St Clair DK, Markesbery WR, Butterfield DA. Proteomic analysis of brain proteins in APP/PS-1 human double mutant knock-in mice with increasing amyloid beta-peptide deposition: insights into the effects of in vivo treatment with N-acetylcysteine as a potential therapeutic intervention in mild cognitive impairment and Alzheimer's disease. **Proteomics** 11: 4243–4256, 2011. doi:[10.1002/pmic.201000523](https://doi.org/10.1002/pmic.201000523).
150. Sultana R, Robinson RA, Di Domenico F, Abdul HM, St Clair DK, Markesbery WR, Cai J, Pierce WM, Butterfield DA. Proteomic identification of specifically carbonylated brain proteins in APP(NLh)/APP(NLh) x PS-1(P264L)/PS-1(P264L) human double mutant knock-in mice model of Alzheimer disease as a function of age. **J Proteomics** 74: 2430–2440, 2011. doi:[10.1016/j.jprot.2011.06.015](https://doi.org/10.1016/j.jprot.2011.06.015).
151. Bader Lange ML, St Clair D, Markesbery WR, Studzinski CM, Murphy MP, Butterfield DA. Age-related loss of phospholipid asymmetry in APP(NLh)/APP(NLh) x PS-1(P264L)/PS-1(P264L) human double mutant knock-in mice: relevance to Alzheimer disease. **Neurobiol Dis** 38: 104–115, 2010. doi:[10.1016/j.nbd.2010.01.004](https://doi.org/10.1016/j.nbd.2010.01.004).
152. Resende R, Moreira PI, Proença T, Deshpande A, Busciglio J, Pereira C, Oliveira CR. Brain oxidative stress in a triple-transgenic mouse model of Alzheimer disease. **Free Radic Biol Med** 44: 2051–2057, 2008. doi:[10.1016/j.freeradbiomed.2008.03.012](https://doi.org/10.1016/j.freeradbiomed.2008.03.012).
153. Siedlak SL, Casadesus G, Webber KM, Pappolla MA, Atwood CS, Smith MA, Perry G. Chronic antioxidant therapy reduces oxidative stress in a mouse model of Alzheimer's disease. **Free Radic Res** 43: 156–164, 2009. doi:[10.1080/10715760802644694](https://doi.org/10.1080/10715760802644694).
154. Mao P, Manczak M, Calkins MJ, Truong Q, Reddy TP, Reddy AP, Shirendeb U, Lo HH, Rabinovitch PS, Reddy PH. Mitochondria-targeted catalase reduces abnormal APP processing, amyloid beta production and BACE1 in a mouse model of Alzheimer's disease: implications for neuroprotection and lifespan extension. **Hum Mol Genet** 21: 2973–2990, 2012. doi:[10.1093/hmg/ddds128](https://doi.org/10.1093/hmg/ddds128).
155. Karuppagounder SS, Xu H, Shi Q, Chen LH, Pedrini S, Pechman D, Baker H, Beal MF, Gandy SE, Gibson GE. Thiamine deficiency induces oxidative stress and exacerbates the plaque pathology in Alzheimer's mouse model. **Neurobiol Aging** 30: 1587–1600, 2009. doi:[10.1016/j.neurobiolaging.2007.12.013](https://doi.org/10.1016/j.neurobiolaging.2007.12.013).
156. Gibson GE, Blass JP. Thiamine-dependent processes and treatment strategies in neurodegeneration. **Antioxid Redox Signal** 9: 1605–1619, 2007. doi:[10.1089/ars.2007.1766](https://doi.org/10.1089/ars.2007.1766).

157. Gibson GE, Luchsinger JA, Cirio R, Chen H, Franchino-Elder J, Hirsch JA, Bettendorff L, Chen Z, Flowers SA, Gerber LM, Grandville T, Schupf N, Xu H, Stern Y, Habeck C, Jordan B, Fonzetti P. Benfotiamine and cognitive decline in Alzheimer's disease: results of a randomized placebo-controlled Phase IIa clinical trial. **J Alzheimers Dis** 78: 989–1010, 2020. doi:10.3233/JAD-200896.
158. Dumont M, Kipiani K, Yu F, Wille E, Katz M, Calingasan NY, Gouras GK, Lin MT, Beal MF. Coenzyme Q10 decreases amyloid pathology and improves behavior in a transgenic mouse model of Alzheimer's disease. **J Alzheimers Dis** 27: 211–223, 2011. doi:10.3233/JAD-2011-110209.
159. Robinson RA, Lange MB, Sultana R, Galvan V, Fombonne J, Gorostiza O, Zhang J, Warriar G, Cai J, Pierce WM, Bredesen DE, Butterfield DA. Differential expression and redox proteomics analyses of an Alzheimer disease transgenic mouse model: effects of the amyloid-beta peptide of amyloid precursor protein. **Neuroscience** 177: 207–222, 2011. doi:10.1016/j.neuroscience.2011.01.005.
160. Sultana R, Robinson RA, Lange MB, Fiorini A, Galvan V, Fombonne J, Baker A, Gorostiza O, Zhang J, Cai J, Pierce WM, Bredesen DE, Butterfield DA. Do proteomics analyses provide insights into reduced oxidative stress in the brain of an Alzheimer disease transgenic mouse model with an M631L amyloid precursor protein substitution and thereby the importance of amyloid-beta-resident methionine 35 in Alzheimer disease pathogenesis? **Antioxid Redox Signal** 17: 1507–1514, 2012. doi:10.1089/ars.2011.4470.
161. Xiang XJ, Song L, Deng XJ, Tang Y, Min Z, Luo B, Wen QX, Li KY, Chen J, Ma YL, Zhu BL, Yan Z, Chen GJ. Mitochondrial methionine sulfoxide reductase B2 links oxidative stress to Alzheimer's disease-like pathology. **Exp Neurol** 318: 145–156, 2019. doi:10.1016/j.expneurol.2019.05.006.
162. Matsuoaka Y, Picciano M, La Francois J, Duff K. Fibrillar beta-amyloid evokes oxidative damage in a transgenic mouse model of Alzheimer's disease. **Neuroscience** 104: 609–613, 2001. doi:10.1016/s0306-4522(01)00115-4.
163. Fan YG, Pang ZQ, Wu TY, Zhang YH, Xuan WQ, Wang Z, Yu X, Li YC, Guo C, Wang ZY. Vitamin D deficiency exacerbates Alzheimer-like pathologies by reducing antioxidant capacity. **Free Radic Biol Med** 161: 139–149, 2020. doi:10.1016/j.freeradbiomed.2020.10.007.
164. Giraldo E, Lloret A, Fuchsberger T, Vina J. Abeta and tau toxicities in Alzheimer's are linked via oxidative stress-induced p38 activation: protective role of vitamin E. **Redox Biol** 2: 873–877, 2014. doi:10.1016/j.redox.2014.03.002.
165. Farr SA, Poon HF, Dogrukul-Ak D, Drake J, Banks WA, Eyerman E, Butterfield DA, Morley JE. The antioxidants alpha-lipoic acid and N-acetylcysteine reverse memory impairment and brain oxidative stress in aged SAMP8 mice. **J Neurochem** 84: 1173–1183, 2003. doi:10.1046/j.1471-4159.2003.01580.x.
166. Poon HF, Farr SA, Thongboonkerd V, Lynn BC, Banks WA, Morley JE, Klein JB, Butterfield DA. Proteomic analysis of specific brain proteins in aged SAMP8 mice treated with alpha-lipoic acid: implications for aging and age-related neurodegenerative disorders. **Neurochem Int** 46: 159–168, 2005. doi:10.1016/j.neuint.2004.07.008.
167. Poon HF, Castegna A, Farr SA, Thongboonkerd V, Lynn BC, Banks WA, Morley JE, Klein JB, Butterfield DA. Quantitative proteomics analysis of specific protein expression and oxidative modification in aged senescence-accelerated-prone 8 mice brain. **Neuroscience** 126: 915–926, 2004. doi:10.1016/j.neuroscience.2004.04.046.
168. Fiorini A, Sultana R, Förster S, Perluigi M, Cenini G, Cini C, Cai J, Klein JB, Farr SA, Niehoff ML, Morley JE, Kumar VB, Butterfield DA. Antisense directed against PS-1 gene decreases brain oxidative markers in aged senescence accelerated mice (SAMP8) and reverses learning and memory impairment: a proteomics study. **Free Radic Biol Med** 65: 1–14, 2013. doi:10.1016/j.freeradbiomed.2013.06.017.
169. Butterfield DA, Poon HF. The senescence-accelerated prone mouse (SAMP8): a model of age-related cognitive decline with relevance to alterations of the gene expression and protein abnormalities in Alzheimer's disease. **Exp Gerontol** 40: 774–783, 2005. doi:10.1016/j.exger.2005.05.007.
170. Poon HF, Farr SA, Banks WA, Pierce WM, Klein JB, Morley JE, Butterfield DA. Proteomic identification of less oxidized brain proteins in aged senescence-accelerated mice following administration of antisense oligonucleotide directed at the Abeta region of amyloid precursor protein. **Brain Res Mol Brain Res** 138: 8–16, 2005. doi:10.1016/j.molbrainres.2005.02.020.
171. Emoto MC, Sato-Akaba H, Hamaue N, Kawanishi K, Koshino H, Shimohama S, Fujii HG. Early detection of redox imbalance in the APPsw/PS1dE9 mouse model of Alzheimer's disease by in vivo electron paramagnetic resonance imaging. **Free Radic Biol Med** 172: 9–18, 2021. doi:10.1016/j.freeradbiomed.2021.05.035.
172. Dixit S, Bernardo A, Walker JM, Kennard JA, Kim GY, Kessler ES, Harrison FE. Vitamin C deficiency in the brain impairs cognition, increases amyloid accumulation and deposition, and oxidative stress in APP/PSEN1 and normally aging mice. **ACS Chem Neurosci** 6: 570–581, 2015. doi:10.1021/cn500308h.
173. Schuessel K, Schäfer S, Bayer TA, Czech C, Pradier L, Muller-Spahn F, Muller WE, Eckert A. Impaired Cu/Zn-SOD activity contributes to increased oxidative damage in APP transgenic mice. **Neurobiol Dis** 18: 89–99, 2005. doi:10.1016/j.nbd.2004.09.003.
174. Rönnbäck A, Pavlov PF, Mansory M, Gonze P, Marliere N, Winblad B, Graff C, Behbahani H. Mitochondrial dysfunction in a transgenic mouse model expressing human amyloid precursor protein (APP) with the Arctic mutation. **J Neurochem** 136: 497–502, 2016. doi:10.1111/jnc.13410.
175. Reddy PH. Abnormal tau, mitochondrial dysfunction, impaired axonal transport of mitochondria, and synaptic deprivation in Alzheimer's disease. **Brain Res** 1415: 136–148, 2011. doi:10.1016/j.brainres.2011.07.052.
176. Perluigi M, Joshi G, Sultana R, Calabrese V, De Marco C, Coccia R, Butterfield DA. In vivo protection by the xanthate tricyclodecan-9-yl-xanthogenate against amyloid beta-peptide (1-42)-induced oxidative stress. **Neuroscience** 138: 1161–1170, 2006. doi:10.1016/j.neuroscience.2005.12.004.
177. Mohammad Abdul H, Butterfield DA. Protection against amyloid beta-peptide (1-42)-induced loss of phospholipid asymmetry in synaptosomal membranes by tricyclodecan-9-xanthogenate (D609) and ferulic acid ethyl ester: implications for Alzheimer's disease. **Biochim Biophys Acta** 1741: 140–148, 2005. doi:10.1016/j.bbadis.2004.12.002.
178. Ansari MA, Joshi G, Huang Q, Opii WO, Abdul HM, Sultana R, Butterfield DA. In vivo administration of D609 leads to protection of subsequently isolated gerbil brain mitochondria subjected to in vitro oxidative stress induced by amyloid beta-peptide and other oxidative stressors: relevance to Alzheimer's disease and other oxidative stress-related neurodegenerative disorders. **Free Radic Biol Med** 41: 1694–1703, 2006. doi:10.1016/j.freeradbiomed.2006.09.002.
179. Li Y, Tian Q, Li Z, Dang M, Lin Y, Hou X. Activation of Nrf2 signaling by sitagliptin and quercetin combination against beta-amyloid induced Alzheimer's disease in rats. **Drug Dev Res** 80: 837–845, 2019. doi:10.1002/ddr.21567.

180. Siddiqui N, Ali J, Parvez S, Zameer S, Najmi AK, Akhtar M. Linagliptin, a DPP-4 inhibitor, ameliorates Aβeta (1-42) peptides induced neurodegeneration and brain insulin resistance (BIR) via insulin receptor substrate-1 (IRS-1) in rat model of Alzheimer's disease. **Neuropharmacology** 195: 108662, 2021. doi:10.1016/j.neuropharm.2021.108662.
181. Sharma S, Verma S, Kapoor M, Saini A, Nehru B. Alzheimer's disease like pathology induced six weeks after aggregated amyloid-beta injection in rats: increased oxidative stress and impaired long-term memory with anxiety-like behavior. **Neurol Res** 38: 838–850, 2016. doi:10.1080/01616412.2016.1209337.
182. Do Carmo S, Cuello AC. Modeling Alzheimer's disease in transgenic rats. **Mol Neurodegener** 8: 37, 2013. doi:10.1186/1750-1326-8-37.
183. Wilson EN, Do Carmo S, Welikovich LA, Hall H, Aguilar LF, Foret MK, Iulita MF, Jia DT, Marks AR, Allard S, Emmerson JT, Ducatenzeiler A, Cuello AC. NP03, a microdose lithium formulation, blunts early amyloid post-plaque neuropathology in McGill-R-Thy1-APP Alzheimer-like transgenic rats. **J Alzheimers Dis** 73: 723–739, 2020. doi:10.3233/JAD-190862.
184. Wilson EN, Do Carmo S, Iulita MF, Hall H, Austin GL, Jia DT, Malcolm JC, Foret MK, Marks AR, Butterfield DA, Cuello AC. Microdose lithium NP03 diminishes pre-plaque oxidative damage and neuroinflammation in a rat model of Alzheimer's-like amyloidosis. **Curr Alzheimer Res** 15: 1220–1230, 2018. doi:10.2174/1567205015666180904154446.
185. Harvey AR, Kendall CL, Syková E. The status and organization of astrocytes, oligodendroglia and microglia in grafts of fetal rat cerebral cortex. **Neurosci Lett** 228: 58–62, 1997. doi:10.1016/s0304-3940(97)00341-8.
186. Bates KA, Martins RN, Harvey AR. Oxidative stress in a rat model of chronic gliosis. **Neurobiol Aging** 28: 995–1008, 2007. doi:10.1016/j.neurobiolaging.2006.05.003.
187. Dowling AL, Head E. Antioxidants in the canine model of human aging. **Biochim Biophys Acta** 1822: 685–689, 2012. doi:10.1016/j.bbadis.2011.09.020.
188. Cotman CW, Head E, Muggenburg BA, Zicker S, Milgram NW. Brain aging in the canine: a diet enriched in antioxidants reduces cognitive dysfunction. **Neurobiol Aging** 23: 809–818, 2002. doi:10.1016/s0197-4580(02)00073-8.
189. Opii WO, Joshi G, Head E, Milgram NW, Muggenburg BA, Klein JB, Pierce WM, Cotman CW, Butterfield DA. Proteomic identification of brain proteins in the canine model of human aging following a long-term treatment with antioxidants and a program of behavioral enrichment: relevance to Alzheimer's disease. **Neurobiol Aging** 29: 51–70, 2008. doi:10.1016/j.neurobiolaging.2006.09.012.
190. Binder LI, Guillozet-Bongaarts AL, Garcia-Sierra F, Berry RW. Tau, tangles, and Alzheimer's disease. **Biochim Biophys Acta** 1739: 216–223, 2005. doi:10.1016/j.bbadis.2004.08.014.
191. Hasegawa M, Watanabe S, Kondo H, Akiyama H, Mann DM, Saito Y, Murayama S. 3R and 4R tau isoforms in paired helical filaments in Alzheimer's disease. **Acta Neuropathol** 127: 303–305, 2014. doi:10.1007/s00401-013-1191-9.
192. Limorenko G, Lashuel HA. Revisiting the grammar of Tau aggregation and pathology formation: how new insights from brain pathology are shaping how we study and target tauopathies. **Chem Soc Rev** 51: 513–565, 2022. doi:10.1039/d1cs00127b.
193. Farías G, Cornejo A, Jimenez J, Guzman L, Maccioni RB. Mechanisms of tau self-aggregation and neurotoxicity. **Curr Alzheimer Res** 8: 608–614, 2011. doi:10.2174/15672051179671258.
194. Mondragón-Rodríguez S, Perry G, Zhu XW, Moreira PI, Acevedo-Aquino MC, Williams S. Phosphorylation of tau protein as the link between oxidative stress, mitochondrial dysfunction, and connectivity failure: implications for Alzheimer's disease. **Oxid Med Cell Longev** 2013: 940603, 2013. doi:10.1155/2013/940603.
195. Alavi Naini SM, Soussi-Yanicostas N. Tau hyperphosphorylation and oxidative stress, a critical vicious circle in neurodegenerative tauopathies? **Oxid Med Cell Longev** 2015: 151979, 2015. doi:10.1155/2015/151979.
196. Feng Y, Xia Y, Yu G, Shu X, Ge H, Zeng K, Wang J, Wang X. Cleavage of GSK-3β by calpain counteracts the inhibitory effect of Ser9 phosphorylation on GSK-3β activity induced by H₂O₂. **J Neurochem** 126: 234–242, 2013. doi:10.1111/jnc.12285.
197. Goedert M, Hasegawa M, Jakes R, Lawler S, Cuenda A, Cohen P. Phosphorylation of microtubule-associated protein tau by stress-activated protein kinases. **FEBS Lett** 409: 57–62, 1997. doi:10.1016/s0014-5793(97)00483-3.
198. Tamagno E, Parola M, Bardini P, Piccini A, Borghi R, Guglielmotto M, Santoro G, Davit A, Danni O, Smith MA, Perry G, Tabaton M. Beta-site APP cleaving enzyme up-regulation induced by 4-hydroxynonenal is mediated by stress-activated protein kinases pathways. **J Neurochem** 92: 628–636, 2005. doi:10.1111/j.1471-4159.2004.02895.x.
199. Sultana R, Boyd-Kimball D, Poon HF, Cai J, Pierce WM, Klein JB, Markesbery WR, Zhou XZ, Lu KP, Butterfield DA. Oxidative modification and down-regulation of Pin1 in Alzheimer's disease hippocampus: a redox proteomics analysis. **Neurobiol Aging** 27: 918–925, 2006. doi:10.1016/j.neurobiolaging.2005.05.005.
200. Butterfield DA, Di Domenico F, Barone E. Elevated risk of type 2 diabetes for development of Alzheimer disease: a key role for oxidative stress in brain. **Biochim Biophys Acta** 1842: 1693–1706, 2014. doi:10.1016/j.bbadis.2014.06.010.
201. Su B, Wang X, Lee HG, Tabaton M, Perry G, Smith MA, Zhu X. Chronic oxidative stress causes increased tau phosphorylation in M17 neuroblastoma cells. **Neurosci Lett** 468: 267–271, 2010. doi:10.1016/j.neulet.2009.11.010.
202. Lovell MA, Xiong S, Xie C, Davies P, Markesbery WR. Induction of hyperphosphorylated tau in primary rat cortical neuron cultures mediated by oxidative stress and glycogen synthase kinase-3. **J Alzheimers Dis** 6: 659–671, 2004. doi:10.3233/jad-2004-6610.
203. Zambrano CA, Egaña JT, Nunez MT, Maccioni RB, Gonzalez-Billault C. Oxidative stress promotes tau dephosphorylation in neuronal cells: the roles of cdk5 and PP1. **Free Radic Biol Med** 36: 1393–1402, 2004. doi:10.1016/j.freeradbiomed.2004.03.007.
204. Cho MH, Kim DH, Choi JE, Chang EJ, Seung Y. Increased phosphorylation of dynamin-related protein 1 and mitochondrial fission in okadaic acid-treated neurons. **Brain Res** 1454: 100–110, 2012. doi:10.1016/j.brainres.2012.03.010.
205. Mahley RW. Apolipoprotein E: cholesterol transport protein with expanding role in cell biology. **Science** 240: 622–630, 1988. doi:10.1126/science.3283935.
206. Strittmatter WJ, Weisgraber KH, Huang DY, Dong LM, Salvesen GS, Pericak-Vance M, Schmechel D, Saunders AM, Goldgaber D, Roses AD. Binding of human apolipoprotein E to synthetic amyloid beta peptide: isoform-specific effects and implications for late-onset Alzheimer disease. **Proc Natl Acad Sci USA** 90: 8098–8102, 1993. doi:10.1073/pnas.90.17.8098.
207. Liu S, Park S, Allington G, Prelli F, Sun Y, Marta-Ariza M, Scholtzova H, Biswas G, Brown B, Verghese PB, Mehta PD, Kwon YU, Wisniewski T. Targeting apolipoprotein E/amyloid beta binding by

- peptoid CPO_Abeta17-21 P ameliorates Alzheimer's disease related pathology and cognitive decline. **Sci Rep** 7: 8009, 2017. doi:[10.1038/s41598-017-08604-8](https://doi.org/10.1038/s41598-017-08604-8).
208. Shi Y, Yamada K, Liddelaw SA, Smith ST, Zhao L, Luo W, et al. ApoE4 markedly exacerbates tau-mediated neurodegeneration in a mouse model of tauopathy. **Nature** 549: 523–527, 2017. doi:[10.1038/nature24016](https://doi.org/10.1038/nature24016).
209. Butterfield DA, Mattson MP. Apolipoprotein E and oxidative stress in brain with relevance to Alzheimer's disease. **Neurobiol Dis** 138: 104795, 2020. doi:[10.1016/j.nbd.2020.104795](https://doi.org/10.1016/j.nbd.2020.104795).
210. Ramassamy C, Averill D, Beffert U, Theroux L, Lussier-Cacan S, Cohn JS, Christen Y, Schoofs A, Davignon J, Poirier J. Oxidative insults are associated with apolipoprotein E genotype in Alzheimer's disease brain. **Neurobiol Dis** 7: 23–37, 2000. doi:[10.1006/nbdi.1999.0273](https://doi.org/10.1006/nbdi.1999.0273).
211. Glodzik-Sobanska L, Pirraglia E, Brys M, de Santi S, Mosconi L, Rich KE, Switalski R, Saint Louis L, Sadowski MJ, Martiniuk F, Mehta P, Pratico D, Zinkowski RP, Blennow K, de Leon MJ. The effects of normal aging and ApoE genotype on the levels of CSF biomarkers for Alzheimer's disease. **Neurobiol Aging** 30: 672–681, 2009. doi:[10.1016/j.neurobiolaging.2007.08.019](https://doi.org/10.1016/j.neurobiolaging.2007.08.019).
212. Caberlotto L, Marchetti L, Lauria M, Scotti M, Parolo S. Integration of transcriptomic and genomic data suggests candidate mechanisms for APOE4-mediated pathogenic action in Alzheimer's disease. **Sci Rep** 6: 32583, 2016. doi:[10.1038/srep32583](https://doi.org/10.1038/srep32583).
213. Yao J, Petanceska SS, Montine TJ, Holtzman DM, Schmidt SD, Parker CA, Callahan MJ, Lipinski WJ, Bisgaier CL, Turner BA, Nixon RA, Martins RN, Ouimet C, Smith JD, Davies P, Laska E, Ehrlich ME, Walker LC, Mathews PM, Gandy S. Aging, gender and APOE isotype modulate metabolism of Alzheimer's Abeta peptides and F-isoprostanes in the absence of detectable amyloid deposits. **J Neurochem** 90: 1011–1018, 2004. doi:[10.1111/j.1471-4159.2004.02532.x](https://doi.org/10.1111/j.1471-4159.2004.02532.x).
214. Brown CM, Wright E, Colton CA, Sullivan PM, Laskowitz DT, Vitek MP. Apolipoprotein E isoform mediated regulation of nitric oxide release. **Free Radic Biol Med** 32: 1071–1075, 2002. doi:[10.1016/s0891-5849\(02\)00803-1](https://doi.org/10.1016/s0891-5849(02)00803-1).
215. Colton CA, Brown CM, Cook D, Needham LK, Xu Q, Czapiga M, Saunders AM, Schmechel DE, Rasheed K, Vitek MP. APOE and the regulation of microglial nitric oxide production: a link between genetic risk and oxidative stress. **Neurobiol Aging** 23: 777–785, 2002. doi:[10.1016/s0197-4580\(02\)00016-7](https://doi.org/10.1016/s0197-4580(02)00016-7).
216. Pedersen WA, Chan SL, Mattson MP. A mechanism for the neuroprotective effect of apolipoprotein E: isoform-specific modification by the lipid peroxidation product 4-hydroxynonenal. **J Neurochem** 74: 1426–1433, 2000. doi:[10.1046/j.1471-4159.2000.0741426.x](https://doi.org/10.1046/j.1471-4159.2000.0741426.x).
217. Montine TJ, Huang DY, Valentine WM, Amarnath V, Saunders A, Weisgraber KH, Graham DG, Strittmatter WJ. Crosslinking of apolipoprotein E by products of lipid peroxidation. **J Neuropathol Exp Neurol** 55: 202–210, 1996. doi:[10.1097/00005072-199602000-00009](https://doi.org/10.1097/00005072-199602000-00009).
218. Lauderback CM, Kanski J, Hackett JM, Maeda N, Kindy MS, Butterfield DA. Apolipoprotein E modulates Alzheimer's Abeta(1-42)-induced oxidative damage to synaptosomes in an allele-specific manner. **Brain Res** 924: 90–97, 2002. doi:[10.1016/s0006-8993\(01\)03228-0](https://doi.org/10.1016/s0006-8993(01)03228-0).
219. Chen HC, Chan DC. Mitochondrial dynamics—fusion, fission, movement, and mitophagy—in neurodegenerative diseases. **Hum Mol Genet** 18: R169–R176, 2009. doi:[10.1093/hmg/ddp326](https://doi.org/10.1093/hmg/ddp326).
220. Yin F, Cadenas E. Mitochondria: the cellular hub of the dynamic coordinated network. **Antioxid Redox Signal** 22: 961–964, 2015. doi:[10.1089/ars.2015.6313](https://doi.org/10.1089/ars.2015.6313).
221. Cenini G, Lloret A, Cascella R. Oxidative stress in neurodegenerative diseases: from a mitochondrial point of view. **Oxid Med Cell Longev** 2019: 2105607, 2019. doi:[10.1155/2019/2105607](https://doi.org/10.1155/2019/2105607).
222. Yu SB, Pekkurnaz G. Mechanisms orchestrating mitochondrial dynamics for energy homeostasis. **J Mol Biol** 430: 3922–3941, 2018. doi:[10.1016/j.jmb.2018.07.027](https://doi.org/10.1016/j.jmb.2018.07.027).
223. Scarpulla RC. Nucleus-encoded regulators of mitochondrial function: integration of respiratory chain expression, nutrient sensing and metabolic stress. **Biochim Biophys Acta** 1819: 1088–1097, 2012. doi:[10.1016/j.bbagr.2011.10.011](https://doi.org/10.1016/j.bbagr.2011.10.011).
224. Pagliarini DJ, Calvo SE, Chang B, Sheth SA, Vafai SB, Ong SE, Walford GA, Sugiana C, Boneh A, Chen WK, Hill DE, Vidal M, Evans JG, Thorburn DR, Carr SA, Mootha VK. A mitochondrial protein compendium elucidates complex I disease biology. **Cell** 134: 112–123, 2008. doi:[10.1016/j.cell.2008.06.016](https://doi.org/10.1016/j.cell.2008.06.016).
225. Lotz C, Lin AJ, Black CM, Zhang J, Lau E, Deng N, Wang Y, Zong NC, Choi JH, Xu T, Liem DA, Korge P, Weiss JN, Hermjakob H, Yates JR 3rd, Apweiler R, Ping P. Characterization, design, and function of the mitochondrial proteome: from organs to organisms. **J Proteome Res** 13: 433–446, 2014. doi:[10.1021/pr400539j](https://doi.org/10.1021/pr400539j).
226. Golpich M, Amini E, Mohamed Z, Azman Ali R, Mohamed Ibrahim N, Ahmadiani A. Mitochondrial dysfunction and biogenesis in neurodegenerative diseases: pathogenesis and treatment. **CNS Neurosci Ther** 23: 5–22, 2017. doi:[10.1111/cns.12655](https://doi.org/10.1111/cns.12655).
227. Lanzillotta C, Di Domenico F, Perluigi M, Butterfield DA. Targeting mitochondria in Alzheimer disease: rationale and perspectives. **CNS Drugs** 33: 957–969, 2019. doi:[10.1007/s40263-019-00658-8](https://doi.org/10.1007/s40263-019-00658-8).
228. Zorzano A, Hernández-Alvarez MI, Sebastian D, Muñoz JP. Mitofusin 2 as a driver that controls energy metabolism and insulin signaling. **Antioxid Redox Signal** 22: 1020–1031, 2015. doi:[10.1089/ars.2014.6208](https://doi.org/10.1089/ars.2014.6208).
229. Cai Q, Jeong YY. Mitophagy in Alzheimer's disease and other age-related neurodegenerative diseases. **Cells** 9: 150, 2020. doi:[10.3390/cells9010150](https://doi.org/10.3390/cells9010150).
230. Agarwal S, Muqit MM. PTEN-induced kinase 1 (PINK1) and Parkin: unlocking a mitochondrial quality control pathway linked to Parkinson's disease. **Curr Opin Neurobiol** 72: 111–119, 2022. doi:[10.1016/j.conb.2021.09.005](https://doi.org/10.1016/j.conb.2021.09.005).
231. Giorgi C, Bouhamida E, Danese A, Prevati M, Pinton P, Patergnani S. Relevance of autophagy and mitophagy dynamics and markers in neurodegenerative diseases. **Biomedicines** 9: 149, 2021. doi:[10.3390/biomedicines9020149](https://doi.org/10.3390/biomedicines9020149).
232. Hunt RJ, Bateman JM. Mitochondrial retrograde signaling in the nervous system. **FEBS Lett** 592: 663–678, 2018. doi:[10.1002/1873-3468.12890](https://doi.org/10.1002/1873-3468.12890).
233. Lanzillotta C, Di Domenico F. Stress responses in down syndrome neurodegeneration: state of the art and therapeutic molecules. **Biomolecules** 11: 266, 2021. doi:[10.3390/biom11020266](https://doi.org/10.3390/biom11020266).
234. Scheff SW, Price DA, Schmitt FA, Mufson EJ. Hippocampal synaptic loss in early Alzheimer's disease and mild cognitive impairment. **Neurobiol Aging** 27: 1372–1384, 2006. doi:[10.1016/j.neurobiolaging.2005.09.012](https://doi.org/10.1016/j.neurobiolaging.2005.09.012).

235. John A, Reddy PH. Synaptic basis of Alzheimer's disease: focus on synaptic amyloid beta, P-tau and mitochondria. **Ageing Res Rev** 65: 101208, 2021. doi:[10.1016/j.arr.2020.101208](https://doi.org/10.1016/j.arr.2020.101208).
236. Reitz C, Mayeux R. Alzheimer disease: epidemiology, diagnostic criteria, risk factors and biomarkers. **Biochem Pharmacol** 88: 640–651, 2014. doi:[10.1016/j.bcp.2013.12.024](https://doi.org/10.1016/j.bcp.2013.12.024).
237. Thibault O, Gant JC, Landfield PW. Expansion of the calcium hypothesis of brain aging and Alzheimer's disease: minding the store. **Ageing Cell** 6: 307–317, 2007. doi:[10.1111/j.1474-9726.2007.00295.x](https://doi.org/10.1111/j.1474-9726.2007.00295.x).
238. Shankar GM, Li S, Mehta TH, Garcia-Munoz A, Shepardson NE, Smith I, Brett FM, Farrell MA, Rowan MJ, Lemere CA, Regan CM, Walsh DM, Sabatini BL, Selkoe DJ. Amyloid-beta protein dimers isolated directly from Alzheimer's brains impair synaptic plasticity and memory. **Nat Med** 14: 837–842, 2008. doi:[10.1038/nm1782](https://doi.org/10.1038/nm1782).
239. Walsh DM, Klyubin I, Fadeeva JV, Cullen WK, Anwyl R, Wolfe MS, Rowan MJ, Selkoe DJ. Naturally secreted oligomers of amyloid beta protein potently inhibit hippocampal long-term potentiation in vivo. **Nature** 416: 535–539, 2002. doi:[10.1038/416535a](https://doi.org/10.1038/416535a).
240. Selkoe DJ. Alzheimer's disease is a synaptic failure. **Science** 298: 789–791, 2002. doi:[10.1126/science.1074069](https://doi.org/10.1126/science.1074069).
241. Yang T, Li S, Xu H, Walsh DM, Selkoe DJ. Large soluble oligomers of amyloid beta-protein from Alzheimer brain are far less neuroactive than the smaller oligomers to which they dissociate. **J Neurosci** 37: 152–163, 2017. doi:[10.1523/JNEUROSCI.1698-16.2016](https://doi.org/10.1523/JNEUROSCI.1698-16.2016).
242. Pham E, Crews L, Ubhi K, Hansen L, Adame A, Cartier A, Salmon D, Galasko D, Michael S, Savas JN, Yates JR, Glabe C, Masliah E. Progressive accumulation of amyloid-beta oligomers in Alzheimer's disease and in amyloid precursor protein transgenic mice is accompanied by selective alterations in synaptic scaffold proteins. **FEBS J** 277: 3051–3067, 2010. doi:[10.1111/j.1742-4658.2010.07719.x](https://doi.org/10.1111/j.1742-4658.2010.07719.x).
243. Klyubin I, Betts V, Welzel AT, Blennow K, Zetterberg H, Wallin A, Lemere CA, Cullen WK, Peng Y, Wisniewski T, Selkoe DJ, Anwyl R, Walsh DM, Rowan MJ. Amyloid beta protein dimer-containing human CSF disrupts synaptic plasticity: prevention by systemic passive immunization. **J Neurosci** 28: 4231–4237, 2008. doi:[10.1523/JNEUROSCI.5161-07.2008](https://doi.org/10.1523/JNEUROSCI.5161-07.2008).
244. Li R, Singh M. Sex differences in cognitive impairment and Alzheimer's disease. **Front Neuroendocrinol** 35: 385–403, 2014. doi:[10.1016/j.yfrne.2014.01.002](https://doi.org/10.1016/j.yfrne.2014.01.002).
245. Podcasy JL, Epperson CN. Considering sex and gender in Alzheimer disease and other dementias. **Dialogues Clin Neurosci** 18: 437–446, 2016. doi:[10.31887/DCNS.2016.18.4/cepperson](https://doi.org/10.31887/DCNS.2016.18.4/cepperson).
246. Tenkora MA, Snyder B, Cunningham RL. Sex-related differences in oxidative stress and neurodegeneration. **Steroids** 133: 21–27, 2018. doi:[10.1016/j.steroids.2017.12.010](https://doi.org/10.1016/j.steroids.2017.12.010).
247. Toro CA, Zhang L, Cao J, Cai D. Sex differences in Alzheimer's disease: understanding the molecular impact. **Brain Res** 1719: 194–207, 2019. doi:[10.1016/j.brainres.2019.05.031](https://doi.org/10.1016/j.brainres.2019.05.031).
248. Medeiros AM, Silva RH. Sex differences in Alzheimer's disease: where do we stand? **J Alzheimers Dis** 67: 35–60, 2019. doi:[10.3233/JAD-180213](https://doi.org/10.3233/JAD-180213).
249. Sumien N, Cunningham JT, Davis DL, Engelland R, Fadeyibi O, Farmer GE, Mabry S, Mensah-Kane P, Trinh OT, Vann PH, Wilson EN, Cunningham RL. Neurodegenerative disease: roles for sex, hormones, and oxidative stress. **Endocrinology** 162: bqab185, 2021. doi:[10.1210/endo/bqab185](https://doi.org/10.1210/endo/bqab185).
250. Dumitrescu L, Mayeda ER, Sharman K, Moore AM, Hohman TJ. Sex differences in the genetic architecture of Alzheimer's disease. **Curr Genet Med Rep** 7: 13–21, 2019. doi:[10.1007/s40142-019-0157-1](https://doi.org/10.1007/s40142-019-0157-1).
251. Pike CJ. Sex and the development of Alzheimer's disease. **J Neurosci Res** 95: 671–680, 2017. doi:[10.1002/jnr.23827](https://doi.org/10.1002/jnr.23827).
252. Radaghdam S, Karamad V, Nourazarian A, Shademan B, Khaki-Khatibi F, Nikanfar M. Molecular mechanisms of sex hormones in the development and progression of Alzheimer's disease. **Neurosci Lett** 764: 136221, 2021. doi:[10.1016/j.neulet.2021.136221](https://doi.org/10.1016/j.neulet.2021.136221).
253. Vegeto E, Villa A, Della Torre S, Crippa V, Rusmini P, Cristofani R, Galbiati M, Maggi A, Poletti A. The role of sex and sex hormones in neurodegenerative diseases. **Endocr Rev** 41: 273–319, 2020. doi:[10.1210/endo/bnz005](https://doi.org/10.1210/endo/bnz005).
254. Yan Y, Dominguez S, Fisher DW, Dong H. Sex differences in chronic stress responses and Alzheimer's disease. **Neurobiol Stress** 8: 120–126, 2018. doi:[10.1016/j.ynstr.2018.03.002](https://doi.org/10.1016/j.ynstr.2018.03.002).
255. Gaignard P, Liere P, Thérond P, Schumacher M, Slama A, Guennoun R. Role of sex hormones on brain mitochondrial function, with special reference to aging and neurodegenerative diseases. **Front Aging Neurosci** 9: 406, 2017. doi:[10.3389/fnagi.2017.00406](https://doi.org/10.3389/fnagi.2017.00406).
256. Vina J, Gambini J, Lopez-Grueso R, Abdelaziz KM, Jove M, Borras C. Females live longer than males: role of oxidative stress. **Curr Pharm Des** 17: 3959–3965, 2011. doi:[10.2174/138161211798764942](https://doi.org/10.2174/138161211798764942).
257. Duong P, Tenkora MAA, Trieu J, McCuiston C, Rybalchenko N, Cunningham RL. Neuroprotective and neurotoxic outcomes of androgens and estrogens in an oxidative stress environment. **Biol Sex Differ** 11: 12, 2020. doi:[10.1186/s13293-020-0283-1](https://doi.org/10.1186/s13293-020-0283-1).
258. Dubal DB. Sex difference in Alzheimer's disease: an updated, balanced and emerging perspective on differing vulnerabilities. **Handb Clin Neurol** 175: 261–273, 2020. doi:[10.1016/B978-0-444-64123-6.00018-7](https://doi.org/10.1016/B978-0-444-64123-6.00018-7).
259. Oveisgharan S, Arvanitakis Z, Yu L, Farfel J, Schneider JA, Bennett DA. Sex differences in Alzheimer's disease and common neuropathologies of aging. **Acta Neuropathol** 136: 887–900, 2018. doi:[10.1007/s00401-018-1920-1](https://doi.org/10.1007/s00401-018-1920-1).
260. Zhu D, Montagne A, Zhao Z. Alzheimer's pathogenic mechanisms and underlying sex difference. **Cell Mol Life Sci** 78: 4907–4920, 2021. doi:[10.1007/s00018-021-03830-w](https://doi.org/10.1007/s00018-021-03830-w).
261. Guo L, Zhong MB, Zhang L, Zhang B, Cai D. Sex differences in Alzheimer's disease: insights from the multiomics landscape. **Biol Psychiatry** 91: 61–71, 2022. doi:[10.1016/j.biopsych.2021.02.968](https://doi.org/10.1016/j.biopsych.2021.02.968).
262. Tsiknia AA, Edland SD, Sundermann EE, Reas ET, Brewer JB, Galasko D, Banks SJ; Alzheimer's Disease Neuroimaging Initiative. Sex differences in plasma p-tau181 associations with Alzheimer's disease biomarkers, cognitive decline, and clinical progression. **Mol Psychiatry** 27: 4314–4322, 2022. doi:[10.1038/s41380-022-01675-8](https://doi.org/10.1038/s41380-022-01675-8).
263. Hu YT, Boonstra J, McGurran H, Stormmesand J, Sluiter A, Balesar R, Verwer R, Swaab D, Bao AM. Sex differences in the neuropathological hallmarks of Alzheimer's disease: focus on cognitively intact elderly individuals. **Neuropathol Appl Neurobiol** 47: 958–966, 2021. doi:[10.1111/nan.12729](https://doi.org/10.1111/nan.12729).
264. Contador J, Pérez-Millan A, Guillén N, Sarto J, Tort-Merino A, Balasa M, Falgàs N, Castellvi M, Borrego-Écija S, Juncà-Parella J, Bosch B, Fernández-Villullas G, Ramos-Campoy O, Antonell A, Bargalló N, Sanchez-Valle R, Sala-Llanch R, Lladó A. Sex

- differences in early-onset Alzheimer's disease. **Eur J Neurol** 29: 3623–3632, 2022. doi:[10.1111/ene.15531](https://doi.org/10.1111/ene.15531).
265. Demetrius LA, Eckert A, Grimm A. Sex differences in Alzheimer's disease: metabolic reprogramming and therapeutic intervention. **Trends Endocrinol Metab** 32: 963–979, 2021. doi:[10.1016/j.tem.2021.09.004](https://doi.org/10.1016/j.tem.2021.09.004).
266. Laws KR, Irvine K, Gale TM. Sex differences in cognitive impairment in Alzheimer's disease. **World J Psychiatry** 6: 54–65, 2016. doi:[10.5498/wjp.v6.i1.54](https://doi.org/10.5498/wjp.v6.i1.54).
267. Claxton A, Baker LD, Wilkinson CW, Trittschuh EH, Chapman D, Watson GS, Cholerton B, Plymate SR, Arbuckle M, Craft S. Sex and ApoE genotype differences in treatment response to two doses of intranasal insulin in adults with mild cognitive impairment or Alzheimer's disease. **J Alzheimers Dis** 35: 789–797, 2013. doi:[10.3233/JAD-122308](https://doi.org/10.3233/JAD-122308).
268. Ferretti MT, Martinkova J, Biskup E, Benke T, Gialdini G, Nedelska Z, Rauen K, Mantua V, Religa D, Hort J, Santuccione Chadha A, Schmidt R. Sex and gender differences in Alzheimer's disease: current challenges and implications for clinical practice: position paper of the Dementia and Cognitive Disorders Panel of the European Academy of Neurology. **Eur J Neurol** 27: 928–943, 2020. doi:[10.1111/ene.14174](https://doi.org/10.1111/ene.14174).
269. Cunningham RL, Singh M, O'Bryant SE, Hall JR, Barber RC. Oxidative stress, testosterone, and cognition among Caucasian and Mexican-American men with and without Alzheimer's disease. **J Alzheimers Dis** 40: 563–573, 2014. doi:[10.3233/JAD-131994](https://doi.org/10.3233/JAD-131994).
270. Pallier PN, Ferrara M, Romagnolo F, Ferretti MT, Soreq H, Cerase A. Chromosomal and environmental contributions to sex differences in the vulnerability to neurological and neuropsychiatric disorders: implications for therapeutic interventions. **Prog Neurobiol** 219: 102353, 2022. doi:[10.1016/j.pneurobio.2022.102353](https://doi.org/10.1016/j.pneurobio.2022.102353).
271. Castellanos DB, Martin-Jimenez CA, Rojas-Rodriguez F, Barreto GE, Gonzalez J. Brain lipidomics as a rising field in neurodegenerative contexts: perspectives with machine learning approaches. **Front Neuroendocrinol** 61: 100899, 2021. doi:[10.1016/j.yfrne.2021.100899](https://doi.org/10.1016/j.yfrne.2021.100899).
272. Farooqui AA. Lipid mediators in the neural cell nucleus: their metabolism, signaling, and association with neurological disorders. **Neuroscientist** 15: 392–407, 2009. doi:[10.1177/1073858409337035](https://doi.org/10.1177/1073858409337035).
273. Wong MW, Braidy N, Poljak A, Pickford R, Thambisetty M, Sachdev PS. Dysregulation of lipids in Alzheimer's disease and their role as potential biomarkers. **Alzheimers Dement** 13: 810–827, 2017. doi:[10.1016/j.jalz.2017.01.008](https://doi.org/10.1016/j.jalz.2017.01.008).
274. Yadav RS, Tiwari NK. Lipid integration in neurodegeneration: an overview of Alzheimer's disease. **Mol Neurobiol** 50: 168–176, 2014. doi:[10.1007/s12035-014-8661-5](https://doi.org/10.1007/s12035-014-8661-5).
275. Yang K, Han X. Lipidomics: techniques, applications, and outcomes related to biomedical sciences. **Trends Biochem Sci** 41: 954–969, 2016. doi:[10.1016/j.tibs.2016.08.010](https://doi.org/10.1016/j.tibs.2016.08.010).
276. Huynh K, Lim WL, Giles C, Jayawardana KS, Salim A, Mellett NA, Smith AA, Olshansky G, Drew BG, Chatterjee P, Martins I, Laws SM, Bush AI, Rowe CC, Villemagne VL, Ames D, Masters CL, Arnold M, Nho K, Saykin AJ, Baillie R, Han X, Kaddurah-Daouk R, Martins RN, Meikle PJ. Concordant peripheral lipidome signatures in two large clinical studies of Alzheimer's disease. **Nat Commun** 11: 5698, 2020. doi:[10.1038/s41467-020-19473-7](https://doi.org/10.1038/s41467-020-19473-7).
277. Wood PL. Lipidomics of Alzheimer's disease: current status. **Alzheimers Res Ther** 4: 5, 2012. doi:[10.1186/alzrt103](https://doi.org/10.1186/alzrt103).
278. Quehenberger O, Dennis EA. The human plasma lipidome. **N Engl J Med** 365: 1812–1823, 2011. doi:[10.1056/NEJMr1104901](https://doi.org/10.1056/NEJMr1104901).
279. Zabel M, Nackenoff A, Kirsch WM, Harrison FE, Perry G, Schrag M. Markers of oxidative damage to lipids, nucleic acids and proteins and antioxidant enzymes activities in Alzheimer's disease brain: a meta-analysis in human pathological specimens. **Free Radic Biol Med** 115: 351–360, 2018. doi:[10.1016/j.freeradbiomed.2017.12.016](https://doi.org/10.1016/j.freeradbiomed.2017.12.016).
280. Penke B, Paragi G, Gera J, Berkecz R, Kovács Z, Crul T, Vigh L. The role of lipids and membranes in the pathogenesis of Alzheimer's disease: a comprehensive view. **Curr Alzheimer Res** 15: 1191–1212, 2018. doi:[10.2174/156720501566618091151716](https://doi.org/10.2174/156720501566618091151716).
281. Whitwell JL, Tosakulwong N, Weigand SD, Graff-Radford J, Ertekin-Taner N, Machulda MM, Duffy JR, Schwarz CG, Senjem ML, Jack CR, Lowe VJ, Josephs KA; Alzheimer's Disease Neuroimaging Initiative. Relationship of APOE, age at onset, amyloid and clinical phenotype in Alzheimer disease. **Neurobiol Aging** 108: 90–98, 2021. doi:[10.1016/j.neurobiolaging.2021.08.012](https://doi.org/10.1016/j.neurobiolaging.2021.08.012).
282. Chen MH, Yang FC, Liang CS. APOE allele testing and Alzheimer disease. **JAMA** 325: 2210–2211, 2021. doi:[10.1001/jama.2021.4919](https://doi.org/10.1001/jama.2021.4919).
283. Meyer MR, Tschanz JT, Norton MC, Welsh-Bohmer KA, Steffens DC, Wyse BW, Breitner JC. APOE genotype predicts when—not whether—one is predisposed to develop Alzheimer disease. **Nat Genet** 19: 321–322, 1998. doi:[10.1038/1206](https://doi.org/10.1038/1206).
284. Malkki H. Alzheimer disease: APOE*epsilon4-associated increase in AD risk linked to phospholipid dysregulation. **Nat Rev Neurol** 11: 610, 2015. doi:[10.1038/nrneurol.2015.180](https://doi.org/10.1038/nrneurol.2015.180).
285. Evans RM, Hui S, Perkins A, Lahiri DK, Poirier J, Farlow MR. Cholesterol and APOE genotype interact to influence Alzheimer disease progression. **Neurology** 62: 1869–1871, 2004. doi:[10.1212/01.wnl.0000125323.15458.3f](https://doi.org/10.1212/01.wnl.0000125323.15458.3f).
286. Proitsi P, Kim M, Whitley L, Simmons A, Sattler M, Velayudhan L, Lupton MK, Soininen H, Kloszewska I, Mecocci P, Tsolaki M, Vellas B, Lovestone S, Powell JF, Dobson RJ, Legido-Quigley C. Association of blood lipids with Alzheimer's disease: a comprehensive lipidomics analysis. **Alzheimers Dement** 13: 140–151, 2017. doi:[10.1016/j.jalz.2016.08.003](https://doi.org/10.1016/j.jalz.2016.08.003).
287. Reitz C, Tang MX, Schupf N, Manly JJ, Mayeux R, Luchsinger JA. Association of higher levels of high-density lipoprotein cholesterol in elderly individuals and lower risk of late-onset Alzheimer disease. **Arch Neurol** 67: 1491–1497, 2010. doi:[10.1001/archneurol.2010.297](https://doi.org/10.1001/archneurol.2010.297).
288. Di Paolo G, Kim TW. Linking lipids to Alzheimer's disease: cholesterol and beyond. **Nat Rev Neurosci** 12: 284–296, 2011. doi:[10.1038/nrn3012](https://doi.org/10.1038/nrn3012).
289. Kao YC, Ho PC, Tu YK, Jou IM, Tsai KJ. Lipids and Alzheimer's disease. **Int J Mol Sci** 21: 1505, 2020. doi:[10.3390/ijms21041505](https://doi.org/10.3390/ijms21041505).
290. Lim WL, Huynh K, Chatterjee P, Martins I, Jayawardana KS, Giles C, Mellett NA, Laws SM, Bush AI, Rowe CC, Villemagne VL, Ames D, Drew BG, Masters CL, Meikle PJ, Martins RN; AIBL research group. Relationships between plasma lipids species, gender, risk factors, and Alzheimer's disease. **J Alzheimers Dis** 76: 303–315, 2020. doi:[10.3233/JAD-191304](https://doi.org/10.3233/JAD-191304).
291. Lim WL, Martins IJ, Martins RN. The involvement of lipids in Alzheimer's disease. **J Genet Genomics** 41: 261–274, 2014. doi:[10.1016/j.jgg.2014.04.003](https://doi.org/10.1016/j.jgg.2014.04.003).
292. Petrovic S, Arsic A, Ristic-Medic D, Cvetkovic Z, Vucic V. Lipid peroxidation and antioxidant supplementation in neurodegenerative diseases: a review of human studies. **Antioxidants (Basel)** 9: 1128, 2020. doi:[10.3390/antiox9111128](https://doi.org/10.3390/antiox9111128).

293. Miller E, Morel A, Saso L, Saluk J. Isoprostanes and neuroprostanes as biomarkers of oxidative stress in neurodegenerative diseases. **Oxid Med Cell Longev** 2014: 572491, 2014. doi:[10.1155/2014/572491](https://doi.org/10.1155/2014/572491).
294. Praticò D. The neurobiology of isoprostanes and Alzheimer's disease. **Biochim Biophys Acta** 1801: 930–933, 2010. doi:[10.1016/j.bbalip.2010.01.009](https://doi.org/10.1016/j.bbalip.2010.01.009).
295. Peña-Bautista C, Baquero M, Vento M, Cháfer-Pericas C. Free radicals in Alzheimer's disease: lipid peroxidation biomarkers. **Clin Chim Acta** 491: 85–90, 2019. doi:[10.1016/j.cca.2019.01.021](https://doi.org/10.1016/j.cca.2019.01.021).
296. Di Domenico F, Coccia R, Butterfield DA, Perluigi M. Circulating biomarkers of protein oxidation for Alzheimer disease: expectations within limits. **Biochim Biophys Acta** 1814: 1785–1795, 2011. doi:[10.1016/j.bbapap.2011.10.001](https://doi.org/10.1016/j.bbapap.2011.10.001).
297. Gamba P, Testa G, Gargiulo S, Staurengi E, Poli G, Leonarduzzi G. Oxidized cholesterol as the driving force behind the development of Alzheimer's disease. **Front Aging Neurosci** 7: 119, 2015., doi:[10.3389/fnagi.2015.00119](https://doi.org/10.3389/fnagi.2015.00119).
298. Butterfield DA, Gu L, Di Domenico F, Robinson RA. Mass spectrometry and redox proteomics: applications in disease. **Mass Spectrom Rev** 33: 277–301, 2014. doi:[10.1002/mas.21374](https://doi.org/10.1002/mas.21374).
299. Butterfield DA, Sultana R. Redox proteomics: understanding oxidative stress in the progression of age-related neurodegenerative disorders. **Expert Rev Proteomics** 5: 157–160, 2008. doi:[10.1586/14789450.5.2.157](https://doi.org/10.1586/14789450.5.2.157).
300. Butterfield DA, Dalle-Donne I. Redox proteomics. **Antioxid Redox Signal** 17: 1487–1489, 2012. doi:[10.1089/ars.2012.4742](https://doi.org/10.1089/ars.2012.4742).
301. Butterfield DA, Perluigi M. Redox proteomics: a key tool for new insights into protein modification with relevance to disease. **Antioxid Redox Signal** 26: 277–279, 2017. doi:[10.1089/ars.2016.6919](https://doi.org/10.1089/ars.2016.6919).
302. Di Domenico F, Perluigi M, Butterfield DA. Redox proteomics in human biofluids: sample preparation, separation and immunochemical tagging for analysis of protein oxidation. **Methods Mol Biol** 1303: 391–403, 2016. doi:[10.1007/978-1-4939-2627-5_23](https://doi.org/10.1007/978-1-4939-2627-5_23).
303. Perluigi M, Di Domenico F, Butterfield DA. Unraveling the complexity of neurodegeneration in brains of subjects with Down syndrome: insights from proteomics. **Proteomics Clin Appl** 8: 73–85, 2014. doi:[10.1002/prca.201300066](https://doi.org/10.1002/prca.201300066).
304. Butterfield DA, Poon HF, St Clair D, Keller JN, Pierce WM, Klein JB, Markesbery WR. Redox proteomics identification of oxidatively modified hippocampal proteins in mild cognitive impairment: insights into the development of Alzheimer's disease. **Neurobiol Dis** 22: 223–232, 2006. doi:[10.1016/j.nbd.2005.11.002](https://doi.org/10.1016/j.nbd.2005.11.002).
305. Sultana R, Boyd-Kimball D, Poon HF, Cai J, Pierce WM, Klein JB, Merchant M, Markesbery WR, Butterfield DA. Redox proteomics identification of oxidized proteins in Alzheimer's disease hippocampus and cerebellum: an approach to understand pathological and biochemical alterations in AD. **Neurobiol Aging** 27: 1564–1576, 2006. doi:[10.1016/j.neurobiolaging.2005.09.021](https://doi.org/10.1016/j.neurobiolaging.2005.09.021).
306. Perluigi M, Sultana R, Cenini G, Di Domenico F, Memo M, Pierce WM, Coccia R, Butterfield DA. Redox proteomics identification of 4-hydroxynonenal-modified brain proteins in Alzheimer's disease: role of lipid peroxidation in Alzheimer's disease pathogenesis. **Proteomics Clin Appl** 3: 682–693, 2009. doi:[10.1002/prca.200800161](https://doi.org/10.1002/prca.200800161).
307. Sultana R, Perluigi M, Newman SF, Pierce WM, Cini C, Coccia R, Butterfield DA. Redox proteomic analysis of carbonylated brain proteins in mild cognitive impairment and early Alzheimer's disease. **Antioxid Redox Signal** 12: 327–336, 2010. doi:[10.1089/ars.2009.2810](https://doi.org/10.1089/ars.2009.2810).
308. Castegna A, Thongboonkerd V, Klein JB, Lynn B, Markesbery WR, Butterfield DA. Proteomic identification of nitrated proteins in Alzheimer's disease brain. **J Neurochem** 85: 1394–1401, 2003. doi:[10.1046/j.1471-4159.2003.01786.x](https://doi.org/10.1046/j.1471-4159.2003.01786.x).
309. Korolainen MA, Goldsteins G, Nyman TA, Alafuzoff I, Koistinaho J, Pirttilä T. Oxidative modification of proteins in the frontal cortex of Alzheimer's disease brain. **Neurobiol Aging** 27: 42–53, 2006. doi:[10.1016/j.neurobiolaging.2004.11.010](https://doi.org/10.1016/j.neurobiolaging.2004.11.010).
310. Sultana R, Poon HF, Cai J, Pierce WM, Merchant M, Klein JB, Markesbery WR, Butterfield DA. Identification of nitrated proteins in Alzheimer's disease brain using a redox proteomics approach. **Neurobiol Dis** 22: 76–87, 2006. doi:[10.1016/j.nbd.2005.10.004](https://doi.org/10.1016/j.nbd.2005.10.004).
311. Di Domenico F, Pupo G, Giraldo E, Badia MC, Monllor P, Lloret A, Schininà ME, Giorgi A, Cini C, Tramutola A, Butterfield DA, Viña J, Perluigi M. Oxidative signature of cerebrospinal fluid from mild cognitive impairment and Alzheimer disease patients. **Free Radic Biol Med** 91: 1–9, 2016. doi:[10.1016/j.freeradbiomed.2015.12.004](https://doi.org/10.1016/j.freeradbiomed.2015.12.004).
312. Korolainen MA, Nyman TA, Nyyssönen P, Hartikainen ES, Pirttilä T. Multiplexed proteomic analysis of oxidation and concentrations of cerebrospinal fluid proteins in Alzheimer disease. **Clin Chem** 53: 657–665, 2007. doi:[10.1373/clinchem.2006.078014](https://doi.org/10.1373/clinchem.2006.078014).
313. Cacciolo A, Di Domenico F, Coccia R, Fiorini A, Cai J, Pierce WM, Mecocci P, Butterfield DA, Perluigi M. Decreased expression and increased oxidation of plasma haptoglobin in Alzheimer disease: insights from redox proteomics. **Free Radic Biol Med** 53: 1868–1876, 2012. doi:[10.1016/j.freeradbiomed.2012.08.596](https://doi.org/10.1016/j.freeradbiomed.2012.08.596).
314. Choi J, Malakowsky CA, Talent JM, Conrad CC, Gracy RW. Identification of oxidized plasma proteins in Alzheimer's disease. **Biochem Biophys Res Commun** 293: 1566–1570, 2002. doi:[10.1016/S0006-291X\(02\)00420-5](https://doi.org/10.1016/S0006-291X(02)00420-5).
315. Yu HL, Chertkow HM, Bergman H, Schipper HM. Aberrant profiles of native and oxidized glycoproteins in Alzheimer plasma. **Proteomics** 3: 2240–2248, 2003. doi:[10.1002/pmic.200300475](https://doi.org/10.1002/pmic.200300475).
316. Sultana R, Reed T, Perluigi M, Coccia R, Pierce WM, Butterfield DA. Proteomic identification of nitrated brain proteins in amnesic mild cognitive impairment: a regional study. **J Cell Mol Med** 11: 839–851, 2007. doi:[10.1111/j.1582-4934.2007.00065.x](https://doi.org/10.1111/j.1582-4934.2007.00065.x).
317. Aluise CD, Robinson RA, Cai J, Pierce WM, Markesbery WR, Butterfield DA. Redox proteomics analysis of brains from subjects with amnesic mild cognitive impairment compared to brains from subjects with preclinical Alzheimer's disease: insights into memory loss in MCI. **J Alzheimers Dis** 23: 257–269, 2011. doi:[10.3233/JAD-2010-101083](https://doi.org/10.3233/JAD-2010-101083).
318. Korolainen MA, Pirttilä T. Cerebrospinal fluid, serum and plasma protein oxidation in Alzheimer's disease. **Acta Neurol Scand** 119: 32–38, 2009. doi:[10.1111/j.1600-0404.2008.01057.x](https://doi.org/10.1111/j.1600-0404.2008.01057.x).
319. Ahmed N, Ahmed U, Thornalley PJ, Hager K, Fleischer G, Münch G. Protein glycation, oxidation and nitration adduct residues and free adducts of cerebrospinal fluid in Alzheimer's disease and link to cognitive impairment. **J Neurochem** 92: 255–263, 2005. doi:[10.1111/j.1471-4159.2004.02864.x](https://doi.org/10.1111/j.1471-4159.2004.02864.x).
320. Conrad CC, Marshall PL, Talent JM, Malakowsky CA, Choi J, Gracy RW. Oxidized proteins in Alzheimer's plasma. **Biochem Biophys Res Commun** 293: 1566–1570, 2002. doi:[10.1006/bbrc.2000.3356](https://doi.org/10.1006/bbrc.2000.3356).
321. Shen L, Chen C, Yang A, Chen Y, Liu Q, Ni J. Redox proteomics identification of specifically carbonylated proteins in the

- hippocampi of triple transgenic Alzheimer's disease mice at its earliest pathological stage. **J Proteomics** 123: 101–113, 2015. doi:[10.1016/j.jprot.2015.04.005](https://doi.org/10.1016/j.jprot.2015.04.005).
322. Sultana R, Mecocci P, Mangialasche F, Cecchetti R, Baglioni M, Butterfield DA. Increased protein and lipid oxidative damage in mitochondria isolated from lymphocytes from patients with Alzheimer's disease: insights into the role of oxidative stress in Alzheimer's disease and initial investigations into a potential biomarker for this dementing disorder. **J Alzheimers Dis** 24: 77–84, 2011. doi:[10.3233/JAD-2011-101425](https://doi.org/10.3233/JAD-2011-101425).
323. Sultana R, Baglioni M, Cecchetti R, Cai J, Klein JB, Bastiani P, Ruggiero C, Mecocci P, Butterfield DA. Lymphocyte mitochondria: toward identification of peripheral biomarkers in the progression of Alzheimer disease. **Free Radic Biol Med** 65: 595–606, 2013. doi:[10.1016/j.freeradbiomed.2013.08.001](https://doi.org/10.1016/j.freeradbiomed.2013.08.001).
324. Hipp MS, Kasturi P, Hartl FU. The proteostasis network and its decline in ageing. **Nat Rev Mol Cell Biol** 20: 421–435, 2019. doi:[10.1038/s41580-019-0101-y](https://doi.org/10.1038/s41580-019-0101-y).
325. Hetz C, Papa FR. The unfolded protein response and cell fate control. **Mol Cell** 69: 169–181, 2018. doi:[10.1016/j.molcel.2017.06.017](https://doi.org/10.1016/j.molcel.2017.06.017).
326. Walter P, Ron D. The unfolded protein response: from stress pathway to homeostatic regulation. **Science** 334: 1081–1086, 2011. doi:[10.1126/science.1209038](https://doi.org/10.1126/science.1209038).
327. Nixon RA. The role of autophagy in neurodegenerative disease. **Nat Med** 19: 983–997, 2013. doi:[10.1038/nm.3232](https://doi.org/10.1038/nm.3232).
328. Perluigi M, Di Domenico F, Butterfield DA. mTOR signaling in aging and neurodegeneration: at the crossroad between metabolism dysfunction and impairment of autophagy. **Neurobiol Dis** 84: 39–49, 2015. doi:[10.1016/j.nbd.2015.03.014](https://doi.org/10.1016/j.nbd.2015.03.014).
329. Keller JN, Dimayuga E, Chen Q, Thorpe J, Gee J, Ding Q. Autophagy, proteasomes, lipofuscin, and oxidative stress in the aging brain. **Int J Biochem Cell Biol** 36: 2376–2391, 2004. doi:[10.1016/j.biocel.2004.05.003](https://doi.org/10.1016/j.biocel.2004.05.003).
330. Hoozemans JJ, Veerhuis R, Van Haastert ES, Rozemuller JM, Baas F, Eikelenboom P, Scheper W. The unfolded protein response is activated in Alzheimer's disease. **Acta Neuropathol** 110: 165–172, 2005. doi:[10.1007/s00401-005-1038-0](https://doi.org/10.1007/s00401-005-1038-0).
331. Tramutola A, Triani F, Di Domenico F, Barone E, Cai J, Klein JB, Perluigi M, Butterfield DA. Poly-ubiquitin profile in Alzheimer disease brain. **Neurobiol Dis** 118: 129–141, 2018. doi:[10.1016/j.nbd.2018.07.006](https://doi.org/10.1016/j.nbd.2018.07.006).
332. Park H, Kang JH, Lee S. Autophagy in neurodegenerative diseases: a hunter for aggregates. **Int J Mol Sci** 21: 3369, 2020. doi:[10.3390/ijms21093369](https://doi.org/10.3390/ijms21093369).
333. Höhn A, Tramutola A, Cascella R. Proteostasis failure in neurodegenerative diseases: focus on oxidative stress. **Oxid Med Cell Longev** 2020: 5497046, 2020. doi:[10.1155/2020/5497046](https://doi.org/10.1155/2020/5497046).
334. Pajares M, Jiménez-Moreno N, Dias IH, Debele B, Vucetic M, Fladmark KE, Basaga H, Ribaric S, Milisav I, Cuadrado A. Redox control of protein degradation. **Redox Biol** 6: 409–420, 2015. doi:[10.1016/j.redox.2015.07.003](https://doi.org/10.1016/j.redox.2015.07.003).
335. Bell MC, Meier SE, Ingram AL, Abisambra JF. PERK-opathies: an endoplasmic reticulum stress mechanism underlying neurodegeneration. **Curr Alzheimer Res** 13: 150–163, 2016. doi:[10.2174/1567205013666151218145431](https://doi.org/10.2174/1567205013666151218145431).
336. Scheper W, Nijholt DA, Hoozemans JJ. The unfolded protein response and proteostasis in Alzheimer disease: preferential activation of autophagy by endoplasmic reticulum stress. **Autophagy** 7: 910–911, 2011. doi:[10.4161/auto.7.8.15761](https://doi.org/10.4161/auto.7.8.15761).
337. Hoozemans JJ, van Haastert ES, Nijholt DA, Rozemuller AJ, Eikelenboom P, Scheper W. The unfolded protein response is activated in pretangle neurons in Alzheimer's disease hippocampus. **Am J Pathol** 174: 1241–1251, 2009. doi:[10.2353/ajpath.2009.080814](https://doi.org/10.2353/ajpath.2009.080814).
338. Cuanalo-Contreras K, Mukherjee A, Soto C. Role of protein misfolding and proteostasis deficiency in protein misfolding diseases and aging. **Int J Cell Biol** 2013: 638083, 2013. doi:[10.1155/2013/638083](https://doi.org/10.1155/2013/638083).
339. Roussel BD, Kruppa AJ, Miranda E, Crowther DC, Lomas DA, Marciniak SJ. Endoplasmic reticulum dysfunction in neurological disease. **Lancet Neurol** 12: 105–118, 2013. doi:[10.1016/S1474-4422\(12\)70238-7](https://doi.org/10.1016/S1474-4422(12)70238-7).
340. Ron D, Hubbard SR. How IRE1 reacts to ER stress. **Cell** 132: 24–26, 2008. doi:[10.1016/j.cell.2007.12.017](https://doi.org/10.1016/j.cell.2007.12.017).
341. Suh JH, Shenvi SV, Dixon BM, Liu H, Jaiswal AK, Liu RM, Hagen TM. Decline in transcriptional activity of Nrf2 causes age-related loss of glutathione synthesis, which is reversible with lipoic acid. **Proc Natl Acad Sci USA** 101: 3381–3386, 2004. doi:[10.1073/pnas.0400282101](https://doi.org/10.1073/pnas.0400282101).
342. Di Domenico F, Pupo G, Tramutola A, Giorgi A, Schininà ME, Coccia R, Head E, Butterfield DA, Perluigi M. Redox proteomics analysis of HNE-modified proteins in Down syndrome brain: clues for understanding the development of Alzheimer disease. **Free Radic Biol Med** 71: 270–280, 2014. doi:[10.1016/j.freeradbiomed.2014.03.027](https://doi.org/10.1016/j.freeradbiomed.2014.03.027).
343. Mulder MP, Witting KF, Ovaa H. Cracking the ubiquitin code: the ubiquitin toolbox. **Curr Issues Mol Biol** 37: 1–20, 2020. doi:[10.21775/cimb.037.001](https://doi.org/10.21775/cimb.037.001).
344. Raynes R, Pomatto LC, Davies KJ. Degradation of oxidized proteins by the proteasome: distinguishing between the 20S, 26S, and immunoproteasome proteolytic pathways. **Mol Aspects Med** 50: 41–55, 2016. doi:[10.1016/j.mam.2016.05.001](https://doi.org/10.1016/j.mam.2016.05.001).
345. Jung T, Grune T. The proteasome and the degradation of oxidized proteins: Part I—structure of proteasomes. **Redox Biol** 1: 178–182, 2013. doi:[10.1016/j.redox.2013.01.004](https://doi.org/10.1016/j.redox.2013.01.004).
346. Jung T, Höhn A, Grune T. The proteasome and the degradation of oxidized proteins: Part II—protein oxidation and proteasomal degradation. **Redox Biol** 2: 99–104, 2014. doi:[10.1016/j.redox.2013.12.008](https://doi.org/10.1016/j.redox.2013.12.008).
347. Höhn TJ, Grune T. The proteasome and the degradation of oxidized proteins: Part III—Redox regulation of the proteasomal system. **Redox Biol** 2: 388–394, 2014. doi:[10.1016/j.redox.2013.12.029](https://doi.org/10.1016/j.redox.2013.12.029).
348. Tramutola A, Di Domenico F, Barone E, Perluigi M, Butterfield DA. It is all about (U)biqutin: role of altered ubiquitin-proteasome system and UCHL1 in Alzheimer disease. **Oxid Med Cell Longev** 2016: 2756068, 2016. doi:[10.1155/2016/2756068](https://doi.org/10.1155/2016/2756068).
349. Weng FL, He L. Disrupted ubiquitin proteasome system underlying tau accumulation in Alzheimer's disease. **Neurobiol Aging** 99: 79–85, 2021. doi:[10.1016/j.neurobiolaging.2020.11.015](https://doi.org/10.1016/j.neurobiolaging.2020.11.015).
350. Cao J, Zhong MB, Toro CA, Zhang L, Cai D. Endo-lysosomal pathway and ubiquitin-proteasome system dysfunction in Alzheimer's disease pathogenesis. **Neurosci Lett** 703: 68–78, 2019. doi:[10.1016/j.neulet.2019.03.016](https://doi.org/10.1016/j.neulet.2019.03.016).
351. Tai HC, Serrano-Pozo A, Hashimoto T, Frosch MP, Spires-Jones TL, Hyman BT. The synaptic accumulation of hyperphosphorylated tau oligomers in Alzheimer disease is associated with dysfunction of the ubiquitin-proteasome system. **Am J Pathol** 181: 1426–1435, 2012. doi:[10.1016/j.ajpath.2012.06.033](https://doi.org/10.1016/j.ajpath.2012.06.033).

352. Lopez Salon M, Pasquini L, Besio Moreno M, Pasquini JM, Soto E. Relationship between beta-amyloid degradation and the 26S proteasome in neural cells. **Exp Neurol** 180: 131–143, 2003. doi:[10.1016/S0014-4886\(02\)00060-2](https://doi.org/10.1016/S0014-4886(02)00060-2).
353. Almeida CG, Takahashi RH, Gouras GK. Beta-amyloid accumulation impairs multivesicular body sorting by inhibiting the ubiquitin-proteasome system. **J Neurosci** 26: 4277–4288, 2006. doi:[10.1523/JNEUROSCI.5078-05.2006](https://doi.org/10.1523/JNEUROSCI.5078-05.2006).
354. Cecarini V, Ding Q, Keller JN. Oxidative inactivation of the proteasome in Alzheimer's disease. **Free Radic Res** 41: 673–680, 2007. doi:[10.1080/10715760701286159](https://doi.org/10.1080/10715760701286159).
355. Gkekas I, Gioran A, Boziki MK, Grigoriadis N, Chondrogianni N, Petrakis S. Oxidative stress and neurodegeneration: interconnected processes in polyQ diseases. **Antioxidants** 10: 1450, 2021. doi:[10.3390/antiox10091450](https://doi.org/10.3390/antiox10091450).
356. Lefaki M, Papaevgeniou N, Chondrogianni N. Redox regulation of proteasome function. **Redox Biol** 13: 452–458, 2017. doi:[10.1016/j.redox.2017.07.005](https://doi.org/10.1016/j.redox.2017.07.005).
357. Di Domenico F, Coccia R, Cocciolo A, Murphy MP, Cenini G, Head E, Butterfield DA, Giorgi A, Schinina ME, Mancuso C, Cini C, Perluigi M. Impairment of proteostasis network in Down syndrome prior to the development of Alzheimer's disease neuropathology: redox proteomics analysis of human brain. **Biochim Biophys Acta** 1832: 1249–1259, 2013. doi:[10.1016/j.bbadis.2013.04.013](https://doi.org/10.1016/j.bbadis.2013.04.013).
358. Di Domenico F, Zuliani I, Tramutola A. Shining a light on defective autophagy by proteomics approaches: implications for neurodegenerative illnesses. **Expert Rev Proteomics** 16: 951–964, 2019. doi:[10.1080/14789450.2019.1691919](https://doi.org/10.1080/14789450.2019.1691919).
359. Li Q, Liu Y, Sun M. Autophagy and Alzheimer's disease. **Cell Mol Neurobiol** 37: 377–388, 2017. doi:[10.1007/s10571-016-0386-8](https://doi.org/10.1007/s10571-016-0386-8).
360. Di Domenico F, Tramutola A, Foppoli C, Head E, Perluigi M, Butterfield DA. mTOR in Down syndrome: role in A beta and tau neuropathology and transition to Alzheimer disease-like dementia. **Free Radic Biol Med** 114: 94–101, 2018. doi:[10.1016/j.freeradbiomed.2017.08.009](https://doi.org/10.1016/j.freeradbiomed.2017.08.009).
361. Perluigi M, Di Domenico F, Barone E, Butterfield DA. mTOR in Alzheimer disease and its earlier stages: links to oxidative damage in the progression of this dementing disorder. **Free Radic Biol Med** 169: 382–396, 2021. doi:[10.1016/j.freeradbiomed.2021.04.025](https://doi.org/10.1016/j.freeradbiomed.2021.04.025).
362. Tramutola A, Lanzillotta C, Di Domenico F. Targeting mTOR to reduce Alzheimer-related cognitive decline: from current hits to future therapies. **Expert Rev Neurother** 17: 33–45, 2017. doi:[10.1080/14737175.2017.1244482](https://doi.org/10.1080/14737175.2017.1244482).
363. Di Domenico F, Tramutola A, Perluigi M. Cathepsin D as a therapeutic target in Alzheimer's disease. **Expert Opin Ther Targets** 20: 1393–1395, 2016. doi:[10.1080/14728222.2016.1252334](https://doi.org/10.1080/14728222.2016.1252334).
364. Nixon RA. Autophagy, amyloidogenesis and Alzheimer disease. **J Cell Sci** 120: 4081–4091, 2007. doi:[10.1242/jcs.019265](https://doi.org/10.1242/jcs.019265).
365. Nixon RA, Yang DS. Autophagy failure in Alzheimer's disease—locating the primary defect. **Neurobiol Dis** 43: 38–45, 2011. doi:[10.1016/j.nbd.2011.01.021](https://doi.org/10.1016/j.nbd.2011.01.021).
366. Nixon RA, Wegiel J, Kumar A, Yu WH, Peterhoff C, Cataldo A, Cuervo AM. Extensive involvement of autophagy in Alzheimer disease: an immuno-electron microscopy study. **J Neuropathol Exp Neurol** 64: 113–122, 2005. doi:[10.1093/jnen/64.2.113](https://doi.org/10.1093/jnen/64.2.113).
367. Heras-Sandoval D, Pérez-Rojas JM, Hernández-Damián J, Pedraza-Chaverri J. The role of PI3K/AKT/mTOR pathway in the modulation of autophagy and the clearance of protein aggregates in neurodegeneration. **Cell Signal** 26: 2694–2701, 2014. doi:[10.1016/j.cellsig.2014.08.019](https://doi.org/10.1016/j.cellsig.2014.08.019).
368. Perluigi M, Pupo G, Tramutola A, Cini C, Coccia R, Barone E, Head E, Butterfield DA, Di Domenico F. Neuropathological role of PI3K/Akt/mTOR axis in Down syndrome brain. **Biochim Biophys Acta** 1842: 1144–1153, 2014. doi:[10.1016/j.bbadis.2014.04.007](https://doi.org/10.1016/j.bbadis.2014.04.007).
369. Tramutola A, Triplett JC, Di Domenico F, Niedowicz DM, Murphy MP, Coccia R, Perluigi M, Butterfield DA. Alteration of mTOR signaling occurs early in the progression of Alzheimer disease (AD): analysis of brain from subjects with pre-clinical AD, amnesic mild cognitive impairment and late-stage AD. **J Neurochem** 133: 739–749, 2015. doi:[10.1111/jnc.13037](https://doi.org/10.1111/jnc.13037).
370. Spilman P, Podlutskaya N, Hart MJ, Debnath J, Gorostiza O, Bredesen D, Richardson A, Strong R, Galvan V. Inhibition of mTOR by rapamycin abolishes cognitive deficits and reduces amyloid-beta levels in a mouse model of Alzheimer's disease. **PLoS One** 5: e9979, 2010. doi:[10.1371/journal.pone.0009979](https://doi.org/10.1371/journal.pone.0009979).
371. Majumder S, Richardson A, Strong R, Oddo S. Inducing autophagy by rapamycin before, but not after, the formation of plaques and tangles ameliorates cognitive deficits. **PLoS One** 6: e25416, 2011. doi:[10.1371/journal.pone.0025416](https://doi.org/10.1371/journal.pone.0025416).
372. Tramutola A, Lanzillotta C, Barone E, Arena A, Zuliani I, Mosca L, Blarmino C, Butterfield DA, Perluigi M, Di Domenico F. Intranasal rapamycin ameliorates Alzheimer-like cognitive decline in a mouse model of Down syndrome. **Transl Neurodegener** 7: 28, 2018. doi:[10.1186/s40035-018-0133-9](https://doi.org/10.1186/s40035-018-0133-9).
373. Di Domenico F, Tramutola A, Barone E, Lanzillotta C, Defever O, Arena A, Zuliani I, Foppoli C, Iavarone F, Vincenzoni F, Castagnola M, Butterfield DA, Perluigi M. Restoration of aberrant mTOR signaling by intranasal rapamycin reduces oxidative damage: Focus on HNE-modified proteins in a mouse model of down syndrome. **Redox Biol** 23: 101162, 2019. doi:[10.1016/j.redox.2019.101162](https://doi.org/10.1016/j.redox.2019.101162).
374. Bordini M, Berg MJ, Mohan PS, Peterhoff CM, Alldred MJ, Che S, Ginsberg SD, Nixon RA. Autophagy flux in CA1 neurons of Alzheimer hippocampus: increased induction overburdens failing lysosomes to propel neuritic dystrophy. **Autophagy** 12: 2467–2483, 2016. doi:[10.1080/15548627.2016.1239003](https://doi.org/10.1080/15548627.2016.1239003).
375. Lee JH, Yang DS, Goulbourne CN, Im E, Stavrides P, Pensalfini A, Chan H, Bouchet-Marquis C, Bleiwas C, Berg MJ, Huo C, Peddy J, Pawlik M, Levy E, Rao M, Staufenbiel M, Nixon RA. Faulty autolysosome acidification in Alzheimer's disease mouse models induces autophagic build-up of Aβ in neurons, yielding senile plaques. **Nat Neurosci** 25: 688–701, 2022. doi:[10.1038/s41593-022-01084-8](https://doi.org/10.1038/s41593-022-01084-8).
376. Di Domenico F, Head E, Butterfield DA, Perluigi M. Oxidative stress and proteostasis network: culprit and casualty of Alzheimer's-like neurodegeneration. **Adv Geriatrics** 2014: 527518, 2014. doi:[10.1155/2014/527518](https://doi.org/10.1155/2014/527518).
377. Filomeni G, De Zio D, Cecconi F. Oxidative stress and autophagy: the clash between damage and metabolic needs. **Cell Death Differ** 22: 377–388, 2015. doi:[10.1038/cdd.2014.150](https://doi.org/10.1038/cdd.2014.150).
378. Chang YT, Chang WN, Tsai NW, Huang CC, Kung CT, Su YJ, Lin WC, Cheng BC, Su CM, Chiang YF, Lu CH. The roles of biomarkers of oxidative stress and antioxidant in Alzheimer's disease: a systematic review. **Biomed Res Int** 2014: 182303, 2014. doi:[10.1155/2014/182303](https://doi.org/10.1155/2014/182303).
379. D'Alessandro A, Zolla L. The SODyssey: superoxide dismutases from biochemistry, through proteomics, to oxidative stress, aging

- and nutraceuticals. **Expert Rev Proteomics** 8: 405–421, 2011. doi:[10.1586/epr.11.13](https://doi.org/10.1586/epr.11.13).
380. Aksenov MY, Markesbery WR. Changes in thiol content and expression of glutathione redox system genes in the hippocampus and cerebellum in Alzheimer's disease. **Neurosci Lett** 302: 141–145, 2001. doi:[10.1016/S0304-3940\(01\)01636-6](https://doi.org/10.1016/S0304-3940(01)01636-6).
381. Butterfield DA, Perluigi M, Sultana R. Oxidative stress in Alzheimer's disease brain: new insights from redox proteomics. **Eur J Pharmacol** 545: 39–50, 2006. doi:[10.1016/j.ejphar.2006.06.026](https://doi.org/10.1016/j.ejphar.2006.06.026).
382. Dodson M, Castro-Portuguez R, Zhang DD. NRF2 plays a critical role in mitigating lipid peroxidation and ferroptosis. **Redox Biol** 23: 101107, 2019. doi:[10.1016/j.redox.2019.101107](https://doi.org/10.1016/j.redox.2019.101107).
383. Lim JL, Wilhelmus MM, de Vries HE, Drukarch B, Hoozemans JJ, van Horssen J. Antioxidative defense mechanisms controlled by Nrf2: state-of-the-art and clinical perspectives in neurodegenerative diseases. **Arch Toxicol** 88: 1773–1786, 2014. doi:[10.1007/s00204-014-1338-z](https://doi.org/10.1007/s00204-014-1338-z).
384. Villavicencio-Tejo F, Olesen MA, Aránguiz A, Quintanilla RA. Activation of the Nrf2 pathway prevents mitochondrial dysfunction induced by caspase-3 cleaved tau: implications for Alzheimer's disease. **Antioxidants (Basel)** 11: 515, 2022. doi:[10.3390/antiox11030515](https://doi.org/10.3390/antiox11030515).
385. Barone E, Di Domenico F, Sultana R, Coccia R, Mancuso C, Perluigi M, Butterfield DA. Heme oxygenase-1 posttranslational modifications in the brain of subjects with Alzheimer disease and mild cognitive impairment. **Free Radic Biol Med** 52: 2292–2301, 2012. doi:[10.1016/j.freeradbiomed.2012.03.020](https://doi.org/10.1016/j.freeradbiomed.2012.03.020).
386. Di Domenico F, Barone E, Mancuso C, Perluigi M, Cocciolo A, Mecocci P, Butterfield DA, Coccia R. HO-1/BVR-a system analysis in plasma from probable Alzheimer's disease and mild cognitive impairment subjects: a potential biochemical marker for the prediction of the disease. **J Alzheimers Dis** 32: 277–289, 2012. doi:[10.3233/JAD-2012-121045](https://doi.org/10.3233/JAD-2012-121045).
387. Howarth C, Gleeson P, Attwell D. Updated energy budgets for neural computation in the neocortex and cerebellum. **J Cereb Blood Flow Metab** 32: 1222–1232, 2012. doi:[10.1038/jcbfm.2012.35](https://doi.org/10.1038/jcbfm.2012.35).
388. Bedse G, Di Domenico F, Serviddio G, Cassano T. Aberrant insulin signaling in Alzheimer's disease: current knowledge. **Front Neurosci** 9: 204, 2015. doi:[10.3389/fnins.2015.00204](https://doi.org/10.3389/fnins.2015.00204).
389. Tang BL. Glucose, glycolysis, and neurodegenerative diseases. **J Cell Physiol** 235: 7653–7662, 2020. doi:[10.1002/jcp.29682](https://doi.org/10.1002/jcp.29682).
390. Kellar D, Craft S. Brain insulin resistance in Alzheimer's disease and related disorders: mechanisms and therapeutic approaches. **Lancet Neurol** 19: 758–766, 2020. doi:[10.1016/S1474-4422\(20\)30231-3](https://doi.org/10.1016/S1474-4422(20)30231-3).
391. Arnold SE, Arvanitakis Z, Macauley-Rambach SL, Koenig AM, Wang HY, Ahima RS, Craft S, Gandy S, Buettner C, Stoeckel LE, Holtzman DM, Nathan DM. Brain insulin resistance in type 2 diabetes and Alzheimer disease: concepts and conundrums. **Nat Rev Neurol** 14: 168–181, 2018. doi:[10.1038/nrneurol.2017.185](https://doi.org/10.1038/nrneurol.2017.185).
392. De Felice FG, Lourenco MV, Ferreira ST. How does brain insulin resistance develop in Alzheimer's disease? **Alzheimers Dement** 10: S26–S32, 2014. doi:[10.1016/j.jalz.2013.12.004](https://doi.org/10.1016/j.jalz.2013.12.004).
393. Yin F, Sancheti H, Patil I, Cadenas E. Energy metabolism and inflammation in brain aging and Alzheimer's disease. **Free Radic Biol Med** 100: 108–122, 2016. doi:[10.1016/j.freeradbiomed.2016.04.200](https://doi.org/10.1016/j.freeradbiomed.2016.04.200).
394. Perluigi M, Barone E, Di Domenico F, Butterfield DA. Aberrant protein phosphorylation in Alzheimer disease brain disturbs pro-survival and cell death pathways. **Biochim Biophys Acta** 1862: 1871–1882, 2016. doi:[10.1016/j.bbadis.2016.07.005](https://doi.org/10.1016/j.bbadis.2016.07.005).
395. Pinho TS, Verde DM, Correia SC, Cardoso SM, Moreira PI. O-GlcNAcylation and neuronal energy status: Implications for Alzheimer's disease. **Ageing Res Rev** 46: 32–41, 2018. doi:[10.1016/j.arr.2018.05.003](https://doi.org/10.1016/j.arr.2018.05.003).
396. Tramutola A, Sharma N, Barone E, Lanzillotta C, Castellani A, Iavarone F, Vincenzoni F, Castagnola M, Butterfield DA, Gaetani S, Cassano T, Perluigi M, Di Domenico F. Proteomic identification of altered protein O-GlcNAcylation in a triple transgenic mouse model of Alzheimer's disease. **Biochim Biophys Acta Mol Basis Dis** 1864: 3309–3321, 2018. doi:[10.1016/j.bbadis.2018.07.017](https://doi.org/10.1016/j.bbadis.2018.07.017).
397. Di Domenico F, Pupo G, Giraldo E, Lloret A, Badia MC, Schinina ME, Giorgi A, Butterfield DA, Vina J, Perluigi M. Autoantibodies profile in matching CSF and serum from AD and aMCI patients: potential pathogenic role and link to oxidative damage. **Curr Alzheimer Res** 13: 112–122, 2016. doi:[10.2174/1567205013666151218131424](https://doi.org/10.2174/1567205013666151218131424).
398. Reed TT, Sultana R, Butterfield DA. Redox proteomics of oxidatively modified brain proteins in mild cognitive impairment. In: *Neuroproteomics*, edited by Alzate O. Boca Raton, FL: CRC Press, 2010.
399. Bubber P, Haroutunian V, Fisch G, Blass JP, Gibson GE. Mitochondrial abnormalities in Alzheimer brain: mechanistic implications. **Ann Neurol** 57: 695–703, 2005. doi:[10.1002/ana.20474](https://doi.org/10.1002/ana.20474).
400. Biessels GJ, Strachan MW, Visseren FL, Kappelle LJ, Whitmer RA. Dementia and cognitive decline in type 2 diabetes and prediabetic stages: towards targeted interventions. **Lancet Diabetes Endocrinol** 2: 246–255, 2014. doi:[10.1016/S2213-8587\(13\)70088-3](https://doi.org/10.1016/S2213-8587(13)70088-3).
401. Dragicevic N, Mamcarz M, Zhu Y, Buzzeo R, Tan J, Arendash GW, Bradshaw PC. Mitochondrial amyloid-beta levels are associated with the extent of mitochondrial dysfunction in different brain regions and the degree of cognitive impairment in Alzheimer's transgenic mice. **J Alzheimers Dis** 20, Suppl 2: S535–S550, 2010. doi:[10.3233/JAD-2010-100342](https://doi.org/10.3233/JAD-2010-100342).
402. Barone E, Di Domenico F, Cassano T, Arena A, Tramutola A, Lavecchia MA, Coccia R, Butterfield DA, Perluigi M. Impairment of biliverdin reductase-A promotes brain insulin resistance in Alzheimer disease: a new paradigm. **Free Radic Biol Med** 91: 127–142, 2016. doi:[10.1016/j.freeradbiomed.2015.12.012](https://doi.org/10.1016/j.freeradbiomed.2015.12.012).
403. Barone E, Di Domenico F, Perluigi M, Butterfield DA. The interplay among oxidative stress, brain insulin resistance and AMPK dysfunction contribute to neurodegeneration in type 2 diabetes and Alzheimer disease. **Free Radic Biol Med** 176: 16–33, 2021. doi:[10.1016/j.freeradbiomed.2021.09.006](https://doi.org/10.1016/j.freeradbiomed.2021.09.006).
404. Kummer MP, Hülsmann C, Hermes M, Axt D, Heneka MT. Nitric oxide decreases the enzymatic activity of insulin degrading enzyme in APP/PS1 mice. **J Neuroimmune Pharmacol** 7: 165–172, 2012. doi:[10.1007/s11481-011-9339-7](https://doi.org/10.1007/s11481-011-9339-7).
405. Barone E, Di Domenico F, Cenini G, Sultana R, Cini C, Preziosi P, Perluigi M, Mancuso C, Butterfield DA. Biliverdin reductase-A protein levels and activity in the brains of subjects with Alzheimer disease and mild cognitive impairment. **Biochim Biophys Acta** 1812: 480–487, 2011. doi:[10.1016/j.bbadis.2011.01.005](https://doi.org/10.1016/j.bbadis.2011.01.005).
406. Triani F, Tramutola A, Di Domenico F, Sharma N, Butterfield DA, Head E, Perluigi M, Barone E. Biliverdin reductase-A impairment links brain insulin resistance with increased Aβ production in an animal model of aging: implications for Alzheimer disease. **Biochim Biophys Acta Mol Basis Dis** 1864: 3181–3194, 2018. doi:[10.1016/j.bbadis.2018.07.005](https://doi.org/10.1016/j.bbadis.2018.07.005).

407. Delikkaya B, Moriel N, Tong M, Gallucci G, de la Monte SM. Altered expression of insulin-degrading enzyme and regulator of calcineurin in the rat intracerebral streptozotocin model and human apolipoprotein E-epsilon4-associated Alzheimer's disease. **Alzheimers Dement (Amst)** 11: 392–404, 2019. doi:[10.1016/j.dadm.2019.03.004](https://doi.org/10.1016/j.dadm.2019.03.004).
408. Vekrellis K, Ye Z, Qiu WQ, Walsh D, Hartley D, Chesneau V, Rosner MR, Selkoe DJ. Neurons regulate extracellular levels of amyloid beta-protein via proteolysis by insulin-degrading enzyme. **J Neurosci** 20: 1657–1665, 2000. doi:[10.1523/JNEUROSCI.20-05-01657.2000](https://doi.org/10.1523/JNEUROSCI.20-05-01657.2000).
409. Gibbs PE, Lerner-Marmarosh N, Poulin A, Farah E, Maines MD. Human biliverdin reductase-based peptides activate and inhibit glucose uptake through direct interaction with the kinase domain of insulin receptor. **FASEB J** 28: 2478–2491, 2014. doi:[10.1096/fj.13-247015](https://doi.org/10.1096/fj.13-247015).
410. Sharma N, Tramutola A, Lanzillotta C, Arena A, Blarmino C, Cassano T, Butterfield DA, Di Domenico F, Perluigi M, Barone E. Loss of biliverdin reductase-A favors Tau hyper-phosphorylation in Alzheimer's disease. **Neurobiol Dis** 125: 176–189, 2019. doi:[10.1016/j.nbd.2019.02.003](https://doi.org/10.1016/j.nbd.2019.02.003).
411. Barone E, Tramutola A, Triani F, Calcagnini S, Di Domenico F, Ripoli C, Gaetani S, Grassi C, Butterfield DA, Cassano T, Perluigi M. Biliverdin reductase-A mediates the beneficial effects of intranasal insulin in Alzheimer disease. **Mol Neurobiol** 56: 2922–2943, 2019. doi:[10.1007/s12035-018-1231-5](https://doi.org/10.1007/s12035-018-1231-5).
412. Lanzillotta C, Zuliani I, Vasavda C, Snyder SH, Paul BD, Perluigi M, Di Domenico F, Barone E. BVR-A deficiency leads to autophagy impairment through the dysregulation of AMPK/mTOR axis in the brain—implications for neurodegeneration. **Antioxidants (Basel)** 9: 671, 2020. doi:[10.3390/antiox9080671](https://doi.org/10.3390/antiox9080671).
413. Brewer GJ. Epigenetic oxidative redox shift (EORS) theory of aging unifies the free radical and insulin signaling theories. **Exp Gerontol** 45: 173–179, 2010. doi:[10.1016/j.exger.2009.11.007](https://doi.org/10.1016/j.exger.2009.11.007).
414. Jang YC, Van Remmen H. The mitochondrial theory of aging: insight from transgenic and knockout mouse models. **Exp Gerontol** 44: 256–260, 2009. doi:[10.1016/j.exger.2008.12.006](https://doi.org/10.1016/j.exger.2008.12.006).
415. Harman D. Free radical theory of aging: effect of free radical reaction inhibitors on the mortality rate of male LAF mice. **J Gerontol** 23: 476–482, 1968. doi:[10.1093/geronj/23.4.476](https://doi.org/10.1093/geronj/23.4.476).
416. Miller ER 3rd, Pastor-Barriuso R, Dalal D, Riemersma RA, Appel LJ, Guallar E. Meta-analysis: high-dosage vitamin E supplementation may increase all-cause mortality. **Ann Intern Med** 142: 37–46, 2005. doi:[10.7326/0003-4819-142-1-200501040-00110](https://doi.org/10.7326/0003-4819-142-1-200501040-00110).
417. Al-Regaiey KA, Masternak MM, Bonkowski MS, Panici JA, Kopchick JJ, Bartke A. Effects of caloric restriction and growth hormone resistance on insulin-related intermediates in the skeletal muscle. **J Gerontol A Biol Sci Med Sci** 62: 18–26, 2007. doi:[10.1093/gerona/62.1.18](https://doi.org/10.1093/gerona/62.1.18).
418. Heilbronn LK, de Jonge L, Frisard MI, DeLany JP, Larson-Meyer DE, Rood J, Nguyen T, Martin CK, Volaufova J, Most MM, Greenway FL, Smith SR, Deutsch WA, Williamson DA, Ravussin E; Pennington CALERIE Team. Effect of 6-month calorie restriction on biomarkers of longevity, metabolic adaptation, and oxidative stress in overweight individuals: a randomized controlled trial. **JAMA** 295: 1539–1548, 2006. doi:[10.1001/jama.295.13.1539](https://doi.org/10.1001/jama.295.13.1539).
419. Parihar MS, Kunz EA, Brewer GJ. Age-related decreases in NAD(P)H and glutathione cause redox declines before ATP loss during glutamate treatment of hippocampal neurons. **J Neurosci Res** 86: 2339–2352, 2008. doi:[10.1002/jnr.21679](https://doi.org/10.1002/jnr.21679).
420. Parihar MS, Brewer GJ. Simultaneous age-related depolarization of mitochondrial membrane potential and increased mitochondrial reactive oxygen species production correlate with age-related glutamate excitotoxicity in rat hippocampal neurons. **J Neurosci Res** 85: 1018–1032, 2007. doi:[10.1002/jnr.21218](https://doi.org/10.1002/jnr.21218).
421. Wang SC, Oelze B, Schumacher A. Age-specific epigenetic drift in late-onset Alzheimer's disease. **PLoS One** 3: e2698, 2008. doi:[10.1371/journal.pone.0002698](https://doi.org/10.1371/journal.pone.0002698).
422. Jones TT, Brewer GJ. Critical age-related loss of cofactors of neuron cytochrome C oxidase reversed by estrogen. **Exp Neurol** 215: 212–219, 2009. doi:[10.1016/j.expneurol.2008.09.011](https://doi.org/10.1016/j.expneurol.2008.09.011).
423. Rimbert V, Boirie Y, Bedu M, Hocquette JF, Ritz P, Morio B. Muscle fat oxidative capacity is not impaired by age but by physical inactivity: association with insulin sensitivity. **FASEB J** 18: 737–739, 2004. doi:[10.1096/fj.03-1104fje](https://doi.org/10.1096/fj.03-1104fje).
424. Figueiredo PA, Powers SK, Ferreira RM, Amado F, Appell HJ, Duarte JA. Impact of lifelong sedentary behavior on mitochondrial function of mice skeletal muscle. **J Gerontol A Biol Sci Med Sci** 64: 927–939, 2009. doi:[10.1093/gerona/glp066](https://doi.org/10.1093/gerona/glp066).
425. Anderson RM, Weindruch R. Metabolic reprogramming in dietary restriction. **Interdiscip Top Gerontol** 35: 18–38, 2007. doi:[10.1159/000096554](https://doi.org/10.1159/000096554).
426. Aliev G, Liu J, Shenk JC, Fischbach K, Pacheco GJ, Chen SG, Obrenovich ME, Ward WF, Richardson AG, Smith MA, Gasimov E, Perry G, Ames BN. Neuronal mitochondrial amelioration by feeding acetyl-L-carnitine and lipoic acid to aged rats. **J Cell Mol Med** 13: 320–333, 2009. doi:[10.1111/j.1582-4934.2008.00324.x](https://doi.org/10.1111/j.1582-4934.2008.00324.x).
427. Brewer GJ. Why vitamin E therapy fails for treatment of Alzheimer's disease. **J Alzheimers Dis** 19: 27–30, 2010. doi:[10.3233/JAD-2010-1238](https://doi.org/10.3233/JAD-2010-1238).
428. Mullaart E, Boerrigter ME, Ravid R, Swaab DF, Vijg J. Increased levels of DNA breaks in cerebral cortex of Alzheimer's disease patients. **Neurobiol Aging** 11: 169–173, 1990. doi:[10.1016/0197-4580\(90\)90542-8](https://doi.org/10.1016/0197-4580(90)90542-8).
429. Caron MC, Sharma AK, O'Sullivan J, Myler LR, Ferreira MT, Rodrigue A, Coulombe Y, Ethier C, Gagné JP, Langelier MF, Pascal JM, Finkelstein IJ, Hendzel MJ, Poirier GG, Masson JY. Poly(ADP-ribose) polymerase-1 antagonizes DNA resection at double-strand breaks. **Nat Commun** 10: 2954, 2019. doi:[10.1038/s41467-019-10741-9](https://doi.org/10.1038/s41467-019-10741-9).
430. Mecocci P, MacGarvey U, Beal MF. Oxidative damage to mitochondrial DNA is increased in Alzheimer's disease. **Ann Neurol** 36: 747–751, 1994. doi:[10.1002/ana.410360510](https://doi.org/10.1002/ana.410360510).
431. Mao P, Reddy PH. Aging and amyloid beta-induced oxidative DNA damage and mitochondrial dysfunction in Alzheimer's disease: implications for early intervention and therapeutics. **Biochim Biophys Acta** 1812: 1359–1370, 2011. doi:[10.1016/j.bbadis.2011.08.005](https://doi.org/10.1016/j.bbadis.2011.08.005).
432. Lyras L, Cairns NJ, Jenner A, Jenner P, Halliwell B. An assessment of oxidative damage to proteins, lipids, and DNA in brain from patients with Alzheimer's disease. **J Neurochem** 68: 2061–2069, 1997. doi:[10.1046/j.1471-4159.1997.68052061.x](https://doi.org/10.1046/j.1471-4159.1997.68052061.x).
433. Mecocci P, Polidori MC, Cherubini A, Ingegni T, Mattioli P, Catani M, Rinaldi P, Cecchetti R, Stahl W, Senin U, Beal MF. Lymphocyte oxidative DNA damage and plasma antioxidants in Alzheimer disease. **Arch Neurol** 59: 794–798, 2002. doi:[10.1001/archneur.59.5.794](https://doi.org/10.1001/archneur.59.5.794).
434. Kadioglu E, Sardas S, Aslan S, Isik E, Esat Karakaya A. Detection of oxidative DNA damage in lymphocytes of patients with

- Alzheimer's disease. **Biomarkers** 9: 203–209, 2004. doi:[10.1080/13547500410001728390](https://doi.org/10.1080/13547500410001728390).
435. Wang J, Xiong S, Xie C, Markesbery WR, Lovell MA. Increased oxidative damage in nuclear and mitochondrial DNA in Alzheimer's disease. **J Neurochem** 93: 953–962, 2005. doi:[10.1111/j.1471-4159.2005.03053.x](https://doi.org/10.1111/j.1471-4159.2005.03053.x).
436. Coskun PE, Beal MF, Wallace DC. Alzheimer's brains harbor somatic mtDNA control-region mutations that suppress mitochondrial transcription and replication. **Proc Natl Acad Sci USA** 101: 10726–10731, 2004. doi:[10.1073/pnas.0403649101](https://doi.org/10.1073/pnas.0403649101).
437. Coskun PE, Wyrembak J, Derbereva O, Melkonian G, Doran E, Lott IT, Head E, Cotman CW, Wallace DC. Systemic mitochondrial dysfunction and the etiology of Alzheimer's disease and down syndrome dementia. **J Alzheimers Dis** 20, Suppl 2: S293–S310, 2010. doi:[10.3233/JAD-2010-100351](https://doi.org/10.3233/JAD-2010-100351).
438. Nunomura A, Tamaoki T, Tanaka K, Motohashi N, Nakamura M, Hayashi T, Yamaguchi H, Shimohama S, Lee HG, Zhu X, Smith MA, Perry G. Intraneuronal amyloid beta accumulation and oxidative damage to nucleic acids in Alzheimer disease. **Neurobiol Dis** 37: 731–737, 2010. doi:[10.1016/j.nbd.2009.12.012](https://doi.org/10.1016/j.nbd.2009.12.012).
439. Nunomura A, Perry G, Pappolla MA, Wade R, Hirai K, Chiba S, Smith MA. RNA oxidation is a prominent feature of vulnerable neurons in Alzheimer's disease. **J Neurosci** 19: 1959–1964, 1999. doi:[10.1523/JNEUROSCI.19-06-01959.1999](https://doi.org/10.1523/JNEUROSCI.19-06-01959.1999).
440. Nunomura A, Perry G, Hirai K, Aliev G, Takeda A, Chiba S, Smith MA. Neuronal RNA oxidation in Alzheimer's disease and Down's syndrome. **Ann NY Acad Sci** 893: 362–364, 1999. doi:[10.1111/j.1749-6632.1999.tb07855.x](https://doi.org/10.1111/j.1749-6632.1999.tb07855.x).
441. Weidner AM, Bradley MA, Beckett TL, Niedowicz DM, Dowling AL, Matveev SV, LeVine H 3rd, Lovell MA, Murphy MP. RNA oxidation adducts 8-OHG and 8-OHA change with Abeta42 levels in late-stage Alzheimer's disease. **PLoS One** 6: e24930, 2011. doi:[10.1371/journal.pone.0024930](https://doi.org/10.1371/journal.pone.0024930).
442. Zhao J, Yue D, Zhou Y, Jia L, Wang H, Guo M, Xu H, Chen C, Zhang J, Xu L. The role of microRNAs in Abeta deposition and tau phosphorylation in Alzheimer's disease. **Front Neurol** 8: 342, 2017. doi:[10.3389/fneur.2017.00342](https://doi.org/10.3389/fneur.2017.00342).
443. Li JJ, Dolios G, Wang R, Liao FF. Soluble beta-amyloid peptides, but not insoluble fibrils, have specific effect on neuronal microRNA expression. **PLoS One** 9: e90770, 2014. doi:[10.1371/journal.pone.0090770](https://doi.org/10.1371/journal.pone.0090770).
444. Wang WX, Rajeev BW, Stromberg AJ, Ren N, Tang G, Huang Q, Rigoutsos I, Nelson PT. The expression of microRNA miR-107 decreases early in Alzheimer's disease and may accelerate disease progression through regulation of beta-site amyloid precursor protein-cleaving enzyme 1. **J Neurosci** 28: 1213–1223, 2008. doi:[10.1523/JNEUROSCI.5065-07.2008](https://doi.org/10.1523/JNEUROSCI.5065-07.2008).
445. de la Monte SM, Wands JR. Molecular indices of oxidative stress and mitochondrial dysfunction occur early and often progress with severity of Alzheimer's disease. **J Alzheimers Dis** 9: 167–181, 2006. doi:[10.3233/jad-2006-9209](https://doi.org/10.3233/jad-2006-9209).
446. Dorszewska J, Oczkowska A, Suwalska M, Rozycka A, Florczak-Wyspianska J, Dezor M, Lianeri M, Jagodzinski PP, Kowalczyk MJ, Prendecki M, Kozubski W. Mutations in the exon 7 of Trp53 gene and the level of p53 protein in double transgenic mouse model of Alzheimer's disease. **Folia Neuropathol** 52: 30–40, 2014. doi:[10.5114/fn.2014.41742](https://doi.org/10.5114/fn.2014.41742).
447. Chen L, Zhang R, Li P, Liu Y, Qin K, Fa ZQ, Liu YJ, Ke YQ, Jiang XD. P53-induced microRNA-107 inhibits proliferation of glioma cells and down-regulates the expression of CDK6 and Notch-2. **Neurosci Lett** 534: 327–332, 2013. doi:[10.1016/j.neulet.2012.11.047](https://doi.org/10.1016/j.neulet.2012.11.047).
448. Prendecki M, Florczak-Wyspianska J, Kowalska M, Ilkowski J, Grzelak T, Bialas K, Kozubski W, Dorszewska J. APOE genetic variants and apoE, miR-107 and miR-650 levels in Alzheimer's disease. **Folia Neuropathol** 57: 106–116, 2019. doi:[10.5114/fn.2019.84828](https://doi.org/10.5114/fn.2019.84828).
449. Kim J, Yoon H, Chung DE, Brown JL, Belmonte KC, Kim J. miR-186 is decreased in aged brain and suppresses BACE1 expression. **J Neurochem** 137: 436–445, 2016. doi:[10.1111/jnc.13507](https://doi.org/10.1111/jnc.13507).
450. Antonarakis SE, Skotko BG, Rafii MS, Strydom A, Pape SE, Bianchi DW, Sherman SL, Reeves RH. Down syndrome. **Nat Rev Dis Primers** 6: 9, 2020. doi:[10.1038/s41572-019-0143-7](https://doi.org/10.1038/s41572-019-0143-7).
451. Antonarakis SE. Down syndrome and the complexity of genome dosage imbalance. **Nat Rev Genet** 18: 147–163, 2017. doi:[10.1038/nrg.2016.154](https://doi.org/10.1038/nrg.2016.154).
452. Pelleri MC, Cicchini E, Locatelli C, Vitale L, Caracausi M, Piovesan A, Rocca A, Poletti G, Seri M, Strippoli P, Cocchi G. Systematic reanalysis of partial trisomy 21 cases with or without Down syndrome suggests a small region on 21q22.13 as critical to the phenotype. **Hum Mol Genet** 25: 2525–2538, 2016. doi:[10.1093/hmg/ddw116](https://doi.org/10.1093/hmg/ddw116).
453. Mentis AF. Epigenomic engineering for Down syndrome. **Neurosci Biobehav Rev** 71: 323–327, 2016. doi:[10.1016/j.neubiorev.2016.09.012](https://doi.org/10.1016/j.neubiorev.2016.09.012).
454. Gardiner K, Herault Y, Lott IT, Antonarakis SE, Reeves RH, Dierssen M. Down syndrome: from understanding the neurobiology to therapy. **J Neurosci** 30: 14943–14945, 2010. doi:[10.1523/JNEUROSCI.3728-10.2010](https://doi.org/10.1523/JNEUROSCI.3728-10.2010).
455. Antonarakis SE, Epstein CJ. The challenge of Down syndrome. **Trends Mol Med** 12: 473–479, 2006. doi:[10.1016/j.molmed.2006.08.005](https://doi.org/10.1016/j.molmed.2006.08.005).
456. Fortea J, Zaman SH, Hartley S, Rafii MS, Head E, Carmona-Iragui M. Alzheimer's disease associated with Down syndrome: a genetic form of dementia. **Lancet Neurol** 20: 930–942, 2021. doi:[10.1016/S1474-4422\(21\)00245-3](https://doi.org/10.1016/S1474-4422(21)00245-3).
457. Lott IT, Head E. Dementia in Down syndrome: unique insights for Alzheimer disease research. **Nat Rev Neurol** 15: 135–147, 2019. doi:[10.1038/s41582-018-0132-6](https://doi.org/10.1038/s41582-018-0132-6).
458. Stagni F, Giacomini A, Guidi S, Ciani E, Bartesaghi R. Timing of therapies for Down syndrome: the sooner, the better. **Front Behav Neurosci** 9: 265, 2015. doi:[10.3389/fnbeh.2015.00265](https://doi.org/10.3389/fnbeh.2015.00265).
459. Bush A, Beal N. Risk factors for dementia in people with Down syndrome: issues in assessment and diagnosis. **Am J Ment Retard** 109: 83–97, 2004. doi:[10.1352/0895-8017\(2004\)109<83:RFFDIP>2.0.CO;2](https://doi.org/10.1352/0895-8017(2004)109<83:RFFDIP>2.0.CO;2).
460. Ghezzi A, Salvioli S, Solimando MC, Palmieri A, Chiostergi C, Scurti M, Lomartire L, Bedetti F, Cocchi G, Folio D, Pipitone E, Rovatti P, Zamberletti J, Gomiero T, Castellani G, Franceschi C. Age-related changes of adaptive and neuropsychological features in persons with Down syndrome. **PLoS One** 9: e113111, 2014. doi:[10.1371/journal.pone.0113111](https://doi.org/10.1371/journal.pone.0113111).
461. Zigman WB, Schupf N, Sersen E, Silverman W. Prevalence of dementia in adults with and without Down syndrome. **Am J Ment Retard** 100: 403–412, 1996.
462. Wiseman FK, Al-Janabi T, Hardy J, Karmiloff-Smith A, Nizetic D, Tybulewicz VL, Fisher EM, Strydom A. A genetic cause of Alzheimer disease: mechanistic insights from Down syndrome. **Nat Rev Neurosci** 16: 564–574, 2015. doi:[10.1038/nrn3983](https://doi.org/10.1038/nrn3983).
463. Doran E, Keator D, Head E, Phelan MJ, Kim R, Totoiu M, Barrio JR, Small GW, Potkin SG, Lott IT. Down syndrome, partial trisomy 21,

- and absence of Alzheimer's disease: the role of APP. **J Alzheimers Dis** 56: 459–470, 2017. doi:[10.3233/JAD-160836](https://doi.org/10.3233/JAD-160836).
464. Jiang Y, Sato Y, Im E, Berg M, Bordini M, Darji S, Kumar A, Mohan PS, Bandyopadhyay U, Diaz A, Cuervo AM, Nixon RA. Lysosomal dysfunction in Down syndrome is APP-dependent and mediated by APP-beta CTF (C99). **J Neurosci** 39: 5255–5268, 2019. doi:[10.1523/JNEUROSCI.0578-19.2019](https://doi.org/10.1523/JNEUROSCI.0578-19.2019).
465. Cataldo AM, Peterhoff CM, Troncoso JC, Gomez-Isla T, Hyman BT, Nixon RA. Endocytic pathway abnormalities precede amyloid beta deposition in sporadic Alzheimer's disease and Down syndrome: differential effects of APOE genotype and presenilin mutations. **Am J Pathol** 157: 277–286, 2000. doi:[10.1016/S0002-9440\(10\)64538-5](https://doi.org/10.1016/S0002-9440(10)64538-5).
466. Botté A, Potier MC. Focusing on cellular biomarkers: the endo-lysosomal pathway in Down syndrome. **Prog Brain Res** 251: 209–243, 2020. doi:[10.1016/bs.pbr.2019.10.002](https://doi.org/10.1016/bs.pbr.2019.10.002).
467. Botté A, Laine J, Xicota L, Heiligenstein X, Fontaine G, Kasri A, Rivals I, Goh P, Faklaris O, Cossec JC, Morel E, Rebillat AS, Nizetic D, Raposo G, Potier MC. Ultrastructural and dynamic studies of the endosomal compartment in Down syndrome. **Acta Neuropathol Commun** 8: 89, 2020. doi:[10.1186/s40478-020-00956-z](https://doi.org/10.1186/s40478-020-00956-z).
468. Busciglio J, Pelsman A, Wong C, Pigino G, Yuan M, Mori H, Yankner BA. Altered metabolism of the amyloid beta precursor protein is associated with mitochondrial dysfunction in Down's syndrome. **Neuron** 33: 677–688, 2002. doi:[10.1016/S0896-6273\(02\)00604-9](https://doi.org/10.1016/S0896-6273(02)00604-9).
469. Lemere CA, Blusztajn JK, Yamaguchi H, Wisniewski T, Saido TC, Selkoe DJ. Sequence of deposition of heterogeneous amyloid beta-peptides and APO E in Down syndrome: implications for initial events in amyloid plaque formation. **Neurobiol Dis** 3: 16–32, 1996. doi:[10.1006/nbdi.1996.0003](https://doi.org/10.1006/nbdi.1996.0003).
470. Leverenz JB, Raskind MA. Early amyloid deposition in the medial temporal lobe of young Down syndrome patients: a regional quantitative analysis. **Exp Neurol** 150: 296–304, 1998. doi:[10.1006/exnr.1997.6777](https://doi.org/10.1006/exnr.1997.6777).
471. Head E, Ances B. Biomarkers in Down syndrome can help us understand Alzheimer's disease. **Lancet** 395: 1951–1953, 2020. doi:[10.1016/S0140-6736\(20\)30916-8](https://doi.org/10.1016/S0140-6736(20)30916-8).
472. Lemoine L, Ledreux A, Mufson EJ, Perez SE, Simic G, Doran E, Lott I, Carroll S, Bharani K, Thomas S, Gilmore A, Hamlett ED, Nordberg A, Granholm AC. Regional binding of tau and amyloid PET tracers in Down syndrome autopsy brain tissue. **Mol Neurodegener** 15: 68, 2020. doi:[10.1186/s13024-020-00414-3](https://doi.org/10.1186/s13024-020-00414-3).
473. Rafii MS. Tau PET imaging for staging of Alzheimer's disease in Down syndrome. **Dev Neurobiol** 79: 711–715, 2019. doi:[10.1002/dneu.22658](https://doi.org/10.1002/dneu.22658).
474. Rafii MS, Lukic AS, Andrews RD, Brewer J, Rissman RA, Strother SC, Wernick MN, Pennington C, Mobley WC, Ness S, Matthews DC; Down Syndrome Biomarker Initiative, Alzheimer's Disease Neuroimaging Initiative. PET imaging of tau pathology and relationship to amyloid, longitudinal MRI, and cognitive change in Down syndrome: results from the Down Syndrome Biomarker Initiative (DSBI). **J Alzheimers Dis** 60: 439–450, 2017. doi:[10.3233/JAD-170390](https://doi.org/10.3233/JAD-170390).
475. Wegiel J, Flory M, Kuchna I, Nowicki K, Wegiel J, Ma SY, Zhong N, Bobrowicz TW, de Leon M, Lai F, Silverman WP, Wisniewski T. Developmental deficits and staging of dynamics of age associated Alzheimer's disease neurodegeneration and neuronal loss in subjects with Down syndrome. **Acta Neuropathol Commun** 10: 2, 2022. doi:[10.1186/s40478-021-01300-9](https://doi.org/10.1186/s40478-021-01300-9).
476. Fortea J, Vilaplana E, Carmona-Iragui M, Benejam B, Videla L, Barroeta I, Fernández S, Altuna M, Pegueroles J, Montal V, Valldeneu S, Giménez S, González-Ortiz S, Muñoz L, Estelles T, Illan-Gala I, Belbin O, Camacho V, Wilson LR, Annus T, Osorio RS, Videla S, Lehmann S, Holland AJ, Alcolea D, Clarimon J, Zaman SH, Blesa R, Lleó A. Clinical and biomarker changes of Alzheimer's disease in adults with Down syndrome: a cross-sectional study. **Lancet** 395: 1988–1997, 2020. doi:[10.1016/S0140-6736\(20\)30689-9](https://doi.org/10.1016/S0140-6736(20)30689-9).
477. Wiseman FK, Pulford LJ, Barkus C, Liao F, Portelius E, Webb R, Chávez-Gutiérrez L, Cleverley K, Noy S, Sheppard O, Collins T, Powell C, Sarell CJ, Rickman M, Choong X, Tosh JL, Sigmanporia C, Whittaker HT, Stewart F, Szaruga M, London Down Syndrome Consortium, Murphy MP, Blennow K, de Strooper B, Zetterberg H, Bannerman D, Holtzman DM, Tybulewicz VL, Fisher EM. Trisomy of human chromosome 21 enhances amyloid-beta deposition independently of an extra copy of APP. **Brain** 141: 2457–2474, 2018. doi:[10.1093/brain/awy159](https://doi.org/10.1093/brain/awy159).
478. Gomez W, Morales R, Maracaja-Coutinho V, Parra V, Nassif M. Down syndrome and Alzheimer's disease: common molecular traits beyond the amyloid precursor protein. **Aging (Albany NY)** 12: 1011–1033, 2020. doi:[10.18632/aging.102677](https://doi.org/10.18632/aging.102677).
479. Rujeedawa T, Carrillo Félez E, Clare IC, Fortea J, Strydom A, Rebillat AS, Coppus A, Levin J, Zaman SH. The clinical and neuropathological features of sporadic (late-onset) and genetic forms of Alzheimer's disease. **J Clin Med** 10: 4582, 2021. doi:[10.3390/jcm10194582](https://doi.org/10.3390/jcm10194582).
480. Ryan NS, Nicholas JM, Weston PS, Liang Y, Lashley T, Guerreiro R, Adamson G, Kenny J, Beck J, Chavez-Gutierrez L, de Strooper B, Revesz T, Holton J, Mead S, Rossor MN, Fox NC. Clinical phenotype and genetic associations in autosomal dominant familial Alzheimer's disease: a case series. **Lancet Neurol** 15: 1326–1335, 2016. doi:[10.1016/S1474-4422\(16\)30193-4](https://doi.org/10.1016/S1474-4422(16)30193-4).
481. Lott IT. Neurological phenotypes for Down syndrome across the life span. **Prog Brain Res** 197: 101–121, 2012. doi:[10.1016/B978-0-444-54299-1.00006-6](https://doi.org/10.1016/B978-0-444-54299-1.00006-6).
482. Cenini G, Dowling AL, Beckett TL, Barone E, Mancuso C, Murphy MP, Levine H 3rd, Lott IT, Schmitt FA, Butterfield DA, Head E. Association between frontal cortex oxidative damage and beta-amyloid as a function of age in Down syndrome. **Biochim Biophys Acta** 1822: 130–138, 2012. doi:[10.1016/j.bbadis.2011.10.001](https://doi.org/10.1016/j.bbadis.2011.10.001).
483. Di Domenico F, Pupo G, Mancuso C, Barone E, Paolini F, Arena A, Blarmino C, Schmitt FA, Head E, Butterfield DA, Perluigi M. BACH1 overexpression in Down syndrome correlates with the alteration of the HO-1/BVR-a system: insights for transition to Alzheimer's disease. **J Alzheimers Dis** 44: 1107–1120, 2015. doi:[10.3233/JAD-141254](https://doi.org/10.3233/JAD-141254).
484. Halliwell B. Reactive oxygen species in living systems: source, biochemistry, and role in human disease. **Am J Med** 91: 14S–22S, 1991. doi:[10.1016/0002-9343\(91\)90279-7](https://doi.org/10.1016/0002-9343(91)90279-7).
485. Murphy MP. How mitochondria produce reactive oxygen species. **Biochem J** 417: 1–13, 2009. doi:[10.1042/BJ20081386](https://doi.org/10.1042/BJ20081386).
486. Valenti D, Manente GA, Moro L, Marra E, Vacca RA. Deficit of complex I activity in human skin fibroblasts with chromosome 21 trisomy and overproduction of reactive oxygen species by mitochondria: involvement of the cAMP/PKA signalling pathway. **Biochem J** 435: 679–688, 2011. doi:[10.1042/BJ20101908](https://doi.org/10.1042/BJ20101908).
487. Conti A, Fabbrini F, D'Agostino P, Negri R, Greco D, Genesio R, D'Armiento M, Olla C, Paladini D, Zannini M, Nitsch L. Altered expression of mitochondrial and extracellular matrix genes in the heart of human fetuses with chromosome 21 trisomy. **BMC Genomics** 8: 268, 2007. doi:[10.1186/1471-2164-8-268](https://doi.org/10.1186/1471-2164-8-268).

488. Busciglio J, Yankner BA. Apoptosis and increased generation of reactive oxygen species in Down's syndrome neurons in vitro. **Nature** 378: 776–779, 1995. doi:[10.1038/378776a0](https://doi.org/10.1038/378776a0).
489. Zana M, Janka Z, Kálmán J. Oxidative stress: a bridge between Down's syndrome and Alzheimer's disease. **Neurobiol Aging** 28: 648–676, 2007. doi:[10.1016/j.neurobiolaging.2006.03.008](https://doi.org/10.1016/j.neurobiolaging.2006.03.008).
490. Pallardó FV, Degan P, d'Ischia M, Kelly FJ, Zatterale A, Calzone R, Castello G, Fernandez-Delgado R, Dunster C, Lloret A, Manini P, Pisanti MA, Vuttariello E, Pagano G. Multiple evidence for an early age pro-oxidant state in Down syndrome patients. **Biogerontology** 7: 211–220, 2006. doi:[10.1007/s10522-006-9002-5](https://doi.org/10.1007/s10522-006-9002-5).
491. Nunomura A, Perry G, Aliev G, Hirai K, Takeda A, Balraj EK, Jones PK, Ghanbari H, Wataya T, Shimohama S, Chiba S, Atwood CS, Petersen RB, Smith MA. Oxidative damage is the earliest event in Alzheimer disease. **J Neuropathol Exp Neurol** 60: 759–767, 2001. doi:[10.1093/jnen/60.8.759](https://doi.org/10.1093/jnen/60.8.759).
492. Barone E, Head E, Butterfield DA, Perluigi M. HNE-modified proteins in Down syndrome: Involvement in development of Alzheimer disease neuropathology. **Free Radic Biol Med** 111: 262–269, 2017. doi:[10.1016/j.freeradbiomed.2016.10.508](https://doi.org/10.1016/j.freeradbiomed.2016.10.508).
493. Perluigi M, Butterfield DA. Oxidative stress and Down syndrome: a route toward Alzheimer-like dementia. **Curr Gerontol Geriatr Res** 2012: 724904, 2012. doi:[10.1155/2012/724904](https://doi.org/10.1155/2012/724904).
494. Head E, Lott IT, Wilcock DM, Lemere CA. Aging in Down syndrome and the development of Alzheimer's disease neuropathology. **Curr Alzheimer Res** 13: 18–29, 2016. doi:[10.2174/1567205012666151020114607](https://doi.org/10.2174/1567205012666151020114607).
495. Zis P, Dickinson M, Shende S, Walker Z, Strydom A. Oxidative stress and memory decline in adults with Down syndrome: longitudinal study. **J Alzheimers Dis** 31: 277–283, 2012. doi:[10.3233/JAD-2012-120073](https://doi.org/10.3233/JAD-2012-120073).
496. Gulesserian T, Engidawork E, Fountoulakis M, Lubec G. Antioxidant proteins in fetal brain: superoxide dismutase-1 (SOD-1) protein is not overexpressed in fetal Down syndrome. **J Neural Transm Suppl** 61: 71–84, 2001. doi:[10.1007/978-3-7091-6262-0_6](https://doi.org/10.1007/978-3-7091-6262-0_6).
497. Pagnotta S, Tramutola A, Barone E, Di Domenico F, Pittalà V, Salerno L, Folgiero V, Caforio M, Locatelli F, Petrini S, Butterfield DA, Perluigi M. CAPE and its synthetic derivative VP961 restore BACH1/NRF2 axis in Down syndrome. **Free Radic Biol Med** 183: 1–13, 2022. doi:[10.1016/j.freeradbiomed.2022.03.006](https://doi.org/10.1016/j.freeradbiomed.2022.03.006).
498. Panagaki T, Lozano-Montes L, Janickova L, Zuhra K, Szabo MP, Majtan T, Rainer G, Maréchal D, Herault Y, Szabo C. Overproduction of hydrogen sulfide, generated by cystathionine beta-synthase, disrupts brain wave patterns and contributes to neurobehavioral dysfunction in a rat model of down syndrome. **Redox Biol** 51: 102233, 2022. doi:[10.1016/j.redox.2022.102233](https://doi.org/10.1016/j.redox.2022.102233).
499. Szabo C. The re-emerging pathophysiological role of the cystathionine-beta-synthase-hydrogen sulfide system in Down syndrome. **FEBS J** 287: 3150–3160, 2020. doi:[10.1111/febs.15214](https://doi.org/10.1111/febs.15214).
500. Paul BD, Snyder SH, Kashfi K. Effects of hydrogen sulfide on mitochondrial function and cellular bioenergetics. **Redox Biol** 38: 101772, 2021. doi:[10.1016/j.redox.2020.101772](https://doi.org/10.1016/j.redox.2020.101772).
501. Valenti D, Braidly N, De Rasmio D, Signorile A, Rossi L, Atanasov AG, Volpicella M, Henrion-Caupe A, Nabavi SM, Vacca RA. Mitochondria as pharmacological targets in Down syndrome. **Free Radic Biol Med** 114: 69–83, 2018. doi:[10.1016/j.freeradbiomed.2017.08.014](https://doi.org/10.1016/j.freeradbiomed.2017.08.014).
502. Head E, Lott IT. Down syndrome and beta-amyloid deposition. **Curr Opin Neurol** 17: 95–100, 2004. doi:[10.1097/00019052-200404000-00003](https://doi.org/10.1097/00019052-200404000-00003).
503. Mehta PD, Capone G, Jewell A, Freedland RL. Increased amyloid beta protein levels in children and adolescents with Down syndrome. **J Neurol Sci** 254: 22–27, 2007. doi:[10.1016/j.jns.2006.12.010](https://doi.org/10.1016/j.jns.2006.12.010).
504. Anandatheerthavarada HK, Biswas G, Robin MA, Avadhani NG. Mitochondrial targeting and a novel transmembrane arrest of Alzheimer's amyloid precursor protein impairs mitochondrial function in neuronal cells. **J Cell Biol** 161: 41–54, 2003. doi:[10.1083/jcb.200207030](https://doi.org/10.1083/jcb.200207030).
505. Simón AM, Schiapparelli L, Salazar-Colocho P, Cuadrado-Tejedor M, Escribano L, López de Maturana R, Del Río J, Perez-Mediavilla A, Frechilla D. Overexpression of wild-type human APP in mice causes cognitive deficits and pathological features unrelated to Abeta levels. **Neurobiol Dis** 33: 369–378, 2009. doi:[10.1016/j.nbd.2008.11.005](https://doi.org/10.1016/j.nbd.2008.11.005).
506. Kann O, Kovacs R. Mitochondria and neuronal activity. **Am J Physiol Cell Physiol** 292: C641–C657, 2007. doi:[10.1152/ajpcell.00222.2006](https://doi.org/10.1152/ajpcell.00222.2006).
507. Xavier JM, Rodrigues CM, Sola S. Mitochondria: major regulators of neural development. **Neuroscientist** 22: 346–358, 2016. doi:[10.1177/1073858415585472](https://doi.org/10.1177/1073858415585472).
508. Valenti D, de Bari L, de Filippis B, Henrion-Caupe A, Vacca RA. Mitochondrial dysfunction as a central actor in intellectual disability-related diseases: an overview of Down syndrome, autism, Fragile X and Rett syndrome. **Neurosci Biobehav Rev** 46 Pt 2: 202–217, 2014. doi:[10.1016/j.neubiorev.2014.01.012](https://doi.org/10.1016/j.neubiorev.2014.01.012).
509. Valenti D, Tullo A, Caratozzolo MF, Merafina RS, Scartezzini P, Marra E, Vacca RA. Impairment of F1F0-ATPase, adenine nucleotide translocator and adenylate kinase causes mitochondrial energy deficit in human skin fibroblasts with chromosome 21 trisomy. **Biochem J** 431: 299–310, 2010. doi:[10.1042/BJ20100581](https://doi.org/10.1042/BJ20100581).
510. Kim SH, Vlkolinsky R, Cairns N, Fountoulakis M, Lubec G. The reduction of NADH ubiquinone oxidoreductase 24- and 75-kDa subunits in brains of patients with Down syndrome and Alzheimer's disease. **Life Sci** 68: 2741–2750, 2001. doi:[10.1016/s0024-3205\(01\)01074-8](https://doi.org/10.1016/s0024-3205(01)01074-8).
511. Valenti D, de Bari L, de Rasmio D, Signorile A, Henrion-Caupe A, Contestabile A, Vacca RA. The polyphenols resveratrol and epigallocatechin-3-gallate restore the severe impairment of mitochondria in hippocampal progenitor cells from a Down syndrome mouse model. **Biochim Biophys Acta** 1862: 1093–1104, 2016. doi:[10.1016/j.bbdis.2016.03.003](https://doi.org/10.1016/j.bbdis.2016.03.003).
512. Valenti D, Stagni F, Emili M, Guidi S, Bartsaghi R, Vacca RA. Impaired brain mitochondrial bioenergetics in the Ts65Dn mouse model of Down syndrome is restored by neonatal treatment with the polyphenol 7,8-dihydroxyflavone. **Antioxidants (Basel)** 11: 62, 2021. doi:[10.3390/antiox11010062](https://doi.org/10.3390/antiox11010062).
513. Vacca RA, Bawari S, Valenti D, Tewari D, Nabavi SF, Shirooie S, Sah AN, Volpicella M, Braidly N, Nabavi SM. Down syndrome: neurobiological alterations and therapeutic targets. **Neurosci Biobehav Rev** 98: 234–255, 2019. doi:[10.1016/j.neubiorev.2019.01.001](https://doi.org/10.1016/j.neubiorev.2019.01.001).
514. Helguera P, Seiglie J, Rodriguez J, Hanna M, Helguera G, Busciglio J. Adaptive downregulation of mitochondrial function in down syndrome. **Cell Metab** 17: 132–140, 2013. doi:[10.1016/j.cmet.2012.12.005](https://doi.org/10.1016/j.cmet.2012.12.005).
515. Coskun P, Helguera P, Nemati Z, Bohannon RC, Thomas J, Samuel SE, Argueta J, Doran E, Wallace DC, Lott IT, Busciglio J. Metabolic and growth rate alterations in lymphoblastic cell lines discriminate

- between Down syndrome and Alzheimer's disease. **J Alzheimers Dis** 55: 737–748, 2017. doi:10.3233/JAD-160278.
516. Wall SB, Oh JY, Diers AR, Landar A. Oxidative modification of proteins: an emerging mechanism of cell signaling. **Front Physiol** 3: 369, 2012. doi:10.3389/fphys.2012.00369.
517. Butterfield DA, Di Domenico F, Swomley AM, Head E, Perluigi M. Redox proteomics analysis to decipher the neurobiology of Alzheimer-like neurodegeneration: overlaps in Down's syndrome and Alzheimer's disease brain. **Biochem J** 463: 177–189, 2014. doi:10.1042/BJ20140772.
518. Rueda Revilla N, Martínez-Cué C. Antioxidants in Down syndrome: from preclinical studies to clinical trials. **Antioxidants (Basel)** 9: 692, 2020. doi:10.3390/antiox9080692.
519. Dierssen M, Fructuoso M, Martínez de Lagrán M, Perluigi M, Barone E. Down syndrome is a metabolic disease: altered insulin signaling mediates peripheral and brain dysfunctions. **Front Neurosci** 14: 670, 2020. doi:10.3389/fnins.2020.00670.
520. Mattson MP, Gleichmann M, Cheng A. Mitochondria in neuroplasticity and neurological disorders. **Neuron** 60: 748–766, 2008. doi:10.1016/j.neuron.2008.10.010.
521. Jurcau A. The role of natural antioxidants in the prevention of dementia—where do we stand and future perspectives. **Nutrients** 13: 282, 2021. doi:10.3390/nu13020282.
522. Di Domenico F, Barone E, Perluigi M, Butterfield DA. Strategy to reduce free radical species in Alzheimer's disease: an update of selected antioxidants. **Expert Rev Neurother** 15: 19–40, 2015. doi:10.1586/14737175.2015.955853.
523. Mielech A, Puścion-Jakubik A, Markiewicz-Żukowska R, Socha K. Vitamins in Alzheimer's disease—review of the latest reports. **Nutrients** 12: 3458, 2020. doi:10.3390/nu12113458.
524. Tadokoro K, Ohta Y, Inufusa H, Loon AF, Abe K. Prevention of cognitive decline in Alzheimer's disease by novel antioxidant supplements. **Int J Mol Sci** 21: 1974, 2020. doi:10.3390/ijms21061974.
525. Forman HJ, Zhang H. Targeting oxidative stress in disease: promise and limitations of antioxidant therapy. **Nat Rev Drug Discov** 20: 689–709, 2021. doi:10.1038/s41573-021-00233-1.
526. Mecocci P, Boccardi V, Cecchetti R, Bastiani P, Scamosci M, Ruggiero C, Baroni M. A long journey into aging, brain aging, and Alzheimer's disease following the oxidative stress tracks. **J Alzheimers Dis** 62: 1319–1335, 2018. doi:10.3233/JAD-170732.
527. Andrieu S, Coley N, Lovestone S, Aisen PS, Vellas B. Prevention of sporadic Alzheimer's disease: lessons learned from clinical trials and future directions. **Lancet Neurol** 14: 926–944, 2015. doi:10.1016/S1474-4422(15)00153-2.
528. Talebi M, Kakouri E, Talebi M, Tarantilis PA, Farkhondeh T, Ilgün S, Pourbagher-Shahri AM, Samarghandian S. Nutraceuticals-based therapeutic approach: recent advances to combat pathogenesis of Alzheimer's disease. **Expert Rev Neurother** 21: 625–642, 2021. doi:10.1080/14737175.2021.1923479.
529. Persson T, Popescu BO, Cedazo-Minguez A. Oxidative stress in Alzheimer's disease: why did antioxidant therapy fail? **Oxid Med Cell Longev** 2014: 427318, 2014. doi:10.1155/2014/427318.
530. Gabbianelli R, Damiani E. Epigenetics and neurodegeneration: role of early-life nutrition. **J Nutr Biochem** 57: 1–13, 2018. doi:10.1016/j.jnutbio.2018.01.014.
531. Cremonini AL, Caffa I, Cea M, Nencioni A, Odetti P, Monacelli F. Nutrients in the prevention of Alzheimer's disease. **Oxid Med Cell Longev** 2019: 9874159, 2019. doi:10.1155/2019/9874159.
532. Hsiao YH, Kuo JR, Chen SH, Gean PW. Amelioration of social isolation-triggered onset of early Alzheimer's disease-related cognitive deficit by N-acetylcysteine in a transgenic mouse model. **Neurobiol Dis** 45: 1111–1120, 2012. doi:10.1016/j.nbd.2011.12.031.
533. Head E, Murphey HL, Dowling AL, McCarty KL, Bethel SR, Nitz JA, Pleiss M, Vanrooyen J, Grossheim M, Smiley JR, Murphy MP, Beckett TL, Pagani D, Bresch F, Hendrix C. A combination cocktail improves spatial attention in a canine model of human aging and Alzheimer's disease. **J Alzheimers Dis** 32: 1029–1042, 2012. doi:10.3233/JAD-2012-120937.
534. Ishrat T, Parveen K, Khan MM, Khuwaja G, Khan MB, Yousuf S, Ahmad A, Shrivastav P, Islam F. Selenium prevents cognitive decline and oxidative damage in rat model of streptozotocin-induced experimental dementia of Alzheimer's type. **Brain Res** 1281: 117–127, 2009. doi:10.1016/j.brainres.2009.04.010.
535. Pinton S, da Rocha JT, Gai BM, Prigol M, da Rosa LV, Nogueira CW. Neuroprotector effect of p,p'-methoxyl-diphenyl diselenide in a model of sporadic dementia of Alzheimer's type in mice: contribution of antioxidant mechanism. **Cell Biochem Funct** 29: 235–243, 2011. doi:10.1002/cbf.1741.
536. Lovell MA, Xiong S, Lyubartseva G, Markesbery WR. Organoselenium (Sel-Plex diet) decreases amyloid burden and RNA and DNA oxidative damage in APP/PS1 mice. **Free Radic Biol Med** 46: 1527–1533, 2009. doi:10.1016/j.freeradbiomed.2009.03.008.
537. Corcoran NM, Martin D, Hutter-Paier B, Windisch M, Nguyen T, Nheu L, Sundstrom LE, Costello AJ, Hovens CM. Sodium selenate specifically activates PP2A phosphatase, dephosphorylates tau and reverses memory deficits in an Alzheimer's disease model. **J Clin Neurosci** 17: 1025–1033, 2010. doi:10.1016/j.jocn.2010.04.020.
538. van Eersel J, Ke YD, Liu X, Delerue F, Kril JJ, Götz J, Ittner LM. Sodium selenate mitigates tau pathology, neurodegeneration, and functional deficits in Alzheimer's disease models. **Proc Natl Acad Sci USA** 107: 13888–13893, 2010. doi:10.1073/pnas.1009038107.
539. Song G, Zhang Z, Wen L, Chen C, Shi Q, Zhang Y, Ni J, Liu Q. Selenomethionine ameliorates cognitive decline, reduces tau hyperphosphorylation, and reverses synaptic deficit in the triple transgenic mouse model of Alzheimer's disease. **J Alzheimers Dis** 41: 85–99, 2014. doi:10.3233/JAD-131805.
540. Zhang ZH, Wu QY, Zheng R, Chen C, Chen Y, Liu Q, Hoffmann PR, Ni JZ, Song GL. Selenomethionine mitigates cognitive decline by targeting both tau hyperphosphorylation and autophagic clearance in an Alzheimer's disease mouse model. **J Neurosci** 37: 2449–2462, 2017. doi:10.1523/JNEUROSCI.3229-16.2017.
541. Yan JJ, Cho JY, Kim HS, Kim KL, Jung JS, Huh SO, Suh HW, Kim YH, Song DK. Protection against beta-amyloid peptide toxicity in vivo with long-term administration of ferulic acid. **Br J Pharmacol** 133: 89–96, 2001. doi:10.1038/sj.bjpp.0704047.
542. Perluigi M, Joshi G, Sultana R, Calabrese V, De Marco C, Coccia R, Cini C, Butterfield DA. In vivo protective effects of ferulic acid ethyl ester against amyloid-beta peptide 1-42-induced oxidative stress. **J Neurosci Res** 84: 418–426, 2006. doi:10.1002/jnr.20879.
543. Mori T, Koyama N, Guillot-Sestier MV, Tan J, Town T. Ferulic acid is a nutraceutical beta-secretase modulator that improves behavioral impairment and Alzheimer-like pathology in transgenic mice. **PLoS One** 8: e55774, 2013. doi:10.1371/journal.pone.0055774.

544. Yan JJ, Jung JS, Kim TK, Hasan MA, Hong CW, Nam JS, Song DK. Protective effects of ferulic acid in amyloid precursor protein plus presenilin-1 transgenic mouse model of Alzheimer disease. **Biol Pharm Bull** 36: 140–143, 2013. doi:[10.1248/bpb.b12-00798](https://doi.org/10.1248/bpb.b12-00798).
545. Mori T, Koyama N, Tan J, Segawa T, Maeda M, Town T. Combined treatment with the phenolics (-)-epigallocatechin-3-gallate and ferulic acid improves cognition and reduces Alzheimer-like pathology in mice. **J Biol Chem** 294: 2714–2731, 2019. doi:[10.1074/jbc.RA118.004280](https://doi.org/10.1074/jbc.RA118.004280).
546. Matsubara E, Bryant-Thomas T, Pacheco Quinto J, Henry TL, Poeggeler B, Herbert D, Cruz-Sanchez F, Chyan YJ, Smith MA, Perry G, Shoji M, Abe K, Leone A, Grundke-Ikbal I, Wilson GL, Ghiso J, Williams C, Refolo LM, Pappolla MA, Chain DG, Neria E. Melatonin increases survival and inhibits oxidative and amyloid pathology in a transgenic model of Alzheimer's disease. **J Neurochem** 85: 1101–1108, 2003. doi:[10.1046/j.1471-4159.2003.01654.x](https://doi.org/10.1046/j.1471-4159.2003.01654.x).
547. Feng Z, Chang Y, Cheng Y, Zhang BL, Qu ZW, Qin C, Zhang JT. Melatonin alleviates behavioral deficits associated with apoptosis and cholinergic system dysfunction in the APP 695 transgenic mouse model of Alzheimer's disease. **J Pineal Res** 37: 129–136, 2004. doi:[10.1111/j.1600-079X.2004.00144.x](https://doi.org/10.1111/j.1600-079X.2004.00144.x).
548. Gunasingh MJ, Philip JE, Ashok BS, Kirubakaran R, Jebaraj WC, Davis GD, Vignesh S, Dhondyuthapani S, Jayakumar R. Melatonin prevents amyloid protofibrillar induced oxidative imbalance and biogenic amine catabolism. **Life Sci** 83: 96–102, 2008. doi:[10.1016/j.lfs.2008.05.011](https://doi.org/10.1016/j.lfs.2008.05.011).
549. Shen YX, Wei W, Yang J, Liu C, Dong C, Xu SY. Improvement of melatonin to the learning and memory impairment induced by amyloid beta-peptide 25–35 in elder rats. **Acta Pharmacol Sin** 22: 797–803, 2001.
550. Shen YX, Xu SY, Wei W, Sun XX, Liu LH, Yang J, Dong C. The protective effects of melatonin from oxidative damage induced by amyloid beta-peptide 25–35 in middle-aged rats. **J Pineal Res** 32: 85–89, 2002. doi:[10.1034/j.1600-079x.2002.1819.x](https://doi.org/10.1034/j.1600-079x.2002.1819.x).
551. Ling ZQ, Tian Q, Wang L, Fu ZQ, Wang XC, Wang Q, Wang JZ. Constant illumination induces Alzheimer-like damages with endoplasmic reticulum involvement and the protection of melatonin. **J Alzheimers Dis** 16: 287–300, 2009. doi:[10.3233/JAD-2009-0949](https://doi.org/10.3233/JAD-2009-0949).
552. Yang X, Yang Y, Fu Z, Li Y, Feng J, Luo J, Zhang Q, Wang Q, Tian Q. Melatonin ameliorates Alzheimer-like pathological changes and spatial memory retention impairment induced by calyculin A. **J Psychopharmacol** 25: 1118–1125, 2011. doi:[10.1177/0269881110367723](https://doi.org/10.1177/0269881110367723).
553. Garcia-Mesa Y, Gimenez-Llort L, Lopez LC, Venegas C, Cristofol R, Escames G, Acuna-Castroviejo D, Sanfeliu C. Melatonin plus physical exercise are highly neuroprotective in the 3xTg-AD mouse. **Neurobiol Aging** 33: 1124.e13–1129, 2012. doi:[10.1016/j.neurobiolaging.2011.11.016](https://doi.org/10.1016/j.neurobiolaging.2011.11.016).
554. Yang X, Yang Y, Li G, Wang J, Yang ES. Coenzyme Q10 attenuates beta-amyloid pathology in the aged transgenic mice with Alzheimer presenilin 1 mutation. **J Mol Neurosci** 34: 165–171, 2008. doi:[10.1007/s12031-007-9033-7](https://doi.org/10.1007/s12031-007-9033-7).
555. Elipenahli C, Stack C, Jainuddin S, Gerges M, Yang L, Starkov A, Beal MF, Dumont M. Behavioral improvement after chronic administration of coenzyme Q10 in P301S transgenic mice. **J Alzheimers Dis** 28: 173–182, 2012. doi:[10.3233/JAD-2011-11190](https://doi.org/10.3233/JAD-2011-11190).
556. McManus MJ, Murphy MP, Franklin JL. The mitochondria-targeted antioxidant MitoQ prevents loss of spatial memory retention and early neuropathology in a transgenic mouse model of Alzheimer's disease. **J Neurosci** 31: 15703–15715, 2011. doi:[10.1523/JNEUROSCI.0552-11.2011](https://doi.org/10.1523/JNEUROSCI.0552-11.2011).
557. Young ML, Franklin JL. The mitochondria-targeted antioxidant MitoQ inhibits memory loss, neuropathology, and extends lifespan in aged 3xTg-AD mice. **Mol Cell Neurosci** 101: 103409, 2019. doi:[10.1016/j.mcn.2019.103409](https://doi.org/10.1016/j.mcn.2019.103409).
558. Lim GP, Chu T, Yang F, Beech W, Frautschy SA, Cole GM. The curry spice curcumin reduces oxidative damage and amyloid pathology in an Alzheimer transgenic mouse. **J Neurosci** 21: 8370–8377, 2001. doi:[10.1523/JNEUROSCI.21-21-08370.2001](https://doi.org/10.1523/JNEUROSCI.21-21-08370.2001).
559. Begum AN, Jones MR, Lim GP, Morihara T, Kim P, Heath DD, Rock CL, Pruitt MA, Yang F, Hudspeth B, Hu S, Faull KF, Teter B, Cole GM, Frautschy SA. Curcumin structure-function, bioavailability, and efficacy in models of neuroinflammation and Alzheimer's disease. **J Pharmacol Exp Ther** 326: 196–208, 2008. doi:[10.1124/jpet.108.137455](https://doi.org/10.1124/jpet.108.137455).
560. Garcia-Alloza M, Borrelli LA, Rozkalne A, Hyman BT, Bacskaí BJ. Curcumin labels amyloid pathology in vivo, disrupts existing plaques, and partially restores distorted neurites in an Alzheimer mouse model. **J Neurochem** 102: 1095–1104, 2007. doi:[10.1111/j.1471-4159.2007.04613.x](https://doi.org/10.1111/j.1471-4159.2007.04613.x).
561. Hamaguchi T, Ono K, Murase A, Yamada M. Phenolic compounds prevent Alzheimer's pathology through different effects on the amyloid-beta aggregation pathway. **Am J Pathol** 175: 2557–2565, 2009. doi:[10.2353/ajpath.2009.090417](https://doi.org/10.2353/ajpath.2009.090417).
562. Feng H-L, Dang H-ZD, Fan H, Chen X-P, Rao Y-X, Ren Y, Yang J-D, Shi J, Wang P-W, Tian J-Z. Curcumin ameliorates insulin signalling pathway in brain of Alzheimer's disease transgenic mice. **Int J Immunopathol Pharmacol** 29: 734–741, 2016. doi:[10.1177/0394632016659494](https://doi.org/10.1177/0394632016659494).
563. Feng HL, Fan H, Dang HZ, Chen XP, Ren Y, Yang JD, Wang PW. Neuroprotective effect of curcumin to Aβ of double transgenic mice with Alzheimer's disease. **Zhongguo Zhong Yao Za Zhi** 39: 3846–3849, 2014.
564. Wang P, Su C, Feng H, Chen X, Dong Y, Rao Y, Ren Y, Yang J, Shi J, Tian J, Jiang S. Curcumin regulates insulin pathways and glucose metabolism in the brains of APPswe/PS1dE9 mice. **Int J Immunopathol Pharmacol** 30: 25–43, 2017. doi:[10.1177/0394632016688025](https://doi.org/10.1177/0394632016688025).
565. Yanagisawa D, Ibrahim NF, Taguchi H, Morikawa S, Hirao K, Shirai N, Sogabe T, Tooyama I. Curcumin derivative with the substitution at C-4 position, but not curcumin, is effective against amyloid pathology in APP/PS1 mice. **Neurobiol Aging** 36: 201–210, 2015. doi:[10.1016/j.neurobiolaging.2014.07.041](https://doi.org/10.1016/j.neurobiolaging.2014.07.041).
566. Valera E, Dargusch R, Maher PA, Schubert D. Modulation of 5-lipoxygenase in proteotoxicity and Alzheimer's disease. **J Neurosci** 33: 10512–10525, 2013. doi:[10.1523/JNEUROSCI.5183-12.2013](https://doi.org/10.1523/JNEUROSCI.5183-12.2013).
567. Liu ZJ, Li ZH, Liu L, Tang WX, Wang Y, Dong MR, Xiao C. Curcumin attenuates beta-amyloid-induced neuroinflammation via activation of peroxisome proliferator-activated receptor-gamma function in a rat model of Alzheimer's disease. **Front Pharmacol** 7: 261, 2016. doi:[10.3389/fphar.2016.00261](https://doi.org/10.3389/fphar.2016.00261).
568. Parikh A, Kathawala K, Li J, Chen C, Shan Z, Cao X, Zhou XF, Garg S. Curcumin-loaded self-nanomicellizing solid dispersion system: part II: in vivo safety and efficacy assessment against behavior deficit in Alzheimer disease. **Drug Deliv Transl Res** 8: 1406–1420, 2018. doi:[10.1007/s13346-018-0570-0](https://doi.org/10.1007/s13346-018-0570-0).
569. Hagl S, Heinrich M, Kocher A, Schiborr C, Frank J, Eckert GP. Curcumin micelles improve mitochondrial function in a mouse model of Alzheimer's disease. **J Prev Alzheimers Dis** 1: 80–83, 2014. doi:[10.14283/jpad.2014.2](https://doi.org/10.14283/jpad.2014.2).
570. Fidelis EM, Savall AS, da Luz Abreu E, Carvalho F, Teixeira FE, Haas SE, Bazanella Sampaio T, Pinton S. Curcumin-loaded nanocapsules

- reverses the depressant-like behavior and oxidative stress induced by beta-amyloid in mice. **Neuroscience** 423: 122–130, 2019. doi:10.1016/j.neuroscience.2019.09.032.
571. Huang HC, Zheng BW, Guo Y, Zhao J, Zhao JY, Ma XW, Jiang ZF. Antioxidative and neuroprotective effects of curcumin in an Alzheimer's disease rat model co-treated with intracerebroventricular streptozotocin and subcutaneous D-galactose. **J Alzheimers Dis** 52: 899–911, 2016. doi:10.3233/JAD-150872.
572. Sadegh Malvajerdi S, Izadi Z, Azadi A, Kurd M, Derakhshankhah H, Sharifzadeh M, Akbari Javar H, Hamidi M. Neuroprotective potential of curcumin-loaded nanostructured lipid carrier in an animal model of Alzheimer's disease: behavioral and biochemical evidence. **J Alzheimers Dis** 69: 671–686, 2019. doi:10.3233/JAD-190083.
573. Arendash GW, Schleif W, Rezai-Zadeh K, Jackson EK, Zacharia LC, Cracchiolo JR, Shippy D, Tan J. Caffeine protects Alzheimer's mice against cognitive impairment and reduces brain beta-amyloid production. **Neuroscience** 142: 941–952, 2006. doi:10.1016/j.neuroscience.2006.07.021.
574. Dall'Igna OP, Fett P, Gomes MW, Souza DO, Cunha RA, Lara DR. Caffeine and adenosine A(2a) receptor antagonists prevent beta-amyloid (25-35)-induced cognitive deficits in mice. **Exp Neurol** 203: 241–245, 2007. doi:10.1016/j.expneurol.2006.08.008.
575. Arendash GW, Mori T, Cao C, Mamcarz M, Runfeldt M, Dickson A, Rezai-Zadeh K, Tane J, Citron BA, Lin X, Echeverria V, Potter H. Caffeine reverses cognitive impairment and decreases brain amyloid-beta levels in aged Alzheimer's disease mice. **J Alzheimers Dis** 17: 661–680, 2009. doi:10.3233/JAD-2009-1087.
576. Laurent C, Eddarkaoui S, Derisbourg M, Leboucher A, Demeyer D, Carrier S, Schneider M, Hamdane M, Müller CE, Buée L, Blum D. Beneficial effects of caffeine in a transgenic model of Alzheimer's disease-like tau pathology. **Neurobiol Aging** 35: 2079–2090, 2014. doi:10.1016/j.neurobiolaging.2014.03.027.
577. Stazi M, Lehmann S, Sakib MS, Pena-Centeno T, Büschgens L, Fischer A, Weggen S, Wirths O. Long-term caffeine treatment of Alzheimer mouse models ameliorates behavioural deficits and neuron loss and promotes cellular and molecular markers of neurogenesis. **Cell Mol Life Sci** 79: 55, 2021. doi:10.1007/s00018-021-04062-8.
578. Porquet D, Casadesús G, Bayod S, Vicente A, Canudas AM, Vilaplana J, Pelegrí C, Sanfeliu C, Camins A, Pallàs M, del Valle J. Dietary resveratrol prevents Alzheimer's markers and increases life span in SAMP8. **Age (Dordr)** 35: 1851–1865, 2013. doi:10.1007/s11357-012-9489-4.
579. Broderick TL, Rasool S, Li R, Zhang Y, Anderson M, Al-Nakkash L, Plochocki JH, Geetha T, Babu JR. Neuroprotective effects of chronic resveratrol treatment and exercise training in the 3xTg-AD mouse model of Alzheimer's disease. **Int J Mol Sci** 21: 7337, 2020. doi:10.3390/ijms21197337.
580. Huang TC, Lu KT, Wo YY, Wu YJ, Yang YL. Resveratrol protects rats from Abeta-induced neurotoxicity by the reduction of iNOS expression and lipid peroxidation. **PLoS One** 6: e29102, 2011. doi:10.1371/journal.pone.0029102.
581. Wang DM, Li SQ, Wu WL, Zhu XY, Wang Y, Yuan HY. Effects of long-term treatment with quercetin on cognition and mitochondrial function in a mouse model of Alzheimer's disease. **Neurochem Res** 39: 1533–1543, 2014. doi:10.1007/s11064-014-1343-x.
582. Sabogal-Guáqueta AM, Muñoz-Manco JI, Ramírez-Pineda JR, Lamprea-Rodríguez M, Osorio E, Cardona-Gómez GP. The flavonoid quercetin ameliorates Alzheimer's disease pathology and protects cognitive and emotional function in aged triple transgenic Alzheimer's disease model mice. **Neuropharmacology** 93: 134–145, 2015. doi:10.1016/j.neuropharm.2015.01.027.
583. Paula PC, Angelica Maria SG, Luis CH, Gloria Patricia CG. Preventive effect of quercetin in a triple transgenic Alzheimer's disease mice model. **Molecules** 24: 2287, 2019. doi:10.3390/molecules24122287.
584. Karimipour M, Rahbarghazi R, Tayefi H, Shimia M, Ghanadian M, Mahmoudi J, Bagheri HS. Quercetin promotes learning and memory performance concomitantly with neural stem/progenitor cell proliferation and neurogenesis in the adult rat dentate gyrus. **Int J Dev Neurosci** 74: 18–26, 2019. doi:10.1016/j.ijdevneu.2019.02.005.
585. Ebrahimpour S, Esmaeili A, Beheshti S. Effect of quercetin-conjugated superparamagnetic iron oxide nanoparticles on diabetes-induced learning and memory impairment in rats. **Int J Nanomedicine** 13: 6311–6324, 2018. doi:10.2147/IJN.S177871.
586. Sun D, Li N, Zhang W, Zhao Z, Mou Z, Huang D, Liu J, Wang W. Design of PLGA-functionalized quercetin nanoparticles for potential use in Alzheimer's disease. **Colloids Surf B Biointerfaces** 148: 116–129, 2016. doi:10.1016/j.colsurfb.2016.08.052.
587. Veinbergs I, Mallory M, Sagara Y, Masliah E. Vitamin E supplementation prevents spatial learning deficits and dendritic alterations in aged apolipoprotein E-deficient mice. **Eur J Neurosci** 12: 4541–4546, 2000.
588. Sung S, Yao Y, Uryu K, Yang H, Lee VM, Trojanowski JQ, Praticò D. Early vitamin E supplementation in young but not aged mice reduces Abeta levels and amyloid deposition in a transgenic model of Alzheimer's disease. **FASEB J** 18: 323–325, 2004. doi:10.1096/fj.03-0961je.
589. Jhoo JH, Kim HC, Nabeshima T, Yamada K, Shin EJ, Jhoo WK, Kim W, Kang KS, Jo SA, Woo JI. Beta-amyloid (1-42)-induced learning and memory deficits in mice: involvement of oxidative burdens in the hippocampus and cerebral cortex. **Behav Brain Res** 155: 185–196, 2004. doi:10.1016/j.bbr.2004.04.012.
590. Tiwari V, Kuhad A, Bishnoi M, Chopra K. Chronic treatment with tocotrienol, an isoform of vitamin E, prevents intracerebroventricular streptozotocin-induced cognitive impairment and oxidative-nitrosative stress in rats. **Pharmacol Biochem Behav** 93: 183–189, 2009. doi:10.1016/j.pbb.2009.05.009.
591. Figueiredo CP, Bicca MA, Latini A, Prediger RD, Medeiros R, Calixto JB. Folic acid plus alpha-tocopherol mitigates amyloid-beta-induced neurotoxicity through modulation of mitochondrial complexes activity. **J Alzheimers Dis** 24: 61–75, 2011. doi:10.3233/JAD-2010-101320.
592. Thakurta IG, Banerjee P, Bagh MB, Ghosh A, Sahoo A, Chattopadhyay S, Chakrabarti S. Combination of N-acetylcysteine, alpha-lipoic acid and alpha-tocopherol substantially prevents the brain synaptosomal alterations and memory and learning deficits of aged rats. **Exp Gerontol** 50: 19–25, 2014. doi:10.1016/j.exger.2013.11.008.
593. Thakurta IG, Chattopadhyay M, Ghosh A, Chakrabarti S. Dietary supplementation with N-acetyl cysteine, alpha-tocopherol and alpha-lipoic acid reduces the extent of oxidative stress and proinflammatory state in aged rat brain. **Biogerontology** 13: 479–488, 2012. doi:10.1007/s10522-012-9392-5.
594. Wang SW, Yang SG, Liu W, Zhang YX, Xu PX, Wang T, Ling TJ, Liu RT. Alpha-tocopherol quinine ameliorates spatial memory deficits by reducing beta-amyloid oligomers, neuroinflammation and oxidative stress in transgenic mice with Alzheimer's disease. **Behav Brain Res** 296: 109–117, 2016. doi:10.1016/j.bbr.2015.09.003.
595. Ibrahim NF, Yanagisawa D, Durani LW, Hamezah HS, Damanhuri HA, Wan Ngah WZ, Tsuji M, Kiuchi Y, Ono K, Tooyama I.

- Tocotrienol-rich fraction modulates amyloid pathology and improves cognitive function in AbetaPP/PS1 mice. **J Alzheimers Dis** 55: 597–612, 2017. doi:[10.3233/JAD-160685](https://doi.org/10.3233/JAD-160685).
596. Milgram NW, Zicker SC, Head E, Muggenburg BA, Murphey H, Ikeda-Douglas CJ, Cotman CW. Dietary enrichment counteracts age-associated cognitive dysfunction in canines. **Neurobiol Aging** 23: 737–745, 2002. doi:[10.1016/s0197-4580\(02\)00020-9](https://doi.org/10.1016/s0197-4580(02)00020-9).
597. Milgram NW, Head E, Zicker SC, Ikeda-Douglas CJ, Murphey H, Muggenburg B, Siwak C, Tapp D, Cotman CW. Learning ability in aged beagle dogs is preserved by behavioral enrichment and dietary fortification: a two-year longitudinal study. **Neurobiol Aging** 26: 77–90, 2005. doi:[10.1016/j.neurobiolaging.2004.02.014](https://doi.org/10.1016/j.neurobiolaging.2004.02.014).
598. Quinn JF, Bussiere JR, Hammond RS, Montine TJ, Henson E, Jones RE, Stackman RW Jr. Chronic dietary alpha-lipoic acid reduces deficits in hippocampal memory of aged Tg2576 mice. **Neurobiol Aging** 28: 213–225, 2007. doi:[10.1016/j.neurobiolaging.2005.12.014](https://doi.org/10.1016/j.neurobiolaging.2005.12.014).
599. Farr SA, Price TO, Banks WA, Ercal N, Morley JE. Effect of alpha-lipoic acid on memory, oxidation, and lifespan in SAMP8 mice. **J Alzheimers Dis** 32: 447–455, 2012. doi:[10.3233/JAD-2012-120130](https://doi.org/10.3233/JAD-2012-120130).
600. Sancheti H, Kanamori K, Patil I, Díaz Brinton R, Ross BD, Cadenas E. Reversal of metabolic deficits by lipoic acid in a triple transgenic mouse model of Alzheimer's disease: a ¹³C NMR study. **J Cereb Blood Flow Metab** 34: 288–296, 2014. doi:[10.1038/jcbfm.2013.196](https://doi.org/10.1038/jcbfm.2013.196).
601. Hou TT, Yang HY, Wang W, Wu QQ, Tian YR, Jia JP. Sulforaphane Inhibits the generation of amyloid-beta oligomer and promotes spatial learning and memory in Alzheimer's disease (PS1V97L) transgenic mice. **J Alzheimers Dis** 62: 1803–1813, 2018. doi:[10.3233/JAD-171110](https://doi.org/10.3233/JAD-171110).
602. Wang W, Wei C, Quan M, Li T, Jia J. Sulforaphane reverses the amyloid-beta oligomers induced depressive-like behavior. **J Alzheimers Dis** 78: 127–137, 2020. doi:[10.3233/JAD-200397](https://doi.org/10.3233/JAD-200397).
603. Zhang R, Miao QW, Zhu CX, Zhao Y, Liu L, Yang J, An L. Sulforaphane ameliorates neurobehavioral deficits and protects the brain from amyloid beta deposits and peroxidation in mice with Alzheimer-like lesions. **Am J Alzheimers Dis Other Dement** 30: 183–191, 2015. doi:[10.1177/1533317514542645](https://doi.org/10.1177/1533317514542645).
604. Zhang R, Zhang J, Fang L, Li X, Zhao Y, Shi W, An L. Neuroprotective effects of sulforaphane on cholinergic neurons in mice with Alzheimer's disease-like lesions. **Int J Mol Sci** 15: 14396–14410, 2014. doi:[10.3390/ijms150814396](https://doi.org/10.3390/ijms150814396).
605. Zhang JZ, Zhang R, Zhan ZP, Li XH, Zhou FY, Xing AP, Jiang CM, Chen YQ, An L. Beneficial effects of sulforaphane treatment in Alzheimer's disease may be mediated through reduced HDAC1/3 and increased P75NTR expression. **Front Aging Neurosci** 9: 121, 2017. doi:[10.3389/fnagi.2017.00121](https://doi.org/10.3389/fnagi.2017.00121).
606. Kim HV, Kim HY, Ehrlich HY, Choi SY, Kim DJ, Kim Y. Amelioration of Alzheimer's disease by neuroprotective effect of sulforaphane in animal model. **Amyloid** 20: 7–12, 2013. doi:[10.3109/13506129.2012.751367](https://doi.org/10.3109/13506129.2012.751367).
607. Lee S, Choi BR, Kim J, LaFerla FM, Park JH, Han JS, Lee KW, Kim J. Sulforaphane upregulates the heat shock protein co-chaperone CHIP and clears amyloid-beta and tau in a mouse model of Alzheimer's disease. **Mol Nutr Food Res** 62: e1800240, 2018. doi:[10.1002/mnfr.201800240](https://doi.org/10.1002/mnfr.201800240).
608. Bahn G, Park JS, Yun UJ, Lee YJ, Choi Y, Park JS, Baek SH, Choi BY, Cho YS, Kim HK, Han J, Sul JH, Baik SH, Lim J, Wakabayashi N, Bae SH, Han JW, Arumugam TV, Mattson MP, Jo DG. NRF2/ARE pathway negatively regulates BACE1 expression and ameliorates cognitive deficits in mouse Alzheimer's models. **Proc Natl Acad Sci USA** 116: 12516–12523, 2019. doi:[10.1073/pnas.1819541116](https://doi.org/10.1073/pnas.1819541116).
609. Rezai-Zadeh K, Arendash GW, Hou H, Fernandez F, Jensen M, Runfeldt M, Shytle RD, Tan J. Green tea epigallocatechin-3-gallate (EGCG) reduces beta-amyloid mediated cognitive impairment and modulates tau pathology in Alzheimer transgenic mice. **Brain Res** 1214: 177–187, 2008. doi:[10.1016/j.brainres.2008.02.107](https://doi.org/10.1016/j.brainres.2008.02.107).
610. Lee YJ, Choi DY, Yun YP, Han SB, Oh KW, Hong JT. Epigallocatechin-3-gallate prevents systemic inflammation-induced memory deficiency and amyloidogenesis via its anti-neuroinflammatory properties. **J Nutr Biochem** 24: 298–310, 2013. doi:[10.1016/j.jnutbio.2012.06.011](https://doi.org/10.1016/j.jnutbio.2012.06.011).
611. Biasibetti R, Tramontina AC, Costa AP, Dutra MF, Quincozes-Santos A, Nardin P, Bernardi CL, Wartchow KM, Lunardi PS, Goncalves CA. Green tea (-)epigallocatechin-3-gallate reverses oxidative stress and reduces acetylcholinesterase activity in a streptozotocin-induced model of dementia. **Behav Brain Res** 236: 186–193, 2013. doi:[10.1016/j.bbr.2012.08.039](https://doi.org/10.1016/j.bbr.2012.08.039).
612. Jia N, Han K, Kong JJ, Zhang XM, Sha S, Ren GR, Cao YP. (-)-Epigallocatechin-3-gallate alleviates spatial memory impairment in APP/PS1 mice by restoring IRS-1 signaling defects in the hippocampus. **Mol Cell Biochem** 380: 211–218, 2013. doi:[10.1007/s11010-013-1675-x](https://doi.org/10.1007/s11010-013-1675-x).
613. Chang X, Rong C, Chen Y, Yang C, Hu Q, Mo Y, Zhang C, Gu X, Zhang L, He W, Cheng S, Hou X, Su R, Liu S, Dun W, Wang Q, Fang S. (-)-Epigallocatechin-3-gallate attenuates cognitive deterioration in Alzheimer's disease model mice by upregulating neprilysin expression. **Exp Cell Res** 334: 136–145, 2015. doi:[10.1016/j.yexcr.2015.04.004](https://doi.org/10.1016/j.yexcr.2015.04.004).
614. Sultana R, Newman S, Mohammad-Abdul H, Keller JN, Butterfield DA. Protective effect of the xanthate, D609, on Alzheimer's amyloid beta-peptide (1-42)-induced oxidative stress in primary neuronal cells. **Free Radic Res** 38: 449–458, 2004. doi:[10.1080/1071576042000206478](https://doi.org/10.1080/1071576042000206478).
615. Sultana R, Newman SF, Abdul HM, Cai J, Pierce WM, Klein JB, Merchant M, Butterfield DA. Protective effect of D609 against amyloid-beta1-42-induced oxidative modification of neuronal proteins: redox proteomics study. **J Neurosci Res** 84: 409–417, 2006. doi:[10.1002/jnr.20876](https://doi.org/10.1002/jnr.20876).
616. Joshi G, Sultana R, Perluigi M, Butterfield DA. In vivo protection of synaptosomes from oxidative stress mediated by Fe²⁺/H₂O₂ or 2,2-azobis-(2-amidinopropane) dihydrochloride by the glutathione mimetic tricyclodecan-9-yl-xanthogenate. **Free Radic Biol Med** 38: 1023–1031, 2005. doi:[10.1016/j.freeradbiomed.2004.12.027](https://doi.org/10.1016/j.freeradbiomed.2004.12.027).
617. Loeff M, Schrauzer GN, Walach H. Selenium and Alzheimer's disease: a systematic review. **J Alzheimers Dis** 26: 81–104, 2011. doi:[10.3233/JAD-2011-110414](https://doi.org/10.3233/JAD-2011-110414).
618. Pinton S, Brüning CA, Oliveira CE, Prigol M, Nogueira CW. Therapeutic effect of organoselenium dietary supplementation in a sporadic dementia of Alzheimer's type model in rats. **J Nutr Biochem** 24: 311–317, 2013. doi:[10.1016/j.jnutbio.2012.06.012](https://doi.org/10.1016/j.jnutbio.2012.06.012).
619. Sultana R, Ravagna A, Mohammad-Abdul H, Calabrese V, Butterfield DA. Ferulic acid ethyl ester protects neurons against amyloid beta-peptide(1-42)-induced oxidative stress and neurotoxicity: relationship to antioxidant activity. **J Neurochem** 92: 749–758, 2005. doi:[10.1111/j.1471-4159.2004.02899.x](https://doi.org/10.1111/j.1471-4159.2004.02899.x).
620. Kanski J, Aksenova M, Stoyanova A, Butterfield DA. Ferulic acid antioxidant protection against hydroxyl and peroxy radical oxidation in synaptosomal and neuronal cell culture systems in vitro:

- structure-activity studies. **J Nutr Biochem** 13: 273–281, 2002. doi:[10.1016/s0955-2863\(01\)00215-7](https://doi.org/10.1016/s0955-2863(01)00215-7).
621. Scapagnini G, Butterfield DA, Colombrina C, Sultana R, Pascale A, Calabrese V. Ethyl ferulate, a lipophilic polyphenol, induces HO-1 and protects rat neurons against oxidative stress. **Antioxid Redox Signal** 6: 811–818, 2004. doi:[10.1089/ars.2004.6.811](https://doi.org/10.1089/ars.2004.6.811).
622. Trombino S, Cassano R, Ferrarelli T, Barone E, Picci N, Mancuso C. Trans-ferulic acid-based solid lipid nanoparticles and their antioxidant effect in rat brain microsomes. **Colloids Surf B Biointerfaces** 109: 273–279, 2013. doi:[10.1016/j.colsurfb.2013.04.005](https://doi.org/10.1016/j.colsurfb.2013.04.005).
623. Ali T, Kim MO. Melatonin ameliorates amyloid beta-induced memory deficits, tau hyperphosphorylation and neurodegeneration via PI3/Akt/GSK3beta pathway in the mouse hippocampus. **J Pineal Res** 59: 47–59, 2015. doi:[10.1111/jpi.12238](https://doi.org/10.1111/jpi.12238).
624. Beal MF. Mitochondrial dysfunction and oxidative damage in Alzheimer's and Parkinson's diseases and coenzyme Q10 as a potential treatment. **J Bioenerg Biomembr** 36: 381–386, 2004. doi:[10.1023/B:JOBB.0000041772.74810.92](https://doi.org/10.1023/B:JOBB.0000041772.74810.92).
625. Noori T, Dehpour AR, Sureda A, Sobarzo-Sanchez E, Shirooie S. Role of natural products for the treatment of Alzheimer's disease. **Eur J Pharmacol** 898: 173974, 2021. doi:[10.1016/j.ejphar.2021.173974](https://doi.org/10.1016/j.ejphar.2021.173974).
626. Silvestro S, Sindona C, Bramanti P, Mazzon E. A state of the art of antioxidant properties of curcuminoids in neurodegenerative diseases. **Int J Mol Sci** 22: 3168, 2021. doi:[10.3390/ijms22063168](https://doi.org/10.3390/ijms22063168).
627. Parikh A, Kathawala K, Song Y, Zhou XF, Garg S. Curcumin-loaded self-nanomicellizing solid dispersion system: part I: development, optimization, characterization, and oral bioavailability. **Drug Deliv Transl Res** 8: 1389–1405, 2018. doi:[10.1007/s13346-018-0543-3](https://doi.org/10.1007/s13346-018-0543-3).
628. Zhang F, Wang H, Wu Q, Lu Y, Nie J, Xie X, Shi J. Resveratrol protects cortical neurons against microglia-mediated neuroinflammation. **Phytother Res** 27: 344–349, 2013. doi:[10.1002/ptr.4734](https://doi.org/10.1002/ptr.4734).
629. Ansari MA, Abdul HM, Joshi G, Opii WO, Butterfield DA. Protective effect of quercetin in primary neurons against Abeta(1-42): relevance to Alzheimer's disease. **J Nutr Biochem** 20: 269–275, 2009. doi:[10.1016/j.jnutbio.2008.03.002](https://doi.org/10.1016/j.jnutbio.2008.03.002).
630. Katebi S, Esmaeili A, Ghaedi K, Zarrabi A. Superparamagnetic iron oxide nanoparticles combined with NGF and quercetin promote neuronal branching morphogenesis of PC12 cells. **Int J Nanomedicine** 14: 2157–2169, 2019. doi:[10.2147/IJN.S191878](https://doi.org/10.2147/IJN.S191878).
631. Ebrahimpour S, Zakeri M, Esmaeili A. Crosstalk between obesity, diabetes, and Alzheimer's disease: Introducing quercetin as an effective triple herbal medicine. **Ageing Res Rev** 62: 101095, 2020. doi:[10.1016/j.arr.2020.101095](https://doi.org/10.1016/j.arr.2020.101095).
632. Dhawan S, Kapil R, Singh B. Formulation development and systematic optimization of solid lipid nanoparticles of quercetin for improved brain delivery. **J Pharm Pharmacol** 63: 342–351, 2011. doi:[10.1111/j.2042-7158.2010.01225.x](https://doi.org/10.1111/j.2042-7158.2010.01225.x).
633. Moreno L, Puerta E, Suárez-Santiago JE, Santos-Magalhães NS, Ramirez MJ, Irache JM. Effect of the oral administration of nanoencapsulated quercetin on a mouse model of Alzheimer's disease. **Int J Pharm** 517: 50–57, 2017. doi:[10.1016/j.ijpharm.2016.11.061](https://doi.org/10.1016/j.ijpharm.2016.11.061).
634. Yatin SM, Varadarajan S, Butterfield DA. Vitamin E prevents Alzheimer's amyloid beta-peptide (1-42)-induced neuronal protein oxidation and reactive oxygen species production. **J Alzheimers Dis** 2: 123–131, 2000. doi:[10.3233/jad-2000-2212](https://doi.org/10.3233/jad-2000-2212).
635. Qi XL, Xiu J, Shan KR, Xiao Y, Gu R, Liu RY, Guan ZZ. Oxidative stress induced by beta-amyloid peptide(1-42) is involved in the altered composition of cellular membrane lipids and the decreased expression of nicotinic receptors in human SH-SY5Y neuroblastoma cells. **Neurochem Int** 46: 613–621, 2005. doi:[10.1016/j.neuint.2005.02.007](https://doi.org/10.1016/j.neuint.2005.02.007).
636. Gugliandolo A, Bramanti P, Mazzon E. Role of Vitamin E in the treatment of Alzheimer's disease: evidence from animal models. **Int J Mol Sci** 18: 2504, 2017. doi:[10.3390/ijms18122504](https://doi.org/10.3390/ijms18122504).
637. Yao Y, Chinnici C, Tang H, Trojanowski JQ, Lee VM, Praticò D. Brain inflammation and oxidative stress in a transgenic mouse model of Alzheimer-like brain amyloidosis. **J Neuroinflammation** 1: 21, 2004. doi:[10.1186/1742-2094-1-21](https://doi.org/10.1186/1742-2094-1-21).
638. Niki E. Interaction of ascorbate and alpha-tocopherol. **Ann NY Acad Sci** 498: 186–199, 1987. doi:[10.1111/j.1749-6632.1987.tb23761.x](https://doi.org/10.1111/j.1749-6632.1987.tb23761.x).
639. Keum YS. Regulation of the Keap1/Nrf2 system by chemopreventive sulforaphane: implications of posttranslational modifications. **Ann NY Acad Sci** 1229: 184–189, 2011. doi:[10.1111/j.1749-6632.2011.06092.x](https://doi.org/10.1111/j.1749-6632.2011.06092.x).
640. Schepici G, Bramanti P, Mazzon E. Efficacy of sulforaphane in neurodegenerative diseases. **Int J Mol Sci** 21: 8637, 2020. doi:[10.3390/ijms21228637](https://doi.org/10.3390/ijms21228637).
641. Singh NA, Mandal AK, Khan ZA. Potential neuroprotective properties of epigallocatechin-3-gallate (EGCG). **Nutr J** 15: 60, 2016. doi:[10.1186/s12937-016-0179-4](https://doi.org/10.1186/s12937-016-0179-4).
642. He M, Liu MY, Wang S, Tang QS, Yao WF, Zhao HS, Wei MJ. Research on EGCG improving the degenerative changes of the brain in AD model mice induced with chemical drugs. **Zhong Yao Cai** 35: 1641–1644, 2012.
643. Smith A, Giunta B, Bickford PC, Fountain M, Tan J, Shytle RD. Nanolipidic particles improve the bioavailability and alpha-secretase inducing ability of epigallocatechin-3-gallate (EGCG) for the treatment of Alzheimer's disease. **Int J Pharm** 389: 207–212, 2010. doi:[10.1016/j.ijpharm.2010.01.012](https://doi.org/10.1016/j.ijpharm.2010.01.012).
644. Chan A, Paskavitz J, Remington R, Rasmussen S, Shea TB. Efficacy of a vitamin/nutriceutical formulation for early-stage Alzheimer's disease: a 1-year, open-label pilot study with an 16-month caregiver extension. **Am J Alzheimers Dis Other Dement** 23: 571–585, 2008. doi:[10.1177/1533317508325093](https://doi.org/10.1177/1533317508325093).
645. Remington R, Bechtel C, Larsen D, Samar A, Doshanjh L, Fishman P, Luo Y, Smyers K, Page R, Morrell C, Shea TB. A phase II randomized clinical trial of a nutritional formulation for cognition and mood in Alzheimer's disease. **J Alzheimers Dis** 45: 395–405, 2015. doi:[10.3233/JAD-142499](https://doi.org/10.3233/JAD-142499).
646. Remington R, Bechtel C, Larsen D, Samar A, Page R, Morrell C, Shea TB. Maintenance of cognitive performance and mood for individuals with Alzheimer's disease following consumption of a nutraceutical formulation: a one-year, open-label study. **J Alzheimers Dis** 51: 991–995, 2016. doi:[10.3233/JAD-151098](https://doi.org/10.3233/JAD-151098).
647. Sano M, Ernesto C, Thomas RG, Klauber MR, Schafer K, Grundman M, Woodbury P, Growdon J, Cotman DW, Pfeiffer E, Schneider LS, Thal LJ. A controlled trial of selegiline, alpha-tocopherol, or both as treatment for Alzheimer's disease. **N Engl J Med** 336: 1216–1222, 1997. doi:[10.1056/NEJM199704243361704](https://doi.org/10.1056/NEJM199704243361704).
648. Zandi PP, Anthony JC, Khachaturian AS, Stone SV, Gustafson D, Tschanz JT, Norton MC, Welsh-Bohmer KA, Breitner JC; Cache County Study Group. Reduced risk of Alzheimer disease in users of antioxidant vitamin supplements: the Cache County Study. **Arch Neurol** 61: 82–88, 2004. doi:[10.1001/archneur.61.1.82](https://doi.org/10.1001/archneur.61.1.82).

649. Galasko DR, Peskind E, Clark CM, Quinn JF, Ringman JM, Jicha GA, Cotman C, Cottrell B, Montine TJ, Thomas RG, Aisen P; Alzheimer's Disease Cooperative Study. Antioxidants for Alzheimer disease: a randomized clinical trial with cerebrospinal fluid biomarker measures. **Arch Neurol** 69: 836–841, 2012. doi:10.1001/archneurol.2012.85.
650. Lloret A, Badia MC, Mora NJ, Pallardó FV, Alonso MD, Viña J. Vitamin E paradox in Alzheimer's disease: it does not prevent loss of cognition and may even be detrimental. **J Alzheimers Dis** 17: 143–149, 2009. doi:10.3233/JAD-2009-1033.
651. Dysken MW, Guarino PD, Vertrees JE, Asthana S, Sano M, Llorente M, et al. Vitamin E and memantine in Alzheimer's disease: clinical trial methods and baseline data. **Alzheimers Dement** 10: 36–44, 2014. doi:10.1016/j.jalz.2013.01.014.
652. Dysken MW, Sano M, Asthana S, Vertrees JE, Pallaki M, Llorente M, et al. Effect of vitamin E and memantine on functional decline in Alzheimer disease: the TEAM-AD VA cooperative randomized trial. **JAMA** 311: 33–44, 2014. doi:10.1001/jama.2013.282834.
653. Petersen RC, Thomas RG, Grundman M, Bennett D, Doody R, Ferris S, Galasko D, Jin S, Kaye J, Levey A, Pfeiffer E, Sano M, van Dyck CH, Thal LJ; Alzheimer's Disease Cooperative Study Group. Vitamin E and donepezil for the treatment of mild cognitive impairment. **N Engl J Med** 352: 2379–2388, 2005. doi:10.1056/NEJMoa050151.
654. Kryscio RJ, Abner EL, Caban-Holt A, Lovell M, Goodman P, Darke AK, Yee M, Crowley J, Schmitt FA. Association of antioxidant supplement use and dementia in the Prevention of Alzheimer's Disease by Vitamin E and Selenium Trial (PREADVISE). **JAMA Neurol** 74: 567–573, 2017. doi:10.1001/jamaneurol.2016.5778.
655. Ding XH, Kryscio RJ, Turner J, Jicha GA, Cooper G, Caban-Holt A, Schmitt FA, Abner EL. Self-reported sleep apnea and dementia risk: findings from the Prevention of Alzheimer's Disease with Vitamin E and Selenium Trial. **J Am Geriatr Soc** 64: 2472–2478, 2016. doi:10.1111/jgs.14393.
656. Abner EL, Kryscio RJ, Caban-Holt AM, Schmitt FA. Baseline subjective memory complaints associate with increased risk of incident dementia: the PREADVISE trial. **J Prev Alzheimers Dis** 2: 11–16, 2015.
657. Naeini AM, Elmadfa I, Djazayeri A, Barekatain M, Ghazvini MR, Djalali M, Feizi A. The effect of antioxidant vitamins E and C on cognitive performance of the elderly with mild cognitive impairment in Isfahan, Iran: a double-blind, randomized, placebo-controlled trial. **Eur J Nutr** 53: 1255–1262, 2014. doi:10.1007/s00394-013-0628-1.
658. Bianchetti A, Rozzini R, Trabucchi M. Effects of acetyl-L-carnitine in Alzheimer's disease patients unresponsive to acetylcholinesterase inhibitors. **Curr Med Res Opin** 19: 350–353, 2003. doi:10.1080/03007995.2020.12102584.
659. Kim JW, Byun MS, Yi D, Lee JH, Jeon SY, Jung G, Lee HN, Sohn BK, Lee JY, Kim YK, Shin SA, Sohn CH, Lee DY; KBASE Research Group. Coffee intake and decreased amyloid pathology in human brain. **Transl Psychiatry** 9: 270, 2019. doi:10.1038/s41398-019-0604-5.
660. Baum L, Lam CW, Cheung SK, Kwok T, Lui V, Tsoh J, Lam L, Leung V, Hui E, Ng C, Woo J, Chiu HF, Goggins WB, Zee BC, Cheng KF, Fong CY, Wong A, Mok H, Chow MS, Ho PC, Ip SP, Ho CS, Yu XW, Lai CY, Chan MH, Szeto S, Chan IH, Mok V. Six-month randomized, placebo-controlled, double-blind, pilot clinical trial of curcumin in patients with Alzheimer disease. **J Clin Psychopharmacol** 28: 110–113, 2008. doi:10.1097/jcp.0b013e318160862c.
661. Baum L, Cheung SK, Mok VC, Lam LC, Leung VP, Hui E, Ng CC, Chow M, Ho PC, Lam S, Woo J, Chiu HF, Goggins W, Zee B, Wong A, Mok H, Cheng WK, Fong C, Lee JS, Chan MH, Szeto SS, Lui VW, Tsoh J, Kwok TC, Chan IH, Lam CW. Curcumin effects on blood lipid profile in a 6-month human study. **Pharmacol Res** 56: 509–514, 2007. doi:10.1016/j.phrs.2007.09.013.
662. Small GW, Siddarth P, Li Z, Miller KJ, Ercoli L, Emerson ND, Martinez J, Wong KP, Liu J, Merrill DA, Chen ST, Henning SM, Satyamurthy N, Huang SC, Heber D, Barrio JR. Memory and brain amyloid and tau effects of a bioavailable form of curcumin in non-demented adults: a double-blind, placebo-controlled 18-month trial. **Am J Geriatr Psychiatry** 26: 266–277, 2018. doi:10.1016/j.jagp.2017.10.010.
663. Ringman JM, Frautschy SA, Teng E, Begum AN, Bardens J, Beigi M, Gyls KH, Badmaev V, Heath DD, Apostolova LG, Porter V, Vanek Z, Marshall GA, Hellemann G, Sugar C, Masterman DL, Montine TJ, Cummings JL, Cole GM. Oral curcumin for Alzheimer's disease: tolerability and efficacy in a 24-week randomized, double blind, placebo-controlled study. **Alzheimers Res Ther** 4: 43, 2012. doi:10.1186/alzrt146.
664. Cox KH, Pipingas A, Scholey AB. Investigation of the effects of solid lipid curcumin on cognition and mood in a healthy older population. **J Psychopharmacol** 29: 642–651, 2015. doi:10.1177/0269881114552744.
665. Zhu CW, Grossman H, Neugroschl J, Parker S, Burden A, Luo X, Sano M. A randomized, double-blind, placebo-controlled trial of resveratrol with glucose and malate (RGM) to slow the progression of Alzheimer's disease: a pilot study. **Alzheimers Dement (NY)** 4: 609–616, 2018. doi:10.1016/j.trci.2018.09.009.
666. Moussa C, Hebron M, Huang X, Ahn J, Rissman RA, Aisen PS, Turner RS. Resveratrol regulates neuro-inflammation and induces adaptive immunity in Alzheimer's disease. **J Neuroinflammation** 14: 1, 2017. doi:10.1186/s12974-016-0779-0.
667. Ide K, Yamada H, Takuma N, Kawasaki Y, Harada S, Nakase J, Ukawa Y, Sagesaka YM. Effects of green tea consumption on cognitive dysfunction in an elderly population: a randomized placebo-controlled study. **Nutr J** 15: 49, 2016. doi:10.1186/s12937-016-0168-7.
668. Lloret A, Esteve D, Monllor P, Cervera-Ferri A, Lloret A. The effectiveness of vitamin E treatment in Alzheimer's disease. **Int J Mol Sci** 20, 2019. doi:10.3390/ijms20040879.
669. Montgomery SA, Thal LJ, Amrein R. Meta-analysis of double blind randomized controlled clinical trials of acetyl-L-carnitine versus placebo in the treatment of mild cognitive impairment and mild Alzheimer's disease. **Int Clin Psychopharmacol** 18: 61–71, 2003. doi:10.1097/00004850-200303000-00001.
670. Wightman EL, Haskell CF, Forster JS, Veasey RC, Kennedy DO. Epigallocatechin gallate, cerebral blood flow parameters, cognitive performance and mood in healthy humans: a double-blind, placebo-controlled, crossover investigation. **Hum Psychopharmacol** 27: 177–186, 2012. doi:10.1002/hup.1263.
671. Scholey A, Downey LA, Ciorciari J, Pipingas A, Nolidin K, Finn M, Wines M, Catchlove S, Terrens A, Barlow E, Gordon L, Stough C. Acute neurocognitive effects of epigallocatechin gallate (EGCG). **Appetite** 58: 767–770, 2012. doi:10.1016/j.appet.2011.11.016.
672. Vemuri P, Lesnick TG, Przybelski SA, Knopman DS, Roberts RO, Lowe VJ, Kantarci K, Senjem ML, Gunter JL, Boeve BF, Petersen RC, Jack CR Jr. Effect of lifestyle activities on Alzheimer disease biomarkers and cognition. **Ann Neurol** 72: 730–738, 2012. doi:10.1002/ana.23665.
673. Liyanage SI, Vilekar P, Weaver DF. Nutrients in Alzheimer's disease: the interaction of diet, drugs and disease. **Can J Neurol Sci** 46: 23–34, 2019. doi:10.1017/cjn.2018.353.

674. Viña J, Sanz-Ros J. Alzheimer's disease: only prevention makes sense. **Eur J Clin Invest** 48: e13005, 2018. doi:[10.1111/eci.13005](https://doi.org/10.1111/eci.13005).
675. Ngandu T, Lehtisalo J, Solomon A, Levälähti E, Ahtiluoto S, Antikainen R, Bäckman L, Hänninen T, Jula A, Laatikainen T, Lindstrom J, Mangialasche F, Paajanen T, Pajala S, Peltonen M, Rauramaa R, Stigsdotter-Neely A, Strandberg T, Tuomilehto J, Soininen H, Kivipelto M. A 2 year multidomain intervention of diet, exercise, cognitive training, and vascular risk monitoring versus control to prevent cognitive decline in at-risk elderly people (FINGER): a randomised controlled trial. **Lancet** 385: 2255–2263, 2015. doi:[10.1016/S0140-6736\(15\)60461-5](https://doi.org/10.1016/S0140-6736(15)60461-5).
676. Soininen H, Solomon A, Visser PJ, Hendrix SB, Blennow K, Kivipelto M, Hartmann T; LipiDiDiet clinical study group. 24-Month intervention with a specific multinutrient in people with prodromal Alzheimer's disease (LipiDiDiet): a randomised, double-blind, controlled trial. **Lancet Neurol** 16: 965–975, 2017. doi:[10.1016/S1474-4422\(17\)30332-0](https://doi.org/10.1016/S1474-4422(17)30332-0).
677. Moll van Charante EP, Richard E, Eurelings LS, van Dalen JW, Ligthart SA, van Bussel EF, Hoevenaer-Blom MP, Vermeulen M, van Gool WA. Effectiveness of a 6-year multidomain vascular care intervention to prevent dementia (preDIVA): a cluster-randomised controlled trial. **Lancet** 388: 797–805, 2016. doi:[10.1016/S0140-6736\(16\)30950-3](https://doi.org/10.1016/S0140-6736(16)30950-3).
678. Grosso G, Mistretta A, Frigiola A, Gruttadauria S, Biondi A, Basile F, Vitaglione P, D'Orazio N, Galvano F. Mediterranean diet and cardiovascular risk factors: a systematic review. **Crit Rev Food Sci Nutr** 54: 593–610, 2014. doi:[10.1080/10408398.2011.596955](https://doi.org/10.1080/10408398.2011.596955).
679. Solfrizzi V, Frisardi V, Seripa D, Logroscino G, Imbimbo BP, D'Onofrio G, Addante F, Sancarlo D, Cascavilla L, Pilotto A, Panza F. Mediterranean diet in predementia and dementia syndromes. **Curr Alzheimer Res** 8: 520–542, 2011. doi:[10.2174/156720511796391809](https://doi.org/10.2174/156720511796391809).
680. Knight A, Bryan J, Murphy K. Is the Mediterranean diet a feasible approach to preserving cognitive function and reducing risk of dementia for older adults in Western countries? New insights and future directions. **Ageing Res Rev** 25: 85–101, 2016. doi:[10.1016/j.arr.2015.10.005](https://doi.org/10.1016/j.arr.2015.10.005).
681. Scarmeas N, Stern Y, Mayeux R, Luchsinger JA. Mediterranean diet, Alzheimer disease, and vascular mediation. **Arch Neurol** 63: 1709–1717, 2006. doi:[10.1001/archneur.63.12.noc60109](https://doi.org/10.1001/archneur.63.12.noc60109).
682. Trichopoulou A, Costacou T, Bamia C, Trichopoulos D. Adherence to a Mediterranean diet and survival in a Greek population. **N Engl J Med** 348: 2599–2608, 2003. doi:[10.1056/NEJMoa025039](https://doi.org/10.1056/NEJMoa025039).
683. Solfrizzi V, Panza F. Mediterranean diet and cognitive decline: a lesson from the whole-diet approach: what challenges lie ahead? **J Alzheimers Dis** 39: 283–286, 2014. doi:[10.3233/JAD-130831](https://doi.org/10.3233/JAD-130831).
684. Gardener S, Gu Y, Rainey-Smith SR, Keogh JB, Clifton PM, Mathieson SL, Taddei K, Mondal A, Ward VK, Scarmeas N, Barnes M, Ellis KA, Head R, Masters CL, Ames D, Macaulay SL, Rowe CC, Szoek C, Martins RN; AIBL Research Group. Adherence to a Mediterranean diet and Alzheimer's disease risk in an Australian population. **Transl Psychiatry** 2: e164, 2012. doi:[10.1038/tp.2012.91](https://doi.org/10.1038/tp.2012.91).
685. Scarmeas N, Stern Y, Tang MX, Mayeux R, Luchsinger JA. Mediterranean diet and risk for Alzheimer's disease. **Ann Neurol** 59: 912–921, 2006. doi:[10.1002/ana.20854](https://doi.org/10.1002/ana.20854).
686. Gu YA, Luchsinger JA, Stern Y, Scarmeas N. Mediterranean diet, inflammatory and metabolic biomarkers, and risk of Alzheimer's disease. **J Alzheimers Dis** 22: 483–492, 2010. doi:[10.3233/JAD-2010-100897](https://doi.org/10.3233/JAD-2010-100897).
687. Morris MC, Tangney CC, Wang YM, Sacks FM, Barnes LL, Bennett DA, Aggarwal NT. MIND diet slows cognitive decline with aging. **Alzheimers Dement** 11: 1015–1022, 2015. doi:[10.1016/j.jalz.2015.04.011](https://doi.org/10.1016/j.jalz.2015.04.011).
688. Scarmeas N, Stern Y, Mayeux R, Manly JJ, Schupf N, Luchsinger JA. Mediterranean diet and mild cognitive impairment. **Arch Neurol** 66: 216–225, 2009. doi:[10.1001/archneurol.2008.536](https://doi.org/10.1001/archneurol.2008.536).
689. Féart C, Samieri C, Rondeau V, Amieva H, Portet F, Dartigues JF, Scarmeas N, Barberger-Gateau P. Adherence to a Mediterranean diet, cognitive decline, and risk of dementia. **JAMA** 302: 638–648, 2009. doi:[10.1001/jama.2009.1146](https://doi.org/10.1001/jama.2009.1146).
690. Martinez-Lapiscina EH, Clavero P, Toledo E, Estruch R, Salas-Salvado J, San Julián B, Sanchez-Tainta A, Ros E, Valls-Pedret C, Martinez-Gonzalez MA. Mediterranean diet improves cognition: the PREDIMED-NAVARRA randomised trial. **J Neurol Neurosurg Psychiatry** 84: 1318–1325, 2013. doi:[10.1136/jnnp-2012-304792](https://doi.org/10.1136/jnnp-2012-304792).
691. Valls-Pedret C, Sala-Vila A, Serra-Mir M, Corella D, de la Torre R, Martínez-González MA, Martínez-Lapiscina EH, Fito M, Perez-Heras A, Salas-Salvado J, Estruch R, Ros E. Mediterranean diet and age-related cognitive decline: a randomized clinical trial. **JAMA Intern Med** 175: 1094–1103, 2015. doi:[10.1001/jamainternmed.2015.1668](https://doi.org/10.1001/jamainternmed.2015.1668).
692. Knight A, Bryan J, Wilson C, Hodgson JM, Davis CR, Murphy KJ. The Mediterranean diet and cognitive function among healthy older adults in a 6-month randomised controlled trial: the MedLey Study. **Nutrients** 8: 579, 2016. doi:[10.3390/nu8090579](https://doi.org/10.3390/nu8090579).
693. Marseglia A, Xu W, Fratiglioni L, Fabbri C, Berendsen AA, Bialecka-Debek A, Jennings A, Gillings R, Meunier N, Caumon E, Fairweather-Tait S, Pietruszka B, De Groot L, Santoro A, Franceschi C. Effect of the NU-AGE diet on cognitive functioning in older adults: a randomized controlled trial. **Front Physiol** 9: 349, 2018. doi:[10.3389/fphys.2018.00349](https://doi.org/10.3389/fphys.2018.00349).
694. Abbatecola AM, Russo M, Barbieri M. Dietary patterns and cognition in older persons. **Curr Opin Clin Nutr Metab Care** 21: 10–13, 2018. doi:[10.1097/MCO.0000000000000434](https://doi.org/10.1097/MCO.0000000000000434).
695. Tangney CC, Li H, Wang YM, Barnes L, Schneider JA, Bennett DA, Morris MC. Relation of DASH- and Mediterranean-like dietary patterns to cognitive decline in older persons. **Neurology** 83: 1410–1416, 2014. doi:[10.1212/WNL.0000000000000884](https://doi.org/10.1212/WNL.0000000000000884).
696. Morris MC, Tangney CC, Wang YM, Sacks FM, Bennett DA, Aggarwal NT. MIND diet associated with reduced incidence of Alzheimer's disease. **Alzheimers Dement** 11: 1007–1014, 2015. doi:[10.1016/j.jalz.2014.11.009](https://doi.org/10.1016/j.jalz.2014.11.009).
697. Blumenthal JA, Babyak MA, Hinderliter A, Watkins LL, Craighead L, Lin PH, Caccia C, Johnson J, Waugh R, Sherwood A. Effects of the DASH diet alone and in combination with exercise and weight loss on blood pressure and cardiovascular biomarkers in men and women with high blood pressure: the ENCORE study. **Arch Intern Med** 170: 126–135, 2010. doi:[10.1001/archinternmed.2009.470](https://doi.org/10.1001/archinternmed.2009.470).
698. Smith PJ, Blumenthal JA, Babyak MA, Craighead L, Welsh-Bohmer KA, Browndyke JN, Strauman TA, Sherwood A. Effects of the dietary approaches to stop hypertension diet, exercise, and caloric restriction on neurocognition in overweight adults with high blood pressure. **Hypertension** 55: 1331–1338, 2010. doi:[10.1161/HYPERTENSIONAHA.109.146795](https://doi.org/10.1161/HYPERTENSIONAHA.109.146795).
699. Hinderliter AL, Sherwood A, Craighead LW, Lin PH, Watkins L, Babyak MA, Blumenthal JA. The long-term effects of lifestyle change on blood pressure: One-year follow-up of the ENCORE study. **Am J Hypertens** 27: 734–741, 2014. doi:[10.1093/ajh/hpt183](https://doi.org/10.1093/ajh/hpt183).

700. Longo VD, Mattson MP. Fasting: molecular mechanisms and clinical applications. **Cell Metab** 19: 181–192, 2014. doi:[10.1016/j.cmet.2013.12.008](https://doi.org/10.1016/j.cmet.2013.12.008).
701. Birringer M. Hormetics: dietary triggers of an adaptive stress response. **Pharm Res** 28: 2680–2694, 2011. doi:[10.1007/s11095-011-0551-1](https://doi.org/10.1007/s11095-011-0551-1).
702. Van Cauwenberghe C, Vandendriessche C, Libert C, Vandenbroucke RE. Caloric restriction: beneficial effects on brain aging and Alzheimer's disease. **Mamm Genome** 27: 300–319, 2016. doi:[10.1007/s00335-016-9647-6](https://doi.org/10.1007/s00335-016-9647-6).
703. Golbidi S, Daiber A, Korac B, Li H, Essop MF, Laher I. Health benefits of fasting and caloric restriction. **Curr Diab Rep** 17: 123, 2017. doi:[10.1007/s11892-017-0951-7](https://doi.org/10.1007/s11892-017-0951-7).
704. Brandhorst S, Choi IY, Wei M, Cheng CW, Sedrakyan S, Navarrete G, Dubeau L, Yap LP, Park R, Vinciguerra M, Di Biase S, Mirzaei H, Mirisola MG, Childress P, Ji LY, Groshen S, Penna F, Odetti P, Perin L, Conti PS, Ikeno Y, Kennedy BK, Cohen P, Morgan TE, Dorff TB, Longo VD. A periodic diet that mimics fasting promotes multi-system regeneration, enhanced cognitive performance, and health-span. **Cell Metab** 22: 86–99, 2015. doi:[10.1016/j.cmet.2015.05.012](https://doi.org/10.1016/j.cmet.2015.05.012).
705. Mattson MP, Moehl K, Ghena N, Schmaedick M, Cheng AW. Intermittent metabolic switching, neuroplasticity and brain health. **Nat Rev Neurosci** 19: 63–80, 2018. doi:[10.1038/nrn.2017.156](https://doi.org/10.1038/nrn.2017.156).
706. Singh R, Manchanda S, Kaur T, Kumar S, Lakhanpal D, Lakhman SS, Kaur G. Middle age onset short-term intermittent fasting dietary restriction prevents brain function impairments in male Wistar rats. **Biogerontology** 16: 775–788, 2015. doi:[10.1007/s10522-015-9603-y](https://doi.org/10.1007/s10522-015-9603-y).
707. Fontán-Lozano A, Sáez-Cassanelli JL, Inda MC, Santos-Arteaga MD, Sierra-Domínguez SA, Lopez-Lluch G, Delgado-García JM, Carrion AM. Caloric restriction increases learning consolidation and facilitates synaptic plasticity through mechanisms dependent on NR2B subunits of the NMDA receptor. **J Neurosci** 27: 10185–10195, 2007. doi:[10.1523/JNEUROSCI.2757-07.2007](https://doi.org/10.1523/JNEUROSCI.2757-07.2007).
708. Li L, Wang Z, Zuo Z. Chronic intermittent fasting improves cognitive functions and brain structures in mice. **PLoS One** 8: e66069, 2013. doi:[10.1371/journal.pone.0066069](https://doi.org/10.1371/journal.pone.0066069).
709. Halagappa VK, Guo Z, Pearson M, Matsuoaka Y, Cutler RG, Laferla FM, Mattson MP. Intermittent fasting and caloric restriction ameliorate age-related behavioral deficits in the triple-transgenic mouse model of Alzheimer's disease. **Neurobiol Dis** 26: 212–220, 2007. doi:[10.1016/j.nbd.2006.12.019](https://doi.org/10.1016/j.nbd.2006.12.019).
710. Patel NV, Gordon MN, Connor KE, Good RA, Engelman RW, Mason J, Morgan DG, Morgan TE, Finch CE. Caloric restriction attenuates Abeta-deposition in Alzheimer transgenic models. **Neurobiol Aging** 26: 995–1000, 2005. doi:[10.1016/j.neurobiolaging.2004.09.014](https://doi.org/10.1016/j.neurobiolaging.2004.09.014).
711. Müller L, Power Guerra N, Stenzel J, Rühlmann C, Lindner T, Krause BJ, Vollmar B, Teipel S, Kuhla A. Long-Term Caloric Restriction Attenuates beta-Amyloid Neuropathology and Is Accompanied by Autophagy in APPswe/PS1delta9 Mice. **Nutrients** 13: 985, 2021. doi:[10.3390/nu13030985](https://doi.org/10.3390/nu13030985).
712. Wang J, Ho L, Qin W, Rocher AB, Seror I, Humala N, Maniar K, Dolios G, Wang R, Hof PR, Pasinetti GM. Caloric restriction attenuates beta-amyloid neuropathology in a mouse model of Alzheimer's disease. **FASEB J** 19: 659–661, 2005. doi:[10.1096/fj.04-3182je](https://doi.org/10.1096/fj.04-3182je).
713. Harvie MN, Pegington M, Mattson MP, Frystyk J, Dillon B, Evans G, Cuzick J, Jebb SA, Martin B, Cutler RG, Son TG, Maudsley S, Carlson OD, Egan JM, Flyvbjerg A, Howell A. The effects of intermittent or continuous energy restriction on weight loss and metabolic disease risk markers: a randomized trial in young overweight women. **Int J Obes (Lond)** 35: 714–727, 2011. doi:[10.1038/ijo.2010.171](https://doi.org/10.1038/ijo.2010.171).
714. Trepanowski JF, Kroeger CM, Barnosky A, Klempel MC, Bhutani S, Hodge KK, Gabel K, Freels S, Rigdon J, Rood J, Ravussin E, Varady KA. Effect of alternate-day fasting on weight loss, weight maintenance, and cardioprotection among metabolically healthy obese adults: a randomized clinical trial. **JAMA Intern Med** 177: 930–938, 2017. doi:[10.1001/jamainternmed.2017.0936](https://doi.org/10.1001/jamainternmed.2017.0936).
715. Witte AV, Fobker M, Gellner R, Knecht S, Flöel A. Caloric restriction improves memory in elderly humans. **Proc Natl Acad Sci USA** 106: 1255–1260, 2009. doi:[10.1073/pnas.0808587106](https://doi.org/10.1073/pnas.0808587106).
716. Bayer-Carter JL, Green PS, Montine TJ, VanFossen B, Baker LD, Watson GS, Bonner LM, Callaghan M, Leverenz JB, Walter BK, Tsai E, Plymate SR, Postupna N, Wilkinson CW, Zhang J, Lampe J, Kahn SE, Craft S. Diet intervention and cerebrospinal fluid biomarkers in amnesic mild cognitive impairment. **Arch Neurol** 68: 743–752, 2011. doi:[10.1001/archneurol.2011.125](https://doi.org/10.1001/archneurol.2011.125).
717. Mahalakshmi B, Maurya N, Lee SD, Kumar VB. Possible neuroprotective mechanisms of physical exercise in neurodegeneration. **Int J Mol Sci** 21: 5895, 2020. doi:[10.3390/ijms21165895](https://doi.org/10.3390/ijms21165895).
718. Erickson KI, Gildengers AG, Butters MA. Physical activity and brain plasticity in late adulthood. **Dialogues Clin Neurosci** 15: 99–108, 2013. doi:[10.31887/DCNS.2013.15.1/kerickson](https://doi.org/10.31887/DCNS.2013.15.1/kerickson).
719. Ionescu-Tucker A, Cotman CW. Emerging roles of oxidative stress in brain aging and Alzheimer's disease. **Neurobiol Aging** 107: 86–95, 2021. doi:[10.1016/j.neurobiolaging.2021.07.014](https://doi.org/10.1016/j.neurobiolaging.2021.07.014).
720. Marosi K, Bori Z, Hart N, Sárga L, Koltai E, Radák Z, Nyakas C. Long-term exercise treatment reduces oxidative stress in the hippocampus of aging rats. **Neuroscience** 226: 21–28, 2012. doi:[10.1016/j.neuroscience.2012.09.001](https://doi.org/10.1016/j.neuroscience.2012.09.001).
721. Moreno-Villanueva M, Kramer A, Hammes T, Venegas-Carro M, Thumm P, Bürkle A, Gruber M. Influence of acute exercise on DNA repair and PARP activity before and after irradiation in lymphocytes from trained and untrained individuals. **Int J Mol Sci** 20: 2999, 2019. doi:[10.3390/ijms20122999](https://doi.org/10.3390/ijms20122999).
722. García-Mesa Y, Colie S, Corpas R, Cristófol R, Comellas F, Nebreda AR, Giménez-Llort L, Sanfeliu C. Oxidative stress is a central target for physical exercise neuroprotection against pathological brain aging. **J Gerontol A Biol Sci Med Sci** 71: 40–49, 2016. doi:[10.1093/gerona/glv005](https://doi.org/10.1093/gerona/glv005).
723. Lu Y, Dong Y, Tucker D, Wang R, Ahmed ME, Brann D, Zhang Q. Treadmill exercise exerts neuroprotection and regulates microglial polarization and oxidative stress in a streptozotocin-induced rat model of sporadic Alzheimer's disease. **J Alzheimers Dis** 56: 1469–1484, 2017. doi:[10.3233/JAD-160869](https://doi.org/10.3233/JAD-160869).
724. Melo KP, Silva CM, Almeida MF, Chaves RS, Marcourakis T, Cardoso SM, Demasi M, Netto LE, Ferrari MF. Mild exercise differently affects proteostasis and oxidative stress on motor areas during neurodegeneration: a comparative study of three treadmill running protocols. **Neurotox Res** 35: 410–420, 2019. doi:[10.1007/s12640-018-9966-3](https://doi.org/10.1007/s12640-018-9966-3).
725. Lin TW, Shih YH, Chen SJ, Lien CH, Chang CY, Huang TY, Chen SH, Jen CJ, Kuo YM. Running exercise delays neurodegeneration in amygdala and hippocampus of Alzheimer's disease (APP/PS1) transgenic mice. **Neurobiol Learn Mem** 118: 189–197, 2015. doi:[10.1016/j.nlm.2014.12.005](https://doi.org/10.1016/j.nlm.2014.12.005).
726. Li BX, Liang F, Ding XY, Yan QW, Zhao YC, Zhang XL, Bai YD, Huang T, Xu B. Interval and continuous exercise overcome memory deficits related to beta-amyloid accumulation through modulating

- mitochondrial dynamics. **Behav Brain Res** 376: 112171, 2019. doi:[10.1016/j.bbr.2019.112171](https://doi.org/10.1016/j.bbr.2019.112171).
727. Lee HJ, Don Kim K. Effect of physical activity on cognition and daily living activities of the elderly with mild dementia. **J Phys Ther Sci** 30: 428–433, 2018. doi:[10.1589/jpts.30.428](https://doi.org/10.1589/jpts.30.428).
728. Lavie CJ, Church TS, Milani RV, Earnest CP. Impact of physical activity, cardiorespiratory fitness, and exercise training on markers of inflammation. **J Cardiopulm Rehabil Prev** 31: 137–145, 2011. doi:[10.1097/HCR.0b013e3182122827](https://doi.org/10.1097/HCR.0b013e3182122827).
729. Alomari MA, Khabour OF, Alzoubi KH, Alzubi MA. Forced and voluntary exercises equally improve spatial learning and memory and hippocampal BDNF levels. **Behav Brain Res** 247: 34–39, 2013. doi:[10.1016/j.bbr.2013.03.007](https://doi.org/10.1016/j.bbr.2013.03.007).
730. Yerokhin V, Anderson-Hanley C, Hogan MJ, Dunnam M, Huber D, Osborne S, Shulan M. Neuropsychological and neurophysiological effects of strengthening exercise for early dementia: a pilot study. **Neuropsychol Dev Cogn B Aging Neuropsychol Cogn** 19: 380–401, 2012. doi:[10.1080/13825585.2011.628378](https://doi.org/10.1080/13825585.2011.628378).
731. Aguiar P, Monteiro L, Feres A, Gomes I, Melo A. Rivastigmine transdermal patch and physical exercises for Alzheimer's disease: a randomized clinical trial. **Curr Alzheimer Res** 11: 532–537, 2014. doi:[10.2174/1567205011666140618102224](https://doi.org/10.2174/1567205011666140618102224).
732. Holthoff VA, Marschner K, Scharf M, Steding J, Meyer S, Koch R, Donix M. Effects of physical activity training in patients with Alzheimer's dementia: results of a pilot RCT study. **PLoS One** 10: e0121478, 2015. doi:[10.1371/journal.pone.0121478](https://doi.org/10.1371/journal.pone.0121478).
733. Frederiksen KS, Larsen CT, Hasselbalch SG, Christensen AN, Høgh P, Wermuth L, Andersen BB, Siebner HR, Garde E. A 16-Week aerobic exercise intervention does not affect hippocampal volume and cortical thickness in mild to moderate Alzheimer's disease. **Front Aging Neurosci** 10: 293, 2018. doi:[10.3389/fnagi.2018.00293](https://doi.org/10.3389/fnagi.2018.00293).
734. Morris JK, Vidoni ED, Johnson DK, Van Sciver A, Mahnken JD, Honea RA, Wilkins HM, Brooks WM, Billinger SA, Swerdlow RH, Burns JM. Aerobic exercise for Alzheimer's disease: a randomized controlled pilot trial. **PLoS One** 12: e0170547, 2017. doi:[10.1371/journal.pone.0170547](https://doi.org/10.1371/journal.pone.0170547).
735. Pedrinolla A, Venturelli M, Fonte C, Munari D, Benetti MV, Rudi D, Tamburin S, Muti E, Zanolla L, Smania N, Schena F. Exercise training on locomotion in patients with Alzheimer's disease: a feasibility study. **J Alzheimers Dis** 61: 1599–1609, 2018. doi:[10.3233/JAD-170625](https://doi.org/10.3233/JAD-170625).
736. Gaitán JM, Boots EA, Dougherty RJ, Oh JM, Ma Y, Edwards DF, Christian BT, Cook DB, Okonkwo OC. brain glucose metabolism, cognition, and cardiorespiratory fitness following exercise training in adults at risk for Alzheimer's disease. **Brain Plast** 5: 83–95, 2019. doi:[10.3233/BPL-190093](https://doi.org/10.3233/BPL-190093).
737. Gaitán JM, Moon HY, Stremlau M, Dubal DB, Cook DB, Okonkwo OC, van Praag H. Effects of aerobic exercise training on systemic biomarkers and cognition in late middle-aged adults at risk for Alzheimer's disease. **Front Endocrinol (Lausanne)** 12: 660181, 2021. doi:[10.3389/fendo.2021.660181](https://doi.org/10.3389/fendo.2021.660181).
738. Castellano CA, Paquet N, Dionne IJ, Imbeault H, Langlois F, Croteau E, Tremblay S, Fortier M, Matte JJ, Lacombe G, Fülöp T, Bocti C, Cunnane SC. A 3-month aerobic training program improves brain energy metabolism in mild Alzheimer's disease: preliminary results from a neuroimaging study. **J Alzheimers Dis** 56: 1459–1468, 2017. doi:[10.3233/JAD-161163](https://doi.org/10.3233/JAD-161163).
739. Maltais M, Rolland Y, Vellas B, Hai PE, Armaingaud D, Cestac P, Rouch L, Cesari M, de Souto Barreto P. Effect of exercise on behavioral symptoms and pain in patients with dementia living in nursing homes. **Am J Alzheimers Dis Other Dement** 34: 89–94, 2019. doi:[10.1177/1533317518803773](https://doi.org/10.1177/1533317518803773).
740. Enette L, Vogel T, Merle S, Valard-Guiguet AG, Ozier-Lafontaine N, Neviere R, Leully-Joncart C, Fanon JL, Lang PO. Effect of 9 weeks continuous vs. interval aerobic training on plasma BDNF levels, aerobic fitness, cognitive capacity and quality of life among seniors with mild to moderate Alzheimer's disease: a randomized controlled trial. **Eur Rev Aging Phys Act** 17: 2, 2020. doi:[10.1186/s11556-019-0234-1](https://doi.org/10.1186/s11556-019-0234-1).
741. Kaufman CS, Honea RA, Pleen J, Lepping RJ, Watts A, Morris JK, Billinger SA, Burns JM, Vidoni ED. Aerobic exercise improves hippocampal blood flow for hypertensive apolipoprotein E4 carriers. **J Cereb Blood Flow Metab** 41: 2026–2037, 2021. doi:[10.1177/0271678X21990342](https://doi.org/10.1177/0271678X21990342).
742. Butterfield DA. Oxidative stress in brain in amnesic mild cognitive impairment. **Antioxidants (Basel)** 12: 462, 2023. doi:[10.3390/antiox12020462](https://doi.org/10.3390/antiox12020462).
743. Kovacs GG, Budka H. Current concepts of neuropathological diagnostics in practice: neurodegenerative diseases. **Clin Neuropathol** 29: 271–288, 2010. doi:[10.5414/npp29271](https://doi.org/10.5414/npp29271).
744. Butterfield DA, Favia M, Spera I, Campanella A, Lanza M, Castegna A. Metabolic features of brain function with relevance to clinical features of Alzheimer and Parkinson diseases. **Molecules** 27: 951, 2022. doi:[10.3390/molecules27030951](https://doi.org/10.3390/molecules27030951).
745. Rummel NG, Butterfield DA. Altered metabolism in Alzheimer disease brain: role of oxidative stress. **Antioxid Redox Signal** 36: 1289–1305, 2022. doi:[10.1089/ars.2021.0177](https://doi.org/10.1089/ars.2021.0177).
746. Johnson EC, Carter EK, Dammer EB, Duong DM, Gerasimov ES, Liu Y, Liu J, Betarbet R, Ping L, Yin L, Serrano GE, Beach TG, Peng J, De Jager PL, Haroutunian V, Zhang B, Gaiteri C, Bennett DA, Gearing M, Wingo TS, Wingo AP, Lah JJ, Levey AI, Seyfried NT. Large-scale deep multi-layer analysis of Alzheimer's disease brain reveals strong proteomic disease-related changes not observed at the RNA level. **Nat Neurosci** 25: 213–225, 2022. doi:[10.1038/s41593-021-00999-y](https://doi.org/10.1038/s41593-021-00999-y).
747. Deribe YL, Pawson T, Dikic I. Post-translational modifications in signal integration. **Nat Struct Mol Biol** 17: 666–672, 2010. doi:[10.1038/nsmb.1842](https://doi.org/10.1038/nsmb.1842).
748. Zhou R, Xu B. Insight of the cytotoxicity of the aggregates of peptides or aberrant proteins: a meta-analysis. **PLoS One** 9: e95759, 2014. doi:[10.1371/journal.pone.0095759](https://doi.org/10.1371/journal.pone.0095759).
749. Perluigi M, Barone E. Aberrant protein networks in Alzheimer disease. **Nat Rev Neurol** 18: 255–256, 2022. doi:[10.1038/s41582-022-00648-3](https://doi.org/10.1038/s41582-022-00648-3).
750. Butterfield DA, Boyd-Kimball D, Reed TT. Cellular stress response (hormesis) in response to bioactive nutraceuticals with relevance to Alzheimer disease. **Antioxid Redox Signal** 38: 643–669, 2023. doi:[10.1089/ars.2022.0214](https://doi.org/10.1089/ars.2022.0214).
751. Shea D, Colasurdo E, Smith A, Paschall C, Jayadev S, Keene CD, Galasko D, Ko A, Li G, Peskind E, Daggett V. SOBA: development and testing of a soluble oligomer binding assay for detection of amyloidogenic toxic oligomers. **Proc Natl Acad Sci USA** 119: e2213157119, 2022. doi:[10.1073/pnas.2213157119](https://doi.org/10.1073/pnas.2213157119).